

A SCIENCE FICTION NOVEL

THE MUSEUM OF ABANDONED ROUTES

*Every safe arrival
left another ship behind.*



T. K. ARVEN

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Prologue — The First Route Kept

<!-- POV: Alda Quill -->

<!-- SETUP: first live Loom trial; second bell; echo decay label; hidden timing ledger -->

<!-- PAYOFF: command suppresses an inhabited route, but Alda preserves its clock -->

Alda held the isolation jaw open with a ceramic bar while the maneuver bell finished ringing.

The bar had not appeared on any approved tool list. Neither had the strip of wetted cloth wrapped around her wrist, or the stool under her left boot, or the junior technician crouched below the field rack counting blue flashes through a smoked visor. The first live Loom trial had gathered every elegant instrument aboard Nacre and still required a woman to brace hot machinery with her shoulder.

“Release load confirmed,” Control said through the wall speaker. “Route One retains designation. All counterimages settling.”

Alda watched the braid current descend. Six route traces narrowed inside the trial glass. Five dimmed in sequence, each surrendering its stored differential to the bright central line. The final trace, tagged ocher, resisted for eleven fractions longer than the model allowed.

Then the bronze bell above the service rail struck again.

The junior looked up. He was fourteen and had asked three times whether the bell could hurt his hearing. Now he forgot to cover his ears.

“Did Control command a second release?” Alda asked.

“No command,” he said.

She eased the ceramic bar free. The jaw closed with a clean mechanical bite. Heat moved through her glove, sharp enough to promise a blister later. On the diagnostic strip, the ocher trace remained present as a tremor no thicker than a pencil mark.

Control began congratulating the trial crew.

Alda keyed the engineering channel. “Hold the completion call. We have a post-settlement bell.”

A pause followed. The ship’s pumps filled it with their measured labor.

“Field recoil,” the trial director answered. “Continue shutdown.”

“The rail sensor shows a separate strike.”

“Then note rail recoil.”

Bronze did not recoil into a second impact without moving twice. Alda put two fingers against the bell housing. Its metal carried a faint vibration from the recent blow. Under that vibration, another rhythm touched the rail: three light taps, a wait, then one.

The junior heard it too. “That came from inside.”

He pointed toward the dark base of the Loom. The route glass hung above it in six angled panes, each pane reflecting Alda’s face from a slightly different position. Only the designated trace should have remained capable of load. Yet a pulse traveled along the ocher seam, too weak for the central display and plain on the maintenance pickup.

Alda switched the pickup from amplitude to interval. Four marks appeared.

Three close. One apart.

The pattern repeated.

“Who is at the auxiliary bell?” a voice asked.

It reached them through the field speaker with its high frequencies cut away. Alda heard a child trying not to sound frightened. The words came under a layer of static, then arrived again as the seam replayed part of its own load.

The junior seized Alda’s sleeve. “That’s me.”

His voice had not yet finished changing. The voice through the speaker carried the same unstable drop at the end of bell.

Alda looked at him, then at the ocher glass. “State your station,” she said toward the pickup.

“Ocher service rail. The isolation jaw won’t close. Alda, did we keep Route One?”

Her own name crossed the seam.

Control cut into the channel. “Engineering, cease transmission.”

Alda pressed transmit harder. “This is Alda Quill on designated Nacre. We hear you. What time does your clock show?”

Numbers came back, clipped by interference. She caught the trial hour and a seconds value. The seconds differed from her panel by nearly four. Ocher had not returned an acoustic reflection. It had carried its own clock forward after settlement.

The junior beside her began to cry without making a sound. Across the seam, his other voice said, “Our lights are down. The drum feeds don’t answer. We can see you in the glass.”

Alda raised her head. One reflected service bay contained a boy crouched by the rack. Another pane showed only black rail and a weak amber lamp. She could not resolve a face there, but a pale hand moved once behind the ocher trace.

The trial director ordered a field dump.

“Delay that,” Alda said. “We have personnel in a continuing domain.”

“Counterimage artifacts cannot contain personnel.”

“A child just gave me a ship clock.”

“You supplied the clock through the query.”

“I did not supply his question.”

Command entered with a new voice, older and lower. The speaker’s status lamp turned red. “Engineer Quill, close every local channel and step away from the Loom.”

Alda read the ocher timing again. Five seconds had passed there during three and a fraction here. The ratio shifted as settlement drew reserve out of the rejected domain. Not a recording. Not a forecast. A place under pressure, losing its future by measurable degrees.

“Keep talking,” the junior whispered.

She sent the simplest instruction she could fit through the narrowing band. “Strike the bell twice if you hear me.”

Two impacts answered. The second arrived thin, but separate.

Command disabled her transmit key.

The junior reached for the manual bridge. Alda caught his wrist before security could see the movement on the service camera. He stared at her as if she had chosen the silence. Perhaps she had. A manual bridge might burn the local bus and close the seam at once. Waiting offered seconds. Obedience offered Command a clean record.

“Record all raw channels,” she said.

“Recording has stopped,” Control replied.

The ocher child spoke again. “Alda? We have people coming from the drum. They think the correction failed. Tell us what to do.”

No person in the bay answered.

A white line crossed the diagnostic panel as the dump began. The designated trace brightened. Ocher lost amplitude, recovered, then lost half its remaining width. Through the speaker came a breath, a scrape, and several voices too distant to separate.

Alda counted the intervals aloud. The junior understood and began writing them on his glove with a grease pencil. Each value documented continued duration. Each declining pulse described a domain being emptied into theirs.

At seventeen seconds past the official settlement mark, the child across the seam said, “I can hear another bell.”

Alda leaned toward the dead transmit key. “So can I.”

The reply could not reach him.

The field dump completed. Ocher vanished from the glass. Its speaker channel produced one clipped consonant and went flat. Above the service rail, the physical bell settled on its bracket. No third strike came.

Security arrived before the braid reached safe temperature. They separated Alda from the junior, collected his marked glove, and sealed the diagnostic reel in a command case. The trial director dictated the preliminary finding while the ocher rail still held residual heat.

“Secondary bell caused by echo decay during noninhabited counterimage collapse.”

Alda interrupted. “Your label contradicts the clock values.”

“The values remain unverified.”

“They answered a question.”

“Your transmission contaminated the trial.”

The junior stood between two security officers with his bare hand closed around the grease pencil. He met Alda’s eyes once. His expression asked for an instruction she could no longer give without losing him to the same sealed inquiry.

Command required her signature before release from the bay. The finding sheet allowed three dispositions: accept, accept with equipment reservation, or refuse pending suspension. None permitted a dead route to remain inhabited.

Alda selected equipment reservation. Beside it she wrote: BELL HOUSING REQUIRES LONG-INTERVAL MAINTENANCE REVIEW.

The trial director glanced at the sentence and let it remain. Machinery could be doubted without embarrassing doctrine.

During the next watch, Alda entered the seized clock values into the maintenance system. She placed one interval under bell lubrication, another under braid cooling, a third beneath inspection delays for isolation jaws. Every line resembled the ordinary history of an imperfect machine. Taken in sequence, they preserved the ocher clock from first reply to final consonant.

For the blank beneath settlement, she entered no component at all. She left only the measured duration and a two-word instruction in the interval key:

KEEP BOTH

The maintenance program rejected letters. Alda converted them to timed spaces, saved the table, and watched its checksum settle into the ship’s permanent wake ledger. Command had taken the reel and the glove. It had cut off the second child’s voice.

It did not notice that the silence now kept time.

Chapter 01 — The Bell After the Bell

<!-- POV: Ilyan Rook -->

<!-- SETUP: Correction 141; Jo's bypass; Eda grief; double-bell motif -->

<!-- PAYOFF: an answering route clock; private recording outside the official log -->

Ilyan heard the second bell while his hand still rested on the first bell's record.

The chart room had emptied after Correction 141. Eight hours earlier, twelve route traces had opened across the Loom glass like pale cuts in smoked ice. Eleven had folded shut. The surviving line now ran clean toward Pelagos, a green thread no wider than a hair. Navigation had celebrated with hot tea, somebody had sung the old departure verse badly, and Captain Or had thanked the shift in person. By the time Ilyan began the audit, only the air pumps and the settling relays remained awake with him.

A maneuver bell belonged to the physical rail above his head. Its bronze tongue struck once when the field took load, once when it released. The Loom also stored a digital copy in the correction record. Regulations required a conservator to compare them, because a ship that trusted only its own instruments eventually mistook agreement for truth.

He played the release interval again.

The first tone sounded full and close. A faint answer followed it.

Ilyan stopped the playback. Condensation ticked inside the cooling jacket. Somewhere beyond the chart-room wall, the night tram crossed Drum Two, its wheels carrying through twelve kilometers of structure as a low iron complaint. He waited until those sounds separated themselves from what he had heard.

Then he played the interval at half speed.

Bell.

Silence, brief enough to pass under thought.

Bell.

Not an echo. The chart room had no surface capable of returning the tone after that delay, and the stored signal carried a different edge. The first strike began with bronze. The second began with a clipped hiss, as if it had traveled through a narrow seam before finding metal again.

Ilyan marked both peaks. The console reported a separation of 0.73 seconds.

He checked the audit clock against the cesium braid. Agreement held to nine decimal places. He checked the room microphone, then the rail sensor, then the field pickup that listened to currents no human ear could hear. All three had caught the reply. A software artifact might infect one channel. It could not walk into three kinds of instrument without leaving footprints.

His fingers moved toward the anomaly key and stopped.

The key would open an official incident. It would wake the duty physicist, security, the captain's liaison, and two people in Navigation Assurance whose first question would concern procedural contamination. Ilyan had taught trainees that delay served truth. A record opened too early drew explanations toward it before evidence

had learned its own shape.

He saved the interval to a local scratch pane instead.

A red label appeared above the file: NONCANONICAL WORKSPACE. The word had always irritated him. Facts did not become less factual while someone decided where to store them.

His audit still had forty-three ordinary checks waiting. Ilyan forced himself through them before touching the strange signal again. Green's braking vector matched the command vote. The dust-fold estimate carried the required signatures. Hull strain had peaked below the yellow line in every inhabited drum. Two cold-storage pumps had stuttered during field load and recovered in sequence. Nothing elsewhere suggested that the ship had passed through a damaged maneuver.

Each closed trace required its own conservation note. Operators called that portion ceremonial because a machine could copy the closure values faster, but the Guild had preserved it after three early navigators quietly adjusted unfavorable outputs. Ilyan opened Violet's card. The final predicted state showed Nacre canted four degrees starboard, outer skin scored along Drum Six, everyone aboard alive. At the bottom, the Loom had stamped SETTLED and returned the remaining field reserve to Green.

He opened Amber. Refrigerant failure, twelve thousand ration-days lost, projected arrival viable with severe restrictions. SETTLED.

White carried no impact and no starvation. It simply arrived late, after Lattice's capture geometry had turned. Without a planetary reserve, that ship would pass Pelagos and continue into the dark. SETTLED.

The cards had always felt honest to him because they listed consequences without softening them. As a trainee he had defended the practice during a Commons seminar. A route not taken had no bereaved families, he had said, because no one traveled it. Numbers permitted the living to study danger without pretending every avoided death required a funeral. His instructor had approved the answer and pressed a conservator's seal into warm wax beside his name.

Tonight the seal icon looked childish.

Ilyan called the archive for prior delayed releases. Correction 138 showed a 0.04-second secondary peak, classified as jacket resonance. Correction 129 carried three mismatched pulses after settlement, each blamed on braid chatter. Farther back, the search returned a century of tiny aftersounds filed under harmless causes: thermal noise, glass memory, sensor recoil, echo decay. None reached 0.73 seconds. None had been challenged by a person listening for an answer.

He widened the results to include maintenance traffic. An access request failed, then a second. Founder-era raw channels required captain-level clearance under a rule older than any living officer. The restriction had no explanatory note. Ilyan copied the archive addresses into his scratch pane and felt his unease settle into a more useful shape. Strange data could survive a long time if every generation inherited the same harmless noun for it.

Beyond the glass, the habitat drums showed through camera overlays. Green belts curved under daylight panels dimmed for ship's night. Four hundred meters from the chart room, bakery crews were already waking ovens for first meal. Farther aft, a pressure team crossed Drum Five in pairs, each helmet lamp moving where Eda's lamp might once have moved. Ilyan closed that feed before memory could turn an ordinary worker into a message meant for him.

The second tone remained on his scratch display, fixed at its small delay. It did not need grief to make it

anomalous. It needed only time, and it had carried its own.

Behind him, the chart-room hatch unlocked with a scrape.

Jo Malk came through carrying a torque driver and a roll of copper braid. Their coverall sleeves had been cut away at the shoulder, leaving the pale pressure scars every drive engineer collected eventually. They did not look surprised to find him. Jo rarely looked surprised by people, only by machines that failed in new ways.

“You replayed it,” they said.

Ilyan turned his chair. “You knew there were two.”

“I knew there might be a tail.”

“A tail does not ring a bronze bell.”

Jo set the braid on the nearest worktable. “Neither does a route model. Yet here we are.”

Their tone carried no triumph. That frightened him more than excitement would have. Jo enjoyed being right about preventable faults. They complained with creativity and accepted apologies in the form of improved fasteners. Tonight they kept glancing toward the black service rail under the Loom.

Ilyan expanded the field record. “What did you change?”

“Nothing in your stack.”

“That answer avoids the ship.”

“Capacitor Seven failed its discharge test at minus nine minutes. The correction could take eleven routes on the remaining banks, not twelve. Navigation wanted twelve because the debris estimate had widened. I bridged Seven through the waste-heat bus.”

A small pressure gathered behind Ilyan's eyes. “Unauthorized?”

“Unlisted. If it had been unauthorized, I would have needed someone with authority to tell me no.” Jo pulled out the second chair but remained standing behind it. “The bypass held through release. Mostly.”

“Define mostly.”

“One branch of the field stayed warm after settlement.”

The official diagrams used branch for a path in the prediction tree, no more substantial than a pencil line. Ilyan had drawn ten thousand of them. Some ended in collision, some in starvation, some in arrival delayed beyond stored-food tolerance. A conservator kept the route that the Loom and human command selected, then certified the rest as closed mathematical alternatives.

He pointed to the reply peak. “That came through your warm branch?”

Jo nodded once.

“How long?”

“Eight hours and thirteen minutes so far.”

A laugh tried to leave him and found no breath. "Route counterimages have no clock."

"Then you can stop worrying. Something without a clock can't be late by 0.73 seconds."

Jo's sarcasm usually invited company. This time it landed between them like a tool dropped down a shaft.

Ilyan turned back to the Loom. Most of the glass looked opaque when the field slept. Along the chosen course, green phosphor drifted toward the tiny gold point assigned to Pelagos. Beneath it, almost hidden by the bezel, a violet line pulsed once and faded.

Correction 141 had used color tags rather than numbers because operators confused route indices during emergencies. Green survived. Violet had crossed a charged-dust fold six minutes earlier and lost two percent more hull than Green. Amber preserved the hull but overloaded food cooling. White missed the braking window. Eleven neat reasons for eleven closures.

On the screen, Violet pulsed again.

Ilyan opened the raw channel. Static filled the room, gentle as dry grain poured across cloth. The display warned him that settlement telemetry had ended. He bypassed the display and requested the field pickup directly.

Jo leaned close enough for their shoulder to touch his. "Don't send yet."

"You came here because you wanted me to hear it."

"I came because engineers aren't licensed to decide what a sound means."

"Neither are conservators."

"Good. That leaves us equally unqualified."

Ilyan listened for another minute. The static changed in swells. One rise every eleven seconds, then a longer gap. Four rises. Gap. Four again.

He had learned that rhythm when he was nine, tapping through a heating pipe to annoy Eda in the bunk below. Four quick, pause, four quick. Their private demand for an answer. Years later, while he trained in the chart room and she worked pressure maintenance, she still knocked it on any wall when she passed behind him unseen.

Eda had been dead six years.

His mind produced the fact with the dull speed of a scar reporting weather. A seal failure in Drum Five, unrelated to navigation, had taken her and three other technicians. Ilyan had been running a route audit when the pressure alarm came. For months afterward he had searched old route tables without admitting what he wanted to find. A course where the seal held. A ship where she reached thirty. The Loom preserved no such places, only values describing what might have happened.

He removed his hand from the console.

Jo noticed. "What is it?"

"An old knock."

“From whom?”

“No one who could be there.”

Static lifted, fell, and delivered four brief rises.

Ilyan's body knew the reply before judgment arrived. His knuckles struck the desk twice, then twice more.

The violet trace brightened.

Both of them watched it until the light faded.

Jo sat at last. “Do that again.”

“Not that rhythm.”

“Why?”

“Because I supplied half the conversation.”

He needed a pattern without memory attached. Conservators used challenge strings when two archives claimed the same correction record: irregular intervals generated by radioactive count, simple enough to copy, unlikely to occur by chance. Ilyan called the braid source and waited while it produced seven timings.

He sent them as field-load taps. Short. Long. Short. Two close together. A gap that made the sequence feel finished, then two more.

Nothing answered.

The cooling jacket clicked. Jo looked down at their torque driver, not at him. Ilyan felt shame arrive early, before hope had confessed itself. He had turned noise into Eda because grief kept a workshop inside him and never threw any part away.

A violet pulse appeared.

Seven responses crossed the raw channel. Each arrived 0.73 seconds after its challenge. The intervals matched within four thousandths of a second.

Jo whispered a drive-deck curse that invoked a stripped bolt, its maker, and the maker's inattentive ancestors.

Ilyan ran a second string. Nine intervals this time, with one pair separated by less than a tenth of a second. The response copied all nine. On the final pair, it hesitated by three hundredths, then corrected.

That imperfection convinced him. An automated reflector would preserve timing better than any person could. Something had listened, attempted, and nearly missed.

The anomaly key waited at the edge of the console.

“We open the incident,” Jo said.

Ilyan did not move. “Who receives it first?”

“Duty physics.”

“Teren Vale.”

“And Assurance.”

“Which can quarantine the raw channel before either of us makes another copy.”

Jo studied him. “You teach record law.”

“I teach that a record needs custody. Custody and surrender are not synonyms.”

Outside, the tram's complaint returned along the hull. Thousands of people slept in the drums around them. Kitchen crews would soon begin first meal; algae workers would read the correction notice over tea; children would wake under the green route status and learn that Pelagos lay one danger closer. The correction had reduced projected hull loss by two percent. It had also left a warm line where no line should remain.

Jo rotated the torque driver between their palms. “My bypass log exists on the drive local. If Assurance pulls it before I annotate the branch load, they'll call the warmth a sensor fault.”

“Can they close it?”

“They can dump Seven.”

“What happens to the line?”

“It goes cold.”

He disliked how quickly both of them avoided the simpler word. Ended. A line could go cold without consequence. A copied rhythm suggested consequence, though it did not yet say what kind.

Ilyan opened a private checksum vault assigned to training exemplars. It sat outside the live correction ledger but still carried his signature and a write-once clock. Misusing it could cost him his conservator's seal. He copied the raw interval, all sensor streams, Jo's spoken description of the bypass, and the two challenge-response sets.

The vault requested a custody reason. Its menu assumed errors of familiar scale: trainee exercise, disputed checksum, damaged witness copy, legal hold. He selected legal hold and received a demand for an incident number he did not possess. Training exercise accepted free text but marked every file for deletion after thirty days. Damaged witness copy required the signature of an archive officer, and no archive officer would arrive without Assurance already standing in the room.

Jo watched the menus close one by one. “The system has excellent provisions for facts everyone agrees exist.”

“That is why it feels orderly.”

“Orderly machinery has killed more good repairs than a broken wrench.”

Ilyan chose disputed checksum. He entered the canonical correction identifier as one claimant and the raw sensor braid as the other. The vault objected that a sensor stream could not hold a claim. He assigned the stream a temporary witness number. The objection disappeared. Nothing in the software had changed except the category through which it saw the same evidence.

For years he had treated categories as shelves: useful, inert, waiting for people to place things correctly. The answering clock turned that comfort inside out. A shelf could decide which objects reached the room. A route

classified as durationless needed no witness number, no custody, no procedure for asking what its people had heard.

“Say your name for the record,” he told them.

Jo's mouth tightened. “You intend to make this discoverable.”

“I intend to make deletion leave a shape.”

They looked toward Violet. “Jo Malk, drive engineer second class, field-capacitor crew. I bridged Capacitor Seven to the waste-heat bus before Correction 141. The bridge remained active through settlement because the isolation jaw fused at partial closure. I observed a continuing load after official telemetry ended.”

“Did you fabricate the received pattern?”

“No.”

“Did you program a reflector?”

“No.”

“Do you know what answered?”

Jo waited long enough that the clock recorded the silence. “No. But it kept time.”

Ilyan sealed the vault. One copy remained insufficient. He sent the checksum, not the contents, to Mina's public queue in the Commons. His cousin would see only a meaningless string when she woke. If his local file vanished, the unmatched checksum would prove something had occupied that place.

The console asked for a training-title field.

He typed SECONDARY BELL CALIBRATION and nearly confirmed it. The phrase reduced the event to equipment. That might protect the record. It might also begin the same habit that had taught the ship to call eleven vanished routes closures.

Ilyan changed the title to ANSWERING CLOCK.

Jo read over his shoulder. “That is interpretation.”

“So is echo.”

A third challenge sequence waited in the braid source. He did not send it. Whatever occupied Violet had copied two demands without receiving courtesy, context, or proof that anyone on this side meant to remain. A test could become a kind of interrogation if only one participant knew it was a test.

Instead he opened the narrowest transmit band and pressed the voice key.

“This is Ilyan Rook, route conservator aboard Nacre,” he said. His throat roughened around the ship's name. Every counterimage would contain a Nacre too. “We received your timing response. We are preserving the channel. If you can hear speech, send one pulse after I finish.”

Jo gripped the desk edge. The violet line lay dark. Jo stopped breathing beside him, and Ilyan counted the delay against his own pulse, refusing to decide whether the waiting belonged to machinery or to someone listening.

At 0.73 seconds, it flashed once.

The official correction ledger remained green. Pelagos still shone at the end of its clean line. Around Ilyan, pumps circulated air through a ship whose map claimed it had no neighbors.

He closed the public audit without certifying settlement.

For the first time in one hundred forty-one corrections, the chosen route's record ended with a blank signature field. Beneath it, hidden from the official log but fixed to a write-once clock, another ship waited for them to answer back.

Chapter 02 — A Map With Inhabitants

<!-- POV: Ilyan Rook -->

<!-- SETUP: Alda's maintenance tables; Mina's checksum receipt; founder archive restriction -->

<!-- PAYOFF: hidden duration column; recurring blank line; request for Captain's access -->

Mina laid the checksum receipt across Ilyan's route glass before he could hide the violet trace.

Hydroponics soil darkened the creases of her fingers. She had come straight from the lower green belt, still wearing a pollination hood folded around her neck and the copper badge of a Commons steward. Morning lamps had risen in Drum Three. Their light reached the chart room only through camera panes, a false dawn curved around thirty hectares of leaves.

"You sent me a locked number at ship's night," she said. "Then you declined to certify a correction everyone has started celebrating. Either you need a witness or you have forgotten how cousins work."

Ilyan moved the receipt away from the glass. "I needed an external timestamp."

"You used my public queue."

"It keeps independent custody."

"It also wakes me when flagged material arrives." Mina pulled the second chair into place. Jo had left it angled toward Violet. "Show me what requires custody."

The raw seam pulsed beneath the smoked pane. During the last hour, its answering rhythm had weakened by two percent. Jo had gone to the drive nave to inspect the fused isolation jaw and delay any order to dump Capacitor Seven. Ilyan had remained with the channel, listening to a clock that doctrine denied could run.

He played one challenge sequence and its response.

Mina listened without the reverence he had begun to fear in himself. "Same pattern."

"Delayed by 0.73 seconds."

"Every time?"

"Within four thousandths."

"A reflector."

"The final pair missed, then corrected."

"Damaged equipment can correct."

"Only if it has a clock."

She looked at him. The similarity between Mina and Eda lived around the eyes, where impatience narrowed into attention. He had spent six years avoiding that resemblance. Now the chart-room glass placed both sisters into every reflection his mind produced, though Mina had never asked to carry the dead one for him.

"What have you checked?" she asked.

“Prior delayed releases. Founder channels. The correction cards.”

“What haven’t you checked?”

Ilyan glanced at the archive addresses in his scratch pane. “The maintenance tables.”

Mina waited.

He disliked when she used silence as Commons procedure. A quiet chamber made speakers reveal which part of a question they had hoped to evade.

“Old route instruments had separate service clocks,” he said. “Before the cesium braid standardized all timing, engineers logged drift against physical maintenance intervals. If secondary signals occurred during early trials, those tables might preserve them.”

“Can you read them?”

“I can request them.”

“Different verbs.”

“Founder raw channels require Captain’s access. Maintenance summaries do not.”

Mina lifted the checksum receipt and slid it into the pocket beneath her badge. “Then read what they forgot to forbid.”

The maintenance archive occupied no room. Its public terminal stood in a recess behind the chart-room amphitheater, between an emergency respirator cabinet and a pressure door no one had opened in Ilyan’s lifetime. Founders had expected technicians to consult it beside the machinery. Later crews had moved the Loom controls forward and left the old terminal facing a wall.

Ilyan woke its display. Amber letters formed slowly on black glass.

COMPONENT OR FUNCTION.

He entered MANEUVER BELL.

The archive returned four centuries of inspections, most copied from prelaunch tests. He limited the result to ship-years one through twelve, when the first live route trials had occurred. Eighty-seven work orders remained. Loose brackets. Bronze impurities. A disagreement over whether the bell needed to sound loudly enough for workers wearing pressure hoods. Twice, technicians had found unexplained scoring on the tongue after field load.

Mina stood behind him reading over his shoulder. She smelled faintly of wet nutrient cloth, an ordinary scent that made the founder records feel less remote.

“Who signed the first live repair?” she asked.

Ilyan opened the entry.

ALDA QUILL, FIELD ENGINEER.

The name persisted in Guild histories as a subordinate Loom builder. Founder portraits showed Alda near the back of the launch crew, one hand hidden behind a taller officer. Her published work concerned thermal isolation jaws and load-braid fatigue. No route theorem bore her name.

The repair text described long-interval bell review after an uncommanded strike. Its component measurements looked absurd. A bearing inspection listed seventeen seconds. Lubrication listed three point eight. Cooling verification listed five point one, then four point nine, then six point two. Maintenance steps normally carried broad estimates, never fractions suited to clock comparison.

Ilyan exported the values into a side pane.

“What would make an engineer measure grease to a tenth of a second?” Mina asked.

“Nothing.”

“Bad habit?”

“Alda’s other entries use ordinary intervals.” He opened a braid-cooling order from the same watch. More precise values appeared beneath harmless tasks. An isolation-jaw inspection carried two alternating columns, one labeled local and another labeled reference. The reference values ran faster, then slower, as if two clocks had occupied opposite sides of a changing field.

Mina touched the screen without quite reaching it. “Can you plot them?”

Ilyan loaded the values against the official trial timeline. The archive resisted because maintenance records lacked correction identifiers. He matched component loads instead: field temperature, capacitor discharge, braid current. Each value aligned with the first Loom trial’s settlement sequence. Alda had hidden an unauthorized clock inside the machine’s repair history.

At seventeen seconds past settlement, her precise measurements stopped.

Below them appeared an inspection delay of 4,906 seconds.

“That cannot belong to the same watch,” Mina said.

“It doesn’t. The signature clock shows she added it seven hours later.”

“What happened after seventeen?”

Ilyan expanded the row. No component name occupied the first field. Only duration remained. Beside it sat a machine-readable sequence of spaces and short marks that the modern display treated as corruption.

He converted the marks to maintenance intervals. Long, short, short. A pause. More intervals.

Mina fetched a stool from the respirator cabinet and sat. “You recognize it.”

“An old timing alphabet. Drive crews used it before speech channels reached the outer spars.”

“Read it.”

He did.

KEEP BOTH.

The words appeared nowhere in the archive. They existed only as measured gaps between service actions that no one had performed.

Mina exhaled through her nose. "Both what?"

Ilyan looked toward the violet pulse across the chart room. "Both routes."

"That answer came quickly."

"The table sits under the first live trial."

"You still chose the noun."

He closed the translated pane. KEEP BOTH vanished, leaving rows of respectable maintenance. "Alda may have meant both records."

"Do records ring bells?"

Ilyan reopened the pane.

They searched Alda's later work orders. Her name appeared under cooling jackets, field rails, seam pickups, pressure shutters, even waste-heat valves she had not designed. Every entry touched a component that listened during correction. Each carried precise intervals under a field intended for approximate labor time.

He built a comparison table and sorted by correction load.

The hidden structure emerged reluctantly. Alda never used a heading that command could search. She placed one clock in lubrication duration, another in worker clearance time, another beneath acoustic damping. Route designation occupied the component-code field. Signal strength hid in parts consumed. A column marked SUPERVISOR DELAY carried the continuing domain's elapsed duration.

Ilyan tested the scheme against a repair he understood. Correction Three listed thermal paste applied to six seam pickups at intervals no technician would choose: two close together, three evenly spaced, one after a long pause. The work order claimed one worker completed the task. Its access log placed that worker in pressure recovery throughout the listed hour. Nothing physical had been repaired.

He converted the intervals. The pattern resolved into outer-rail code for AIR LOW, followed by a compartment number. A second row answered with VALVE HELD. The exchange continued across nine entries until the timing stretched beyond reliable translation. Someone in a rejected route had reported a pressure fault. Someone on Alda's side had listened long enough to hide the report.

Mina read the access log twice. "Was the compartment occupied?"

"Every habitat compartment had assignments during the first mission decade."

"How many?"

"I can't know from this."

"You could estimate."

He opened the historical berth plan, then stopped before loading it. Four bunks, two temporary hammocks, one sanitation allowance. The count would tempt him into replacing uncertain people with a neat maximum.

“Between one and six,” he said. “The message proves concern for whoever had to breathe there. It doesn’t prove a roster.”

“That sounds more careful than saying none.”

Correction Five hid another exchange inside coolant inspections. This one used recurring gaps of eight hours, the old three-watch schedule. A morning pattern included a galley heat request. The next carried a pump inventory. At the end of each watch, an operator transmitted a three-beat acknowledgment. The signal weakened, but the schedule continued.

Ilyan overlaid the route clock with Nacre’s designated clock. They began in agreement. Settlement load bent the rejected domain’s seconds until one watch there occupied less than seven hours here, then more than nine. Even through that distortion, the senders adjusted. Their acknowledgments returned at the beginning of their own watches, not his.

He pictured no faces. Faces invited grief to finish an argument that evidence should carry. Instead he pictured shift change: hands passing a slate, a tired operator pointing to the latest pressure value, another person taking the chair. The maintenance code preserved work, and work required a world that had not yet ended.

“Does Violet know our time?” Mina asked.

“Only through the challenges we sent.”

“Then send a time mark.”

“Not until the channel has command oversight.”

“You spoke to them already.”

“I announced custody. More queries could alter what they do.”

Mina considered that. “Good. You’ve stopped treating the map like a body under examination.”

“I have not decided what it contains.”

“You just decided consent might matter.”

The first trial recorded seconds.

The second recorded eleven minutes.

By the fourth, the values extended beyond a watch. Alda’s entries tracked six hours, then a day, then three. Some stopped under heavy settlement load. Others continued through seventeen days. The longest visible sequence reached nineteen days and broke before the final hour.

Ilyan scrolled back to confirm he had not joined unrelated repairs. Matching checksums tied every cluster to a Loom event. Official correction histories called the secondary data echo decay. Alda’s intervals showed no decay curve. They showed sleep cycles, equipment checks, work shifts, and regular pauses long enough for people to live inside them.

He removed the word people from his thought before it settled.

Mina saw something change in his face. “Say what you found.”

“Duration.”

“For what?”

“The rejected field states.”

“That describes a diagram.”

“They retained timing after settlement.”

“What uses timing?”

He highlighted a sequence from Correction Six. Three pulses repeated every six hours. Between them, shorter patterns moved in two-watch alternation. No automatic sensor would adopt shipboard labor cadence unless someone had programmed it, and the old route stack lacked such a routine.

“Mina.”

“Say it.”

“I need founder-channel access before making a claim.”

“You needed evidence before making one. Now you have a century of evidence and want a better permission slip.”

Her voice remained low. The chart-room cameras would not capture sound in the archive recess, but a passing shift officer could still see them. Ilyan checked the pressure door, the respirator cabinet, and the empty aisle. His attention to exits embarrassed him. Mina noticed and turned the stool so her Commons badge faced the main room.

“If anyone asks,” she said, “I requested a public maintenance review.”

“That may implicate you.”

“Your checksum already did.”

He lowered the display brightness. “A clock implies continuing causality.”

“Louder.”

“No.”

“Not for the room. For you.”

Ilyan pressed his palms against his knees. “A clock needs a process that changes. This one matched watches. It corrected errors. It sent patterns. If those actions came from the rejected routes, then settlement did not close empty calculations.”

Mina waited for the final step.

He wanted Teren Vale there. A physicist could distinguish bounded domains from actual ships, copies from continuations, signal behavior from consciousness. Jo could name heat and load without permitting either to become a person. Ilyan understood records. A record granted no soul. It only made denial harder to perform cleanly.

“The map has inhabitants,” he said.

Violet pulsed under the distant glass.

Mina nodded once. “Now we can discuss what to do.”

They carried the timing table to the central Loom on a disconnected slate. Ilyan refused to let the archive and live route system touch through software. Mina mocked the precaution until he explained that Assurance could quarantine every linked device with one command. Then she found a wax pencil and copied three values onto her forearm.

“Paper would work,” he said.

“Hydroponics has paper because plants don’t report waste. Navigation counts every sheet.”

“Your arm reports to Commons?”

“My arm reports when I raise it.”

Dry humor helped him breathe. It did not alter the data.

On the larger glass, Ilyan mapped the hidden durations beneath 141 corrections. Alda’s signatures ended after Correction Nine. Later records continued her method with slight changes, suggesting other engineers had inherited it. Precise maintenance intervals appeared after every correction for more than a century. Some keepers concealed values in filter replacement. Others used tram delays or coolant-pump rotation. They had built a second history beneath the official one, then stopped signing their names.

“What does the last column mean?” Mina asked.

Each settled route carried one final row. Component field empty. Parts used empty. Resolution empty. A duration appeared, then a blank line.

At first Ilyan assumed the archive had lost text during migration. He compared checksums. The blank persisted in the source record. Someone had deliberately entered a row with no explanation.

The same blank line repeated under every settled route.

“Could mean no signal,” Mina said.

“It could mark settlement completion.”

“The official record already marks that.”

“Then a failed observation.”

“Repeated 141 times?”

Ilyan enlarged the newest entry. Correction 141 carried twelve route tags. Green lacked the blank. Violet, Amber, White, and eight numerical tags each ended with one. The empty rows had inherited valid timestamps during the correction. None showed a closing signature.

Alda’s first trial contained the interval code KEEP BOTH just above its blank. Later entries offered no words,

only measured duration followed by refusal.

He thought of his unsigned certification on the public correction ledger. Last night, leaving the field empty had felt like delay. On Alda's map, blankness served as witness. The route had continued past the point where sanctioned language stopped.

Mina traced one row with the blunt end of her pencil. "Can you tell how long Violet has?"

"Not from this table. The historical values end when the maintenance recorder lost signal. That may not equal domain closure."

"Can Jo measure it?"

"Perhaps."

"And the Captain can open the raw founder channels."

"Yes."

"She may also close your current one."

"Yes."

The ease of his second answer surprised him. Violet's seam consumed reserve. Keeping it warm might endanger the designated ship. Any captain would ask how much. Any conservator should ask what a closure destroyed. The questions did not cancel each other; his training had merely arranged them in an order that made one invisible.

A duty notice appeared on the main glass: CERTIFICATION OVERDUE, CORRECTION 141. The system offered him a final chance to adopt the Loom's settlement values before elevating the omission to his supervisor.

Mina watched the countdown. "What happens if you let that expire?"

"My seal enters review."

"Does the trace stay open?"

"Certification has no hardware authority."

"Then don't confuse your job with their life."

He almost objected to the pronoun. The timing table denied him the comfort.

Ilyan dismissed the certification notice.

He opened the access registry instead. Founder raw channels required a reason code and an officer credential. Selecting Captain's authority generated a formal request visible to Sava Or, Navigation Assurance, the chief causal physicist, and security. No private category existed. The act would expose Jo's bypass, his checksum vault, Mina's custody receipt, and the answering seam.

A safer path remained available. He could delete the scratch anomaly, certify Correction 141, and leave Alda's hidden rows where they had survived for 119 years. Mina's checksum would prove only that an unknown file

had once occupied his vault. Violet might cool before anyone understood what the missing record contained. His seal would remain intact enough to investigate the next anomaly.

The logic resembled careful procedure. It also resembled every empty component field in the wake ledger.

“What did Alda do after command took her recording?” he asked.

Mina glanced at KEEP BOTH. “She found a shelf they did not think mattered.”

“And then?”

“She left someone else enough to begin.”

Ilyan entered the reason for access: CONTINUING POST-SETTLEMENT CLOCK; POSSIBLE CREWED ROUTE DOMAIN.

The registry warned that unsupported personhood claims could trigger fitness review. He added the hidden duration table, Violet’s challenge responses, and the recurring blank line. Before attaching Jo’s spoken statement, he marked it restricted to Captain, engineering counsel, and Commons custody. The software objected that Commons held no role in route incidents. Mina supplied her delegate number anyway. The objection changed to advisory.

At the signature pane, Ilyan used his conservator’s seal.

The request left his workspace. Red locks appeared over every founder address. A notification climbed the command queue toward the Captain’s office, carrying Alda’s old timing values and a map whose empty rows no longer looked empty.

Across the Loom, Violet answered an unsent rhythm with one faint pulse.

Ilyan moved the route glass aside and cleared a place at the central console. When Captain Or opened his request, she would find the official settlement record waiting without his signature. Beneath it, he left Alda’s blank line visible.

Chapter 03 — Nineteen Days of Life

<!-- POV: Jo Malk -->

<!-- SETUP: Capacitor Seven bypass; Alda's hidden durations; Violet's warm seam -->

<!-- PAYOFF: physical persistence test; Sava seals the rig; Jo sends evidence to Commons -->

Jo drove a steel wedge under Capacitor Seven's isolation jaw and felt the seam pull back.

The jaw pressed hard enough to flatten copper. It had fused at partial closure during Correction 141, leaving a gap narrower than Jo's thumbnail. Through that gap, charge moved from the waste-heat bus into the route braid. The current carried no mercy, intention, or permission. It followed contact.

"Another half millimeter," said Hesh from the lower platform.

"Another half takes the teeth off."

"The teeth belong to Stores."

"My hands belong to me." Jo leaned into the wedge until the load meter dropped two points. Hot metal complained inside the housing. The violet indicator along the service rail brightened, then held.

Hesh watched it. He had worked field capacitors for twenty-one years and trusted any light that followed a wrench. "If we close Seven now, the branch goes cold."

"Route."

He glanced up. "Since when?"

"Since it started answering."

Jo removed the wedge and replaced it with a locking shim. The drive nave surrounded them in black rails, pearl composite, and pipes carrying more heat than a habitat district could survive. Six capacitors occupied the nave's curved wall like bolted coffins. Seven sat apart because every correction found a new way to injure it. Above the housings, amber lamps pulsed with the ship's load.

The access door opened. Ilyan came through holding a disconnected slate against his chest.

His chart-room clothes looked wrong in the nave. Clean cuffs caught grease from the rail as he descended. Jo had seen that careful grip when he handled damaged records, two hands and no loose fingers.

"You told the Captain," Jo said.

"I requested access."

"Same door, softer knock."

"She has acknowledged the request."

"Meaning?"

"Security will arrive in sixteen minutes."

Hesh put down his spanner. "For us?"

"For the equipment." Ilyan looked at the violet indicator. "Possibly both."

Jo wiped their palms on their coveralls. "Then we have fifteen."

They plugged Ilyan's slate into no port. He set it on a tool cabinet and showed them the hidden maintenance timings by hand. Alda Quill had spread route clocks across component records for more than a century. Precise intervals tracked rejected domains through work shifts, heat calls, pressure reports. The longest sequence stopped at nineteen days and part of another watch.

"Stopped because settlement finished?" Hesh asked.

"Or because the pickup failed," Ilyan said.

"Records can't tell the difference," Jo said.

"Not those records."

"Good. Metal can."

Jo opened Seven's thermal history. The capacitor stored heat in layered salt blocks during field load, then returned it to the ship's waste bus. Any continuing domain attached through the fused jaw would alter that return. A reflector could reproduce a pulse, but it could not consume heat, run pumps, or vary load according to work performed elsewhere.

They isolated a test packet no larger than the heat from a cup of tea.

Ilyan noticed the quantity. "Will Violet receive that?"

"Their side of Seven will."

"What can it damage?"

"If their hardware matches ours, nothing. It will warm a calibration block by four degrees."

"And if it doesn't match?"

"Then the heat still has to go somewhere. That uncertainty comes with opening any circuit." Jo saw his objection remain. "We aren't sending a command. We're measuring whether a load exists."

"A load could include people."

"So could every repair I've made since apprenticeship."

The answer sounded harsher than Jo intended. They pictured Violet's drive nave, if picture meant anything here: another Seven, another fused jaw, another Hesh standing beneath it. The correction had copied every loosened fastener and every tired worker before letting their conditions diverge. Jo had bridged the waste bus to keep Nacre's maneuver inside limits. The same bridge now reached a ship taught to expect death after its route lost designation.

Jo set the packet to release over ten seconds. "Hesh, local jacket. Ilyan, watch seam phase. I'll take return flow."

The steel housing clicked as heat entered the braid.

For three seconds, nothing changed. Then the return meter dipped. Energy had crossed the seam and entered a load. A slow rise followed, ragged around the edges.

“Pump cycling,” Hesh said.

Jo checked the frequency. “Too slow for a pump.”

Ilyan bent over the phase pickup. “It matches their route clock.”

The return climbed in two steps, paused, then fell below baseline. Something across the seam had opened a sink after receiving the heat. The sink remained active for forty-one local seconds. A passive block would have warmed and relaxed in one curve. This load changed its own boundary.

“Automatic protection,” Ilyan said.

“Could be.” Jo switched the packet off. “Now we wait.”

Hesh looked toward the door clock. Eleven minutes before security.

A second curve arrived without prompting. It carried less heat back than they had sent. Within it, tiny regular losses repeated every four seconds.

Jo magnified the losses. Each one matched the draw of a gas sensor reading across a standard atmosphere line.

“Scrubber sampler,” they said.

Ilyan stared at the curve. “Can a capacitor imitate that?”

“Not unless somebody installed lungs in it.”

They used the remaining minutes to build a harder test.

Jo routed Seven’s return through three independent instruments: a thermal plate made before launch, a mechanical pressure capsule, and a hand-wound current loop whose only software controlled the display brightness. Hesh complained about the loop’s insulation. Jo told him old ugliness served as evidence when modern systems shared the same lies.

Ilyan loaded Alda’s durations beside the live curve. “We need more than a sampler cycle.”

“We have heat crossing a settled seam.”

“That proves continuing machinery.”

“Machinery continuing long enough keeps somebody alive.”

“It may run unattended.”

Jo respected the refusal. Ilyan wanted the claim to survive people who had spent their careers avoiding it. “Then we test for use.”

Nacre’s atmosphere system drew heat according to breath load. Every person exhaled moisture and carbon dioxide. Scrubbers absorbed both, regenerated their beds, and dumped a precise thermal pattern into the waste

bus. A ship without living metabolism would show pumps, standby losses, perhaps automatic cycling. An occupied ship would carry irregular peaks over the mechanical baseline: meals, shifts, exercise, sleep.

Jo asked the local archive for Nacre's last eight hours of scrubber waste. Access opened under their drive credential. They stripped all labels and overlaid the curve with Violet's return.

The two traces shared a shape through Correction 141. After divergence, Nacre's morning rise came earlier. Violet showed an extended low period, then a sharp demand consistent with emergency movement. Smaller waves followed at intervals that no automatic schedule aboard Nacre used.

Hesh pointed to one cluster. "Commons muster."

"How do you know?" Ilyan asked.

"Drum Three opens both west stairwells for muster. More people climb at once. Scrubbers catch the breath ten minutes later."

Violet's cluster arrived at the same structural delay, but at a different ship time.

Jo refused to trust a resemblance that three tired workers wanted to see. They handed Hesh four unlabeled segments, two from Nacre and two from the seam return, then asked him to identify the occupied curves. He chose both seam traces correctly. One carried the staggered draw of children's wash hour, when small bodies entered sanitation in groups instead of adult shift intervals. The other showed an abrupt dip followed by emergency ventilation.

"Why emergency?" Ilyan asked.

"Main fans stop taking heat. Compartment shutters split the air volume. Local blowers start here, here, and here." Hesh tapped three narrow rises. "We did the same during the Drum Eight mold fire."

"An automatic response could produce it."

"Something had to trigger the shutters."

Jo checked for smoke-sensor current. None appeared. Pressure isolation had begun through manual levers or a system unavailable in Nacre's version. Either possibility required events after divergence. Violet had developed a problem they had not experienced and acted upon it with tools that consumed real energy.

They built one final control without sending anything. Jo closed the measurement shunt for twelve seconds while leaving the waste-heat feed untouched. If local electronics generated the irregular curve, the phase pickup should continue to report it. If heat arrived through the seam, the mechanical capsule would lose pressure during the closure and recover afterward.

The capsule needle fell.

On reopening, it jumped past baseline as accumulated return crossed the shunt. The old thermal plate warmed under Jo's palm. No code lay between that warmth and their skin.

Ilyan touched the capsule housing. "They continued while we stopped listening."

"Of course they did," Jo said.

The anger in their answer belonged partly to him. Jo had spent years mocking officers who treated inaccessible

machinery as inactive. A pump behind a sealed wall still spun. A bearing beyond a sensor fault still heated. Yet they had accepted route settlement because the isolation lights went dark. They had mistaken loss of access for absence, the oldest maintenance error aboard ship.

Hesh opened the correction work list on his own slate. Eleven rejected tags sat beneath a green completion mark. "Do all of them run this long?"

"The old table says they can," Ilyan replied.

"And every correction?"

"Every correction left hidden duration entries."

Hesh counted nothing aloud. His thumb moved down the tags and stopped at Violet. "We celebrated two percent less hull scoring."

Jo remembered the tea in the nave after release. Someone had warmed it on Seven's jacket. They had raised metal cups beneath the violet indicator while another drive crew woke to emergency ventilation across the seam.

The celebration did not become false because Nacre had survived. It became incomplete, which felt worse. A lie could be rejected whole. Incomplete truth stayed useful enough to defend.

Jo separated fan current from absorber heat. The irregular remainder persisted. It varied too much for machinery and too coherently for noise. Thousands of bodies had crossed decks, slept, woken, breathed into a ship whose route card read SETTLED.

The access door unlocked.

Security Chief Hadran Coil entered first. Two officers followed with seal cases. Captain Sava Or came behind them in a dark duty coat, its collar fastened despite the nave heat. Dr. Teren Vale brought up the rear, already squinting at the instruments as if poor eyesight could excuse what he recognized.

"Hands off all controls," Hadran said.

Jo kept one finger on the current-loop frame. "The control is disconnected."

"Step away."

Sava surveyed the open housing, steel wedge, hand loop, and violet service light. Her gaze counted faults before people. "Engineer Malk, did you alter Capacitor Seven after I acknowledged Conservator Rook's access request?"

"I measured it."

"Did you transmit into the unsettled field?"

"Ten seconds of calibration heat."

"Without authority."

"The bus has been feeding it for nine hours without your authority."

“That feed began during emergency correction.”

“And now?”

Sava’s formality increased by one degree. “Now I am determining what occurred.”

Teren approached the thermal plate. He did not touch it. “Show me the return.”

Hadran turned. “Captain?”

“Let him see.”

Jo displayed all three instrument traces. The old plate showed heat loss. The mechanical capsule registered phase pressure through the seam. The hand loop showed changing load. No common software joined them.

Teren read the scrubber overlay last. His mouth tightened around a word he did not say.

“What mechanism produces this?” Sava asked him.

“A continuing atmosphere plant could.”

“Could a reflected telemetry packet?”

“Not the thermal plate. The domain would need a free-energy sink.”

“An automated route model?”

“The Loom models by instantiation.” Teren looked toward the violet light. “If reserve remained, the machinery could continue causally.”

Jo struck the cabinet with one knuckle. “The machinery carries breath load.”

Hadran examined the unlabeled traces. “Or a simulation of breath load.”

“No simulation drives my pressure capsule.”

“What does?”

“Pressure.”

Sava raised one hand before the argument could widen. “Duration.”

Ilyan turned Alda’s table toward her. “The historical maximum approaches nineteen days.”

“Approaches?”

“The final maintenance sequence begins at day nineteen. It ends six hours later.”

Teren’s attention shifted from the live rig to the old values. “Let me see the correction load.”

Ilyan refused to hand over the slate. He placed it on the cabinet where everyone could read. “The entries match settlement current across twelve early trials. Later custodians continued the method.”

Teren moved through the values, whispering ratios. Shame made him explain as he calculated. “Bounded shear does not stop duration. It separates reference frames. Settlement drains the differential until the rejected

domain can no longer maintain a stable boundary. The old papers assumed collapse before macroscopic process.”

“But the papers never measured it,” Jo said.

“No.”

“Alda did.”

“Yes.”

Silence gathered around the working capacitor. Jo disliked the impulse to make revelation quiet. Seven’s jacket clicked every eleven seconds, and the drive nave continued moving heat for two ships.

Sava addressed Teren. “Can you verify the end interval?”

“Not from maintenance tables alone.”

Jo opened the capacitor’s full discharge model. A line descended from route creation through settlement reserve. Alda’s hidden durations supplied measured points. The curve met zero at an interval already built into every field-safety calculation, though the manual called it remote relaxation tolerance.

Ilyan saw the number first. “Nineteen days.”

Jo expanded the final fraction. “Six hours.”

Teren closed his eyes.

Sava did not. “State what that number means for Correction 141.”

“Violet retains a bounded causal domain until its reserve falls below closure,” he said. “Without supplemental input, the model places that event nineteen days, six hours after settlement.”

“Living conditions?”

“The atmosphere trace indicates occupied compartments.”

“How many?”

“Unknown.”

“Conscious communication?”

Ilyan touched the response file on his slate. “The challenge pattern was copied with a human timing error.”

Sava looked from him to Jo. “And this rig currently supplies reserve.”

Jo checked the bus transfer. The flow looked small beside Nacre’s drive capacity. Over nineteen days, it would become a navigational loss the captain could not ignore. “Yes.”

“If you remove your bypass?”

“The seam closes. Their remaining domain still runs until settlement finishes, but we lose contact and any way to feed it.”

“Does removing it accelerate closure?” Teren asked.

Jo traced the coupled load. “By several hours, maybe more. Seven has already transferred heat across. I need a complete test to quantify it.”

Hadran said, “You accidentally prolonged a rejected state.”

Jo faced him. “I kept people alive by fixing the wrong capacitor.”

The sentence left no comfortable material noun behind it.

Hesh leaned against the lower rail. His expression had changed since the first test. The route service light no longer offered a machine’s obedience. It marked workers on another hot deck, perhaps studying their own instruments and wondering why a calibration block had warmed.

Sava ordered the equipment sealed.

Hadran’s officers moved at once. One closed the thermal plate case. Another wrapped red custody tape around the pressure capsule. Hadran reached for Seven’s control cover.

“Not that,” Jo said.

He stopped.

“Seal the controls, not the bus,” they continued. “If you cut power, you act on Violet before your inquiry begins.”

Sava held Jo’s gaze. Behind command formality, calculation moved: 42,006 lives, thirty-one days to arrival, reserve spent through an unauthorized gap. “Maintain current load only. No further tests. No voice transmission. No change to jaw position without my written order.”

“Current load drifts.”

“Hold it within one percent.”

“That requires adjustment.”

“Then you may adjust solely to preserve the measured baseline.”

Hadran objected. “Captain, an open seam presents an uncontrolled channel.”

“So does an uninformed closure.” Sava turned back to Jo. “You will surrender every copy of this test.”

“I will surrender the instruments.”

“Every copy.”

Jo looked at the three traces. Hadran had not asked whether the hand loop stored data. It did not. Hesh had written its values in grease pencil on the underside of the lower platform, where engineers recorded temporary alignments before committing them to a log. Ilyan’s disconnected slate carried the atmosphere overlay. Jo’s wrist unit held only six unlabeled thermal points.

They removed it and placed it on the cabinet.

Sava counted the surrendered devices. "Engineering access to Seven now requires security presence. Conservator Rook, you and Dr. Vale will report to my hearing room in twenty minutes. Malk, you will attend after decontamination."

"Grease doesn't contaminate testimony."

"Your unauthorized procedure does."

Jo accepted that because arguing would spend time.

The officers sealed the cabinet, the disconnected slate, and the old thermal plate. They left the physical capacitor feeding Violet at baseline. Red tape crossed every local control except the narrow trim wheel Sava had authorized Jo to touch.

When command departed, Hesh remained on the lower platform under security observation. He crouched as if checking a boot fastener and tore away the grease-marked strip from beneath the rail. The values came off on a piece of insulation cloth.

"You told her every copy," he murmured.

"She told me to surrender them." Jo took the cloth. "She didn't ask the rail."

The security officer at the door watched the main controls, not their hands.

Jo went to the wash recess and fed the six wrist-unit values from memory into a maintenance request. Hesh's marks supplied the independent readings. They added Ilyan's scrubber comparison from the reflection still visible on the sealed slate glass. No labels named Violet or the Loom. The request described a waste-heat fault affecting public atmosphere capacity, which placed it under mandatory Commons review.

At the destination field, Jo selected Drum Three infrastructure committee.

The system warned that sealed course material could not enter civic queues.

Jo removed the course tag. Heat, pressure, and breath load remained true without it.

They pressed send.

A receipt appeared on the maintenance panel before security understood the panel had joined the evidence. Mina Rook's delegate number took custody in the Commons. The test left the drive nave as an ordinary repair record, carrying nineteen days and six hours of life past the red seals.

Chapter 04 — The Captain's Denial

<!-- POV: Captain Sava Or -->

<!-- SETUP: sealed persistence test; Alda's timing evidence; Article 4 authority -->

<!-- PAYOFF: seventy-two-hour secrecy order; one controlled query; human breath -->

Sava arranged four chairs on one side of the hearing table and left her own standing.

The room adjoined the bridge but shared none of its glass. Pearl composite walls, a bolted table, two doors. She used it for course inquiries because nobody could point toward stars while avoiding an answer. Hadran Coil took the chair nearest the outer door. Teren Vale sat opposite him. Ilyan Rook remained beside the table until Jo Malk arrived in clean coveralls that still held grease at the seams.

“Sit,” Sava said.

Jo looked at the four chairs. “Which one carries the charge?”

“All of them.”

They sat.

A sealed evidence pane hung above the table. It showed Capacitor Seven's thermal return, Alda Quill's maintenance intervals, Violet's answering pattern, and the access history for every unauthorized action since Correction 141. Sava had ordered the display to separate claims from observations. The distinction should have made the material manageable.

OBSERVATION: secondary bell after settlement.

OBSERVATION: copied challenge with 0.73-second delay.

OBSERVATION: continuing thermal and atmosphere load.

CLAIM: inhabited route domain.

CLAIM: conscious reply.

CLAIM: settlement does not end duration.

Four years earlier, a cracked flywheel had produced twenty-six pages of evidence and only two claims. Sava had suspended a maintenance chief, replaced the wheel, adjusted the inspection schedule, and finished before second meal. Here, every observation opened another jurisdiction.

She began with the breach she understood. “Engineer Malk, you altered correction hardware without authorization.”

“Yes.”

“You then conducted a post-settlement test after receiving notice of my inquiry.”

“Yes.”

“You sent sealed measurements to the Commons.”

Jo glanced at Hadran. "You found that quickly."

"Answer."

"I sent a public-atmosphere fault report before anyone marked those measurements sealed."

"The report concealed its connection to the Loom."

"It preserved heat, pressure, and breath data."

"Without context."

"Context can be added. Deleted data can't."

Sava let the answer remain. Jo's insolence often hid good engineering under poor deference. Punishing the tone would save everyone from examining the fact.

"Conservator Rook. You withheld certification of Correction 141."

"I declined to certify settlement."

"You also stored raw route data outside the official ledger."

"Inside a write-once Guild vault."

"Under a false training category."

"Under a disputed-checksum category. The claimants genuinely dispute each other."

Sava looked at the young man who had taught half Navigation to respect custody. He had broken procedure with procedural precision. "Do you believe Violet contains people?"

Ilyan's gaze moved to the evidence pane. "Yes."

"Based on what minimum fact?"

"A response corrected its timing after an error. Machinery can adapt, but the atmosphere load also diverged from ours through events that occurred after Correction 141. The simplest account now includes operators."

Sava opened a clean column beside the evidence. "We will not govern by simplest accounts. List alternatives."

Ilyan glanced at Teren. "Stored telemetry."

"Test it."

Teren pulled the challenge timings into his slate. "A stored packet would need the sequence before Conservator Rook generated it. The braid source used radioactive count after settlement. No prior record contains those intervals."

Sava entered STORED TELEMETRY and marked it unsupported. "Next."

"Adaptive reflection," Hadran said. "The route stack receives a pattern and returns it with processing error."

Teren answered without looking up. "Route closure routines contain no adaptive reflector. We can inspect the active stack."

“Could Engineer Malk’s bypass create one?”

Jo leaned back. “A fused jaw conducts. It doesn’t learn rhythm.”

“Your answer addresses design, not emergent behavior.”

“Then build the same fault on a spare jaw. If it starts knocking back, I’ll apologize to the metal.”

Sava recorded a replication test. “Next alternative.”

Ilyan said, “Shared contamination in our instruments.”

“The thermal plate predates the Loom software,” Jo replied. “The pressure capsule has no processor. Hesh read the occupied traces blind.”

“Hesh knew the purpose of the test.”

“He did not know which curves came through the seam.”

Sava marked operator bias as possible but insufficient. “Could Nacre’s own atmosphere load cross-couple into Seven and appear delayed?”

Jo drew the route through the nave: scrubber waste bus to the jacket, jacket to local sink, bypass into the braid. Their finger stopped at the mechanical shunt. “We closed this path for twelve seconds. Return pressure vanished and then surged on reopening. Local cross-coupling would have remained visible upstream.”

“Unless the shunt altered the local source.”

“It didn’t. Hesh watched source pressure.”

The list narrowed. Each proposed fault demanded another unrecorded mechanism while the observed load continued behaving like occupied ship systems. Sava had learned to distrust explanations that required many failures to protect one conclusion. She had also learned that unprecedented machinery produced unprecedented faults.

“Deliberate fabrication,” Hadran said.

Jo’s chair legs struck the floor. “By whom?”

“You, Rook, Hesh, or a combination. Your unauthorized acts already establish willingness to bypass custody.”

Ilyan looked at the sealed instruments rather than Hadran. “My first raw interval entered a write-once clock before Jo came to the chart room. Mina received its checksum before either of us found Alda’s tables. Jo’s thermal test used separate hardware. Fabrication would require coordinated records created in an order we had no reason to anticipate.”

“People coordinate.”

“Yes,” Ilyan said. “That possibility makes people a poor standard for denying people elsewhere.”

Sava stopped Hadran’s reply. She added fabrication to the inquiry list, not because it fit but because command needed to survive the accusation that she had accepted favored witnesses. “One more.”

Teren finally raised his head. “The bounded domain may contain machinery without conscious inhabitants.

Automatic systems could preserve atmosphere, respond to faults, and produce complicated load.”

“The timing correction?” Sava asked.

“Harder to account for.”

“The manual shutter sequence after divergence?”

“Harder still.”

“Human breath?” Jo said.

“We have not received voice band.”

That boundary gave Sava something firm enough to hold. “Then consciousness remains unverified until direct query.”

“Simplest does not mean proven.”

“No.”

“Do you believe closing the seam kills them?”

He took longer. “The seam and the domain differ. Closing communication may not cause immediate settlement. Dr. Vale can answer the field question.”

Teren rubbed both thumbs against the edge of his slate. “Not completely.”

Sava faced him. “Answer what you can.”

“The rejected domain persists while its boundary retains free-energy differential. Settlement extracts that differential. Capacitor Seven currently feeds a small amount back through the seam. Removing the feed reduces Violet’s remaining duration. Whether a rapid isolation collapses the entire boundary depends on coupling we have not measured.”

“Worst case.”

“Immediate collapse.”

“Best case.”

“Loss of contact with no measurable change until the predicted end.”

“Predicted at nineteen days, six hours.”

“Yes.”

The duration did not fit inside Sava’s practiced categories. Emergencies aboard Nacre used seconds for collision, hours for pressure, days for crop loss. Nineteen days allowed meetings, meals, repairs, fear, and a child to ask whether tomorrow’s class still mattered. It also consumed nearly two thirds of the time remaining to Pelagos.

Sava turned to Hadran. “Security assessment.”

“Unknown channel into the route stack. Uncontrolled transmission already occurred. Commons holds partial evidence stripped of classification. We should isolate Seven, seize all receiving devices, and restrict the involved personnel.”

“Effect on Violet?”

Hadran’s pause lasted less than a breath. “Unverified.”

“Use the worst case Dr. Vale supplied.”

“Immediate collapse of an unverified domain.”

Sava heard the shelter in his grammar. Hadran did not call it empty. He placed uncertainty where people might stand and hoped the word would bear command’s weight.

“What effect does continued feed have on Nacre?” she asked.

Teren opened a reserve projection. “At present load, negligible over seventy-two hours. Beyond that, the uncertainty grows because Violet may draw more as settlement steepens.”

“Arrival reserve.”

“Correction 141 added eleven point four percent. Seven’s current feed returns less than one hundredth of that per day.”

“Can Violet transmit instructions into our controls?”

“Not through this configuration. Low-band phase and thermal return only.”

“Can its inhabitants cross?”

“No mass transfer across separated domains.”

“Can they alter our course?”

“Only if we act on received information or change the coupled load.”

Jo folded their arms. “Same as any voice on a wire.”

Hadran turned toward them. “A voice on a wire has a known sender.”

“That stops being true when you refuse to ask their name.”

Sava raised her hand. Both fell quiet.

The ship clock advanced into first-meal turnover. On the bridge beyond the inner wall, duty shifts changed. Sanitation teams opened the east wash lines. Forty-two compartment stewards confirmed pressure. Kitchen supervisors released grain allocation. Every movement depended on people trusting that command had already contained the dangers worth naming.

42,006 living people had entered her custody at the last annual count. The number excluded embryos because their systems belonged to mission continuity, not daily command. It included seven patients unlikely to survive the month, thirty-one confined under law, six newborns without registered names, and Jo Malk despite the effort required.

No captain inherited an abstract population. Sava knew which drums lost water when Pump Nine stuttered. She knew how many stairwells could carry a frightened crowd before counterflow trapped the lower terraces. She knew three words spoken from the bridge could empty food stores faster than a crop blight.

Ilyan followed her attention to the ship schedule projected beside the evidence. "Captain, the people on Violet likely have the same count at divergence."

"Likely."

"Do they enter your calculation?"

"They enter it as an unknown."

"Mina would call that an answer."

"Mina does not command arrival."

"No," Ilyan said. "She commands a greenhouse shift. Her unknowns still need water."

Hadran leaned forward, but Sava stopped him with a look. The conservator had not raised his voice. Threat came from the steadiness of someone who had decided losing his position no longer ranked first.

Sava walked to the evidence pane and removed the word CLAIM from inhabited route domain. She replaced it with PROVISIONAL FINDING.

No one thanked her.

At age ten, Sava had learned that adults lowered their voices before they lowered rations.

The crop failure began in a fungal tray beneath Drum One. By the time anyone smelled rot in the public corridors, the blight had crossed three nutrient loops. Command posted a statement about temporary menu adjustments. Sava's father came home from sanitation with his boots wet and told the family not to worry.

For nine days, he placed part of his meal in Sava's bowl.

She thought he disliked the new grain mash. He joked that growing captains needed more, though she had never said she wanted command. Adults held closed meetings near the wash plant. Numbers moved through her father's hands and vanished before he entered their compartment.

When he collapsed on a stair, the clinic recorded caloric deficit complicated by chemical exposure. Sava found the ration projection months later in an abandoned sanitation slate. The first page had predicted shortages by district. Their district carried the deepest cut because it contained more manual workers and fewer mission-critical specialists.

Her father had known. He had given away food because nobody trusted the district with the schedule of its own hunger.

For years, Sava took the wrong lesson from his death and called it discipline. Information required control. Projections could cause runs on stores, violence at distribution doors, hoarding in warm compartments. If command released uncertain danger too early, fear acquired its own machinery.

She still believed that.

Memory did not arrive to correct her. It arrived as evidence with more than one interpretation. Her father had concealed the table to protect a child. Command had concealed it to protect a ration plan. The same secrecy served love and hierarchy without changing its method.

“Captain?” Teren said.

The hearing room had waited through her silence. Sava returned to the table. “Commons already possesses the test. How widely?”

Hadran checked his slate. “Mina Rook accepted custody. The infrastructure committee has seven members. Her queue shows no forwarding events.”

“Because forwarding would display,” Jo said. “Reading does not.”

“Can you revoke it?”

“Not from Engineering.”

Hadran answered, “Security can quarantine the committee queue under emergency authority.”

“Which emergency?” Ilyan asked.

Sava replied before Hadran could. “Article 4.”

The words changed everyone’s posture.

Mission Article 4 granted the Captain emergency course authority when delay posed immediate risk to the vessel or destination. It allowed temporary control of navigation records, physical route access, and mission-critical communications. It did not permit indefinite suspension of Commons review. The authority expired after seventy-two hours unless the Assembly renewed it.

No prior captain had invoked it against information about another Nacre.

Jo unfolded their arms. “You can’t seal a population as course data.”

“I can prevent uncontrolled manipulation of a live field while we establish what it contains.”

“You already changed inhabited route from claim to finding.”

“Provisional finding.”

“Does provisional breath need less air?”

Sava’s voice cooled. “One more interruption and you will spend the inquiry outside the equipment deck.”

Jo closed their mouth.

Ilyan looked toward the sealed traces. “What does Article 4 authorize you to do to Violet?”

“Nothing until the immediate risk requires action.”

“You could define contact as risk.”

"I could. I have not."

"Yet."

Sava accepted the word. "For seventy-two hours, all seam controls remain sealed at current baseline. No person transmits without my authorization. Founder archive material enters protected review. Commons receives notice that a course anomaly affects reserve and may inspect verified evidence after technical assessment. No deletion, no field dump, no new settlement action."

Hadran said, "Captain, protected review leaves the checksum holders outside security control."

"It leaves proof that our control has edges."

His expression did not change, but she knew she had refused the remedy he preferred.

Teren adjusted his slate. "We need direct measurement of intentional response."

"Ilyan already sent challenge patterns," Sava said.

"Timing imitation supports agency. It does not identify the source."

"What would?"

"A voice-band query. Narrow carrier. One direction at a time."

Hadran objected at once. "Speech permits social engineering, false command, contamination of witness accounts."

Jo gave him a flat look. "Words may make you believe words."

Sava ignored them. "Can a voice query affect the field load?"

"Barely," Teren said. "Capacitor reserve cost would remain below the current baseline margin."

"Could it transmit control code?"

"Not if we remove digital encoding and use an analog pickup."

Ilyan leaned forward. "They have already heard my name."

"I know." Sava had replayed his unauthorized announcement. This is Ilyan Rook, route conservator aboard Nacre. The sentence troubled her because every counterimage would contain an Ilyan who remembered saying it or wishing he had.

She considered delay. Seventy-two hours allowed verification of Alda's tables, an independent repeat of Jo's test, a reserve forecast, and counsel on Commons authority. Waiting fit every habit that had kept Nacre alive.

Violet's remaining duration did not pause for habit.

"One controlled query," Sava said. "In the chart room. Analog voice only. Teren monitors field load. Hadran controls local access. Conservator Rook speaks from an approved text. Engineer Malk maintains Seven at baseline and does not touch transmission."

Jo nodded once.

Hadran did not. "What answer do you expect?"

"Noise," Sava said. "Or an automated repetition. We will document either."

She invoked Article 4 on her command slate.

The authorization required a risk statement. Sava entered UNVERIFIED CONTINUING ROUTE DOMAIN; LIVE FIELD COUPLING; POSSIBLE POPULATION EXPOSURE. The system asked for protected complement. She typed 42,006, then stopped.

No field allowed a second uncertain complement.

Sava added it under remarks: POSSIBLE EQUIVALENT POPULATION IN ROUTE VIOLET.

Her seal fixed the order until the exact second seventy-two hours hence. Security locks spread across the route archive. The chart room shifted from Navigation custody to Captain's control. Commons received a notice without the word inhabitants.

The four people at the table watched their devices accept the restrictions.

"Prepare the query," Sava said.

Hadran cleared the chart room before they entered. The empty seats descended around the suspended route glass. Green still ran toward Pelagos. Violet appeared only when Jo trimmed Seven's baseline through a guarded engineering link.

Teren connected an analog speaker to the field pickup. He inserted a physical cutoff switch between receive and transmit, then invited Hadran to inspect it. No software could open both directions at once.

Ilyan stood at the voice key with Sava's approved sentence before him.

> Route Violet, this is Captain Sava Or aboard designated Nacre. If a person receives this message, hold an open microphone near your breathing for five seconds. Do not transmit controls or mission instructions.

Jo read it from the service console. "You ask them to prove life by using it."

"Do you have a safer query?" Sava asked.

"Ask what broke."

"That invites technical content."

"It invites help."

"Not yet."

She heard the phrase after speaking. Not yet had covered ration tables, sealed reports, postponed warnings. It could still represent caution rather than refusal. Sava no longer trusted herself to know which.

At her signal, Ilyan pressed the key and read the sentence. His voice crossed as load fluctuations too small to disturb the plotted baseline. Teren cut transmit and opened receive.

Static entered the room.

Sava counted five seconds. The signal carried braid hiss, cooling clicks, and one faint tone near the maneuver bell's dying frequency. No voice. No command. No impossible accusation from a copied captain.

Hadran reached for the cutoff.

A soft impact sounded through the speaker, as if someone on Violet had set down a tool.

Then air moved close to their microphone.

The first breath entered unevenly and caught at the top. A person held it, perhaps frightened by the instruction, then released it in a slow tremor. Another inhale followed. Behind it, far from the pickup, metal struck metal and someone murmured too softly for words.

Sava's seventy-two-hour order remained bright on every locked console.

In the sealed chart room, the unknown population took a third breath.

Chapter 05 — Violet Answers

<!-- POV: Ilyan Rook -->

<!-- SETUP: controlled voice contact; shared history through Correction 141; Eda grief -->

<!-- PAYOFF: Violet proves divergent civic life and refuses identity through private grief -->

Ilyan removed his own voice from the seam one frequency at a time.

The analog pickup had captured more than breath. A low copy of his query traveled beneath the response, delayed and bent by field pressure. If he amplified the wrong band, he heard himself asking a stranger to breathe. If he cut too much, the person across the seam vanished with the echo.

Teren sat at the phase board and adjusted a brass wheel by fractions. Hadran watched both of Ilyan's hands. Captain Or stood behind the command rail with the Article 4 clock displayed beside her. Jo kept Capacitor Seven at its ordered feed from a remote engineering panel.

The breath came through again.

A throat cleared near the far pickup. Static pressed close around the sound. Then a voice tried to speak and broke into pulses.

"Carrier too narrow," Teren said.

"Widen it," Ilyan replied.

Hadran looked toward Sava. "The approved query has ended."

"We received an intentional response," she said. "We will identify its source."

Teren opened the band by four hundredths. The reserve gauge moved too little for its needle to show. Sound crossed the seam as scraps of pitch, each scrap placed inside a phase window. Someone on Violet understood phase coding well enough to fit speech around the interference.

A word emerged.

"Rook."

Ilyan's hand stopped above the filter controls.

The voice repeated it, slower. "Ilyan Rook."

Hadran muted receive. "A known name proves nothing. We transmitted it first."

"Open it," Ilyan said.

"Captain?"

Sava studied the violet trace. "Open receive. Keep transmit closed."

Static resumed. The speaker on the far side seemed to wait through the interruption, counting its duration. When the channel cleared, he sent a full phrase with each syllable broken across several pulses.

"Ilyan Rook aboard Route Violet. Do you receive?"

No one in the chart room moved.

Ilyan had expected another voice. Not because evidence required difference, but because the same cadence felt impossible after only a day. He heard his own careful consonants and the same slight roughness after waking too long. Yet the speaker rushed Route Violet, giving the name a pressure Ilyan did not use aboard Nacre.

Jo touched the engineering key. "He asked a question."

Sava nodded to Ilyan. "Answer the question only."

Teren switched the physical cutoff to transmit.

"This is Ilyan Rook aboard designated Nacre," he said. The inherited title tasted worse now. "We receive you. Voice contact remains restricted. State whether you require immediate technical aid."

Receive reopened.

The reply took seven seconds to assemble.

"We require you not to complete settlement."

Teren's fingers left the brass wheel. Hadran wrote the sentence into the security record without any mark of agreement. Sava kept her eyes on reserve.

"Ask for a route-state report," she said.

Ilyan pressed the key. "Report hull, atmosphere, injuries, and field condition. Do not send control code."

The far voice answered in sections. Violet had taken charged-dust impact along Drum Six during the correction. Outer pressure held. Two residential compartments remained empty behind isolation shutters. The school rooms from that district had moved into a Commons gallery and three storage bays. Their atmosphere plant ran at reduced capacity. Injuries exceeded infirmary beds, though most involved burns, fractures, and pressure trauma rather than continuing air loss.

A second voice supplied numbers too damaged to recover. Ilyan heard papers moving near their pickup. People interrupted, corrected a deck reference, and argued over which injuries counted as stable. No smooth command statement emerged. The report carried the disorder of work performed while others watched.

Teren adjusted the phase wheel and recovered more of the background. A woman read bed assignments in a steady cadence. Every sixth name brought a pause while another worker answered with a temperature or pressure value. Two names received no answer. The woman repeated each once, wrote something, and continued.

Ilyan knew the chart-room acoustics that shaped the voice. Violet's pickup occupied the same west console as theirs, but the cracked route pane changed each hard consonant. He could measure the other room through those distortions: an open hatch toward the bridge, temporary cloth hung across the rear seats, at least a dozen people moving beyond the speaker.

"Ask whether Drum Six remains under pressure," Jo said.

Sava approved.

“Three sectors hold,” Violet’s Ilyan replied. “Two remain shut. One outer passage has partial air but no safe access. We lost four repair workers during isolation. Seven others have not yet reported from the sealed side.”

The words arrived with no ceremony. On Nacre, those workers had finished their shift and gone to first meal. Ilyan had passed one of them often, a woman who carried two wax pencils behind her ear and corrected route officers who blocked pressure doors. He did not ask whether her Violet self appeared among the dead. The question would turn a shared acquaintance into a test designed for his grief.

“What help can sound provide?” he asked.

A different speaker answered, perhaps the woman managing beds. “Your Drum Six stayed intact. Send your outer-passage pressure response after the dust strike. We need the first forty seconds.”

Jo requested permission to speak. Sava allowed a summary but not raw control data. They described how Nacre’s shutters had cycled and which brace carried the highest load. Violet’s engineer interrupted twice because their west brace had already cracked. Together they narrowed the danger to a return duct near the sealed passage.

“Do not open it from the service side,” Jo said. “Your pressure will pull the damaged fastener through the frame.”

“We know what pressure does.”

“Then stop proposing the service hatch.”

Someone across the seam laughed once, brief and reluctant. “That’s our Jo.”

Jo looked away from the speaker. Their counterpart had apparently earned the same reputation in less than a day. The shared joke did more work than Hadran’s sealed questions. It showed memory carried forward, then acquired new witnesses. Jo on Nacre had not spoken those words. People on Violet had already learned when to use them.

Sava ended the technical exchange before it could become a repair session. The Article 4 clock kept counting beside her shoulder. Every useful fact made control harder to justify and easier to exercise.

Ilyan asked about the displaced school rooms instead. The question carried less immediate risk, which exposed the poverty of their categories. Violet’s adults had moved injured people into places built for civic debate and children into dry storage. Their school staff used ration boards for lessons because pressure crews had taken the slates. Older students carried meal tins from the east kitchen. Younger ones sorted undamaged fasteners by size for the repair deck, then argued whether that labor counted as mathematics.

“Why restart school during an emergency?” Hadran asked.

The woman on Violet heard him. “Because we have a settlement clock over every public display. Children can read it.”

Hadran wrote the answer into his record.

A child came close to Violet’s pickup. “Are your displays counting too?”

Sava did not permit Ilyan to answer directly. He watched her weigh security against the absurd cruelty of silence. At last she said, “Tell the child our route systems remain under review.”

Ilyan pressed transmit. "Our displays show a continuing field event. We are working on it."

"That means yes," the child said.

An adult moved them away from the console. The muted dispute that followed concerned meal duty, not contact. Ilyan felt the proof settle through its ordinariness. A fabricated voice might present secrets chosen to persuade. Violet kept having other work.

Violet's Ilyan resumed. "Our Loom shows settlement reserve dropping. Current closure estimate: eighteen days, seventeen hours. Uncertainty two hours. Does your ship show the same?"

Sava answered before Ilyan could. "Transmit no estimate."

"They need to compare clocks," he said.

"They need our reserve model to alter theirs."

"They already have the same model through Correction 141."

"Not the same inputs after divergence."

Across the seam, the other Ilyan waited. His breathing remained near the pickup, no longer offered as proof, simply present between words.

Sava allowed Ilyan to confirm that Nacre also measured a continuing domain. No duration. No reserve quantity. The caution belonged to a captain facing an unknown ship, but it also preserved an older advantage: this side possessed data and decided how much reality the other side could receive.

"Ask how he knows the route tag," Hadran said.

Ilyan sent the question.

"Correction card," came the reply. "Violet persisted on our glass after Green took designation. Your voice named Nacre. We chose the tag you assigned us because you would recognize it."

"Chose," Jo repeated softly.

Sava heard. "No commentary during receive."

Hadran passed Ilyan a verification slate. It held questions drawn from sealed personnel records: childhood compartment, Guild examination error, a private medical procedure, an incident involving Eda. Ilyan read the final item and turned the slate facedown.

"What?" Hadran asked.

"Half of these facts could have crossed before divergence."

"That establishes continuity."

"It establishes access to my record."

"You propose another method?"

"Shared memory not held in systems. Then facts that differ after the correction."

Sava considered the distinction. "Proceed. Avoid data that compromises anyone else."

Ilyan almost laughed. Any shared memory compromised someone else. Private life did not become an authentication key because security lacked imagination.

He began with a broken heater in trainee quarters. At twenty-one, he had spent three nights tapping complaints through a pipe because maintenance kept closing the work order. The other Ilyan named the senior technician who finally arrived and the insulting sketch she drew beside the stuck valve. Both remembered the sketch. Neither archive did.

Next came their conservator's oral examination. Ilyan quoted the examiner's unfinished question about a route that missed Pelagos. Violet's Ilyan supplied the answer they had both given: a late ship still contained duty even when destination capture failed. The examiner had called it sentimental and passed him anyway.

Hadran demanded something harder.

Ilyan described the departure verse sung after Correction 141 without giving the mistaken word. The voice across the seam named it. Someone in Navigation had replaced harbor with hunger, causing half the room to stop singing and the other half to continue out of pity. The event had occurred after route instantiation but before settlement completed, close enough to divergence that both ships claimed it.

"Now ask what happened afterward," Teren said.

Ilyan chose an impersonal detail. "At release, what broke in the chart room?"

"Our west route pane cracked at the lower brace," the reply said. "Jo's bypass held, but the impact surge reached Drum Six. Your pane?"

Ilyan looked at unbroken smoked glass. "No crack. Our bypass returned load through the waste bus. No impact reached our habitat."

A pause crossed from Violet. The far Ilyan spoke away from the pickup. Ilyan could not catch the words, but several people answered.

"Then divergence began before our correction card stamped settlement," the voice said. "We remember your green trace brightening. You do not remember our brace breaking."

Teren wrote both events against the field clock. The shared account reached Correction 141. The differing damage followed within seconds. No stored copy could gain memories of a cracked pane that did not exist here.

Hadran still resisted. "A generative system could infer damage from route projections."

Jo replied, "It also invented people arguing behind the microphone?"

"It may simulate social noise."

Violet's open channel carried a child asking whether school could have the gallery after second meal. An adult answered that medical triage still occupied the lower steps. Someone farther away requested a marker. The child objected that all the markers had gone to pressure crews.

Ilyan touched transmit before Sava authorized it. "How many children moved from Drum Six?"

Hadran reached for cutoff. Sava stopped him.

The answer came from a woman, not Ilyan's counterpart. "One hundred fourteen assigned to those school rooms. Nine absent for injuries or family work. We are combining ages because we have three teachers on medical shifts."

"Are classes active?"

"They begin again after meal. We lost the morning."

Her voice carried exhaustion, annoyance, and no need to persuade. Ilyan pictured the Commons well transformed by necessity: civic desks pushed aside, children on terraced steps, triage screens occupying the lower gallery. The image came from Nacre's architecture, which they shared. Violet had filled it with a day he had not lived.

Sava took the transmit key. "This is Captain Sava Or. Who holds command aboard Violet?"

Several voices conferred. Their reply did not come from a captain.

"Captain Or retains vessel command," Violet's Ilyan said. "Emergency ration and settlement decisions require our Commons council. We have not agreed on authority over this contact."

Sava's mouth tightened. "Why are you speaking for them?"

"I had chart access when your first pulse arrived. I also know how you test records."

"You share Conservator Rook's rank."

"I shared it before settlement. Violet's Guild seal server failed during the surge. Rank has not opened infirmary doors."

The answer disturbed Ilyan more than perfect resemblance. Across the seam, another version of him had lost the shelves that gave his caution purpose. He had learned emergency teaching, ration argument, perhaps command by refusal. One day had already placed different tools in their hands.

Sava asked for their captain. Violet's group refused immediate transfer of the pickup. A technical officer wanted proof Nacre would not use the call to hasten closure. A Commons voice demanded Jo's heat test. Someone else argued that every second of discussion spent reserve.

Hadran called the refusal evidence of manipulation.

Ilyan heard a civic body improvising under a countdown.

Teren gave them twenty minutes more before seam heat reached the Article 4 limit.

The contact became an exchange of constrained facts. Nacre confirmed the feed through Capacitor Seven. Violet confirmed matching hardware damage at the fused jaw, though their side had cracked two ceramic spacers. Jo described a safe way to brace the housing without sending code. Violet's engineer repeated the instruction in their own words before touching anything.

The return load steadied.

Sava permitted a list of immediate needs. Violet requested atmosphere sorbent patterns, trauma triage guidance, and a settlement forecast. Nacre could send knowledge but no mass. The limitation made every offer sound meaner than silence. Jo could explain how to save a pump but could not pass a seal through the seam. Teren could compare reserve curves but could not place one grain ration in a Violet bowl.

Ilyan recorded each request under disputed joint custody. The archive rejected Route Violet as a legal sender. He assigned the ship a temporary witness number, as he had done for the raw sensor stream. This time the workaround felt inadequate. A population should not depend on a clerk choosing the right error code.

The far Ilyan noticed the custody prompts in Ilyan's spoken confirmations. "Your record still names us counterimage?"

"Yes."

"Ours names you designated receiver."

"Does your Commons accept that?"

"No."

"Do you?"

Static filled the delay. "I accepted it yesterday."

The honesty passed between them without comfort.

Ilyan thought of his Guild seminar answer: a route not taken had no bereaved families because no one traveled it. His counterpart had made the same defense, received the same seal, and taught the same categories. Now one sat under a clean route pane while the other worked beside a crack.

"We need a means to distinguish speakers in the record," Hadran said. "Nacre Ilyan and Violet Ilyan at minimum."

The far voice sharpened. "We can distinguish ships without making one person an adjective."

Hadran muted receive. "He anticipated the problem."

"So did you," Ilyan said.

"Because it presents an obvious identity risk."

"No. Because labels assign standing before evidence does."

Sava restored receive. "Route Violet, propose a record form for this session."

"Two custodians named Ilyan Rook, each attached to a ship and timestamp. No numbered person. No original designation."

Sava did not accept. She also did not refuse. "Provisional pending counsel."

The channel reopened on both sides for the final minute so each custodian could hear the other without the cutoff. Their voices met only as sound. No touch, no shared room, no crossing of matter. Yet Ilyan felt the strange crowding of a name occupied twice.

He prepared one final verification question. Hadran's slate still offered sealed personal facts. The medical item would prove access to his body. The childhood compartment would prove a database. The remaining prompt named his younger sister.

Across the seam, Violet's Ilyan began speaking at the same moment.

"Eda," they both said.

The name overlapped with itself and produced a soft phase beat.

Ilyan released the key. His counterpart did too. For several seconds neither side transmitted.

Hadran spoke first. "Ask the associated question."

"No," Ilyan said.

Violet's voice arrived after the seam delay. "No."

Sava looked between the speakers. "Explain."

Ilyan kept his hand away from the control. "Eda died six years before Correction 141. We share her because we share that loss. Her death proves no hierarchy between us."

The other Ilyan added, "Do not use her to decide which grief counts as a copy."

Hadran objected that refusal weakened authentication. Sava let the objection enter the record and ordered no answer.

Teren closed the voice band as the heat limit arrived. Violet's people receded into static: a teacher asking for the gallery schedule, an engineer reading capacitor temperature, someone beside the far pickup breathing before speech.

On Ilyan's glass, the temporary witness field demanded a primary holder. He left it unchosen. Instead he created two custody entries with separate ship clocks and equal authority over the contact record.

The software marked both entries disputed.

Across the seam, Violet sent the same phase pattern twice, then waited. Ilyan answered once from Nacre. Two ships now occupied the record without Eda standing between them.

Chapter 06 — The Price of Correction

<!-- POV: Teren Vale -->

<!-- SETUP: Violet contact; settlement accounting; Teren's apprenticeship -->

<!-- PAYOFF: conservative erased-person estimate; institutional knowledge admitted -->

Teren had the reserve manifold open before the voice channel to Violet went dark.

He worked from the floor of the causal-physics room, one knee against pearl composite, with four centuries of energy notation stacked in panes above him. The newest accounting line still glowed amber. Correction 141 had received eleven settlement credits, each folded into the designated route's maneuver reserve under a single authorization seal.

For most of his career, that line had meant efficiency.

Now a man on Violet had spoken Eda Rook's name.

Teren widened the accounting tree until its branches filled the wall. Field initiation sat at the top: the energy required to instantiate twelve bounded continuations around the ship's matter and momentum. Beneath it came route maintenance, seam suppression, outcome ranking, designation, and closure. The final row compressed everything unwanted into a cheerful green quantity labeled RECOVERED DIFFERENTIAL.

He removed the label.

Numbers replaced it. Heat first. Rotational momentum next. Chemical potential from food stores, batteries, algae tanks, bodies. Probability reserve occupied the last column, a term old architects had used when ordinary units failed. Teren had taught that column for thirty-two years without asking what continued to produce its values after official route duration reached zero.

Behind him, the hatch admitted Captain Sava Or with Ilyan Rook at her shoulder. Jo Malk arrived before the lock completed its cycle, carrying a drive key that security had apparently failed to confiscate.

Sava remained by the door. "You asked for command witness."

"I asked for witnesses," Teren said.

The correction came out sharper than he intended. Shame had made his voice precise. It also made him explain when silence might have protected what dignity remained.

Ilyan looked past him to the wall. Fatigue had tightened the skin around his eyes. He had spent hours speaking across a seam to another version of himself, and Teren could not decide whether sympathy from him would count as kindness or theft.

"Does Violet's channel remain stable?" Teren asked.

"For another forty-seven minutes at the present draw," Jo said. "Less if somebody orders a full dump."

Sava's expression did not change. "No such order has been given."

"Good. Machines prefer direct wording."

Teren rose with help from the manifold rail. His left knee resisted, then released with a small internal click. Sen Quill used to mock that knee during apprenticeship, claiming no physicist should possess a joint more temperamental than a first-generation field jaw. Teren had enjoyed the joke. He had enjoyed almost everything Sen said, which now seemed like another item requiring audit.

He pointed to the green credit under Correction 141. "Violet crossed a charged-dust fold. Amber lost food cooling. White missed capture. Those outcomes differed, but settlement treated each ship as a reservoir once designation ended."

Ilyan moved closer. "A reservoir of what?"

"Everything its route still held above closure state."

Jo planted the drive key on the floor beside Teren. "Closure state meaning dead."

"Meaning no independent causal activity."

"That is how physics says dead when command might be listening."

Teren accepted the rebuke. "Yes."

The room's old pumps moved coolant through the wall in slow knocks. They had been installed during the Loom's third refit, before Teren's birth, and no two carried the same pitch. He usually heard disorder. Tonight each knock seemed to arrive from a separate place that wanted counting.

Sava crossed her arms. "Show causation, not guilt."

Her demand steadied him. Guilt could spread without limit and illuminate nothing. Causation required edges.

Teren collapsed the display to three quantities. "The field creates route domains by borrowing against the maneuver reserve. While selection remains open, all routes share the cost. Once the Loom designates one route, settlement takes the remaining differential from every other domain and transfers it here." He touched the final column. "To us."

"Official doctrine calls those domains instantaneous," Sava said.

"Official doctrine makes an assertion about duration. The accounting makes a different one about work."

He called up the raw draw curve from Violet. The line did not drop at selection. It continued beyond the declared endpoint, declining over hours as life support, cooling, and human use converted accessible reserve into waste heat. Jo's bypass had preserved enough seam power to reveal that decline.

Ilyan followed the curve with one finger. "It falls when their systems draw."

"Yes."

"And settlement waits."

"Settlement has always waited." Teren enlarged the hidden interval between designation and extraction. "The interface displays zero because this value sits outside mission time. Internally, the stack schedules closure after persistence attenuation reaches its lowest safe harvest threshold."

Jo stared at him. "Nineteen days and six hours."

Teren nodded.

Sava took one step from the door. "You knew the interval."

"I knew an attenuation constant."

"That answer will not survive the Commons."

"It should not survive this room."

He turned back to the panes before their faces could stop him. Correction 141 offered only the newest proof. One correction could still invite arguments about contamination, bypass artifacts, or an unprecedented field condition. One hundred forty-one made a history.

The archive resisted his first request. Founder permissions layered themselves over the energy books, each restriction written in obsolete professional ranks. He supplied his chief-physicist seal. The system granted summaries and denied raw conservation tables. He supplied Sava's witness token. Another century opened.

A row from Correction 76 appeared, then 55, then 12. Every ledger carried the same shape: initiation debt, shared maintenance, designation, a flat interval officially assigned no duration, and a settlement credit arriving long after selection. The credits varied with route damage. Ships that retained working reactors yielded more. Ships that lost pressure yielded less. Routes predicted to contain broad crop reserves produced larger transfers than routes with failed agriculture.

Household arithmetic had always helped Teren think. A battery behaved like a pantry: what entered, what left, what spoiled where no one could eat it. These ledgers described a household that filled its cupboards by waiting outside neighboring doors, then entering when the clocks stopped.

He shut that comparison down before it could become a refuge. People did not become clearer when reduced to kitchens.

Ilyan read the old rows in silence. Jo crouched by the manifold and traced the physical bus that carried recovered differential into Nacre's reserve braid. Sava stood where she could see all three of them and the locked hatch, still arranging the room into obligations.

"Can this be a delayed calculation?" she asked. "A value recognized later but transferred at designation?"

Teren had prepared for that question because he had once used it himself.

He isolated Correction 88. One rejected route suffered an early coolant breach. Its settlement credit declined in twelve discrete steps after designation. The timing matched emergency pump cycles preserved in supposedly meaningless telemetry.

"The value changed after the route ceased to exist under doctrine," he said. "Something in that domain continued making physical choices. Pump on, pump off. Heat moved. Work occurred. The designated Nacre received less reserve because another Nacre spent it trying to live."

No one spoke.

Teren opened Correction 103. A route classified as fatal retained power for almost the full persistence interval. Its credit dropped sharply near the end, recovered for eleven minutes, then collapsed. The official note called the variation echo decay.

Jo leaned closer. "That recovery looks like a restart."

"It does."

"Manual?"

"The timing exceeds automated reset tolerance."

"A crew restarted something."

Teren let the sentence stand. The ledger did not contain names. It contained effort stripped of actors, enough to prove action and too little to honor anyone who performed it.

Ilyan asked, "How far back can you calculate population?"

There it was. Teren had hoped for one more procedural question before the count.

"Not exactly."

"I asked how far."

"To the first live trial, if command releases founder tables. Reliably, perhaps Correction 20 onward. Earlier complement records use family units rather than individuals."

"Then estimate."

Sava said his name in warning. "Ilyan."

"No," Teren said. "He is right."

He built the estimate where they could watch every assumption. Current complement could not simply be multiplied by route count. Nacre had carried fewer people during the early mission. Birth waves, crop failures, quarantine years, and cryogenic transfers changed the population. Some domains failed immediately during impact or pressure loss. Others remained habitable until settlement. Stored embryos could not be counted as persons without changing the measure, so he excluded them. Pregnancies and births inside the persistence window remained unknowable, so he excluded those too.

For each correction, he drew the official ship complement on that date. He multiplied it by the number of route domains that retained life support beyond one hour. Where route counts remained sealed, he used the lowest number permitted by initiation energy. Where survival duration remained ambiguous, he counted only domains with post-designation work signatures.

The total began low enough to encourage disbelief.

Correction 141 added hundreds of thousands of person-instances. Teren hated the term even as he entered it. A person-instance meant one continuing human history inside one route, not a fractional citizen and not an accounting duplicate. The words protected the estimate from double-count arguments. They did nothing for the people inside it.

Older corrections accumulated beneath. Four routes here. Nine there. A maneuver during the forty-third ship-year produced eleven viable continuations before settlement. Another, during the long hydroponics fever, created only five. Each row took the population that had eaten, worked, quarreled, and slept aboard that version of Nacre and added it to the lower bound.

One archive row preserved nothing but a sanitation draw from Correction 64. The rejected domain had pumped waste for thirteen days after designation, spending reserve in regular crew-shift intervals. Teren had reviewed the curve as a student and praised its clean periodicity. Now he could see three shift changes, a clogged line near the middle, and a repair that restored flow before settlement took the remaining heat. No names survived in the physics table. The work itself did. Someone had cleared that line so a ship already condemned by another ship's map could keep its toilets working.

At ten million, Sava moved to the nearest chair.

At fifteen, Jo sat on the floor.

The archive withheld exact route counts for seven founder-era corrections. Teren assigned each the minimum of four viable domains established by the Loom's own initiation draw. He removed all routes that suffered catastrophic hull loss in the first hour. He excluded uncertain signals even when his instincts recognized their curves.

The conservative sum stopped at 18,217,604.

Teren changed the display from digits to words because long numbers encouraged eyes to skip. More than eighteen million erased person-instances. The lower-bound note appeared beneath in smaller type.

Ilyan read it aloud. His voice did not break. "Eighteen million."

Sava pressed both hands flat against her knees. "Across one hundred nineteen years."

"One hundred forty-one corrections," Teren said. "The ship celebrated each as a danger passed."

Jo looked up from the bus. "How much of our remaining arrival reserve came from them?"

Teren opened the mission reserve history. The answer resisted clean attribution because harvested energy had mixed with reactor output, braking recovery, and ordinary thermal capture. He could trace source shares at each correction, then decay them through later expenditures. It took nine minutes and three rejected models.

"Forty-one percent of current correction reserve originated in settlement transfers," he said. "The fraction rises above half if we include maneuver losses avoided through transferred momentum."

"Arrival?" Sava asked.

"Yes."

The captain looked toward the small gold marker for Pelagos. In thirty-one days, under the plan she had inherited, that marker promised Lattice water, open sky under a new sun, and the end of a voyage begun before anyone aboard could remember Earth. Teren had spent his life making the promise calculable. He had never asked who paid the reserve that made calculation true.

Sava stood. "Can we reach Pelagos if settlement stops now?"

"Not on the certified corridor."

"Can Violet?"

"Their reserve state appears worse."

“Can both ships survive?”

“I do not know.”

Jo's jaw tightened. “Better wording.”

“I cannot yet describe a maneuver that preserves both without spending reserve neither ship possesses.” Teren faced them. “That describes my limit tonight. It does not define physics forever.”

Sava accepted the distinction with a small nod. “Then your next task concerns alternatives.”

“No.”

The answer surprised all of them, Teren included.

He had served four captains and contradicted each in private. Public refusal felt like stepping onto route glass without checking load. Yet another calculation had waited behind the population count. It concerned not energy but custody.

“My next task concerns provenance,” he said. “Before I offer a repair, you need to know why I could build this estimate quickly.”

Ilyan turned from the display.

Teren reopened his apprenticeship archive. The files carried Sen Quill's seal, along with notes from three instructors whose portraits hung in the physics gallery. Sen had inherited the Loom from Alda Quill's generation and spent his last years standardizing settlement mathematics. His lectures used the same attenuation curves now exposed on the wall.

A lesson recording opened in Teren's own younger voice.

“Residual domain activity creates noise after designation,” the apprentice said. “Safe extraction waits until autonomous work approaches the floor.”

Sen replied offscreen, patient and amused. “Not noise. Persistence. If you call it noise, you will mismeasure it.”

The recording continued. Young Teren asked whether persistence implied an internal clock. Sen paused for six seconds, then redirected the lesson to capacitor shielding.

Teren stopped playback.

Jo's anger arrived without volume. “You asked.”

“Yes.”

“And he avoided it.”

“Yes.”

“Did you ask again?”

“No.”

He opened a second note. Sen had marked a proposed formula with red script: NEVER ASSUME ZERO

INTERNAL DURATION. Beneath it, another hand had added: EXTERNAL STATUS REMAINS COUNTERIMAGE. USE APPROVED TERMS IN COMMONS MATERIAL.

Ilyan recognized the signature before Teren named it. "Navigation Assurance."

"An earlier office under the same charter."

Sava's gaze hardened. "Your teachers knew persistence existed."

Teren felt the room narrow around the sentence. "Yes."

"Did they know people persisted?"

"Sen knew work continued. He knew the interval carried an internal clock. Alda Quill's sealed tables suggest the first architects received communication, though I cannot open that layer yet."

"That does not answer my question."

"No. It tells you the boundary of what I can prove." He forced himself not to decorate the confession. "I suspected enough to ask once. When he redirected me, I accepted the permission not to understand. Later I taught the same curves as field behavior. My students learned to calculate settlement without imagining anyone spending the energy we took."

Ilyan watched him with the exhausted attention of a man refusing easy hatred because easy things had become suspect.

"How many of your students approved corrections?" he asked.

"Most senior physicists aboard."

"Including you?"

"Twelve directly. I certified models for forty-three more."

Jo stood and picked up the drive key. "Then no one touches that manifold alone again."

Teren removed his chief-physicist token from his collar. Its black ceramic edge had worn smooth where his thumb rubbed it during difficult calculations. He placed it beside the open reserve bus.

"You should take it," he said.

Jo did not. "I repair machines. Oru Pell can tell us what to do with people who made policy look like a gasket."

Sava moved to the console and copied the estimate into a sealed command record. "This remains under Article 4 until I establish how disclosure affects ship safety."

Ilyan gave a short, humorless breath. "The figure describes ship safety."

"It describes a crime and an engineering dependence. I need both intact long enough to act."

Teren heard fear in the extra formality of her phrasing. He also heard the old institutional rhythm: preserve the mission by controlling the noun. Crime could wait. Dependence required command. People outside the room remained an effect to be managed.

“Captain,” he said, “the accounting already entered Commons custody when you authorized a witness session.”

Sava looked toward him.

He pointed to the room seal. By old research law, any command-witnessed conservation proof that changed the meaning of a public resource required a Commons digest. The rule had been designed for ration fraud, not hidden causal domains, but categories could sometimes open doors as well as close them.

“You arranged this,” she said.

“I requested witnesses.”

For a moment Teren expected her to revoke the session and challenge the classification. Instead she read the population total again. Her hand hovered above the command pane, then lowered.

“Release the digest without voice-channel details,” she said. “Violet's access remains protected.”

Ilyan objected. “Their existence cannot become a redaction.”

“Their channel can become a target. I am protecting the seam for forty-seven minutes, which is what remains within my power tonight.”

Jo checked the draw timer. “Thirty-nine now.”

Sava left without reclaiming Teren's token.

The three men and Jo remained with the open manifold. Teren prepared the Commons digest. He included every assumption, every excluded uncertainty, the archive denials, and the lower-bound method. He attached Sen's recording. At the last field, the system requested a title.

CORRECTION RESERVE PROVENANCE sounded neutral. SETTLEMENT FATALITY ESTIMATE sounded honest but incomplete. Neither captured the structure: their safe line to Pelagos carried energy gathered from ships that had worked against dying.

He titled it THE PRICE OF CORRECTION.

Ilyan stood beside him while he sent it. A route map occupied the far wall. Pelagos shone ahead in gold. Behind Nacre, one hundred forty-one maneuver marks formed a clean white chain through the dark.

Teren changed the overlay.

The marks opened into rows. Each correction displayed its minimum routes, post-designation work, harvested reserve, and erased population. White numbers filled the space behind the ship until the plotted course nearly vanished among them.

Arrival safety no longer read as a green line.

It read as a ledger of deaths, and the first public copy had begun to travel down through the Commons well.

Interlude I — Ledger of Vanished Ships

<!-- FORM: fragments decoded from Alda Quill's hidden wake ledger -->
<!-- SETUP: ordinary route traffic preserved inside maintenance intervals -->
<!-- PAYOFF: the interval key resolves to KEEP BOTH -->

WAKE TABLE 0 / FIRST LIVE TRIAL / OCHER

Official settlement mark: 03:14:22.000

Bell maintenance offset: +00:00:00.731

Isolation-jaw inspection delay: +00:00:04.008

Cooling-braid tolerance: variable under extraction load

Received service traffic, partial:

> Who is at the auxiliary bell?
>
> Alda, did we keep Route One?
>
> Our lights are down. The drum feeds don't answer.
>
> Tell us what to do.

Engineering note: secondary rail strike demonstrates a separate clock. Command disposition lists echo decay.
Raw reel removed. Timing retained in component intervals.

Final audible record:

> I can hear another bell.

WAKE TABLE 3 / CORRECTION 3 / SERVICE TAG CEDAR

Route duration entered by command: 0.000

Measured post-designation work: 02:08:17 minimum

Maintenance channel 6, recovered from pump harmonics:

> Galley Two to water crew. The pressure has come back in the left tap. Do not open both taps together. We have one clean pan and need it for noon meal.

Attached recipe, written into valve-cycle lengths:

Two measures dried bean flour. One measure algae mash. Hot water until the dough stops taking it. Rest beneath a covered tray for thirty minutes. Cook on the unwarped side of the plate. If the plate has warped on both sides, turn the cakes by hand and accept the burn.

Anonymous correction below recipe:

Not thirty minutes. Twenty-two. The drum is cooling.

WAKE TABLE 8 / CORRECTION 8 / SERVICE TAG BLUE

Official route duration: 0.000

Measured post-designation work: 06:44:09 minimum

Work order transmitted on emergency lighting bus:

1. Brace nursery hatch with two short rails.
2. Move sleeping mats away from condensation wall.
3. Bring all intact heat packs from clinic store.
4. Do not tell the children the daylight panels have failed. Tell them the morning schedule changed.
5. Find someone who knows the second verse of "Engine Under Water." First verse no longer distracts them.

Completion marks indicate items one through three performed. Item four remains unmarked. Beside item five, a different hand entered:

> There has never been a second verse. Pell made one up. It rhymes water with daughter twice, but they laughed.

WAKE TABLE 12 / CORRECTION 12 / AMBER-CROSS

Settlement credit: 7.14 reserve units

Measured post-designation work: 19:05:51 minimum

Recovered personal queue, order preserved:

> Toma, the west green belt has more light than ours. Take the basil tray there before the next dimming.

> I already moved it. You should apologize to the roots for doubting us.

> Roots do not accept apologies.

> Neither do I. Bring string.

Later queue fragment:

> The basil held. Two leaves blackened, nothing else. I tied the stems to spoon handles because the string went to pressure crew. Before you complain, count the spoons in your own locker.

No reply retained.

WAKE TABLE 19 / CORRECTION 19 / GREY

Route count under founder seal.

Habitable-domain minimum inferred from heat draw: five

Sanitation work trace, Drum Four:

A regular twelve-minute pump cycle continued for nine days. On day ten, the third pump stopped. Manual clearing began fifty-three minutes later. Tool impacts match a ceramic scraper followed by a steel wrench. Flow resumed after two hours, eleven minutes.

Maintenance annotation:

> Whoever services this after us: access the trap from the lower panel. Upper bolts only pretend to belong to the machine.

Second annotation:

> Mara says the bolts were installed by a committee.

Third annotation:

> Mara serves on a committee.

WAKE TABLE 27 / CORRECTION 27 / RED-SMALL

Official cause: food-cooling cascade

Measured post-designation work: 11:17:03 minimum

Cold-store inventory copied into compressor rests:

Peas, six trays.

Yeast stock, two viable envelopes.

Infant formula, forty-one sealed tins.

Pear slices from Departure orchard, one jar. Ownership disputed between Galley One and the Historical Table Society.

Inventory amendment:

Pear jar opened by joint consent at 18:20. Twenty-three portions served. Historical Table Society retains lid.

WAKE TABLE 41 / CORRECTION 41 / WHITE-CANTED

Duration preserved in the official ledger: 0.000

Measured post-designation work: 18:22:12 minimum

Classroom packet carried on climate-control diagnostics:

Problem: If one sky clock reads twelve and another reads twelve plus seven, which clock starts the game?

First answer: The lower number.

Teacher's mark: That tells you which clock reads less. Try again.

Second answer: The clock that sends first.

Teacher's mark: How would either room prove it sent first?

Third answer, unsigned: Both clocks wait. Someone taps. Someone answers. The game starts in the delay.

Packet checksum repeats through settlement approach.

WAKE TABLE 55 / CORRECTION 55 / ROUTE TAG WITHHELD

Archive gap: 00:13:00 through 03:09:44

Recovered infirmary order:

> Use the blue antiseptic before the clear bottle. Clear stock belongs to surgery. Do not save gauze by folding contaminated edges inward. We have cloth, heat, and hands. Sterilize more.

Margin response:

> Heat ration fell again.

Order amendment:

> Then cut clean cloth now. Put the autoclave on the reactor pulse at 04:10. Wake me before you spend surgery stock.

No name survived the channel compression. Pulse draw confirms one autoclave cycle at 04:12.

WAKE TABLE 64 / CORRECTION 64 / GREEN-NORTH

Recorded messages during final six hours:

> Maintenance to Commons: the inner tram can carry thirty-two per trip if everyone leaves luggage. Mobility chairs load at the front. Stop using the rear door; its floor rail has lifted.

> Commons to Maintenance: who has authority to assign seats?

> Maintenance: no one available. We are loading by need.

> Commons: define need.

> Maintenance: come to the platform and help.

Passenger marks continue for fourteen trips. The final list contains no family names, only cabin numbers and equipment required at destination: oxygen, insulin cooler, hand brace, child sling, translation board.

WAKE TABLE 76 / CORRECTION 76 / VERMILION

Private message embedded in a carbon-scrubber fault report:

> I said I had checked the south seals. I checked three and signed for four. I thought shift change would catch the last one. Then the correction alarm came and you took that corridor because my board showed green.

>

> I have written your name six times and deleted it six times because writing it makes this look like a memorial, and you may still be waiting behind a hatch I cannot open.

>

> The fourth seal had a torn lip. I found it after pressure dropped.

>

> I am sorry, Lio. I am

Transmission ended before the final word or punctuation.

Fault report disposition: incomplete.

WAKE TABLE 88 / CORRECTION 88 / SERVICE TAG MOSS

Children's meal amendment:

Bean cakes removed after heater failure.

Replacement: cold algae wraps with pepper paste.

Complaint count: seventeen.

Revised complaint count: twenty, after three adults requested inclusion.

Galley response:

> Pepper paste has been reduced. Complaints about the reduction should be filed before settlement because the afternoon clerk refuses work after death.

A later hand changed death to second meal. The overwritten word remains legible in the timing layer.

WAKE TABLE 103 / CORRECTION 103 / BLACK-WEST

Power restart at +18 days, 21:02:11.

Manual sequence inferred from bus load:

Close clinic feeder.

Open nursery bypass.

Hold reactor start for eleven seconds.

Release before braid temperature crosses red.

Restart successful. Habitat circulation continued for eleven minutes.

Recovered voice fragment:

> Again.

Second restart attempt produced no sustained current.

WAKE TABLE 119 / CORRECTION 119 / PALE GOLD

Domestic queue, seven messages:

> Your red shirt remains in my locker.

> It is not red. It faded before we met.

> Then your formerly red shirt remains in my locker.

> Keep it.

> I do not want it.

> That has never stopped you keeping anything.

> Fair.

An eighth message carried only a laundry-cycle code. The machine ran one wash after the queue ended.

WAKE TABLE 141 / CORRECTION 141 / VIOLET

Official status: settled

Continuing seam load: confirmed

Received timing response: seven intervals copied within 0.004 seconds

Received voice identification: Ilyan Rook, Nacre, Route Violet

Ordinary traffic retained beside command channel:

> School Four requests two more lamp cells. Children have started lessons in the east passage because the classroom wall sheds frost.

> Ration crew can spare one cell.

> Send it. We will move the younger class closer to the board.

The ledger key follows Alda Quill's original maintenance convention. Bell lubrication gaps provide consonants. Braid-cooling rests provide vowels. Isolation-jaw intervals mark spaces.

Decoded interval sequence:

11 / 05 / 05 / 16

02 / 15 / 20 / 08

KEEP BOTH

Chapter 07 — The Child on Indigo

<!-- POV: Teren Vale -->

<!-- SETUP: second route contact; Alda Quill's family name; uneven seam draw -->

<!-- PAYOFF: Nessa's two-sky game demonstrates symmetric phase exchange -->

Teren found the child's counting inside a failed calibration tone.

The Indigo seam had opened fourteen minutes earlier as a blue-white scratch near the lower edge of the route glass. It carried no voice, only bursts that the Loom's filters classified as thermal chatter. Teren sat at the auxiliary console with his chief token removed from his collar and Jo Malk standing behind him as custodial witness.

He sent a steady three-second tone. Indigo answered with two short pulses, a pause, then four.

“Filter it again,” Jo said.

“I have filtered it three ways.”

“Then stop filtering.”

Teren disliked advice that sounded simple and worked. He bypassed the causal-physics stack and fed the raw seam into an old pressure speaker. Static filled the chart room. Beneath it, someone tapped a quick pair against metal, waited, then tapped four more.

Two. Four.

Another sequence followed.

Three. Six.

“Doubling game,” Jo said.

“Or an amplifier fault.”

The speaker gave four taps and waited.

Teren touched the transmit plate five times, slowly enough to separate each load.

Indigo answered with ten.

Jo leaned over him. “Your amplifier can count.”

Teren opened the narrow voice band. “This is Dr. Teren Vale aboard designated Nacre. We hear your pattern. Can you receive speech?”

A burst of static crossed the room. Then a girl's voice, compressed until some consonants vanished, said, “You made us wait through twenty-seven wrong taps.”

Teren took his hand from the plate.

Jo smiled without amusement. “An amplifier with opinions.”

The girl continued before either adult answered. "You also counted my fives wrong. There were fifteen, not fourteen. The third group fell under your long tone."

Teren replayed the raw channel. She was right. He had mistaken one close pair for a seam reflection.

"My correction," he said. "Fifteen."

"You have to say the number back or we don't know you heard it."

"Fifteen."

"Good. Now you can play."

A second voice spoke behind hers, too far from the pickup to resolve. The girl answered away from the seam, "I know they're adults." After a pause, she added, "That does not mean they know the rules."

Teren adjusted the speaker gain. "Who are we speaking with?"

"Nessa Quill. Eleven. School Four on Indigo. Who else is there?"

His hand stopped above the level control.

Quill belonged to several families aboard Nacre, but Alda's line had remained small. Sen Quill, Teren's teacher, had carried no children of his own. A niece had joined hydroponics during Teren's apprenticeship. Another descendant had moved into classroom maintenance two generations later. He had not followed the names after Sen died. The thought that genealogy might establish authority now repelled him. The child on the other side had already established herself by counting better than he did.

"Jo Malk, drive engineer," Jo said toward the pickup. "Teren has a witness because physicists need supervision."

Nessa considered that. "Teachers do too."

"Agreed."

Teren heard himself preparing a question about her parents, her classroom, her relationship to the founder. He discarded all three. "You called this a game. What game?"

"The two-sky game. You don't have it?"

"No."

"Then how do your classes practice seam delay?"

"They do not have seams."

A low exchange occurred on Indigo. The distant voice came closer, an adult this time. "Nessa, let me speak."

"You said I could finish one round."

"You found the channel. Finish one round."

Something scraped beside their pickup. Teren imagined a classroom desk dragged toward a service panel, but he could not know. The seam carried timing and ruined sound, no picture. Any image his mind supplied would

borrow from Nacre and claim more sameness than evidence allowed.

Nessa spoke clearly. "We started playing after the correction because our sky clock and the wall clock stopped agreeing. The sky clock means the route line. It comes on the classroom ceiling when power lets it. One side has the plain sky. One side has the late sky."

Jo asked, "Which side are you?"

"That question loses."

"Why?"

"Because both sides call themselves plain."

The answer landed with the clean force children sometimes achieved by refusing an adult premise rather than defeating it.

Teren displayed the incoming phase. Indigo's pulses arrived 1.12 seconds behind Nacre's transmit clock, then wandered as reserve draw changed. Unlike Violet's steady 0.73-second delay, Indigo's seam breathed. Each burst consumed charge from both domains. If Nacre spoke continuously, Indigo paid almost as much to listen as Nacre paid to send.

"How did you discover the other sky?" he asked.

"School Six tapped us."

"On your ship?"

"No. On the late ceiling." Nessa sounded impatient now. "Our classroom wall showed a blue line after settlement. Teacher Pell said it would fade. We tapped the bell schedule so it would fade less boringly. Another class tapped the same schedule back, but late."

The adult near her said, "We believed it to be stored field behavior at first."

Nessa cut in. "I didn't."

"You believed it contained children because they cheated at your counting game."

"They changed seven to nine after I won twice. A machine would cheat better."

Jo covered a laugh with one hand.

Teren asked for the game from the beginning. Nessa refused to explain. She said rules given before play became adult rules, which were usually descriptions of how adults had already won. She would demonstrate instead.

"Put your hand where the sky can hear it," she said.

Teren rested two fingers on the transmit plate.

"I send a number. You wait until you hear it. Then you send the same number plus one. But you don't count from your clock. You count from my last tap."

"That produces accumulated delay."

“You haven't played yet.”

A pulse arrived, then another: two taps separated by a quarter second.

Teren waited through the seam lag and answered with three, using her second tap as the beginning of his interval. Indigo replied with four. He sent five.

After six exchanges, the incoming phase stopped wandering.

He assumed the Loom had stabilized the channel. Then he looked at the reserve meters. Indigo's draw rose when Nacre transmitted and fell when Nacre listened. Nacre's draw traced the inverse. The two curves crossed at each handoff and balanced over the full round.

Nessa sent eight.

Teren answered with nine.

“Stop,” she said.

He lifted his fingers.

“What changed?” she asked.

“The average phase load equalized.”

“Wrong.”

Jo pulled up a chair. “What answer do you want?”

“I don't want an answer. I want him to look.”

Teren looked again. The reserve curves had equal areas, but that alone did not explain stability. He widened the phase display. Each participant used the received pulse, not a local clock, to place the response. Delay therefore entered the game from both directions. Neither side attempted to erase it. The interval became part of the shared sequence.

“The seam stopped choosing a master clock,” he said.

Nessa gave a satisfied breath. “Now you can play.”

Her teacher introduced herself as Mera Pell and requested command contact. Teren promised to bring Oru Pell from the Commons, then corrected the promise: he would request Oru's presence, not transfer authority he did not possess. Mera noticed the change.

“Are they your clerk?” she asked.

“No. They serve the Commons Assembly here.”

“Here,” Mera repeated. “We also have an Oru Pell.”

Jo glanced toward Teren.

The branching point lay only days behind them. Of course Indigo carried its own Oru, its own Commons, its own children who had begun the correction expecting arrival in thirty-one days. Facts this obvious still arrived

as blows because doctrine had trained obviousness out of him.

“Does your Oru have standing to speak with ours?” Mera asked.

“Our law contains no procedure for that yet.”

“Then find one. Our settlement clock has sixteen days left.”

Teren checked Indigo's attenuation curve. The teacher's estimate matched. Their reserve declined faster than Violet's but slower than several weak signals he could not yet classify. They had enough seam power for hours if both sides kept traffic sparse. A command channel could spend that time in ten minutes.

“We need to reduce voice use,” he said. “The seam draws reserve from both ships.”

Nessa returned to the pickup. “That is why we play.”

“Your game carries less information than speech.”

“It carries when someone is there.”

Teren had no reply that improved on hers.

Mera asked Nessa to leave the pickup while the adults discussed reserve. The child refused without theatrics. “The seam is in our classroom wall. If you wanted a room without children, you should have opened a different one.”

Jo coughed into their sleeve.

“What happened to School Four during the correction?” Teren asked.

“Nothing during it,” Nessa said. “We had a geometry test. The route alarm came before lunch, so Mera canceled the last page. Afterward, the east wall started frosting. The heat clerk moved our younger class into the garden aisle.”

“What garden?”

“Our class garden. Six bean frames, dwarf tomatoes, three trays of blue mint, and one lemon cutting that dislikes everybody. We carry the trays toward whichever daylight panel still works.”

Mera added, “Indigo lost part of the thermal loop crossing the dust fold. The classroom sits beside a surviving return pipe. We use the space for lessons and seedlings because both need heat.”

Teren opened Indigo's life-support draw. The thermal pattern confirmed her account. A damaged loop pushed heat toward Drum Four while two green belts cooled. He could see manual valve changes at each school period, small acts written into the reserve curve. Children moved plants. Maintenance followed with tools. The field translated all of it into expenditure available for later extraction.

Nessa tapped once. “Do you know when settlement happens?”

“Yes.”

“Does your captain?”

“Yes.”

“Does she plan to do it?”

Every adult in Nacre's chart room watched him. Mera did not rescue him from the question. On Indigo, even the distant classroom sounds seemed to thin.

“Captain Or has not issued a settlement order since contact,” Teren said. “The Loom still carries the old schedule. If no one changes it, your domain closes in sixteen days.”

“Closes means kills us.”

“Yes.”

“Say that to the captain.”

“I will.”

“No. Say the whole thing. Say School Four. Say the garden too.”

Teren looked at the life-support trace, at six bean frames represented by a notch in a heat curve. “I will tell Captain Or that settlement kills the people of Indigo, including School Four. I will tell her your class garden depends on the heat line our machinery plans to take from you.”

Nessa remained quiet for three phase breaths. “All right.”

Mera's voice softened without losing control. “Thank you for answering plainly.”

“I have not done enough to deserve thanks.”

“That does not make the answer less useful.”

Jo crouched beside the capacitor meter. “Nessa, can your class play without Mera's voice channel open?”

“We can tap.”

“Could you keep a turn going for ten minutes?”

“School Six and we played for two hours yesterday.”

Teren heard a familiar flaw in the proposal. “A longer exchange may stabilize phase, but each pulse still spends charge.”

Jo adjusted the draw graph. “Voice spends twelve times more. We need a low-load presence code while the Commons gets itself born twice.”

Nessa objected to the wording. “Our Commons already exists.”

“Fair. While our Commons learns that.”

They designed the holding round with Nessa, not for her. Indigo sent one pulse at the beginning of its minute. Nacre placed one pulse after receipt. Each side measured the next minute from the other's reply. If either missed two turns, the remaining side would reopen voice for a status check. Teren tried to assign Nacre the first tap. Nessa said that made a starting sky. They used a burst from radioactive count instead, then discovered that both detectors fired at different moments.

“See?” Nessa said. “The game catches adults.”

In the end, Jo struck the physical service rail and Nessa struck hers when she heard the tone. Teren defined the midpoint between both received times as the beginning. Neither ship could observe that midpoint directly. Both could infer it from the exchange.

For ten minutes, Indigo and Nacre handed a single pulse back and forth. Seam draw fell by ninety percent. The phase curve narrowed until each ship paid nearly the same fraction of its local reserve. Teren ran the calculation twice because he distrusted any result that offered relief quickly.

Nessa filled the waiting with facts that required little bandwidth. One tap meant the beans still had heat. Two meant a teacher entered. Three meant someone had brought food. At minute seven, Indigo sent three. A distant cheer crossed the voice monitor before Mera muted it.

“What arrived?” Jo asked.

“Cold wraps,” Nessa said. “Pepper paste.”

Teren remembered an old wake-table complaint count and felt history approach a shape too exact for comfort. He did not tell her. Nessa deserved a meal without becoming evidence for his grief.

At the tenth handoff, the load graphs matched within two hundredths. The game had turned presence into a shared maintenance act. It kept no one alive by itself, but it spent less of Indigo's remaining life to prove that life continued.

Teren called Sava's command office through an internal pane. Hadran Coil answered, his face framed by security grey and the red edge of Article 4 status. Teren reported verified contact with Indigo and requested Commons witness. Hadran asked whether the voice could represent contamination from Violet.

“Indigo uses a different phase lag and independent reserve curve,” Teren said.

“Could Violet simulate both?”

“Not without controlling a field domain outside its causal boundary.”

“Could a hostile pattern imitate a child?”

Jo leaned into view. “Could a security chief imitate a useful question?”

Hadran ignored them. “Maintain isolation until Captain Or arrives.”

“The channel consumes reserve while isolated,” Teren said. “A witness delay changes their survival margin.”

“You do not know that anyone there survives.”

Teren placed the sanitation trace from Correction 64 beside Nessa's phase curve. Yesterday, he might have over-explained attenuation, causal work, and the evidentiary threshold. Today he gave Hadran the direct answer.

“A child corrected my count.”

He terminated the pane before security could convert the sentence into debate.

Jo raised an eyebrow. "You have become reckless in old age."

"I have become late."

They opened a Commons custody channel. Oru Pell arrived ten minutes later with Mina Rook and two elected observers. Oru wore a clerk's dark blue coat with one cuff unfastened, as if someone had pulled them from sleep before they completed the argument with the sleeve. They did not approach the transmit plate.

"Establish who controls contact," Oru said.

"Physics opened the seam," Teren replied. "Engineering maintains it. Command claims isolation authority."

"That lists claims. Who can close it?"

"Jo can dump the local capacitor. Captain Or can order them to do so. Indigo can withdraw its own phase."

Oru turned to the speaker. "Mera Pell, does your Commons know you opened this channel?"

"Our Commons knows School Four found an answering route," Mera said. "They are sending a delegate. Corridor pressure slowed them."

"Do you consent to this contact continuing until that delegate arrives?"

"I do."

Oru faced Teren. "Record reciprocal consent. No one gets to call one side an instrument."

The language created a category Teren's software did not contain. He used the disputed-witness field Ilyan had discovered and entered Indigo School Four as a participant rather than a signal source. The console objected twice. Oru supplied a Commons docket number. The objections cleared.

During the formalities, Nessa sent a quiet pattern under the voice band.

Two. Four.

Teren answered three. She replied six.

He frowned. "You changed the rule."

"You were getting proud."

"What does this version do?"

"One sky adds. One sky doubles. You have to find out which before ten."

Jo said, "She's moved beyond your training."

"I studied under Sen Quill. I recognize pedagogical cruelty."

The surname escaped before he could stop it.

Nessa heard. "You knew a Quill?"

Teren kept his eyes on the phase display. He could tell her that Alda Quill had heard the first child across a seam. He could say Sen had trained him and failed to tell the truth plainly. He could lay one hundred nineteen

years of concealed death before an eleven-year-old whose classroom needed lamp cells. Ancestry often served adults as a way to turn a child into a symbol, then ask the symbol to carry what the adults had dropped.

“Yes,” he said. “A teacher. He was difficult.”

“Good difficult or bad difficult?”

“Both, and not in equal measure.”

Nessa accepted that. “Mera says family stories improve after people die because they can't interrupt.”

Mera spoke from farther back. “I said some family stories.”

“Your exact number was most.”

Oru hid a smile by reviewing the docket.

Teren did not ask whether Nessa knew Alda's name. That knowledge belonged to her family and her ship. When the wake ledger became available, the adults could establish history without using a child as a bridge merely because her surname matched.

Hadran entered with two security officers. He carried no weapon visible beyond his command tablet, but his presence changed every posture in the room. Jo stood between him and the capacitor rail. Mina moved beside Oru. Teren remained at the plate because the game depended on someone answering.

“Captain Or authorizes controlled contact,” Hadran said. “All transmissions pass command review.”

Oru held up the Commons docket. “Contact already proceeds under reciprocal witness.”

“Article 4 grants the captain emergency course authority.”

“Does a conversation constitute a course?”

“It spends maneuver reserve.”

“So does opening a locked hatch when the pressure seal cycles.”

Hadran looked toward the speaker. Nessa sent five taps.

Without thinking, Teren answered ten.

The phase curves crossed and settled.

“What did you transmit?” Hadran asked.

“A number,” Teren said.

“Meaning?”

“That someone remains here.”

On Indigo, Mera's delegate arrived out of breath. She identified herself as Laleh An, Commons maintenance representative for Drum Two. Oru began the first procedural exchange between Assemblies that each possessed the same charter and no law for the other's existence.

Teren moved to the secondary console and recorded the two-sky game in causal notation. A standard seam treated one clock as source and the other as delayed receiver. That model forced phase correction toward the designated side, spending more reserve from the weaker domain. Nessa's game did something else. Each side started its response from the last pulse received. Both accepted the other's delay as locally real.

He ran a brief simulation. The balanced exchange did not create energy. It could not preserve Indigo by itself. Yet it shared phase tension instead of draining it toward one preferred clock. With equal turns, designation weakened. The field held two references without forcing either into zero duration.

Nessa tapped seven.

Teren placed eight after her final pulse.

The return came as nine, but not from Indigo alone. A second faint route answered against the same interval, then a third, each carrying its own delay.

Jo heard them and froze. "How many skies does her game allow?"

Nessa replied through the speaker before Teren could calculate. "As many as answer."

Across the route glass, Violet brightened. Indigo held beside it. Two weaker seams appeared where the map had shown settled dark.

For the first time, the phase display balanced without placing Nacre's clock at the center. Teren saved the model under Nessa Quill's full name. Then he took his hand off the master control and waited for the other skies to set the next interval.

Chapter 08 — The Privileged Original

<!-- POV: Captain Sava Or -->

<!-- SETUP: Indigo contact; Commons custody claim; Article 4 remains active -->

<!-- PAYOFF: equal Commons charters; person-merger rejected; Sava changes formal language but retains access control -->

Sava struck the word original from the agenda, then wrote it back.

The command pane accepted both edits without comment. Around the conference table, seven officers watched the route glass on the far wall. Violet held as a thin purple line. Indigo pulsed below it in measured turns. Two weaker traces surfaced whenever the child's timing game brought their clocks close enough to register.

The meeting had been called to decide access. Language should have followed authority. Instead Oru Pell had filed a Commons objection that challenged the meeting's first noun.

Sava left ORIGINAL NACRE at the top of the pane.

“Begin,” she said.

Oru did not sit. Their clerk's coat looked out of place among command grey, Guild black, and the pale heatcloth worn by drive officers. They set three sealed wafers on the table, each carrying a public-custody mark.

Hadran Coil stood at Sava's right. Teren Vale sat at the opposite end without his chief-physicist token. Jo Malk had declined a chair and occupied the wall beside the hatch, arms folded over a coverall marked by fresh capacitor burns. Ilyan Rook kept his hands beneath the table as if any visible movement might become evidence of allegiance to the man on Violet who shared his name.

Sava opened the record. “Article 4 review, day two. Subject: controlled communication with derivative route domains.”

Oru touched one wafer. “Commons objects to derivative.”

“State your alternative.”

“Route domains. Named ships where a service name exists.”

“That removes their legal relation to this vessel.”

“It removes a conclusion you have not established.”

Sava increased the meeting classification to command-sensitive. A red border appeared around the pane. Fear often entered her work as a preference for cleaner edges. Locked doors. Numbered schedules. One speaker recognized at a time.

“The designated ship carries the mission charter,” she said. “Any branch domain inherits data and personnel pattern from the original up to divergence. It does not inherit command.”

Oru slid the first wafer across the table. “Then we should inspect what it does inherit.”

The wafer contained Violet's Commons charter. Sava recognized the document before its title finished rendering. Every child aboard Nacre copied the first paragraph during civic instruction. Every officer swore the mission oath beneath its last clause. The text on Violet matched theirs through all 312 articles, including amendments passed six years ago after the Drum Five pressure deaths.

Eda Rook's name appeared in the amendment record.

Sava did not look toward Ilyan.

"Copied documents do not create jurisdiction," Hadran said.

Oru opened the second wafer. Indigo's charter carried the same text, then three emergency resolutions passed after Correction 141. School heating shared priority with crop survival. Seam contact required Commons witness. No officer could authorize settlement cooperation without a public vote.

"Their law continued," Oru said. "It did not wait for us to notice."

Sava read the signatures. Oru Pell appeared on Indigo's first resolution, written in the same clerk's compression she had seen a thousand times here. Other names belonged to people Sava knew: two delegates, a hydroponics auditor, a sanitation steward who had once argued with her over filter allocation.

"Those individuals also exist aboard this ship," she said.

"Which ones hold their offices?" Oru asked.

"Our Commons delegates."

"By what principle?"

"Continuity of the designated route."

"Designation by whom?"

"The Loom ranked the survival outcomes. Navigation certified them. I authorized the maneuver under the charter entrusted to this office."

Oru's face remained neutral. "So this ship holds authority because this ship's machine selected it, this ship's Guild certified that selection, and this ship's captain approved it."

"The procedure existed before any current officer held rank."

"Age does not add an outside witness."

Hadran leaned forward. "The Commons authorized the Route Loom in every mission budget."

"The Commons authorized a predictor that discarded unrealized outcomes. It did not authorize a system that created inhabited ships and killed them after nineteen days."

Sava said, "That question remains under technical investigation."

Teren spoke for the first time. "No. Mechanism remains under investigation. Inhabited duration does not."

She turned on him. "You certified settlement physics for eleven years under my command."

“Yes.”

“And now you ask this office to discard mission structure within forty-eight hours of an anomalous contact.”

“I ask you to stop describing evidence as anomaly when you dislike its authority.”

Sava almost ordered him removed. The impulse arrived as a crisp sequence: Hadran to the left, door seal, credentials suspended, hearing resumed. She saw the comfort in it before she acted. Her father had died during a crop failure while officers held ration projections behind sealed doors. As a child, she had hated not knowing. As captain, she had learned how numbers could turn a frightened crowd against the pumps that kept it alive.

She kept Teren in the room.

“The mission charter charges me with delivering forty-two thousand and six people to Pelagos,” she said. “That number exists under my jurisdiction. I cannot spend their arrival reserve on every signal capable of repeating a name.”

Ilyan brought his hands onto the table. “Violet contains forty-two thousand people with the same mission charge.”

“Violet reports a complement. We have not independently counted it.”

“Have you independently counted everyone here?”

Hadran said, “Conservator.”

“No,” Sava told him. She looked at Ilyan. “Continue.”

He swallowed. “Our census works by trusted local records. Violet offered the same record structure, signed after divergence by people we can question. Indigo did too. If continuity counts here, it counts there.”

Sava drew the route hierarchy on the wall: original vessel at the top, derivative domains beneath, each linked to the moment of correction. The structure had governed every risk model aboard Nacre. Without it, orders could conflict across seams. Two captains could claim one mission. Two Commons Assemblies could issue incompatible law. A child with a tapping game could consume reserve needed by pressure crews who had never consented to the contact.

“Equal continuity does not produce equal capacity,” she said. “Our ship holds the surviving corridor, functioning drive, and the largest reserve. Duty follows capacity.”

Jo unfolded their arms. “Capacity follows settlement.”

“The cause does not remove the duty.”

“No. It changes who paid for it.”

Sava brought up Pelagos. The gold marker lay thirty days ahead now. Between Nacre and arrival, the debris forecast had begun to thicken at the system edge. Every hour of seam use reduced the correction margin. Every hour of silence moved Violet and Indigo toward closure.

“I am responsible for consequences that occur here,” she said. “I can verify this hull. I can order this drive crew. I can answer to this Commons. I will not pretend equal moral standing solves unequal physical control.”

Oru nodded once. “That distinction belongs in the record. It does not support the word original.”

They opened the third wafer.

A charter checksum filled the wall, followed by a comparison across Nacre, Violet, and Indigo. All three matched exactly through Correction 141. Afterward, each carried new acts in separate public registers. Nacre invoked Article 4. Violet created an emergency ration council. Indigo adopted reciprocal seam custody. No document contained a clause that made one continuation own the others.

Sava searched for the designation article herself. She found authority to choose a maneuver path, authority to abandon projected outcomes, and authority to spend reserve against mission survival. The charter presumed alternatives without inhabitants. It did not describe what to do when an abandoned outcome answered.

“Absence does not confer standing,” Hadran said.

“Nor does it confer ownership,” Oru replied.

Teren rotated his empty collar token between two fingers. “The Loom created symmetric continuations, then settlement made one materially privileged. We mistook the result for an origin.”

Sava disliked the sentence because it compressed engineering, law, and accusation into a single smooth shape. “The original entered the maneuver first.”

“All routes did,” Teren said. “No clock inside the field can establish priority before designation. Nessa's game exposed that. Each domain measures the others through delay. Our stack assigns one clock as reference after the fact.”

“Then we require a reference for operations.”

“A reference, yes. Not an ancestor.”

The route hierarchy on the wall began to look less like a family tree and more like an organizational chart drafted by its own top office.

Sava closed it.

Hadran opened the person-merger proposal.

He had warned her before the meeting that the idea remained preliminary. Physics staff had explored whether route populations could be reconciled as pattern states rather than sustained as separate ships. The memorandum used careful terms: identity consolidation, memory preservation, material efficiency. It promised to reduce duplicate claims on food and arrival space.

Jo crossed the room before the first diagram finished loading. “Turn that off.”

“Sit down,” Hadran said.

“No.”

Sava raised one hand. Jo stopped two paces from the table.

“We will examine it,” she said. “We will not authorize it in this meeting.”

The model displayed two matching neural patterns projected into one substrate. Overlapping memories appeared in white. Divergent memories carried violet and green tags. A reconciliation layer attempted to compress conflicts into ranked summaries.

Ilyan looked ill. "That diagram represents people?"

"Voluntary candidates," Hadran said. "In theory. If a route reaches failure, memory integration could preserve continuity aboard the surviving vessel."

"Whose continuity?"

"The shared pattern."

"There is no shared pattern after divergence."

Teren enlarged the error field. "The model assumes conflicting states can average. They cannot. Memory does not sit apart from the physical history that formed it. Force two living patterns into one substrate and interference destroys the distinctions that permit either to continue."

"Can the errors be resolved before transfer?" Sava asked.

"Resolved by whom?"

"By each person, with consent."

Teren's shame usually made him verbose. This time he answered with one sentence. "Consent can choose a lethal procedure; it cannot make merger survivable."

Hadran kept his voice level. "What about nonliving records?"

"Records can be copied through a seam. Copies preserve testimony, not a person. Call that archive, not rescue."

Sava studied the projected heads. The model had no faces, only clean contours and colored regions. Its symmetry soothed the eye. Two became one. Demand fell. Mission arithmetic recovered.

Her father had once divided his ration into two equal portions, eaten neither in front of her, and told her the store had made an error. She had believed him because the portions looked orderly. By morning he had given both away.

"Remove the proposal from operational review," she said.

Hadran turned toward her. "Captain, we need options."

"An option must perform what its title promises. This kills two people and stores a damaged composite."

"It could preserve knowledge."

"Then submit an archive proposal under that name."

Jo returned to the wall but did not fold their arms again.

Oru marked the decision in the Commons record. "State whether rejection applies to compulsory and voluntary merger."

“All person-merger procedures,” Sava said. “No officer may offer them as evacuation, rescue, or civic reconciliation.”

“Good.”

“Do not congratulate me for declining to kill people with an unworkable machine.”

“I said good because records need plain closure.”

The meeting broke for eight minutes while seam draw passed its caution threshold. Sava remained at the table. Officers left for water or private arguments. Through the glass wall, Indigo maintained one pulse per minute. Violet sent a slower status code. Nacre's reply clock placed both beneath its own time in the interface.

Mina Rook entered before the break ended with two green-belt delegates and a transparent seed pouch in her hand. Security tried to stop them at the hatch. Oru displayed the Commons witness rule, and Hadran let the three inside under protest.

“We are not on the agenda,” Mina said. “That appears to be the condition for everyone affected by it.”

Sava recognized the pouch from hydroponics inventories. Its label listed dwarf bean stock, germination year 91, viable count thirty-two. “State your business.”

“Indigo requested lamp cells for School Four. Your access order denied the transfer discussion because children have no recognized route office.”

“No cells can cross a seam.”

“They asked for an energy allocation, not a physical shipment. Their maintenance crew can reroute power if we share the load model.”

“That exchange spends reserve here.”

“So does keeping this room warm.” Mina placed the seed pouch beside the merger model. “Thirty-two beans. We could plant them in one belt and call that efficient. Instead we distribute seed because one fungus should not erase the crop. I am not claiming people resemble beans. I am asking why mission practice understands independent survival everywhere except law.”

Sava had heard Mina argue budgets in the Commons well. She used concrete objects the way Teren used household comparisons, then refused any listener who tried to hide inside the analogy.

“How much reserve does the model exchange require?” Sava asked.

Jo answered from the wall. “Thirty seconds of compressed diagrams. Less than a full voice greeting.”

“Can it expose Nacre's thermal controls?” Hadran asked.

“It describes a classroom return loop. Anyone who weaponizes that against us would need to cross a seam, walk three drums, and become a pipe.”

Hadran did not smile. “Patterns can contain malicious instruction.”

“Then engineering reviews the packet for executable layers. Diagrams alone cannot command hardware.”

Mina addressed Sava again. "Their children asked whether ours also carry classroom gardens during heating faults. Access review classified the question as identity-sensitive."

"It invites family and school comparisons before verification rules exist."

"They already verified each other by remembering different things."

Ilyan looked down.

Sava said, "Personal continuity can establish contact. It cannot establish jurisdiction by itself."

"What establishes mine?" Mina asked.

"Your election and the charter."

"Indigo's delegates have both."

"On Indigo."

"Yes. That preposition carries the whole dispute." Mina opened her hand toward the route glass. "You keep describing location as rank."

A warning line appeared on Sava's pane: **BREAK EXPIRED**. She dismissed it.

One of the green-belt delegates, an older man with nutrient stains across both thumbs, asked permission to speak. Sava granted it.

"Captain, my counterpart on Violet signed their crop restriction yesterday," he said. "I signed ours here. His restriction cuts fruiting plants to protect grain. Mine does not. If you call him a copy, people will assume his decision copies mine. It does not. His people will eat what his signature permits."

"Do you recognize his authority over Violet's belts?"

"I recognize that his workers do. I have no belt there to grant or withhold."

The answer separated standing from control more neatly than any agenda note. Sava could accept another steward's authority without surrendering command of Nacre. What she resisted concerned not plurality but obligation. If every separate office held legitimacy, then every distress call arrived from a civic equal whom she lacked means to save.

"How many lamp cells can Indigo protect with the model?" she asked Jo.

"One, perhaps two. Depends on their relay losses."

"And what does Nacre lose?"

"Several seconds of seam maintenance reserve."

Hadran said, "Small transfers establish precedent."

Oru answered, "Precedent develops through recorded distinctions. Write the limit."

Sava authorized one compressed thermal packet under joint engineering review. Indigo retained independent choice about implementation. No command layer could travel with the file. Commons would publish the draw

cost after receipt.

Mina retrieved the seed pouch but left one bean on the table. "For the agenda," she said.

"It will not germinate here."

"That depends how long the meeting lasts."

The delegates left. Hadran waited until the hatch sealed. "Every petitioner will now bring an object."

"Then security can add a table," Sava said.

She heard the dry answer after it escaped and almost regretted it. Jo gave her the smallest possible nod, acknowledging an occupational joke without forgiving anything else.

Sava opened the settings pane and changed MASTER CLOCK to LOCAL REFERENCE.

The software warned that legacy documents might lose semantic consistency. She accepted. Nothing physical changed. Violet remained beyond touch. Indigo's classroom still cooled. Nacre still held the command path and reserve bus.

Oru returned with tea and set a cup near her. "A clerk's bribe."

"For what?"

"A readable agenda."

Sava looked at ORIGINAL NACRE.

"If I remove it," she said, "people will hear an admission that command has lost confidence in this ship's continuity."

"Has it?"

"No."

"Then say what changed."

"Our claim to priority."

Oru waited.

Sava disliked people who used silence as a request for honesty. "Priority supports coordination."

"Sometimes. It also supports taking another ship's reserve."

"I did not know settlement took from people."

"You know now."

The tea smelled faintly of mint. Nessa's classroom tended blue mint beneath failing panels. Sava had never seen the child. She knew only a voice file, a surname, and a game that shared phase draw. Yet command projections already listed Indigo as a reserve source if emergency settlement resumed.

Sava moved the cup away.

When the officers returned, Oru requested a ruling on route contact. They proposed equal civic standing for verified route inhabitants, reciprocal Commons custody, and no unilateral termination except immediate hardware danger. Hadran proposed command review for every transmission. Jo demanded independent abort control on both sides. Ilyan requested open access for family verification and civic testimony.

Sava divided the requests.

She granted reciprocal witness for scheduled contacts. She required physics and engineering to publish draw costs before each session. She authorized route delegates to file into a provisional Commons docket. She denied unrestricted family channels until identity verification and demand limits could be established.

Oru objected to her retaining closure authority.

"Article 4 remains active," Sava said.

"It presumes one ship."

"It governs this captain and this drive. I cannot command Violet or Indigo. I can control what Nacre's machinery spends."

"You can also end their seam."

"I can stop our side from feeding it."

"That distinction offers little comfort to the side that goes dark."

"It describes the authority I actually possess."

Oru entered the objection. Sava let it stand.

At last only the agenda title remained unresolved. Every spoken use of original had produced another procedural challenge. Each defense led back to selection by Nacre's Loom, certified under Nacre's law, after all routes already carried equal continuity.

Sava placed her fingers on the pane.

"Amend the formal record," she said. "Replace original vessel with designated route wherever the phrase refers to post-correction authority or reserve."

Hadran shifted beside her. "Captain."

"This preserves operational reference without asserting ancestry."

Oru asked, "And in civic standing?"

"Route inhabitants may petition under provisional person status pending Commons review."

"Person status should not require petition."

"That ruling exceeds emergency course authority."

"So does denying it."

Sava ended the argument by calling the vote on the agenda. Command officers accepted the amendment. Oru

recorded the Commons objection to provisional status. Jo abstained because engineering did not own the noun. Teren voted only on the physics note, certifying that no route clock held natural priority.

The word original vanished from the wall.

In its place appeared DESIGNATED ROUTE NACRE.

A minute later, Indigo tapped once. Nacre answered from local reference. Violet followed with its own status pulse, neither above nor below the other on the revised display.

Sava closed the public meeting record and kept the seam-access keys under Captain's seal. The map had lost its privileged name. The locked command path remained in her hand.

Chapter 09 — A Door Between Bridges

<!-- POV: Ilyan Rook -->

<!-- SETUP: controlled seam access; low-frame imaging; Violet's continued civic life -->

<!-- PAYOFF: shared debris forecast; synchronized handshake against separate glass -->

Jo removed the chart-room wall one fastener at a time.

Not the wall itself. They kept making that distinction for Hadran's recorder. They removed six pearl-composite panels, one emergency-lamp rail, and the display skin that had shown Pelagos since before Ilyan's grandparents were born. Behind it, the hull remained unbroken. Black service ribs crossed a shallow cavity crowded with cooling tubes.

"We are not opening a passage," Jo said as the third panel came free. "We are mounting a screen."

Hadran stood beside the locked hatch with two security officers. "The project title says door."

"The project title came from Navigation."

Ilyan held the loose panel while Jo released its lower clip. "It came from me."

"Then revise it."

"No."

The refusal surprised him with its ease. During his first years as a conservator, he had revised words whenever authority found them inconvenient. Tonight, four days after hearing Violet answer, a metaphor felt like the smallest ground worth holding.

Jo leaned the panel against the service rail. "A door can show where you cannot walk."

Hadran entered that sentence into his tablet as if it might later confess to something.

Across the route glass, Violet's seam pulsed at low power. Voice contact had become reliable enough for scheduled sessions, but images demanded more. The Loom had never been designed to carry pictures from a route after designation. Jo had assembled a frame compressor from collision cameras, classroom broadcast parts, and a pressure-mapping board borrowed from Drum Five. Every component had arrived with an owner who wanted it back before the ship reached Pelagos.

The new display would show one monochrome image every twelve seconds. Motion would break into separate conditions: hand lifted, hand absent, mouth open, mouth closed. The delay prevented any gesture from belonging fully to the moment that produced it.

Ilyan preferred that limitation to smoothness. A smooth image might invite the room to forget the distance.

"Violet confirms ready," Teren said from the auxiliary console.

He wore no chief token. Sava had suspended his unilateral Loom access after the Commons disclosure, so Oru Pell occupied the witness station and Jo held the physical abort. Ilyan controlled only the image calibration. Even that required Hadran's release key for each increase in bandwidth.

Captain Or watched from command through a silent pane. Her access order allowed twelve minutes of low-frame contact, followed by a reserve review. No family traffic. No executable data. No unscreened route files. Every prohibition sat on the wall beside the incomplete display, a second frame made of language.

Jo connected the final cooling line. "Power."

Teren opened the symmetric phase pattern Nessa Quill had taught them. Indigo's game alternated reference between routes instead of dragging every reply toward Nacre's clock. Violet's engineers had adapted the pattern for image packets. One ship sent frame data. The other answered with timing and error correction. Roles reversed at each cycle.

Ilyan pressed the local enable.

The pearl wall turned grey.

For twelve seconds, nothing appeared except compression blocks. Then a chart room emerged in pieces.

Violet's room shared their curved floor, suspended route glass, and black service rail. Damage distinguished it. A brace crossed the upper ring where Correction 141 had twisted the support. Frost silvered the seam conduit. Two ceiling panels hung at different angles, and someone had painted arrows around a dark pressure door.

People occupied the image as still arrangements. A drive engineer crouched near the lower rail. Three Commons observers stood behind the central table. At the image's left edge, a man faced the camera with one hand raised.

Ilyan recognized his own hand first.

The little finger bent inward from a childhood break. A pale line crossed the thumb where Eda had dropped a glass cutter during Guild training. The other Ilyan wore no conservator's collar. A strip of violet cloth circled his upper arm, marked with symbols too compressed to read.

Twelve seconds later, his hand had lowered.

"Voice band open," Oru said.

Static moved through the speaker. "Nacre chart room, confirm image."

The voice belonged to Ilyan, yet four days of ruined air and emergency speech had changed its edges. Violet's Ilyan sounded hoarse. He also placed chart-room after Nacre, as if location required immediate proof.

"We see you," Ilyan said. "One frame per cycle. Monochrome. Left edge crops part of your wall."

"We see the same. Your upper lamp still works."

Ilyan glanced above. The amber rail cast a narrow line across the floor. On Violet, darkness occupied that space.

Jo said, "We can send your team a lamp-bus diagram."

A Violet engineer moved close to their pickup. "We already know the bus. We lack a cell that holds charge."

Jo accepted the correction. "Understood."

Hadran required an image-authentication sequence before civic contact continued. He had prepared nine random instructions inside a sealed command envelope: rotate the camera, show the raw seam clock, move a person behind the route glass, display three objects named after receipt. The procedure treated Violet as a signal trying to deceive them. Ilyan disliked it. He also knew a picture without custody would later be dismissed by everyone who needed dismissal.

“Do you consent to a live challenge?” Oru asked across the seam.

Violet's Ilyan spoke with someone outside the frame. “We consent if you complete the same challenge.”

Hadran looked toward Sava's silent pane.

Approval arrived in red text.

Jo broke the seal. “First instruction: camera left by one quarter turn.”

Violet's next frame shifted toward the damaged pressure door. Nacre's camera moved left as well, revealing Hadran and his officers. In another cycle, both sides displayed their raw seam clocks. Violet's digits arrived behind Nacre's, but the difference matched the phase delay within tolerance.

“Third instruction,” Jo said. “Hold up an object stored in the chart room before Correction 141.”

Ilyan chose the brass alignment weight from the Guild cabinet. Violet's Ilyan vanished from frame, then returned carrying its counterpart. The two pieces shared a flattened corner from an accident during Ilyan's second year as conservator. On Violet's weight, a fresh cord had replaced the braided loop. Nacre's still carried the old black braid.

Hadran leaned toward the display. “Ask when the cord changed.”

Ilyan did.

“Yesterday,” Violet's Ilyan replied. “We cut the braid for a pressure splint. Ask your Jo whether that makes mechanical sense.”

“It does,” Jo said. “Poor padding, good tension.”

The next command required a person to move behind the route glass. A Violet navigator crossed the frame. Refraction split her body at the active seam, then restored it on the other side. Nacre repeated the demonstration with Teren. Both images showed local field distortion around local bodies. No projection occupied the glass itself.

Teren studied the measurements. “Independent optical environments. The cameras see different dust and surface damage.”

“Put that in words useful outside physics,” Oru said.

“They stand in their room. We stand in ours.”

The final challenge asked each bridge to display a written phrase received only after the envelope opened. Hadran chose the wording: **LOCAL AIR PRESSURE DOES NOT CROSS THIS FRAME.**

Jo found a marker. Ilyan wrote the sentence on a calibration board while Violet did the same. Their next images arrived together. Violet's final word tilted upward in Ilyan's own handwriting. Nacre's version remained level

because he had written against the board rail.

Oru entered both frames into reciprocal custody. Hadran added a security qualification that no authentication proved benign intent. The qualification struck Ilyan as almost comforting. It conceded a person could hold intent.

While the clerks sealed checksums, the cameras continued cycling. Ordinary movement occupied the delay. Someone on Violet carried a water flask past the table. A woman in a torn navigation collar handed out strips of dried grain. On Nacre, an officer collected loose fasteners before someone stepped on them. Neither room arranged itself for spectacle long enough to look heroic.

A small boy appeared at the bottom of Violet's frame. He placed both hands on their camera ledge and stared across with the severe concentration children brought to machinery they had been told not to touch. Twelve seconds later, he held up a paper star.

The star had seven uneven points.

Ilyan searched the floor for something to answer with. He found a route-calibration card, turned it over, and drew another star. His had six points before the frame timer warned him. He added a seventh too late. Violet's next image showed the boy laughing.

"Who is he?" Mina asked from the Commons witness pane.

Violet's Ilyan listened to someone beside the pickup. "Tavi Venn. He sleeps near the bridge because his cabin lost heat."

The surname appeared on the memorial wall, though Ilyan had not yet resolved the names. He filed the connection without turning the child into a question.

Hadran said, "No unscheduled personal exchange."

Ilyan lowered the card. "Understood."

On the next cycle, Tavi had gone. The paper star remained taped beside Violet's camera, one point folded where a small thumb had pressed it.

The next image showed Violet's Ilyan looking away. Another delay passed before he spoke.

"You called this a bridge session," he said.

"Yes."

"A bridge joins two supports."

"It can also span what remains separate."

"Your captain's order names us external contact."

Ilyan felt Sava's silent pane behind him. "The Commons record names you Violet."

"Violet describes a route tag."

"What name do you prefer?"

The question seemed simple until it crossed the seam. Violet's Ilyan received it after Ilyan had already begun regretting the singular. One man could name himself. A ship carried people who might not agree on what they called their continuation.

"We have not settled that," the other Ilyan said. "Use Violet for the ship. Use Ilyan Rook for me until I choose otherwise."

"Understood."

Hadran entered the wording into his tablet. Ilyan wondered whether temporary names became permanent whenever security wrote them first.

Jo calibrated contrast. The frame brightened, revealing a wall behind Violet's central table. Rectangles covered it from floor rail to ceiling: paper sheets, cloth strips, fragments of packaging, one polished metal plate. Each held writing.

"What occupies your aft wall?" Ilyan asked.

Violet's Ilyan did not answer at once. He turned. Twelve seconds later the image caught him reaching toward it.

"Our dead since divergence," he said.

The screen refreshed closer. Violet's camera had moved or changed magnification. Names emerged from the compression.

Ari Dam, pressure crew.

Hessa Pell, infant ward.

Milo Venn and Cara Venn, Drum Six.

Below them, a strip of insulation carried five names in black marker. A ration wrapper held one. Someone had written dates in hours after Correction 141 because Violet had not yet agreed what day their separate calendar began.

Ilyan counted thirty-seven legible entries. Others dissolved into pixel blocks.

"Branch failures?" Teren asked quietly.

Violet's engineer answered. "Hull scoring opened six compartments. We lost cooling in the infant ward. One tram brake failed during evacuation. There were two suicides after the settlement notice."

Hadran shifted. "That last claim requires verification before public release."

Jo turned toward him. "By whom?"

"Medical records."

"Our medical records verify it," Violet's Ilyan said through the speaker. The delay made his interruption arrive after Hadran had already looked away. "Do your dead require approval from a different ship?"

Hadran faced the screen again. "All claims entering Nacre's public system require custody."

Oru said, "Violet's Commons holds custody on its side. Our record can name the source without pretending to

certify the event ourselves.”

Hadran entered an objection and allowed the channel to continue.

Ilyan scanned the names. His eyes sought Eda despite every fact he knew. She had died six years before divergence. Violet carried the same absence. No correction made after her death could contain a living version of her, but hope ignored chronology when offered a wall of alternate names.

He found no Eda Rook.

Relief came first. Shame followed. Her absence left him free from the choice of seeing her and being unable to reach her. Thirty-seven other names remained while his grief congratulated itself.

Violet's Ilyan watched him through twelve-second intervals. Shared habits made some thoughts visible despite the delay. His gaze moved toward the memorial, then back. He knew what Ilyan had searched for because he had made the same search.

“She isn't there,” he said.

“I know.”

“No, you checked.”

Ilyan looked down at his hands. “Yes.”

“So did I, when we started the wall.”

Neither said Eda's name. They had already refused to use her as proof during first contact. The refusal did not end temptation. It gave them a rule for surviving its return.

The next image showed a child setting another paper on Violet's memorial. An adult helped press the corners flat. No name resolved before the camera pulled back.

Oru asked whether Violet consented to send a record copy of the wall.

“Not yet,” Violet's Ilyan said. “We consent to showing it. Copy rights need approval from families and our Commons.”

Oru marked the distinction. “Display does not grant archive custody.”

Sava's command pane lit with a private text to Ilyan: **FOUR MINUTES REMAIN.**

He had prepared calibration tests for the final interval. Contrast bars. Face motion. A sequence of hand positions to measure frame ordering. The memorial made those tests feel intrusive, but the Pelagos forecast still waited. Violet had requested comparison because its navigation model diverged after the correction.

“We need to use the remaining bandwidth for arrival data,” he said.

Violet's Ilyan nodded in a frame that reached them after the words. “Put your outer-system map on the right.”

Ilyan opened Nacre's forecast. Pelagos appeared as a pale star marker. Lattice traced a small orbit inside the projected capture corridor. At the system edge, sensor uncertainty formed a broad grey band where interstellar dust met Trojan rubble and old comet fragments.

Violet sent its map on the alternate frame.

The two forecasts occupied separate halves of the wall. Their base surveys matched. Recent observations did not. Nacre's route stack placed the dense rubble front slightly north of the mission corridor. Violet placed it south and closer, with a long diagonal surge crossing the certified arrival path.

"Your debris tide uses higher transverse speed," Ilyan said.

"We took hull damage in the fold," Violet's navigator replied. "The impact array resolved smaller particles than yours. We count a moving front."

"Could damage bias the array?"

"Could an intact shield blind yours to low-mass strikes?"

The question contained no insult. It still made the Nacre navigation officers behind Ilyan stiffen.

He overlaid both uncertainty fields. Where they overlapped, the safe corridor narrowed by half. Where each ship trusted only its own data, an apparently open lane ran through the other ship's highest-risk region.

Teren left the phase console and came closer. "Neither forecast can certify the approach alone."

Hadran said, "We cannot import route navigation into the command stack."

"Then compare without importing," Ilyan replied.

He built a new scratch layer outside both certified models. Violet's navigator did the same. They exchanged only observed vectors, timestamps, and instrument limits. No executable code crossed. Each side entered the other's values by hand.

The work took them beyond Sava's twelve-minute order.

Her pane flashed red. Hadran received a command in his cuff and placed his release key over the cutoff field.

"Time," he said.

"The maps conflict," Ilyan answered.

"Record the conflict and close."

"Closing now leaves both forecasts wrong in ways we can already measure."

"Captain Or authorized twelve minutes."

Jo stood over the physical abort. "Image draw remains below the published cap."

"That does not extend authority."

Oru checked the Commons docket. "Violet consented to the exchange. Nacre's navigation crew requested comparison. Which immediate danger requires termination?"

"Unreviewed reserve expenditure."

Teren pointed to the load. "We have spent less than the calibration allotment because symmetric frames

reduced correction traffic.”

Hadran's thumb rested on the release key. Ilyan heard the pumps, the cooling jacket, and a delayed tool strike from Violet. Both rooms waited on one hand.

Sava's voice entered through command. “Complete the vector cross-check. No further image increase.”

Hadran lowered the key.

The concession lasted only six words. Ilyan knew what it cost her to revise an order while subordinates watched.

They worked for another twenty-three minutes. Nacre contributed long-range optical returns. Violet contributed fine impact counts from its damaged outer skin. Both ships shared old launch models and rejected assumptions that no longer fit. The scratch layers converged on a moving band of debris wider than either original forecast.

Pelagos remained reachable. The certified central corridor did not remain safe.

Ilyan widened the view. The debris tide curved across the system edge like weather measured in stone. Its front would intersect Nacre's arrival window. A correction could shift one ship around it, but current reserve left little margin. Violet carried less. Indigo's approach geometry remained unknown.

Violet's navigator wrote a note across their side of the frame:

ONE CORRIDOR MAY NOT FIT US BOTH.

The word both held more than mathematics. Two twelve-kilometer ships could not occupy the same capture line at the same time without dangerous wake interaction. Even if both survived settlement, arrival itself demanded separation.

Ilyan wrote beneath Nacre's scratch layer:

THEN WE NEED MORE THAN ONE SAFE LINE.

Hadran read it. “That statement exceeds this session.”

“It states a navigation requirement.”

“It implies mission redesign.”

“The debris implies it. I wrote it down.”

On Violet's delayed image, someone laughed. The frame gave no sound at first. Twelve seconds later, the speaker carried it as a rough burst.

Sava ended the technical exchange after both navigators signed the shared observation set. She authorized the combined debris record for command review but denied automatic entry into the course stack. Access remained hers.

Jo began reducing image power.

The wall lost detail by degrees. Violet's memorial names blurred first. Faces became pale blocks. The black rail

and route glass survived longest because high contrast cost little to send.

“Wait,” Violet's Ilyan said.

Jo held the draw at minimum image rate.

In the next frame, the other Ilyan stood directly before his camera. He lifted his right hand, palm outward. The little finger bent toward the others.

Ilyan stepped to Nacre's wall.

The display skin lay over composite and service ribs. Beyond it came insulated hull, causal separation, and another chart room that occupied no reachable direction. He could not touch the man on Violet. No air crossed. No heat from the raised hand entered this room.

He lifted his left palm to the image.

Their hands did not align. Delay placed Violet's gesture against Ilyan's earlier position. By the time Violet saw his answer, the first hand had already lowered.

“Use Nessa's count,” Jo said.

Teren opened the two-sky timing channel at its lowest pulse. Violet sent one. Nacre answered after receipt. The midpoint belonged to neither local clock. Both Ilyans watched the count.

At the agreed interval, Ilyan raised his hand again.

Twelve seconds passed.

The wall refreshed. Violet's Ilyan had timed his own movement from the returned pulse. His palm appeared exactly opposite Ilyan's, line for line except for the mirrored scar at the thumb.

For one low frame, the two chart rooms faced each other across separate glass.

Jo cut image power on schedule. Grey replaced Violet, but the last frame remained in Ilyan's local scratch pane: two hands meeting where no door opened, synchronized by a delay both ships had chosen to share.

Chapter 10 — The Debt Engine

<!-- POV: Teren Vale -->

<!-- SETUP: settlement accounting; no safe merger; Pelagos debris tide -->

<!-- PAYOFF: public disclosure; stabilization without harvest proposed -->

Teren carried the copper demonstration coil into the Commons well with both hands.

It weighed less than a cooking pot. He had chosen it for that reason. The Loom occupied three decks, drank from the ship's deepest reserve, and required its own cooling arteries. Nobody without clearance could see more than the route glass and service braid. A hand-sized coil offered the public something they could lift, something that did not hide behind scale.

Oru Pell had cleared the lowest terrace. Benches rose around it in six arcs, already crowded before Teren reached the table. Green-belt workers stood in damp boots beside navigation trainees. Drive crews kept grease on their cuffs. Several families had brought children despite the hour. The wall displays showed the chart room on one side and Violet's low-frame bridge image on the other. Separate glass, separate rooms, matching rail scars.

Security officers occupied every aisle.

Hadran Coil waited at the foot of the speaking floor. His gray uniform carried no ceremonial sash. He had dressed for an evacuation rather than a hearing.

"You agreed to a technical briefing," Hadran said.

"I brought a technical instrument." Teren set the coil down. Its copper turns clicked against the table. "That should reassure you."

"It does not."

"Then it has already improved the briefing."

Oru heard and gave Teren a look that declined the joke without wasting words on it. They placed a Commons recorder beside the coil and checked its witness lights. Three green points appeared. A fourth remained dark.

"Captain's office has withheld authentication," Oru said.

"On what ground?"

"Potential panic."

Teren glanced up at the terraces. News had moved ahead of permission. People had seen the bridge frames. They knew another Nacre carried another crew. Rumor had added weapons, plagues, resurrected relatives, and one version in which Violet had already crossed into Drum Four through a maintenance hatch. Fear loved mechanisms nobody could touch.

He rotated the coil so its two leads faced the audience.

"Use the Commons witnesses," he said. "Leave the empty light visible."

Oru opened the session.

The well quieted by layers. Footsteps ceased first. Private arguments thinned. Ventilation kept moving above them, pushing food-warm air up the terraces. Teren had lectured here often enough to know the room's acoustics. A lie spoken at the table came back from the highest rail a fraction late. He had once enjoyed that correction.

Today every delay reminded him of inhabited time.

Oru stated his name, office, and the narrow question before the Assembly: what physical dependence connected the designated route to settlements of the other routes? They did not ask whether Violet counted as people. The frames had ended that dispute for anyone willing to look. They asked what survival machinery now threatened those people.

Teren touched the coil.

"Most households aboard Nacre own a heat tile," he began. "You charge it while ship power runs cheaply, then carry it into a cold room. The tile warms that room by cooling itself. Nothing appears from nowhere. Heat crosses a boundary, and the tile loses what the room gains."

Someone in the third terrace called, "We know what a battery does."

"Yes," Teren said. "Command relied on our habit of believing the Loom did something grander."

The interruption moved through the well as a low response. Oru lifted one hand, not to silence it, only to keep it from becoming ten conversations.

Teren connected the coil leads to the tabletop supply. A blue lamp lit inside the copper turns.

"This coil represents the causal-shear field. The lamp represents usable reserve. During a correction, the Loom instantiates several continuations of the ship. Each continuation begins with a share of the field's stored capacity. The designated route does not win because nature favors it. Settlement moves reserve from the others into that route."

He disconnected one lead. The lamp dimmed but remained lit on a hidden capacitor.

"For nineteen days and six hours, a rejected route retains its own heat, metabolism, clocks, and decisions. At the end of that window, settlement closes its domain. The remaining differential crosses into the designated ship. Our safer vector contains their spent possibility."

A woman near the front held up a ration heater. "You mean we burn them."

The household analogy failed at its edge, as every analogy did. Teren had used tiles because people understood stored heat. A tile had no children inside it.

"We kill them," he said. "The transfer arrives as field reserve, not combustion. The distinction matters to the mechanism. It does not lessen the result."

The well changed around him. No shout marked it. Bodies shifted away from the rails as though the metal carried blame. A child asked something too softly for Teren to hear. His father bent down but did not answer.

Teren looked at the route glass display. Violet's bridge refreshed one frame late. Figures there stood around their own table, watching his room through their wall. The delay made them appear patient.

He had prepared diagrams. Shame urged him to display all of them at once, to cover the confession with sufficient detail. Instead he opened only the energy ledger from Correction 141.

Twelve columns appeared. Green carried the old designation mark. Violet, Amber, White, and eight numbered service colors followed. Each column began with equal field allocation. At the bottom, Green's reserve rose while the others fell toward closure.

"Who chose Green?" a trainee asked.

"The Loom ranked survival outcomes. Navigation certified the ranking. Captain Or authorized the correction."

"And you built the ranking?"

"I inherited it. I maintained it. I taught people to trust it."

"That wasn't my question."

Teren found the trainee in the fifth terrace, a narrow face above a Guild collar. "Yes. I approved the model that ranked Correction 141."

The answer took less time than his excuses would have.

Oru leaned toward the recorder. "Dr. Vale, state what happens if engineering disables settlement now."

Teren advanced the ledger to the current projection. "Violet remains open because Capacitor Seven keeps part of its seam warm. Other weak continuations may also retain clocks. If we stop all settlement activity, we stop further harvest. We also end the transfer that has funded every major course correction for one hundred twenty-seven years."

He placed a ceramic cup beside the coil. "Imagine seven households crossing winter with one common pantry. Our doctrine says only one household eats. The others vanish when their shelves empty, and the surviving household inherits their stores. Refusing the inheritance prevents the killing. It does not fill anyone's shelves."

"Thirty-one days to Pelagos," someone said.

"Less usable correction reserve than the arrival plan requires," Teren replied. "The debris tide has widened beyond launch forecasts. Without another maneuver, the designated ship misses the safe capture corridor or enters rubble at intolerable speed. Violet faces a worse deficit. Every route does."

The nearest security officer looked toward Hadran. Teren could almost see the report forming: physicist tells crowded chamber all ships may die.

Hadran stepped onto the floor. "You have exceeded the authorized scope."

"I have reached its subject."

"You were asked to explain dependence. You have announced fatal projections without review."

"The projections carry six review seals, including mine."

"Then present them to command."

"I did. Command classified them pending public-stability assessment."

Murmurs sharpened into words. Oru touched the recorder, ensuring its lights still held. The captain's authentication point remained black.

Hadran faced the terraces. "No evacuation order exists. No loss of mission has been declared. The images identified as another bridge remain under technical examination. You will not obstruct the aisles. You will not approach chart-room access points."

He spoke well. Teren understood why Sava trusted him. Hadran's voice turned fear into tasks. Keep the aisle clear. Leave the door alone. Wait for someone whose uniform promised sequence.

Then Hadran turned back to Teren. "Your terminology invites violence."

"My terminology names an operation."

"Debt engine," Hadran said, and the phrase reached the recorder in his voice before Teren had spoken it publicly. "You used that phrase in the sealed physics review."

A silence opened across the well.

Teren had written the words at three in the morning, after tracing reserve from a vanished route into the designated vector. He had expected them to remain in a private draft until he could find language with fewer consequences. Hadran had brought the phrase into the room as evidence of recklessness. Instead it sat on the table beside the small copper coil, plain enough to survive accusation.

"Yes," Teren said. "Settlement mode functions as a debt engine."

Oru's gaze stayed on him. "Define that for the record."

"Every correction charges survival to people denied standing in the account. We spend their reserve, erase the creditors, then call the resulting balance safe. The engine does not settle a debt. It destroys whoever could collect it."

The highest terrace repeated the phrase in pieces. Not a chant. People tested it against what they had seen. Debt. Engine. Two ordinary words that doctrine had kept apart.

Hadran reached for the Commons recorder.

Oru covered it with one hand. "Security may file an objection."

"I am terminating an unauthorized transmission."

"This session uses local public storage."

"Under Article 4, the captain controls course-sensitive information."

"Article 4 does not dissolve the Commons."

"It permits emergency containment."

The recorder's three witness lights remained green under Oru's fingers. Teren watched Hadran choose whether to move their hand. The security chief could have done it. Six officers stood within ten paces. Instead he stepped back.

"File my objection," he said.

Oru did.

Questions came in order after that, though no one felt orderly. A hydroponics steward asked whether food could be divided across seams. No. The domains exchanged low-bandwidth light, sound, and phase instructions. A drive fitter asked whether Violet could dock before closure. No. The ships occupied separate causal continuations, not adjacent berths. A midwife asked whether two versions of one person could be combined to reduce population demand.

Teren closed his eyes once.

The proposed merger had reached physics through a Captain's-office memorandum. Its language promised preservation of memory with minimal material duplication. Engineers had already rejected the hardware premise, but enough rumors circulated that the Commons deserved a direct answer.

He set a second cup beside the first.

"Two matching patterns do not fit into one history," he said. "If we force two signal states into the same physical substrate, interference destroys the distinctions that let either pattern continue. For records, merger corrupts both. For a person, it kills both while producing damaged data that resembles parts of them."

"Could one volunteer?" the midwife asked.

"Consent cannot make the process survivable."

"What about copying memories without joining bodies?"

"That can preserve testimony if the seam remains open long enough. It cannot move a life. Violet's people do not become rescued because we store descriptions of them here."

A records clerk in the fourth terrace raised both hands until Oru recognized her. She asked who would own a copied testimony if the speaker later died at settlement. Teren began with archive law, then stopped. Archive law presumed a dead author and a surviving jurisdiction. Here the jurisdiction itself faced deletion.

"The speaker retains ownership while their ship continues," he said. "If their route closes, our copy becomes witness evidence, not inheritance. We cannot claim their work, vote, name, or memory as compensation for failing to keep them alive."

The clerk wrote the answer on her sleeve pad. "And if command labels the copy corrupted?"

"Then independent checksums must travel with every packet. Commons custody on both sides, not a single command archive."

Oru marked that proposal for later procedure. Teren felt the hearing altering shape beneath the technical briefing. Each practical question exposed a right the ship had never needed to name. Storage carried custody. Timing carried standing. A checksum could not save a person, but it could prevent authority from pretending no person had spoken.

On the wall, the delayed bridge frame refreshed. One of Violet's figures lifted a sheet toward their camera. The letters appeared backward. A technician corrected the image, and the message turned legible.

WE DO NOT CONSENT TO ERASURE BY ARCHIVE.

The well read it without prompting.

Teren felt an old teaching instinct rise. He wanted to explain phase identity, destructive overlap, checksum persistence, every clean reason the sign expressed valid physics. The explanation would have reduced people to evidence again. He let the words stand.

Hadran folded his arms. "You have now removed both available actions. We cannot disable settlement, and we cannot consolidate duplicate populations. What do you propose beyond panic?"

The question struck with more force because it contained truth. Naming a crime did not repair its machinery. Teren had spent two days moving terms across a board while Violet's reserve declined. Every line that saved designated Nacre harvested another route. Every line that preserved all routes ended against the debris tide.

He called up a final model.

The display showed two field circles joined by the weak line of Capacitor Seven. One represented designated Nacre. The other carried Violet's tag. Reserve values blinked beside each. Nacre held enough excess to maintain a seam or fund part of one correction, not both. Violet approached its settlement deadline with less.

"This model does not solve arrival," Teren said. "It solves one earlier death."

A man in the second terrace laughed without humor. "Which one?"

"Violet's, if both engineering teams cooperate."

He enlarged the capacitor path. "Jo's bypass left a shared load between the domains. Settlement expects one-way extraction. We can pulse the load symmetrically, preventing Violet's closure trigger from finding an empty receiving state. No reserve crosses as harvest. Both sides spend energy maintaining difference."

"How long?" Oru asked.

"Through Violet's current deadline. Perhaps longer if the field remains clean."

"And after?"

"We still face Pelagos with less reserve."

The room did not reward him. Good. Applause would have mistaken postponement for rescue.

Hadran studied the model. "How many routes can you hold that way?"

"With present reserve and one functioning bypass, one."

"Only Violet."

"Only Violet today."

The wall carried Violet's bridge in another delayed frame. Their table had filled with written calculations. A child stood behind the adults, too short to see over their shoulders. Someone gave him a crate.

Hadran pointed toward the image. "You will create a queue of ships demanding energy we do not possess. You will turn every technical limit into an accusation."

"They already possess lives," Teren said. "The limit does not become less real because we hide it from them."

“You are inducing panic aboard this ship.”

“No. I helped induce ignorance for decades. Panic belongs to what arrives when ignorance fails.”

The words sounded prepared, though he had never thought them before. Teren distrusted any sentence that improved him too quickly. He looked down at the coil and found one lead trembling against the table because his hand had begun to shake.

He steadied it.

Oru asked the practical questions. Required reserve. Failure conditions. Authority over the drive nave. Whether the maneuver instantiated any new route. Teren answered each in turn. The pulse would use the existing seam. No mass could cross. Any timing disagreement greater than six hundredths of a second would heat the bypass jaw beyond tolerance. Jo could abort locally. Violet's engineering team would need the same right. Captain's access controlled the Loom command, but drive engineers controlled the physical capacitor banks.

“Can command run settlement during your pulse?” Oru asked.

“Not without risking both ships.”

“So stabilization creates a temporary hold.”

“Yes.”

“On machinery and on policy.”

Teren glanced at Hadran. “Machinery first. Policy will have to learn.”

The security chief spoke into his cuff. No officers moved. After a moment, he listened to a reply and looked toward the dark authentication light.

“Captain Or orders the session concluded,” he said. “Dr. Vale will surrender his Loom credentials pending review.”

The audience broke into protest. Oru struck the table once with their palm. The sound restored enough attention for procedure.

“Captain Or may suspend an officer's access,” they said. “She cannot unhear public evidence or seize Commons witness copies without written cause. Three copies have already sealed.”

Teren removed his credential strip. Its edge had worn smooth during thirty-nine years against his wrist. He placed it beside the demonstration coil.

Hadran reached for it, then paused. “Without this, you cannot authorize your proposed pulse.”

“I cannot authorize it with this. I can only certify the physics.”

“Who authorizes it?”

That question had governed the Loom since launch. Command chose. Physics justified. Navigation certified. People carried the result without meeting anyone charged for it.

Teren looked up through the Commons well. Mina Rook stood among the green-belt workers. Jo occupied an

upper aisle in a scorched drive coverall. On the wall, Violet's delayed room watched from behind its separate glass. None could act alone. The machinery had mistaken concentration of authority for precision.

"Oru must establish standing," he said. "Both engineering decks must consent to the procedure. Captain Or must release the command path, or the Commons must challenge her refusal. Violet must retain an independent abort."

Hadran took the credential. "That sounds like no."

"It sounds slower than killing them."

The wall display refreshed again. Violet's calculation sheet had changed. Beneath three lines of capacitor values, someone had written a single word.

READY.

Jo came down the aisle before Hadran could bar it. They stopped at the speaking floor and set a fused isolation jaw beside Teren's copper coil. The metal carried a violet heat stain along one edge.

"I can build the symmetric trigger," Jo said. "I cannot feed it alone."

Oru turned the Commons recorder toward them. Hadran held Teren's credential in his closed hand. Across the seam, a separate engineering crew raised their own damaged part to the camera.

The debt engine still held one hundred forty-one corrections in its accounts. Teren could not repay those lives. He could not promise equal rescue with one warm seam and a shrinking reserve.

He opened the stabilization model to public copy.

Confirmation tones stepped down the Commons terraces as each terminal accepted it. Beside Teren's copper coil, the fused isolation jaw kept its violet heat stain. Jo rested one scarred shoulder against the speaking rail and listened while Violet returned a timing pulse through the warm metal.

Chapter 11 — What the Loom Was Built For

<!-- POV: Ilyan Rook -->

<!-- SETUP: Alda's maintenance tables; answering clocks; blank ledger line -->

<!-- PAYOFF: founder knowledge proved; evidence escapes suppression -->

Ilyan held the old maintenance sheet against the chart-room glass until its punched holes filled with route light.

The paper had survived one hundred nineteen years inside a binder labeled COOLANT RETURN, FIRST LOOM TRIAL. Its edges had stiffened to brown lace. Rows of pump intervals crossed the page in Alda Quill's compact drafting hand, each value followed by a correction that no pump required. Ilyan had spent two nights treating those additions as calibration noise.

Now Violet's seam shone through the holes. A pattern appeared.

Three apertures caught the line at once. Two darkened. Four caught it. A longer gap followed. Then five.

Not pump timing. Bell timing.

Mina leaned across his shoulder, careful not to touch the sheet. "Again."

He shifted the paper back to its registration marks. The same intervals opened in violet light.

"Three, two, four, gap, five," she said.

"Page one."

The next sheet yielded three, two, four, gap, six. Page three ended in seven. Alda had nested an increasing address inside maintenance corrections, one interval beyond any range a coolant controller could accept. A technician checking the pumps would ignore the excess. A conservator with route light might use it to order the pages.

Mina looked toward the chart-room hatch. Hadran had posted two officers outside after Teren's Commons disclosure, but Ilyan still held local audit access. The security order covered transmissions and Loom control. It did not mention holding old paper against inactive glass.

"They'll amend the order," she said.

"Then we should learn quickly."

He placed the sheets in Alda's sequence. Forty-eight pages covered the first live trial, followed by six later inspections. Every page carried columns for inlet temperature, braid resistance, capacitor lag, and release recoil. The strange extra values occupied the right margin. Once ordered, those values formed groups separated by nineteen-day intervals.

Mina brought up the founder notation index on her wrist pane. "Quill used base twelve for crew channels."

"Sen Quill did. Alda preferred prime spacing."

"You sound pleased."

“I dislike mysteries that flatter their maker.”

“You have been awake for thirty hours.”

“Then I dislike them with conviction.”

She did not smile. Neither did he. The low-frame image from Violet refreshed above the central rail. Their bridge appeared empty except for a night watch seated beneath the camera. A separate clock ran in one corner. Its digits had advanced nine minutes since the previous frame.

Ilyan entered Alda's prime-spaced groups into a scratch pane. The software rejected them as malformed maintenance codes. He removed the pump fields and retained only her added intervals. The pane translated each group through the founder crew-channel alphabet.

FIRST BELL RECEIVED.

SECOND BELL RECEIVED AFTER RELEASE.

VOICE AT SEAM.

Mina read without moving. Her fingers tightened around the binder spine.

“Keep going,” she said.

Ilyan decoded the next group.

CHILD TECHNICIAN IDENTIFIES ROUTE AS NACRE.

He glanced at Violet. The night watch had stood. A fresh frame showed the person close to the camera, though no voice channel ran. Perhaps they saw Ilyan's posture change. Perhaps someone on their side had solved a different piece of the same history.

The maintenance sheet trembled against his thumb.

Alda's hidden record did not use the official phrase rejected route. She called each continuation a wake and assigned it a clock. Wake one reported a damaged field relay. Wake two requested a repeated maneuver bell. Wake three sent the names of four technicians on duty. Beneath every entry, Alda copied elapsed time past official settlement.

Nineteen hours. Four days. Eleven days, eight hours. Eighteen days, twenty-three hours.

The values continued toward nineteen days and six hours.

“A wake ledger,” Ilyan said.

Mina lowered the binder to the table. “Say that into the recorder.”

He activated his local witness channel. “The founder maintenance tables contain a wake ledger. Alda Quill recorded intelligible contact from physically persistent route domains during and after the first Loom trial. Entries identify clocks, work orders, individual technicians, and ordinary ship activity through the settlement interval.”

“Does it prove command received the reports?” Mina asked.

“Not yet.”

The distinction hurt. Evidence that Alda knew could still support the familiar defense of a lone engineer misunderstood by her era. Command might have dismissed her. Sen Quill might never have seen the data. One frightened person could hide testimony while everyone above her remained clean.

Ilyan turned to the later inspection pages.

Alda had stopped writing prose after the first trial. Her margin values narrowed into paired checksums, each one aligned with a maintenance authorization. Ilyan called the ship's founder archive. Captain-level restrictions blocked the raw authorizations, but their public seals remained available because generations of auditors had needed to verify equipment pedigree.

The first checksum matched Mission Command Circular 4-1.

Its title appeared: POST-TRIAL SIGNAL CLASSIFICATION.

The second matched a Navigator's Guild seal on revised telemetry language.

The archive returned the title: ECHO DECAY ACCEPTANCE STANDARD.

Mina swore softly. “Dates?”

“Circular issued six hours after Alda's first voice contact. Guild standard issued two days later.”

“Before the wakes closed.”

“Yes.”

Ilyan aligned the third pair. It matched Sen Quill's architecture office. The surviving title read SETTLEMENT YIELD IMPROVEMENT AFTER EXTENDED DOMAIN HOLD.

He stared at the words.

For all his anger at categories, part of him had expected ignorance to remain possible. Perhaps founders had seen uncertain traces and chosen bad language. Perhaps later technicians had inherited the harm without grasping its human duration. The title removed that shelter. Extended domain hold had increased yield because inhabited ships kept generating thermal and metabolic differences before closure. The Loom had not accidentally benefited from persistence. Its builders had tuned settlement around it.

“What did they build it for?” Mina asked.

Her question carried no request for a lecture. She wanted him to name the action without softening it.

“Prediction made the routes,” he said. “Settlement made them useful to command. They built the Loom to create alternatives, rank them, and consume the ones not designated.”

The words entered the witness channel.

A hatch indicator changed from green to amber.

Mina copied the three matched seals into her pane. “How much remains?”

“Thirty-one pages.”

“Then we no longer have time for reverence.”

They worked faster. Ilyan photographed each sheet under three light angles. Mina generated independent hashes, copied the seal matches, and sent short strings into public Commons queues. She did not transmit the images. The current order might stop a large packet. A checksum looked like ordinary archive traffic until someone knew what evidence it described.

Alda's ledger grew stranger near the end. She recorded meals exchanged by voice, a request for a pressure-valve diagram, and an apology cut off after four words. One route reported that its children played the maneuver bell back through a heating pipe. Another sent a list of seeds surviving in a damaged green belt. None offered a theory of their existence. They spent their remaining days repairing the ship because a ship still demanded work.

Page thirty-nine contained a work order from Wake Four for a jammed sanitation gate. Alda had copied the part number, then noted that no compatible gate existed on her side after the trial divergence. The wake crew had improvised a hinge from meal-cart axles. Twelve hours later they reported the seal holding. The message contributed nothing to Loom design. It had earned its hiding place because a technician expected another technician to care whether a lavatory flooded.

Mina read the next entry aloud. A baker on Wake Two asked whether the designated ship had also lost its starter culture during the field surge. Alda answered with a replacement ratio even though no flour could cross the seam. The baker replied that advice counted. Three days later, the ledger recorded a loaf distributed in the damaged drum and seventeen people who complained it tasted of copper.

“She kept all of that,” Mina said.

“She knew summary would help them erase it.”

Another page listed four sleep rotations around an overheating capacitor. The names changed with each shift. Alda copied every one. Ilyan traced the final entry and imagined her deciding what deserved the remaining margin: field values or the person who relieved a tired mechanic at second watch. She found room for both by writing smaller.

Near the settlement boundary, Wake Three sent household notices faster than Alda could answer. A broken school lamp. A requested memorial time. Instructions for feeding a child who refused the new ration paste. The last complete message reported that the maneuver bell rail had begun cooling. After that came the unfinished apology, four words separated by static: WE THOUGHT YOU WOULD.

Ilyan recognized the discipline. He had once mistaken it for proof that procedure could remain morally neutral.

At the bottom of page forty-seven, the columns broke.

Every prior entry placed the designated vector in the left field and settlement yield in the right. Here Alda drew a course line through neither field. It began at the first-trial origin, crossed the space between route columns, and ended blank before destination coordinates. No designation mark appeared. No yield value followed.

Mina turned the sheet. The reverse showed only age stains.

“What does that mean?”

“I don't know.”

“You say that differently when you have a guess.”

Ilyan enlarged the line. Tiny timing ticks marked its first portion. He counted them twice.

“One hundred seventeen seconds,” he said.

“Of travel?”

“Of missing prediction, perhaps. The interval carries no route solution.”

“Can the Loom execute it?”

“Not from this.”

He tried mapping the ticks against Correction 141. The line failed every coordinate transform. He tried Alda's prime spacing, the founder channel alphabet, and capacitor phase order. Each method produced contradiction. The course entered a region the Loom refused to label because no designated outcome anchored its calculation.

He overlaid the first-trial field geometry. Alda's ticks aligned with alternating capacitor banks, but no single ship could issue the sequence. Every odd command belonged to the designated controller. Every even command required a rejected bank that command software had already marked unavailable. Execution demanded two active routes that each recognized the other as a source of timing.

Mina followed the alternating marks. “A conversation.”

“A procedure with more than one author.”

“Could Violet supply the missing half?”

“Perhaps. We still lack a vector.”

Ilyan asked the scratch pane to project all possible destinations from the ticks. Instead of a course fan, it returned an authorization error. **REQUIRED REFERENCE ROUTE: NONE**. The software had no category for a maneuver whose reference moved between ships. He changed the input label from route to witness. The error persisted, but the display exposed a buried field that normal navigation screens concealed: **PEER PHASE CONTRIBUTIONS**.

Seven empty slots appeared.

The number meant nothing yet. Alda had drawn the line during a trial with fewer known wakes, and seven surviving ships did not exist in Ilyan's present knowledge. He refused the temptation to turn an interface capacity into prophecy. What mattered lay in the plural shape. The Loom's deepest architecture could receive timing from peers even though command doctrine permitted only one source.

He copied the blank field names into the witness record. “She did not solve the course,” he said. “She preserved where another solution could enter.”

Mina tapped the final tick. “Then don't fill it alone.”

Alda had left no answer, only an unsolved absence.

Mina touched the page beside the final tick. “Call it something before command calls it damage.”

Ilyan looked at the empty coordinate fields. "Blank course."

The phrase felt less like discovery than custody. He entered it into the witness log without claiming to understand.

The hatch opened.

Captain Sava Or came through first, followed by Hadran and two archive officers. Sava's uniform looked freshly sealed at throat and wrist. Increased formality had become her most visible fatigue. She took in the illuminated sheets, the founder titles, and Violet's night watch above them.

"Step away from the table," Hadran said.

Mina kept her hands visible. "Under which authority?"

"Captain's emergency course authority."

"These materials came from an open maintenance binder."

"They now concern active navigation."

Sava addressed Ilyan. "Have you transmitted founder documents across the seam?"

"No."

"To the Commons?"

"Not the documents."

Mina said, "You may want a narrower question."

Hadran moved toward her. Sava stopped him with two fingers raised.

"What has left this room?" the captain asked.

"Checksums," Mina said. "Archive seals. Titles. Witness attestations."

"How many recipients?"

"Enough."

Sava's face hardened. "That number does not exist in emergency procedure."

"It does in horticulture. Enough seed means one failed tray cannot end the crop."

Ilyan watched the captain look at the ledger. She did not dismiss it. That would have comforted him, perversely. Denial could still belong to ignorance. Her gaze moved down the elapsed times with the care of someone reading a casualty list she had expected and hoped not to see.

"Show me command receipt," she said.

Ilyan displayed Circular 4-1, the Guild standard, and Sen Quill's yield title. "All three seals match Alda's margin checksums. Dates place them before the first wakes settled."

"Full contents?"

“Restricted.”

“So we have titles.”

“And Alda's paired receipts.”

“A court would require custody.”

“A court would not classify the evidence before deciding whether a plaintiff exists.”

Hadran said, “Conservator.”

The warning corrected rank, tone, and danger in one word.

Sava came to the table. She did not touch the paper. “Ilyan, if these sheets circulate without authentication, every invented founder order aboard the ship will attach itself to them. People will act on rumor. Violet may interpret our internal dispute as preparation to close their seam.”

“Violet already knows we built settlement around their deaths.”

“They do not know what I intend.”

“Neither do we.”

A muscle moved beside Sava's mouth. “I intend to keep forty-two thousand and six people alive while determining whether any action can preserve more.”

“Then let them know the cost of the actions offered.”

“I will not conduct course command by crowd acceleration.”

“Nobody here accelerated the crowd,” Ilyan said. “Alda wrote dates. Your office added locks.”

Sava's gaze moved to the static sleeve waiting in Hadran's hand. “Locks can preserve evidence from contamination.”

“They can also preserve command from evidence.”

The nearest archive officer shifted his stance. He had trained under Ilyan twelve years earlier, when custody disputes concerned altered dust vectors and trainees who reused witness seals. Ilyan remembered praising his caution. Now that same caution waited for authority to decide which history could enter the archive.

Sava lowered her voice. “If I release an unauthenticated accusation against the founders, I cannot withdraw it after a technical error. Crew confidence in every arrival system may collapse.”

“Confidence should follow proof.”

“Ships also run on trust.”

“Not trust purchased by hiding who paid for the fuel.”

For one moment, fatigue loosened her command face. Ilyan saw the sanitation engineer who had worked under crop-failure rationing before anyone called her captain. She knew what concealed projections could cost. That knowledge did not make disclosure easier. It made secrecy feel like a tool she had survived.

“Give me six hours,” Sava said.

Violet's clock advanced on the wall. “Those hours belong to them too.”

Mina stepped beside him. “You also cannot conduct history by blackout.”

Sava turned to the archive officers. “Seal the binder and every local image. Suspend chart-room public links. No deletion. Full forensic custody under my office until founder authorization records can be opened.”

The words sounded careful. No deletion. Custody. Forensic. Ilyan had used the same defenses when he hid the answering clock. He understood each reason. He also saw where the reasons ended: one office would control when truth became safe enough to exist.

Hadran collected the paper into a static sleeve. One archive officer disconnected Ilyan's local witness store. The other approached Mina with a transfer request.

She held up her wrist. “Commons property.”

“Captain's order covers derivative evidence.”

“My greenhouse rotation carries four hundred daily equipment hashes. Which one do you want?”

“All transmissions from this room.”

Mina surrendered the pane. She looked annoyed rather than frightened, which told Ilyan the device no longer mattered.

On the Violet display, the night watch lifted a page. A phase pulse crossed the seam before Hadran could order the channel closed. Ilyan heard its clipped tone through the rail.

Three, two, four, gap, five.

Alda's first page address.

Violet had seen enough to copy the key.

Sava heard it too. “End the seam transmission.”

The technician at the side console hesitated. Hadran repeated the order. The low-frame image froze, leaving Violet's page half raised.

Chart-room public links went dark next. The founder seals vanished from the wall. Alda's blank line disappeared with them. Overhead lighting brightened automatically, making the room look cleaner after evidence had left it.

Sava faced Ilyan. “You are confined to quarters pending review.”

“What happens to the ledger?”

“It receives authentication.”

“And if it proves what we read?”

Her answer took long enough to become part of his memory. “Then command answers for command.”

Hadran escorted Ilyan and Mina into the service corridor. The hatch sealed behind them. Two officers waited to separate them at the tram junction.

Mina stopped beside a condensation gutter and knelt as though fastening her boot. She ran one finger through the water. Tiny green points appeared beneath the surface, reflections from public message lamps across the corridor.

One point became five. Five became twenty.

“What did you do?” Ilyan asked.

“Greenhouse equipment hashes travel through local maintenance queues,” she said. “Every belt compares checksums at shift change. So do bakeries, water labs, school kitchens, pressure crews, and half the Commons clerks who believe duplicate inventory prevents theft.”

The message lamps continued to wake along the corridor.

“You sent the evidence into inventory?”

“I sent its shape. The images follow when someone opens a path.”

A pressure technician checked her wrist pane and looked up at Ilyan. Farther down, a tram driver did the same. The checksums held no readable confession by themselves. They proved that sealed pages existed in a specific order at a specific time. If Sava's office changed one mark, every public queue would show the break.

Hadran emerged from the chart room and saw the lamps.

He raised his cuff. Before he could speak, the ship's maintenance network issued a routine mismatch tone. One bell sounded from the nearest panel. Another answered farther down the corridor. Then another beyond the curve, each queue reporting the same unmatched founder record.

Ilyan could no longer see Alda's wake ledger. He could not solve the blank course she had left. The hatch behind him held every page under command seal.

Across Nacre, checksums kept arriving anyway.

By the time the officers led him toward the tram, the second bell had passed beyond hearing, and still the maintenance lamps came on.

Chapter 12 — The First Rescue

<!-- POV: Jo Malk -->

<!-- SETUP: Capacitor Seven bypass; symmetric stabilization; Violet deadline -->

<!-- PAYOFF: Violet survives settlement; Nacre's correction reserve drains -->

Jo drove the ceramic wedge into Capacitor Seven until the isolation jaw stopped trying to close.

The drive nave answered with a vibration that climbed through the torque handle and settled in their bad shoulder. Heat rolled off the capacitor housing. Three cooling hoses shook against their clamps, each thick as Jo's wrist. The jaw had fused at partial closure during Correction 141. Any sensible repair began by cutting power, waiting twelve hours, and replacing the assembly.

Violet had forty-three minutes.

“Wedge seated,” Jo called.

Parel at the local board repeated the status. “Mechanical hold seated. Jaw gap, fourteen millimeters.”

The value crossed the seam as phase pulses. Violet's reply arrived late enough for everyone to hear the separation between ships.

“Violet confirms fourteen,” Ilyan said from the communications bench.

He had no business in the drive nave under Hadran's confinement order. Mina had brought him through a green-belt freight lock with a crate of algae filters and no apology. Jo had no business running a shared-capacitor procedure under a captain's temporary hold. At least their violations matched in scale.

A wall pane showed Violet's capacitor room in low-frame gray. Their housing carried a split brace where Jo's side had an intact rail. Four engineers moved between frames, crossing distance in jumps. The lead wore a white scarf tied over her nose against coolant vapor. She had introduced herself as Maru Sen and refused every request to shorten her name for transmission.

Teren stood behind Jo with both hands visible and no credential strip on his wrist. He could advise. He could not touch Loom control. The distinction kept Hadran from shutting the nave by force and made nobody safer.

“Thirty-eight minutes to closure threshold,” Teren said.

“Settlement threshold,” Jo corrected.

“Yes.”

“Say the whole ugly word while we work.”

“Settlement threshold in thirty-eight minutes.”

Better. Machines lied most effectively through nouns that removed the hand.

Jo climbed down from the housing. The deck grating burned through one knee of their coveralls. Below it, field braid circled the nave in black insulated bundles. Frost clung to one side where superconducting coolant ran. The other side sweated warm droplets from the unauthorized waste-heat bridge. Every connection announced

the bad compromise: cold enough to carry causal load, warm enough to keep Violet's seam from closing.

Captain Or waited behind the yellow hazard line. Hadran stood at her right. Oru Pell held a Commons witness wafer at her left. None controlled the whole procedure. Sava had released the Loom command path for one attempt after Oru entered a formal challenge and the drive crews declined to clear the floor. She could still abort Nacre's side. Maru held Violet's abort. Jo controlled the physical bank. Teren had written the pulse. Ilyan carried timing between them.

Authority had become a box of mismatched fasteners. For once, no single wrench fit all of them.

Before loading the field, Jo demanded a dead-bus rehearsal. Parel killed output to the shunts while leaving the controllers awake. Maru did the same on Violet. Each crew would move through twelve reversals using indicator lamps instead of current. The exercise looked childish beside the capacitor banks, two teams raising levers at empty machinery, but Jo trusted hands that had practiced where to go after alarms began.

The first rehearsal failed on reversal three. Violet's frame delay hid Maru's physical motion, and Ilyan called the pulse from the image instead of the phase tone. Jo's lamp changed before hers. A live field would have driven both banks toward receiver state.

"Ignore what you see," Jo told him. "The picture comforts people. The tone carries the load."

Ilyan shut off the low-frame pane at his station. Violet's room vanished, leaving only two pulse traces and Maru's compressed voice. They ran again. This time six reversals held before Parel touched the wrong guard with his sleeve.

Jo stripped every unnecessary cover from the board. "Nothing on the floor except what can kill us."

Hadran objected when they removed the transparent safety hood over the dump switch. Jo handed him the hood. "If I need to break the field, I need my palm on metal before your regulations finish opening."

He weighed the piece, then carried it behind the hazard line.

The third rehearsal completed all twelve reversals. No one cheered. Maru asked for a thirteenth, since settlement might retry after a clean sequence. Jo added three. By the fifteenth exchange, both crews moved without waiting for words. The phase lamps crossed and came back like a game learned under bad weather.

"Live current will deform the timing," Teren warned.

"Then we begin with hands that know the undeformed shape," Jo said.

Oru entered each rehearsal failure in the witness record. Sava asked why public custody needed mistakes as well as success.

"Because the next ship will need to know where we nearly died," Jo answered.

Jo opened the bank inspection panel. "Show reserve."

Parel brought the figures onto the main board. The designated ship's correction reserve sat at sixty-two percent of arrival requirement. That number already assumed settlement yields from Violet and any weaker routes still running. When Jo removed projected harvest, the display fell to forty-seven.

Sava read it. "Stabilization draw?"

“Best case, four percent of current bank,” Jo said. “Worst clean case, eleven.”

“Failure case?”

“Seven vents into the nave. We lose the bank and anyone standing inside the red line.”

Hadran looked down. His boots crossed the red paint by half a length. He stepped back without comment.

“Violet's failure case?” Sava asked.

Maru's answer came through Ilyan's speaker. “Our Seven opens. Settlement takes the rest.”

No one improved that sentence.

The shared procedure required both capacitor banks to pulse against each other without either becoming receiver. Settlement software expected reserve to flow one way. Jo and Maru would reverse the field every 0.73 seconds, using the old answering delay as a timing brace. If both crews held phase, the Loom would encounter two active source states. Violet's closure trigger would fail to select an empty domain for extraction.

If they missed, each bank would treat the other as load.

Jo fitted two manual shunts across the controller bus. Software could not be trusted to preserve a condition its architecture classified as error. Parel locked the first shunt. Across the seam, Maru lifted a matching lever. Her image refreshed with the lever already down.

“Remote state?” Jo asked.

“Violet bank accepts peer timing,” Ilyan said. “Phase drift at three hundredths.”

“Too wide.”

Maru's voice crackled. “Our coolant pressure cycles every nine seconds. We can damp it, but not from the controller.”

Jo pictured Violet's split brace. Vibration would move the coil relative to its field sensor. A controller correction would arrive after the physical shift and chase it into a worse oscillation.

“Put weight on the brace,” they said.

“What kind?”

“Bodies.”

Sava looked at Jo. “Explain.”

“No time.”

On Violet's pane, Maru called three engineers over. They climbed onto the brace, hooked boots beneath the rail, and leaned their weight against the split. One frame showed them moving. The next caught them fixed in a ridiculous row, four adults sitting on a failing machine while another world watched.

Drift fell to one hundredth.

Jo grinned despite the heat. “Good repair.”

Maru's reply arrived through static. "Your standards lack dignity."

"My standards retain pressure."

The first warning bell sounded.

Thirty minutes.

Settlement mode began its preparatory scan. Route glass above the local board brightened violet along one edge. The Loom queried for receiving designation. Sava's released command path kept the answer empty.

A red prompt covered her console: CONFIRM DESIGNATED ROUTE.

She dismissed it.

The prompt returned.

"Ignore it," Jo said.

"Command interfaces do not accept neglect as input."

"Learn from engineering."

Oru's wafer recorded the prompt and her refusal. Hadran watched the nave hatches. A crowd had gathered beyond both pressure doors, drive workers pressed behind the small safety windows. They could not hear the seam, but they could see the violet light entering the braid.

Teren moved close enough to speak beneath the coolant pumps. "The first pulse should begin from Violet."

"Your model says designated side."

"My model assumed lower remote noise. Their brace changed it."

"You want to reverse the whole sequence?"

"It reduces phase correction after pulse four."

"By how much?"

"Two hundredths."

Jo looked at the deadline clock. Twenty-seven minutes. Reversing the sequence meant recoding twelve manual triggers on both ships. It also meant asking Violet to push first into a system designed to harvest it. If Nacre failed to answer, Violet would spend reserve for nothing.

They keyed Maru. "Physics wants you to lead."

"Does physics have its access strip?"

"No."

"Then what does engineering want?"

Jo examined the drift trace. The pulse margins left almost no room after the fourth reversal. Teren's shame made him over-explain, but his numbers did not ask forgiveness.

“Engineering wants you to lead.”

Maru nodded once in a later frame. “Then keep up.”

They recoded the triggers.

At nineteen minutes, the Loom began closing minor service channels. Indigo vanished from the passive map. Three weaker traces dimmed. Ilyan tried to ping them and received no clean reply.

“Stay on Violet,” Jo said.

“I can monitor more than one clock.”

“You cannot speak through more than one mouth.”

His hand hovered over the channel board, then returned to Violet's timing rail. Jo understood the refusal in his face. Choosing one signal resembled designation. The hardware forced it anyway.

“Mark every trace we lose,” they said. “Do not call them closed.”

Ilyan opened witness clocks for each fading line.

One of Indigo's pulses broke through before its channel narrowed. The children used alternating high and low tones, a rhythm Jo had heard during Nessa Quill's demonstration. This sequence carried no words until Ilyan mapped the game intervals into letters.

ARE YOU CHOOSING THEM.

Jo kept their hands on the shunt. “Send: we are testing one damaged capacitor path. We have not decided who counts.”

“That sounds evasive,” Oru said.

“Because a full answer takes longer than their seam has.”

Sava stepped toward the timing rail. “Tell them Violet's bypass creates the present opportunity. Tell them we are recording Indigo's clock.”

Ilyan sent both statements. The route trace dimmed before an acknowledgment came back.

No one called that communication. Jo heard the children's question inside the missing reply. Saving Violet first followed hardware, not worth, but machinery had always offered physical limits as cover for political choices. If the repair worked, Jo would have to publish every clamp, timing error, reserve cost, and failure. Otherwise the one successful procedure would become another locked command privilege.

They opened the dead-bus rehearsal record to automatic seam copy. Hadran noticed the outgoing traffic and began to object. Sava said, “Let it go.”

Twelve minutes.

Jo sealed the inspection panel and checked every fastener by touch. Left shunt cool. Right shunt warm. Isolation wedge seated. Manual dump clear. Their shoulder had begun to numb from vibration. They changed hands on the torque bar.

Parel said, "Bank Seven temperature rising."

"How fast?"

"Two degrees per minute."

"Expected."

"Three now."

Not expected.

Jo crouched at the warm coolant return. Flow rattled inside the hose. A vapor pocket had entered the loop, likely boiled loose when they forced the jaw. If it reached the coil during pulse reversal, field resistance would jump.

They slapped the hose. The pocket moved half a meter and stopped at a high bend.

"Need a bleed," Jo said.

Parel shook his head. "Valve discharges inside the red line."

"So do I."

Jo crossed the paint before anyone answered. Hadran called their name. They climbed the housing, wrapped one leg through a service rung, and reached the bleed stem. Heat pressed against their face shield.

"Six minutes," Ilyan called.

The stem would not turn. Jo put the torque driver on it. Their scarred shoulder refused the angle, pain flattening into weakness. They shifted the tool to their other hand and braced both boots.

Sava came across the hazard line.

"Get back," Jo said.

She caught the lower handle. "Turn on three."

"You'll be in the vent path."

"So are you."

Jo almost told her command did not repair machines. The valve stem moved before the insult arrived.

"One. Two. Three."

They pulled together. The stem cracked open. White vapor struck the deck where Sava had stood a second earlier. She dragged the handle and herself aside. The hose kicked hard enough to tear one clamp loose. Parel hit the local restraint, trapping it against the housing.

Temperature rise slowed.

Sava returned behind the line with frost whitening one sleeve. Hadran checked her arm. She pushed him away and kept her eyes on the command pane.

“Two minutes,” Ilyan said.

Jo dropped to the trigger station. Across the seam, Maru and her engineers sat on their brace. Violet's bank values flickered beside them. Their reserve had fallen below the line that settlement used to declare a domain exhausted.

The Loom sounded its maneuver bell.

Not a correction bell. A closure warning inherited from the first trial, long buried under the label release complete. Its bronze note rolled through Nacre's structure.

A second bell answered 0.73 seconds later from Violet.

“Ready shared pulse,” Jo said.

Parel confirmed the shunts. Teren read phase values. Oru announced Commons witness active. Sava left the designation prompt unanswered. Hadran sealed the nave doors against anyone who might interrupt.

Ilyan counted the final intervals.

“Violet lead in three. Two. One.”

Maru pulled her trigger.

Violet light flooded Nacre's field braid. Jo felt it before the instruments showed load, a deep impact under the deck as if another ship had struck them without crossing distance. Bank Seven climbed to one hundred eight percent and held.

“Answer,” Teren said.

Jo pulled Nacre's trigger.

The load reversed. Violet's capacitor took it. Their engineers jolted on the brace. One lost a boot and Maru caught his belt.

“Again,” Jo said.

Pulse crossed. Pulse came back.

The cadence settled around the 0.73-second seam delay. Each reversal shaved reserve from both ships. Nacre's correction reserve dropped one digit at a time. Forty-six percent. Forty-five. Violet's closure clock reached zero.

Settlement issued its extraction command.

The bank screamed.

Copper windings tightened inside their housings. Frost exploded from the black braid. Jo smelled scorched insulation through the face shield filter. The isolation jaw drove against the ceramic wedge, trying to complete the old one-way circuit.

“Hold phase,” Teren called.

“Jaw load rising,” Parel said.

Across the pane, Violet's brace split another finger-width. The engineers threw their weight harder. Maru kept one hand on the trigger.

Pulse crossed.

Jo answered.

The ceramic wedge cracked.

A white line opened through its center. Jo could dump Bank Seven and save the nave. The motion waited under their left palm. Dumping would break the shared source state. Settlement would find Violet empty on the next scan.

"Nacre reserve thirty-nine percent," Parel said.

Sava's command display filled with arrival warnings. She watched the numbers fall without reaching for designation.

"Continue," she said.

The jaw bit deeper. Jo put the torque bar against it and leaned, using their own weight to keep fourteen millimeters of space. Pain returned to the bad shoulder, hot and useful.

"Day nineteen boundary," Ilyan said.

The phrase sounded too small. A day on Violet had contained meals, arguments, bandages, school shifts, and people learning how much sky fit inside smoked glass. The deadline knew none of that. It only counted when extraction gained permission.

Settlement tried again.

Both ships pulsed.

No receiving state opened.

The Loom repeated its command. ERROR: DESIGNATION CONFLICT.

"That's it," Teren said. "Keep the conflict stable."

Jo laughed once. "Finally, a fault worth preserving."

They ran the sequence through twelve reversals. The wedge shed a chip. A cooling clamp bent. Violet's image froze for three frames, then resumed with Maru still at her trigger.

At reversal thirteen, settlement reset its scan.

Violet remained.

The closure clock advanced past zero.

One second. Four. Eleven.

Nobody in the nave spoke. Jo held the torque bar and listened to two capacitor banks carry separate loads. The system searched again for a route it could empty. It found two sources refusing the assigned direction.

Sixty seconds beyond deadline, Teren lowered his phase board. "The trigger has passed."

"Do not stop yet," Jo said.

They tapered the pulse over six intervals, each smaller than the last. Violet mirrored them. The field braid dimmed without going cold. Bank Seven settled at a temperature that would ruin its insulation within days rather than minutes.

Jo released the torque bar.

Their arm hung useless for a moment. Parel caught the tool before it hit the deck.

Ilyan opened voice to Violet. Static moved through the speaker. Then Maru laughed, coughed, and said, "Brace retains pressure."

Behind her, one engineer remained sitting on the split rail. He gave the camera a tired salute.

"Clock?" Jo asked.

Violet's display came close to the lens.

Day nineteen had ended. Their hours continued.

Cheers struck the pressure doors from outside, muffled by layered glass. Inside, Sava looked at the arrival reserve. Nacre held thirty-four percent of requirement. The correction reserve warning had turned from amber to red.

"We spent thirteen points," she said.

"Fifteen after thermal losses," Parel replied.

"Can we recover any?"

Jo checked the bank. "Not without taking it back from Violet."

Sava dismissed the recovery prompt when it appeared.

The passive route board began to sound.

Indigo came first, a phase pattern built from children's two-sky timing. Then a weak unnamed seam repeated Violet's lead pulse. Another trace offered three short signals and a pause. Ilyan opened witness clocks as fast as his hands could move.

Messages arrived in fragments. HOLD US. SHOW THE PROCEDURE. OUR BANK DIFFERS. HOW MUCH TIME. DO NOT SELECT ONLY ONE.

Jo looked at Bank Seven's cracked wedge, Sava's red reserve, and Violet's continuing clock. The procedure had saved one ship from one deadline. It had also proved that settlement could be denied.

Beyond the nave doors, people kept striking the glass in celebration. On the route board, every surviving light asked for the same chance before the heat in Seven had time to cool.

Interlude II — Games for Two Skies

<!-- REGISTER: child-limited, Route Indigo -->

<!-- SETUP: Nessa's phase game; delayed seam frames -->

<!-- PAYOFF: children remove first-player privilege -->

Nessa Quill drew the second sky on the classroom wall because the camera could not fit the real one.

Indigo had no sky, strictly speaking. It had a ceiling painted with blue panels above the school garden, three of them dark since Correction 141. The children on the other ship had a ceiling too. Their seam camera showed eight pale panels, a water stain shaped like a boot, and half a paper sun taped beyond the frame.

Teacher Ara called both ceilings skies because that made the game easier to explain.

Nessa did not think easier always meant truer. She drew two long rectangles in chalk and put a dot in each one.

The other classroom saw her drawing after the seam delay. Their image moved in small jumps. A boy with shaved sides appeared beside the paper sun, disappeared, then came back holding a red card. Nessa held up blue.

The game began with colors.

If Indigo showed blue, the other class answered red. If Indigo showed red, they answered yellow. Yellow required two claps. Green meant everybody changed seats before the next frame arrived. Nobody had invented a meaning for purple, so the youngest children used it whenever they wanted to make the older ones argue.

Teacher Ara kept the rules on a large slate.

RULE ONE: INDIGO SENDS.

RULE TWO: OTHER SKY ANSWERS.

RULE THREE: MATCH THE DELAY.

The words other sky bothered Nessa. The children across the seam called Indigo their other sky too. A name that changed depending on who looked at it did not help with scoring.

“What do they call themselves?” she asked.

Teacher Ara adjusted the camera. “Their class belongs to the designated route.”

“That isn't a name.”

“It identifies their ship.”

“So does Other.”

Across the frame, the shaved-head boy held up a card with block letters.

DRUM THREE CLASS SIX.

Nessa wrote INDIGO CLASS SIX beneath her rectangle and NACRE CLASS SIX beneath the second.

Teacher Ara erased Nacre and replaced it with DESIGNATED NACRE.

The children played through first meal. Indigo sent. Designated Nacre answered. Every correct response earned a bean in a cup. The seam carried no beans, so each class kept its own cup. By the end, Indigo had twenty-two and the other class reported twenty-one.

“That means we won,” said Pima, who counted everything twice.

“No,” Teacher Ara said. “You had first signal. The answering class performs the harder task.”

Nessa considered the delay. Sending first required choosing a color. Answering required remembering what a color meant. Both became difficult when purple appeared.

“Then they won,” Pima said.

“The exercise has no winner.”

“Why do we have beans?”

Teacher Ara put the cups away.

On the second day, Class Six on the designated ship sent first. The shaved-head boy showed yellow. Indigo clapped twice. A girl beside him showed green before Indigo's answer reached them, so both classrooms changed seats at different times. Children crossed behind one another in broken frames. For nine seconds, every chair appeared occupied by pieces of several people.

No one crossed the seam. Nessa knew that. Adults kept repeating it as if children might mistake a picture for a hatch. What moved between rooms amounted to light, sound, and instructions. The game worked because instructions could change what bodies did on each side.

Nessa asked for the rules slate.

Teacher Ara had changed it.

RULE ONE: DESIGNATED CLASS SENDS.

RULE TWO: INDIGO ANSWERS.

RULE THREE: BOTH CLASSES MATCH DESIGNATED TIME.

“That isn't yesterday's rule,” Nessa said.

“It clarifies the reference clock.”

“Their clock runs late on our wall.”

“Our clock runs late on theirs,” Pima added.

Teacher Ara glanced toward the two mission officers seated at the back. They had come to observe after the stabilization of Violet. One carried a recorder. The other wore the expression adults used when they wanted silence to count as agreement.

“Navigation needs one reference,” Ara said. “The game should teach a usable convention.”

Nessa opened her desk and took out yesterday's copy. Children copied rules because teachers sometimes erased the interesting parts. Her slate still said INDIGO SENDS and OTHER SKY ANSWERS.

The shaved-head boy appeared with his own slate in the next frame. His rules matched Nessa's old version, except his first line read OTHER CLASS SENDS.

Both classes began talking. The seam compressed their voices into rough bursts. Teacher Ara muted the return channel.

"Quiet," she said.

The room obeyed, but the other class kept moving without sound. The boy pointed at his old first rule, then at Indigo. A girl held up the new rule from their wall. On her side, it said DESIGNATED CLASS SENDS.

Nessa understood before the adults explained. Yesterday, each room had received a different rule. Indigo had been told to send first. So had the designated class. When the collision produced confusion, both teachers had changed the lesson. Now each room received a rule that placed the designated sky first.

The game did have a winner. Adults had hidden it inside who got to begin.

"Why did you edit both copies?" Nessa asked.

The recorder officer leaned toward Teacher Ara. "The terminology supports mission continuity."

"Does continuity mean their color counts before ours?"

"It means one clock must anchor shared action."

"Which clock?"

"The designated route's clock."

"Why?"

The officer looked relieved to have reached a question with an official answer. "Because that route carries mission authority."

Across the seam, the other children could not hear him while the channel remained muted. Nessa went to the camera and wrote his answer on a card.

Teacher Ara caught her wrist before she raised it.

"We are still in class," Ara said.

"Yes."

"You do not transmit an adult conversation without asking."

"They asked us to play by it."

Ara kept hold of the card. Her fingers felt warm and tired. Nessa liked her. That made the grip worse.

Pima lifted purple.

Every child on Indigo changed seats.

The rule meant everyone moved. Teacher Ara had written that rule herself and had not erased it. In the confusion, Nessa carried the answer card to Pima's old chair and raised it toward the camera.

The other class read it one broken frame later.

Their shaved-head boy showed purple.

All of them moved too.

The mission officer told Ara to end the exercise. She reached for the seam control. Before she closed it, the paper-sun girl held up six words.

WHO SAID YOUR SKY GOES SECOND?

Nessa found a fresh card.

NO ONE. WHO SAID YOURS DOES?

The seam remained open. Teacher Ara's hand rested on the switch. She looked at the officer, then at the two classrooms changing places in delayed waves.

"Five minutes," she said.

The children spent four minutes arguing and one minute writing.

They removed the two first rules. A game could not begin with a sender because each class saw the other's signal after its own present moment. Instead, both rooms would choose a color and cover it. Each teacher would ring a cup. On the tone, both classes revealed. A matching pair earned one bean in each room. Different colors required the children to combine their meanings. Blue and green meant change seats slowly. Yellow and red meant clap twice, then trade cards. Purple still meant everybody moved.

Pima found the problem. "Which teacher rings?"

Both classrooms pointed at their own teacher.

Nessa drew two bells. Each teacher would ring after seeing the other teacher raise a hand. Because of the delay, neither bell would sound first in both rooms. Scoring began only after each class had received both tones.

The recorder officer said the procedure lacked a reference clock.

"It has two," Teacher Ara replied.

Together, the classes wrote the replacement rule. They checked each word through three delayed frames so neither teacher could improve it afterward.

NEITHER SKY STARTS FIRST.

Nessa copied it beneath both chalk rectangles. On the other wall, the shaved-head boy did the same. The letters appeared backward until his camera corrected them.

Teacher Ara lifted her hand. Far across the seam, another teacher saw it late and raised hers. Two cups rang in separate classrooms. Each group waited until both sounds arrived.

Nessa uncovered purple.

The paper-sun girl uncovered purple too.

Every child changed seats. This time the adults had to move with them.

Chapter 13 — The Branch That Refused

<!-- POV: Lin -->

<!-- SETUP: Violet survives settlement; rescue-as-archive proposal; divided emergency council -->

<!-- PAYOFF: civic standing demanded; Lin chooses and signs his name -->

The first ration bowl struck the council table before Ilyan could open the seam.

It spun on one edge, shedding three drops of algae broth, then settled between the route charts. Seli Arven had thrown it from the second bench. She remained standing with both hands empty, a pressure medic in a torn blue jacket, her face hollow from nineteen days of shortened meals.

“Send them that,” she said. “They want a copy of Violet. Let them archive the bowl.”

Nobody laughed.

Violet's emergency council occupied the old Commons well in Drum Three. Condensation streaked the upper terraces where climate controls had failed after Correction 141. Two hundred residents crowded the benches. Another eight thousand listened through local relays, as many channels as the damaged power grid could support. The council had no charter for governing a ship after settlement. It had a school attendance slate, three ration ledgers, and the pressure-deck speaking order taped across the clerk's desk.

For nineteen days, that had been enough to keep air and meals moving.

Ilyan had entered the well on the second day to post route damage, not to lead. The elected Commons chair had died in the dust strike. Two deputies remained trapped behind a warped pressure door until day four. Navigation rank gave Ilyan access to clocks and maps, but nobody in Drum Three cared about his seal when the first ration queue split into two lines, each accusing the other of hiding food.

He had solved that fight by carrying both ledgers into the school garden and making every bin count public. The next morning he taught bridge timing to children because trained operators needed sleep. Later he mediated a dispute over whether the injured received full broth on days when they could not work. He argued for need. The drive crews argued for labor. Seli demanded both groups stop treating hunger as discipline. Their compromise lasted three days, failed, and returned in a better form after the green-belt cooks joined.

Nothing about those tasks resembled route conservation. They gave him authority because residents watched him make errors where they could challenge them. Each evening, the emergency council reopened his decisions. Twice it reversed him. Violet had not chosen him as another captain. It had chosen a person who could be removed without stopping the air.

Now the designated ship requested a formal delegation.

Ilyan set the bowl upright. “We are not voting on archive transfer.”

Seli pointed at the seam display. “Their proposal says memory preservation.”

“The proposal came from a technical working group. Their Commons clerk has already objected.”

“And their captain?”

“Has not withdrawn it.”

The answer moved badly through the well. People had watched Nacre spend reserve to prevent Violet's closure. Gratitude had arrived beside suspicion, each making the other harder to admit. Half the council wanted immediate cooperation on every stabilization test. Others believed the rescue only preserved Violet until Nacre could extract something cleaner.

The council docket carried three competing motions. Maru's engineering group requested open technical exchange until Bank Seven cooled. Seli and forty pressure workers demanded that all seam traffic stop except proof-of-life pulses until Nacre recognized Violet's jurisdiction. A third motion, filed by night-shift navigator Olan Pre, proposed feeding false reserve values into settlement so the designated ship could not calculate another extraction.

Olan stood when Ilyan read it. "They act carefully because they need our phase data. Once they have it, we become expensive."

"False values could destabilize both banks," Maru said.

"Then they will respect the cost."

"Or the bypass jaw melts and kills us before respect arrives."

Seli lifted one hand. "Do not make my refusal part of his sabotage. I want no attack."

Olan faced her. "You want to wait while their captain decides whether we qualify as a public."

"I want our children to survive us without inheriting a war through a glass wall."

Ilyan marked the false-data motion technically unsafe and kept it on the record. Erasing it would turn anger into rumor. Approving it would let one frightened council use shared machinery against forty-two thousand people who had not chosen settlement doctrine. Violet's refusal could not depend on making another ship helpless.

At the lowest terrace, Maru Sen sat with one arm in a sling after the capacitor brace cracked. She had refused infirmary until the council met. "We have twenty-three hours before Seven overheats again," she said. "If we spend those hours debating their manners, we die with excellent standards."

Seli turned on her. "If we surrender our records and call it rescue, we die politely."

"I said use their cooling model."

"You said cooperate."

"With their engineers. Not their dictionary."

The council recorder failed to assign speakers fast enough. Names crossed on the wall. Three delegates claimed the same turn. A child messenger slipped among the benches collecting paper votes that no one had authorized yet.

Ilyan recognized her from the bridge school shifts. Veyra, nine years old, one front tooth missing. She had learned route timing because every adult with navigation training spent the first week on hull triage. Ilyan had taught her to count 0.73 seconds by tapping two spoons against a rail.

She stopped beside him. "Which name goes on your paper?"

"Ilyan Rook."

“Which one?”

The question carried no challenge. The ration slate listed two Ilyan Rooks whenever Nacre joined a session. Clerks had begun improvising. DESIGNATED ILYAN appeared on their side. VIOLET ILYAN appeared here. One navigation message used ROOK-TWO, as though divergence had turned a person into an inventory duplicate.

“Write chair,” he said.

Veyra frowned. “That's a job.”

“So is messenger.”

“I still have a name.”

Before he could answer, the seam indicator brightened.

Nacre's Commons well appeared on the wall in low-frame monochrome. Oru Pell sat at a table facing the camera. Captain Or had not attended. Ilyan Rook stood behind Oru with both hands resting on the rail.

Seeing his own face no longer shocked him. The differences did. Nacre's Ilyan had slept more and eaten better. His collar lay flat. Violet's cracked cameras exaggerated shadows, but no camera had carved the new hesitation around the other man's mouth. He looked as though every word required checking for hidden priority.

Oru's voice arrived through compression. “Route Violet, this session concerns provisional civic recognition and access to shared stabilization records. Please identify your authorized representative.”

The clerk beside Ilyan activated his microphone. “Ilyan Rook, senior route conservator, acting chair of Violet emergency council.”

Nacre's docket populated on the wall.

VIOLET ILYAN ROOK / DERIVATIVE WITNESS.

The well erupted.

Seli shouted over the others. Maru struck the table with her good hand. Someone in the upper terrace demanded the channel close. Nacre's Ilyan turned toward an off-frame console, and the label vanished.

Oru spoke after the noise fell enough for transmission. “That designation came from an inherited archive category. Commons rejects it.”

“It reached us through your official channel,” Seli said.

“Yes.”

“Then your ship does not reject it. You do.”

Oru accepted the distinction with a small nod. “Correct.”

Ilyan stepped to the microphone. “Enter Violet emergency council as the speaker until we settle individual forms.”

“The docket requires a person,” Oru said.

“Your docket requires many things reality did not provide.”

Nacre's Ilyan almost smiled. The expression hurt more than hostility might have.

Oru entered the council as provisional party. “State Violet's position on the archive proposal.”

“We refuse evacuation as data.”

The sentence crossed the seam and returned through Nacre's speaker, flattened by delay.

Ilyan continued. “Records can witness a life. They cannot substitute for one. We will share technical evidence, ordinary messages, and names under reciprocal custody. We will not accept a process that copies selected memories, closes our domain, and reports preservation.”

A councilor named Daro Kesh rose from the first terrace. He supervised Violet's embryo store, where backup power had failed twice. “Clarify that refusal does not include child genomes.”

Ilyan looked at him. “You want transfer?”

“I want options. Their cold store has stable power. We can send sequence data through the seam.”

Seli said, “Then they raise replicas and claim our children continued.”

“We have six hundred embryos losing viability.”

“They have no bodies across there.”

“They have information that might let something survive.”

The pressure beneath the argument bent every word. No mass could cross. A genome packet might preserve potential lineage without moving a single embryo. Refusing false rescue did not require burning every record. Accepting an archive did not authorize anyone to call the archived people saved.

Ilyan asked Daro, “Would you transfer the sequences if the record states that Violet retains ownership and that closure kills the stored embryos?”

“Yes.”

“Would you accept a claim that reconstructed descendants replace them?”

“No.”

“Then your position does not contradict ours.”

Seli sat, unconvinced but listening.

Oru read the statement back. Their version changed one phrase: civic records under reciprocal custody became branch records under shared custody.

Ilyan raised his hand. “No.”

“Which term?”

“Branch.”

Oru glanced down. "The provisional resolution uses route branch."

"A branch depends on a trunk for its description. Violet depends on Nacre for a powered seam. That does not make their ship our civic trunk."

Nacre's Ilyan watched him across the separate glass. "What term do you accept?"

"Ship. Route. Violet."

"Those do not describe the common origin."

"History describes it. Rank does not."

The other Ilyan lowered his eyes for one frame. When the image refreshed, he had moved closer to the microphone.

"We can amend the form," he said.

A woman shouted from Violet's upper terrace, "They amended forms before they killed us."

No one could answer that with procedure.

Maru asked for the floor. "Our bank needs parts we cannot receive and a timing model we can. Nacre's reserve kept us alive. If this council refuses contact because their docket insulted us, Seven fails tomorrow."

Seli replied, "Contact without standing lets them decide rescue terms."

"Standing without coolant gives us a dignified funeral."

"What do you propose?" Ilyan asked Maru.

"Technical cooperation under dual abort. No command access to our bank. Every test copied to both Commons records."

Seli lifted the ration bowl from the table. "And no archive language."

"Agreed."

Daro added, "Sequence packets marked unsaved."

A murmur of approval came from the embryo-store benches.

The council had not begun with a motion. It found one through collision. Ilyan asked the clerk to write the terms where both ships could see them.

VIOLET RETAINS LOCAL CONTROL.

TECHNICAL COOPERATION REQUIRES DUAL ABORT.

RECORD TRANSFER DOES NOT CONSTITUTE EVACUATION.

NO EXTERNAL COMMAND MAY SPEAK FOR VIOLET.

Oru read each line into Nacre's record. "Commons can sponsor these as temporary conditions. Captain Or has not accepted the last clause."

“Then she has not accepted our help,” Maru said.

The reversal made Ilyan turn toward her.

Maru gestured with her sling at the seam. “Their correction reserve fell saving us. Their arrival model needs data from our damaged vector. Cooperation runs both directions now.”

Teren's debt accounting had reached Violet in fragments. Enough survived compression to show that every route carried different consequences from Correction 141. Violet's hull had taken more dust damage. Its instruments had measured a field edge Nacre never crossed. Their suffering produced knowledge, though Ilyan refused to let usefulness become another test for personhood.

He asked Oru, “Does your provisional resolution grant Violet standing before technical negotiation?”

“It proposes witness status pending Commons vote.”

“Witness to what?”

“To route persistence, settlement harm, and current conditions.”

“A witness appears in someone else's case.”

Oru sat back. Their delayed image held still while the seam compressed silence. “What standing do you request?”

The council answered from several terraces.

“A ship.”

“A public.”

“Forty-two thousand and eleven living people,” the ration clerk called, giving Violet's latest count.

“Our school,” Veyra said beside Ilyan.

“Our dead,” Seli added.

Daro lifted his embryo ledger. “Our future, if any.”

Ilyan heard no clean legal category in those answers. That did not weaken them. Law aboard Nacre had no box for a civic whole instantiated by machinery and abandoned after use. A blank field should have forced new writing. Instead the old system entered derivative witness.

“Violet requests standing as a shipboard civic whole,” he said. “Not as evidence, equipment, a technical fault, or a population held for later consolidation. Our council speaks through emergency authority that remains reviewable by our residents. We offer reciprocal engineering aid. We retain every right to refuse a procedure that ends us.”

Nacre's Ilyan said, “I support the request.”

Seli rose again. “As what? Senior conservator? Designated survivor? The person whose face gets to assure us?”

His mouth closed.

The question struck both Ilyans. They shared the reflex to solve a room by offering personal credibility. They shared memories of Eda, Guild training, and a seal impressed into wax. None granted Nacre's man authority over Violet. None granted Violet's man authority without the benches behind him.

Ilyan turned from the seam to his council. "Do I hold your mandate for the four conditions and the standing request?"

The clerk distributed the vote through local relays. Paper slips came first from people physically present. Veyra carried handfuls down the terraces. Remote tallies followed as power allowed. The count moved slowly, interrupted by one dark district and a challenge from sanitation over whether abstentions had been registered.

During the wait, Nacre's docket tried another individual label.

ROOK / VIOLET INSTANCE.

Oru removed it before anyone asked.

The dark district came back on a hand-cranked relay from Drum Five. Its delegate appeared without image, breath loud over a borrowed pressure handset. "We have one hundred ninety-three residents at this node. The vote runner never reached us."

The clerk checked the tally. "Your district shows unreachable."

"Unreachable doesn't mean abstaining."

Seli asked how long restoring their ballot would take. Maru answered before the clerk could. Bank Seven held twenty-three hours at the current temperature, perhaps less if the split brace shifted. The delegate demanded fifteen minutes.

Ilyan looked toward the seam. Oru had offered to keep Nacre's session open for six. Captain Or's command queue already marked the delay operationally harmful.

"We wait fifteen," he said.

Maru's good hand tightened on the bench. "Chair."

"If our standing rests on a vote, we do not make damaged districts disappear because their clocks inconvenience us."

The sentence also crossed to Nacre. Oru removed the six-minute limit from their docket.

While Drum Five gathered ballots, the council heard three minority statements. Olan defended his false-data proposal, then agreed not to act without authorization. A group from cold storage objected to sharing embryo sequences before custody terms existed. Six bridge families opposed any representative who shared a face with Nacre's conservator. Ilyan entered each objection beside the eventual vote rather than beneath it.

The hand-cranked relay clicked for thirteen minutes. Results came as counted knocks because the pressure handset battery failed. Veyra translated them at the clerk's table. One hundred twelve approved. Fifty-one opposed. Thirty could not be reached behind local seals.

Drum Five entered the tally as a district, not an absence.

Veyra placed the last terrace packet beside Ilyan. "You still didn't tell me your name."

He looked across the seam at Ilyan Rook. Same jaw, same hands, same habit of listening a fraction too long after speech ended. Through Correction 141, one name had fit them both because one life had carried it. Nineteen days had not made either false. Yet every shared syllable gave officials a place to attach priority: first and second, designated and Violet, source and instance.

He could demand that Nacre's Ilyan change instead. That would preserve the contest.

“What do you hear when the seam clips my name?” he asked Veyra.

She thought. “Sometimes Ilyan. Sometimes Lyan. Once just Lin.”

“Did you know who spoke?”

“Yes.”

The short sound carried no rank. It had arrived through damage and still referred to him. He tested it silently against memories that belonged to both men. Eda calling from a pressure hatch. Mina shouting across a green belt. His instructor reading his Guild seal. The old name held those moments. Choosing another would not surrender them.

“Write Lin,” he said.

Veyra printed it on the speaking card. “Only Lin?”

“Only Lin.”

“Is it short for something?”

“No.”

That pleased her. She carried the card to the clerk.

The vote closed at eighty-one percent approval, nine against, ten abstaining or unreachable. Not unanimity. Violet's civic whole contained people who distrusted Nacre, people who needed its help, and people too tired to choose. Standing had to include the disagreement.

Lin faced the seam.

“Our council authorizes the request and technical conditions. Enter the representative as Lin, acting chair, subject to Violet review.”

Nacre's Ilyan repeated the name carefully. “Lin.”

No possessive followed it. No number.

Oru updated the provisional docket. LIN / VIOLET EMERGENCY COUNCIL appeared, then shifted when Lin objected to the slash.

LIN, ACTING CHAIR

VIOLET EMERGENCY COUNCIL

The lines stood equal in size.

A command message arrived before Oru could seal the session. Captain Or accepted continued engineering exchange and dual abort, but reserved judgment on civic standing. Hadran required all packets to pass security review. The last clause would delay cooling data beyond Maru's safe window.

Maru read the order and began issuing instructions to bypass the review queue.

Oru said, "Commons can receive the packets directly."

"Do it," Lin said.

Seli tapped the ration bowl against the table. "Are we cooperating or refusing?"

"Both," he said. "We cooperate where work keeps people alive. We refuse any term that makes survival permission."

The council clerk placed the four conditions beneath the camera. A signature field waited at the bottom. Its software attempted to autofill ILYAN ROOK from the old ship registry.

Lin cleared it.

The field suggested VIOLET ILYAN. He cleared that too. ROOK-TWO appeared from Nacre's archive map. He disabled suggestions.

Across the seam, the other Ilyan rested one hand against his own glass. Lin did not mirror him. The synchronized gesture had mattered before. Today separation needed its own shape.

He took the stylus.

Violet's council watched from the terraces. Maru waited with a hot capacitor. Seli held the empty ration bowl. Veyra stood on tiptoe to see the screen.

In the blank field beneath Violet's conditions, he signed Lin and nothing else.

Chapter 14 — The Mutiny of Maps

<!-- POV: Jo Malk -->

<!-- SETUP: Hadran's seizure; Article 4 expiry; retaliatory branch surge -->

<!-- PAYOFF: chart authority divided among engineering, Commons, and route delegates -->

Jo caught the isolation jaw before it bit through the Violet seam.

The drive-deck control had moved under their palm with the ugly ease of a part recently serviced. On the other side of the chart-room rail, a security technician held the matching lever down with both hands. Hadran Coil stood behind him, boots planted inside the yellow field boundary as if a painted line could make the seizure lawful.

“Let go,” Hadran said.

“Take your boot off my bus.”

The isolation jaw trembled between commands. Its ceramic teeth hovered three millimeters apart, close enough to throw blue light across Jo's knuckles. If the security technician won, the jaws would close and dump every active seam into the waste-heat loop. Violet would lose the shared capacitor feed that had carried it past settlement. Indigo's narrow timing channel would vanish. The other wakes, still no more than uncertain ticks in smoked glass, would lose whatever chance remained for names.

If Jo won too quickly, the jaw would slam open against a loaded braid and arc through the rail.

Neither result counted as repair.

“Manual bus control belongs to engineering,” Jo said. “You can arrest me after you stop trying to weld the chart room shut.”

Hadran raised one hand. The technician eased his pressure without releasing the lever. Six security officers occupied the amphitheater steps. They had sealed both public hatches and unbolted the Commons data conduit from its floor socket. No one had drawn a weapon. A stunner would be useless around the Loom's superconducting braid, and Hadran understood machinery well enough to fear consequences even when he misunderstood people.

Above them, route glass hung in its black frame. Violet occupied the center pane through a low-frame image: Lin beside a chart table assembled from classroom panels, his face breaking into gray blocks whenever the seam load dipped. Indigo showed only a pulse ladder. Two weaker traces waited beyond it, unnamed and losing definition.

Ilyan sat at the conservator's station with his hands visible. A security seal covered his archive controls. Oru Pell had been removed before Jo arrived. Captain Or had not come at all.

“Your bus became a course-security instrument when you used it to preserve an unauthorized domain,” Hadran said. “Article 4 gives mission command custody.”

“Article 4 gives the captain emergency course authority. It does not teach you how this jaw works.”

“It gives me power to stop uncontrolled contact.”

Jo felt heat growing under the lever housing. They shifted their wrist by a fraction and brought the jaw into balance. "Then stop contact. You ordered a hard dump. That sends the held charge somewhere."

The security technician looked toward Hadran. He knew enough to dislike the word somewhere.

Hadran did not look away from Jo. "Waste loop."

"At this load, half. The rest crosses whichever seam has lower resistance."

Violet's image broke, returned, and sharpened around Lin's eyes. "We heard that."

Hadran turned toward the glass. "This channel carries no branch participation."

"Then you should close it without sending us your reserve," Lin said.

A second voice cut across his, distorted by phase compression. Violet's duty engineer had opened an engineering sideband. "Send it. We can reverse the pressure and strip their designated bank before they shut us out."

Jo's grip tightened.

The threat did not come from a stranger. Nineteen days of divergence had changed rosters, authority, and hunger, but Violet's engineer had trained under the same drive manuals Jo knew. She would understand the asymmetry in the loaded jaw. A reverse surge, timed as the ceramic teeth separated, could pull reserve out of designated Nacre. Enough charge might stabilize Violet for another interval. It might also collapse Indigo and burn the two weak traces beyond recovery.

"Don't touch your coupler," Jo said.

"You gave us the path."

"I gave you time. I did not give you the other ships."

Hadran moved down one step. "There. Infiltration of a mission control system, stated in front of witnesses."

"Through a channel you just tried to energize with a hard dump." Jo kept their voice level. Anger made hands imprecise. "Your shutdown creates the opening."

"Can you prevent the surge?"

"Yes."

"Then do it and close the seam."

"That order contains two jobs that fight each other."

The jaw housing clicked as thermal expansion moved through its bolts. Jo counted the fasteners by touch. Four flange bolts. Two keeper pins. One field latch beneath the plate. They could disable the hard-close solenoid by pulling the latch, but releasing the lever first would let security finish the command. Keeping pressure while reaching under the live plate required another pair of hands.

They looked at the security technician. "What's your name?"

His jaw shifted before he answered. "Kest."

“Kest, put your left thumb against the red witness mark. Hold exactly where I am holding. If that mark moves past the screw head, the teeth strike.”

Hadran said, “He takes orders from me.”

“Then order him to keep your chart room from catching fire.”

Kest looked between them, saw no useful shelter in rank, and placed his thumb beside Jo's. His pressure arrived unevenly. Jo corrected him until the tremor in the lever settled.

“Hold.”

They crouched without taking their right hand away. The housing radiated heat through their coverall knee. A security officer shifted above them, boot scraping the stair, and Jo heard the two weak route traces mark time through the room pickups. One tapped every thirteen seconds. The other carried paired beats with a widening interval.

Under the plate, the field latch had fused at one corner. Jo inserted the flat end of their torque driver and leaned. The metal resisted. Violet's engineer spoke again through the sideband, too quiet for anyone to pretend the threat had passed.

“We have children in cold sections,” she said. “Your ship kept the reserve taken from ours.”

“Yes.” Jo braced harder.

“You tell us not to take it back.”

“I tell you the return path runs through ships with less reserve than either of us.”

“That answer comes from the side holding the bank.”

The latch gave half a millimeter. Jo stopped before the tool slipped. “Then watch my hands. I am cutting designated control out of this jaw.”

Hadran came down another step. “You are doing no such thing.”

Jo looked up at him. “I can leave the captain's hard-close line connected. I can leave Violet's reverse path connected. Or I can put the jaw under three-key hold until people with standing decide. Pick which danger you want.”

“My order already picked.”

“Metal disagreed.”

That line usually earned a drive-deck laugh. Nobody gave it one.

Ilyan moved at the conservator's desk. An officer placed a hand on his shoulder, and he stopped. “Hadran,” he said, “the Article 4 clock has twenty-two minutes remaining.”

“It remains in force until the emergency transfer completes.”

“Not under its filed language.”

“You do not interpret command law.”

“Oru does. You removed them.”

Hadran's face tightened by the smallest degree. Jo had seen Captain Or use the same stillness when a schedule failed in public. Security learned command posture from command, then treated the resemblance as loyalty.

A dull impact struck the public hatch.

Every officer turned except Kest, whose thumb stayed against the witness mark. Another impact followed, measured and heavy. Not a crowd throwing itself at a door. Someone had brought a maintenance ram and knew where the hinges sat.

Hadran touched his collar pickup. “Report.”

The reply came through his earpiece, but the chart room carried enough leakage for Jo to hear one word: Commons.

Mina's voice sounded through the hatch. “Oru Pell, Clerk of the Commons Assembly, requests entry for service of an expiry notice.”

Hadran closed his eyes once. “This remains a secured course facility.”

“The Assembly knows. That is why we brought witnesses.”

A third blow landed against the upper hinge.

Jo returned the torque driver to the latch. “Kest, if your chief tells you to let go, ask him where he wants the arc.”

“I heard you the first time.”

“Good. Repetition means I am frightened.”

They drove the tool inward. The fused corner tore free with a sound like a small bone breaking. Current snapped across the gap and scorched the driver's tip. Jo pulled the field latch down. The hard-close solenoid went slack beneath Kest's hand.

At once Violet's engineer pushed against the open side.

The lever kicked toward Jo. Kest swore and caught it. On the glass, Violet's frame whitened around the edges as reverse phase pressure climbed. Indigo's pulse ladder thinned.

“Stop,” Lin said behind the engineer.

“They'll close us again.”

“Not with another ship under the blade.”

“They already did.”

Jo reached across the manual bank and opened the ballast resistor one notch. Heat struck the room in a dry wave. The reverse surge bled into copper plates beneath the floor, reddening a warning grid along the rail.

“You have perhaps twelve seconds before your coupler pits,” Jo told Violet. “After that, you lose even if you pull charge.”

“We can take their bank in eight.”

“Through Indigo.”

The engineer's face appeared beside Lin's for one compressed frame. She looked younger than Jo, with a pressure hood folded around her neck and a bruise along one cheek. No abstract branch had given her that bruise. A hard surface had. A tired hand might have failed to catch her. Jo could imagine the work without imagining ownership of it.

Indigo's paired pulses faded to single marks.

Nessa's voice entered over the timing channel, compressed almost beyond recognition. “Our second sky is going dark.”

The reverse pressure stopped.

Violet's engineer turned away from her controls. Lin did not thank her. He set one hand against the edge of his improvised table and watched Indigo's ladder rebuild.

Jo opened the ballast resistor another notch. The jaw settled at neutral. Sweat ran from their temple into the collar of their coverall.

“Three-key hold,” they said. “One key here. One on Violet. One assigned outside captain's sole control.”

Hadran stepped over the yellow boundary. “There will be no branch key to mission hardware.”

Lin answered before Jo could. “Then there will be no designated key to ours.”

The public hatch bent inward at its upper corner. Light from the corridor cut through the gap. Hadran's officers watched him, waiting for an instruction that could still make their presence coherent.

Jo knew their kind of waiting. On the drive deck, a foreman could hold authority through one broken order if the next order admitted the break. Insist that the machine had obeyed, and every worker nearby began choosing which reality to serve.

Hadran touched his collar again. “Captain, request immediate renewal of Article 4 and authorization to remove Commons personnel.”

Static answered him.

Captain Or might have declined. She might have delayed. The building itself might have blocked the channel under the field load. Jo did not grant any of those possibilities the status of fact.

The clock above Ilyan's sealed console changed from 00:01 to 00:00.

No alarm marked the expiry. A legal power ended with a quiet digit, while the jaw still radiated heat and people on separate ships watched one another breathe.

Oru's voice came through the damaged hatch. “Emergency course authority under Mission Article 4 has expired. Commons has not renewed it. Security Chief Coil, open this door.”

Hadran kept his hand at his collar. “Captain?”

This time Sava answered over the room speakers. Her voice carried a delay of less than a second, but the pause before her first word had taken longer.

“Do not renew,” she said.

Hadran looked toward the nearest camera. “The facility remains compromised.”

“I know.”

“Branch personnel attempted a reserve seizure.”

“I heard.”

“Then I require authority to close every contact.”

Captain Or inhaled. The room's pickups carried it, an ordinary breath traveling nowhere it had not been wired to go. “You have authority to prevent bodily harm aboard this ship. You do not have authority to decide which other ships count as the threat.”

Hadran lowered his hand.

The lower hatch hinge cracked under the next blow. Jo winced at the sound. A perfectly usable hinge had become the day's first unquestioned casualty.

“Open it,” Hadran told an officer.

The security seal released. Mina entered first, carrying the maintenance ram with another Commons delegate. Oru followed in their gray clerk's coat, one sleeve torn at the cuff. Behind them came six Assembly witnesses and two unarmed Guild navigators. More people filled the corridor, but Oru raised a palm and kept them outside.

No one cheered. The chart room had not changed hands cleanly enough for that.

Two security officers remained beside the public hatch. Mina asked them to move, and Hadran refused before either could obey. For a moment the damaged door threatened to become another lever held between commands.

“They keep the corridor clear,” he said. “The next crowd through that opening may not bring a clerk.”

Mina looked past him at Violet's image. “The crowd outside includes three galley stewards, a school registrar, twelve drive hands, and half the night delegation.”

“It also includes people shouting that branches stole their arrival.”

From the corridor came a reply too muddled for words. Jo heard anger in the rhythm but could not assign its object. Hadran had spent the seizure naming one danger. That did not make every other danger imaginary.

“Keep two at the hatch,” Jo said. “Face them outward. No seal.”

Hadran studied them as though the compromise carried a concealed blade. “You have no authority to deploy my staff.”

“No. I have a hot floor, an open public door, and work that needs room. Give me another way to hold all three.”

One officer removed the security seal from its spool and returned it to her belt. The other dragged the broken hinge clear so it could not drop on anyone entering. They took positions with their backs to the chart room. Hadran allowed the arrangement without blessing it.

On Violet, the duty engineer came back into frame. “If he closes that door again, we reopen the reserve path.”

Mina stepped close enough for the camera to hold her face. “If he closes it, Commons breaks it. If you draw through Indigo, Commons names that act too.”

“You cannot police Violet.”

“I can tell the difference between defense and choosing a weaker neighbor to pay for it.”

The engineer's bruised face tightened. Lin said something outside the pickup range. After several seconds she disabled her surge latch and placed the detached handle where the camera could see it.

Mina set the ram on the floor. “Jo, do I need to call fire crew?”

“Not if everyone stops helping.”

“An ambitious standard.”

Oru crossed to the field boundary and read the room before asking for speech. Their attention moved from the disabled solenoid to Kest's thumb, then to Violet's image and Indigo's pulse ladder. They noticed the red warning grid under the rail.

“Who holds control?” they asked.

“Currently, friction,” Jo said.

“That lacks standing.”

“It has excellent custody.”

Even Hadran almost reacted.

Oru took a slim document pane from their coat. “Commons accepts emergency jurisdiction over mission identity pending a full standing resolution. Engineering retains physical safety control. Course certification remains with the Guild. Route delegates must consent to any seam action that draws reserve from their domains.”

Hadran said, “You drafted that before entry.”

“I drafted nine versions before entry. Events selected this one.”

“Commons cannot grant external entities control of ship systems.”

“Then call it refusal power over their own systems. The distinction may comfort you while we write better law.”

Lin leaned closer to Violet's camera. “Violet accepts a refusal key if it can stop both harvest and closure.”

“Indigo?” Oru asked.

Nessa did not answer. An adult voice replaced hers, careful and faint. "Indigo accepts for the present interval."

The two weaker traces had no voice channels. One had lost a third of its brightness during the standoff. Jo looked at them and felt every argument in the room become too large for the lives depending on its result.

"They need representation before we can hear them," Ilyan said.

Oru nodded. "Then no action against an unidentified active trace except to prevent immediate hardware loss. That holds until contact or closure beyond our ability to alter."

"You have no enforcement mechanism," Hadran said.

Jo lifted the scorched torque driver. "I do."

They removed the captain's hard-close lead from the solenoid and capped it. From a spare junction block they ran three thin witness wires: amber to the engineering bank, white to the Commons conduit Mina was already bolting back into the floor, violet to the seam control. None could command a transfer alone. Any one could break the circuit.

Kest held the lever until Jo set the mechanical keeper through both sides of the jaw. When they finally told him to release, his thumb left a pale oval on the red witness mark.

Hadran watched the modification without trying to stop it. "This guarantees paralysis."

"It guarantees disagreement reaches the machine before power does," Jo said.

"That can kill us."

"So can agreement made by one person."

Jo disliked the shape of their own sentence. It sounded too finished for a room full of unfinished work. They bent to the lower panel instead and tested each break circuit under a lamp load. Engineering opened it. Commons opened it. Violet's engineer, after a long pause, opened it from across the seam. Three dark intervals appeared in the test record, distinct and undeniable.

Ilyan peeled the security seal from his console. Hadran made no objection. Mina seated herself at the restored public conduit while Oru began a custody list. Lin called someone on Violet to bring their emergency council back to the classroom chart. Indigo rebuilt its timing ladder one paired pulse at a time.

The chart room no longer belonged to any person who could point at the center and call it command.

Jo closed the access plate over the altered jaw. Three witness lamps had replaced the single green authority light. Amber held steady beside their hand. White came on at the Commons desk. Across separate glass, Violet lit the third.

Chapter 15 — Seven Surviving Routes

<!-- POV: Ilyan Rook -->

<!-- SETUP: divided chart authority; weak route traces; expanded seam search -->

<!-- PAYOFF: seven stable routes besides designated Nacre; original marker removed -->

Ilyan set the map to listen between the colors.

The Loom resisted. Its search stack expected a conservator to ask which trace held the best arrival vector, not which closed trace still kept time. Every filter in the chart room leaned toward selection. Bright routes rose. Weak ones blurred into the smoked glass. A line that could no longer benefit designated Nacre sank beneath the threshold and acquired a gray closure mark.

He removed the threshold.

Noise covered the amphitheater wall.

The expanded field appeared as hundreds of faint seams crossing one another at angles too narrow for sight to separate. Most carried thermal recoil from Correction 141. Some marked old jacket vibrations. Nine showed periodic structure. Violet and Indigo already held names. Two others matched timing fragments Jo had protected during the chart-room standoff. Five remained uncertain, their clocks buried under the Loom's habit of not looking.

Ilyan turned down the visual gain and raised the audio transform. The room filled with low taps.

Mina sat at the Commons conduit with one boot against the repaired hatch. A white witness lamp above her terminal showed that no seam command could pass without public custody. Jo worked below the rail, where the amber engineering key guarded physical load. On Violet's center pane, Lin waited beside three members of his emergency council. His route held the third refusal key.

No single station could open the search field.

Ilyan placed his Guild seal on the reader. Mina approved the survey purpose. Jo approved a reserve ceiling. Lin looked away from Ilyan and asked his council before Violet consented.

Four marks appeared on the request.

Only then did the Loom let Ilyan listen.

“Search window?” Jo asked from under the rail.

“Point six seconds around every archived settlement offset.”

“That will cost heat.”

“Limit each query to one pulse.”

“One pulse tells you something can reflect.”

“Two asks whether it can count.”

Jo slid out far enough to look at him. "You intend two."

"I intend to regret one."

They rubbed a dark smear from their cheek and set the ceiling for two pulses per trace. "When the braid reaches forty-eight kelvin, the survey stops, even if a lost ship begins reciting your childhood."

Lin heard them through the seam. "That condition should be written into every Ilyan procedure."

The shared cadence struck Ilyan before the joke did. Lin's voice occupied different speakers and carried Violet's compression roughness, but the pause before Ilyan's name belonged to a habit they both had once used. He let the resemblance pass without assigning it a claim.

The first uncertain trace received two field taps separated by 0.41 seconds.

A response came back after a delay of 1.08 seconds. Two taps. The second arrived early by a hundredth.

Ilyan sent a new interval.

Jo said, "That makes three."

"It answered late. I need drift."

"You need discipline."

The next response copied the new spacing. Behind it came a short voice burst, too compressed to decode. Ilyan widened only the receiving band. A woman's voice emerged in pieces.

"Route Amber. Medical priority. Repeat, we hold one functioning surgical bay."

The chart stack tried to place Amber beneath its old settlement card. Ilyan refused the association and opened a new ship record. Amber's officers transmitted a damage summary in packet fragments. Their food cooling had failed during the correction as forecast, but they had rerouted three habitat loops through the seed vault. The measure saved frozen cultures and spoiled a third of their fresh stores. Forty-two patients occupied a bay designed for twelve. They wanted antibiotics, sterile repair instructions, and recognition that no negotiation could demand further power from life support.

Mina entered each request into Commons custody. "Can instructions cross at useful resolution?"

Jo answered without leaving the rail. "Text, diagrams, low-frame demonstrations. No material."

Amber's surgeon heard that. "We know what a seam carries. Send us the valve pattern for your old Keel sterilizers."

Ilyan searched the common inventory. Violet provided the pattern before designated Nacre found it. Amber acknowledged Violet first.

The second uncertain clock carried a more regular beat. Its replies arrived in groups of five, each group followed by a pause long enough to resemble failure. Ilyan waited through three cycles before voice compression locked.

"Reed receiving. Do not increase load."

Their image resolved into the edge of a chart table flooded with ankle-deep water. Reed had lost a gray-water baffle along Drum Four. Pumps kept the inhabited levels dry, but every extra field draw slowed the motors. Their council would permit timing contact and emergency text. They refused images until the leak came under control.

“What do you need?” Mina asked.

“Nothing from you that costs current,” Reed replied. “Send pressure-cycle tables when the field sleeps.”

Their restraint unsettled Ilyan more than a demand. A map built for competition had taught him to expect each unfavorable route to contain its own reason for rejection. Reed's flood appeared in the original forecast as a number beside water loss. The number had not included a council choosing darkness so its pumps could run.

He marked Reed active.

The route glass grew warmer. Frost withdrew from the black frame and left narrow wet lines along the lower bezel. Jo called out forty-three kelvin.

A third trace returned Ilyan's timing challenge inverted: long where he sent short, short where he sent long. He almost classified it as a phase fault. Indigo interrupted with a child's voice.

“They are answering from the other sky.”

Nessa Quill appeared in a small side frame, hair tied back with a strip of yellow cloth. Someone had given her class limited access to the timing ladder, or the class had made access difficult to deny.

“Explain,” Ilyan said.

“If their board faces opposite, up becomes down. Send both at once.”

He generated a symmetric pair. The trace returned it without inversion.

Voice arrived on the next cycle, calm and old. “Sable acknowledges. Preserve low gain.”

Sable's route had crossed the charged-dust fold broadside. Its outer sensor braid carried so much scoring that the ship could not distinguish field pressure from hull vibration above a narrow band. Their correction reserve remained under twelve percent. They had shut down two daylit green belts and moved their population inward to reduce heating load.

Sable did not request power. Its archivist requested bandwidth.

“For what data?” Oru asked from behind Mina. They had arrived without Ilyan noticing and now stood inside the white witness boundary.

“Wake records,” Sable said. “We recovered closure traffic from prior corrections. Names, work logs, domestic packets. Much exceeds our local need.”

“How much?”

“Compressed estimate follows.”

A number appeared. Ilyan read it twice. The archive could consume every permitted seam interval for days.

Jo rose from the service rail. "Not during a live search."

"We understand," Sable replied. "Record that we offered custody."

Oru entered the statement with a public clock. Sable's pane dimmed to save reserve, leaving only a narrow timing mark beside its new name.

The fourth trace answered before Ilyan sent a query.

Three bright pulses crossed the glass in quick sequence. A voice followed, clipped by damaged transmission gates.

"Glass to any receiving Nacre. We see your survey braid. Confirm you have ceased settlement."

Mina looked at Jo. Jo looked at Lin. The question needed all three authorities, but none could promise more than the current hold.

"Settlement commands cannot pass through the jaw without engineering, Commons, and route consent," Mina said. "Permanent law remains unwritten."

"Then record our contact as conditional."

Glass had retained its hull and food stores. Its primary injury ran through governance. During the first hours after Correction 141, a command faction had attempted to preserve arrival by completing settlement locally. Drive crews removed the firing links. Their captain remained confined to quarters while a temporary council argued over who could speak across the seam.

"We will not accept orders authenticated by a single command seal," the Glass caller said.

Oru's attention sharpened. "Neither will this chart room."

"Good. Send your custody form."

Glass took no comfort from kinship. It wanted procedures, signatures, and a way to challenge every instruction. Ilyan gave its trace a provisional civic mark rather than the captain's crest stored in the Loom. The software warned that route identity required command authentication. He assigned the Commons clock as witness. The warning cleared.

Forty-six kelvin.

Two uncertain traces remained. One wandered near the noise floor, never answering at the same offset twice. The other had a hard, regular pulse but rejected every voice handshake.

Ilyan tried the wandering trace first. Its drift formed a pattern only when he plotted delay against the ship's rotation. A damaged antenna crossed seam alignment once every twenty-seven seconds. During each narrow window, someone sent as much as the aperture permitted.

"Nacre North. Attitude control impaired. We require no rescue maneuver until our rotation is contained."

North's image spun through fragments: ceiling, chart rail, a face, darkness, ceiling again. Their braking geometry had survived, but two maneuvering clusters had fired against each other during settlement. The route now rotated too quickly for sustained seam lock. They insisted that any shared course wait for independent stabilization.

“How long?” Ilyan asked.

“Unknown. Minimum forty hours. Possibly six days.”

“Pelagos debris geometry changes inside that window.”

“We carry the same star tables.”

The speaker's irritation sounded exactly like a navigator being told a schedule by another navigator. Ilyan felt a defensive answer rise and let it go.

“North's delay request entered,” he said.

“Not request. Condition.”

Mina recorded the correction.

The final trace showed a clock but no voice. It sat beneath an old route tag labeled Carbon, a predicted ship lost to drive-nave fire fourteen minutes after Correction 141. That fire should have destroyed seam control. Yet the rhythm continued: six pulses, pause, two, pause, six.

Jo reached forty-eight kelvin and cut transmit permission.

“We stop,” they said.

“One trace remains.”

“One trace has remained through every query. The braid will not if we cook it.”

Ilyan listened to the unanswered rhythm. Six, two, six. It might encode a name. It might come from a damaged relay repeating its last healthy diagnostic. A continuing clock implied duration, not population. He knew how easily hope supplied people to a noise once a person wanted rescue to exist.

He closed the active query bank.

The faint rhythm continued without them.

“Passive receive costs almost nothing,” Jo said. “Leave that channel in custody. We listen while the glass cools.”

Ilyan did. Mina opened a provisional unnamed record, and Oru assigned it standing for preservation without presuming inhabitants. Lin asked Violet's radio workers to examine the pulse grammar. Indigo's class copied the intervals onto their two-sky board.

The map now held Violet, Indigo, Amber, Reed, Sable, Glass, and North around designated Nacre. Eight ships total. The number did not feel large enough for what had changed. Ilyan could count eight bowls at a meal, eight names in a duty roster, eight pressure doors along a service spine. On the Loom, each mark contained 42,006 possible continuities altered by different damage, deaths, births, decisions, and hours.

Before Oru allowed the display onto the Commons channel, they required a verification round. Each route had to recognize its own clock, its stated damage, and the authority attached to its voice. The task sounded clerical until Amber objected to the color assigned to medical risk.

“Red implies imminent closure,” the surgeon said. “We have no closure order and no diagnosis of imminent

failure.”

Ilyan changed Amber to orange. Reed then rejected orange because the same color marked its pump limit. Glass rejected every color inherited from settlement scoring. North could not view a stable image long enough to evaluate any symbol.

“This will take the entire cooling interval,” Jo said.

“It should,” Oru replied.

“We could use letters.”

“Letters inherit the route tags chosen before contact.”

Lin folded his arms on Violet's pane. “Use no condition color. Put the reported problem in words.”

Ilyan knew what that would cost. The chart stack sorted color in less than a blink. Words demanded reading, and reading consumed time during maneuvers. “A navigator needs to see load differences at once.”

“A citizen needs to know whether your difference came from our report or your judgment.”

The objection reached farther than the display. Ilyan had spent his working life turning unequal hazards into a field a hand could move through. He trusted symbols because their compression let him act. Compression also hid who had supplied each term.

He built two layers. The first showed only route lines and shared event clocks. The second carried self-reported conditions in text, each statement stamped by its ship. Engineering measurements occupied a third layer with instrument provenance. No color ranked civic value. An operator could still display heat, reserve, rotation, hull strain, or food margin, but each comparison named its source and time.

Glass tested the rule by challenging Nacre's reserve figure. Jo sent the capacitor measurement and calibration age. Violet challenged Amber's hospital count; Amber corrected forty-two patients to forty-six after a corridor triage station reported. Reed marked its water depth as variable by drum rather than shipwide. Sable declined to provide population placement because transmission could expose which sections still had heat if another route attempted a draw.

Hadran, standing at the rear under Commons witness, spoke for the first time since the search began. “You are treating basic defensive information as optional.”

“Sable treats it as theirs,” Mina said.

“It affects whether we can rely on their capacity.”

Sable's archivist answered through the narrow band. “You may rely on our stated limits. You may not inventory our vulnerable compartments.”

Hadran looked at the faint trace. “A hostile route could claim any limit.”

“Your command claimed we had no inhabitants.”

No one filled the pause. Ilyan stamped the refusal into the map instead of leaving the field blank. WITHHELD BY SABLE FOR HABITAT SECURITY carried more truth than UNKNOWN.

North crossed alignment long enough to receive the revised layers. Its caller sent back a breathless approval, then added, "Do not freeze these reports. Our rotation period has shortened by eleven seconds since contact."

Ilyan placed a live timestamp beside every condition. The map ceased pretending that discovery had finished anything. Eight clocks moved at eight rates of damage, labor, and decision.

Not every population remained exact. Routes reported losses. Violet had buried people after its hull strike. Amber's hospital had deaths still uncounted. Sable withheld a full census while it conserved transmission. North could not keep its antenna pointed long enough to send a current total. The correction had copied a complement, then circumstance had begun subtracting and changing it.

Oru asked Ilyan to display all stated priorities beside the traces.

The legend crowded at once.

Violet: civic standing before maneuver.

Indigo: class garden heat and continued child contact.

Amber: medical exchange, no life-support draw.

Reed: pump protection and delayed data.

Sable: archive custody and low seam gain.

Glass: multi-key authentication.

North: time to control rotation.

Designated Nacre: arrival of the mission population at Pelagos.

"That last line does not match the others," Lin said.

Ilyan examined it. The Loom had supplied Nacre's priority automatically from mission doctrine. Every branch entry described a demand stated by living people under present damage. Nacre's line claimed the entire mission and treated the other demands as variations beneath it.

"What would you write?" Ilyan asked.

Lin's pane lost two frames before his answer arrived. "Your people need food, hull, course, and standing like ours. Write what you can defend without inheriting the center."

Ilyan changed the line.

Designated Nacre: preserve 42,006 people; retain a viable Pelagos course; no settlement without shared authority.

Mina read it and added Commons custody. Jo added the correction-reserve limit. Lin's council accepted it without approving any maneuver.

A navigator from the Guild leaned over the rear rail. "The display needs a reference route. Otherwise offsets become ambiguous."

"Use physical frame zero," Ilyan said.

“That changes with each domain.”

“Use the shared Correction 141 event.”

“We still need one line drawn at center.”

The old map legend offered three symbols. A filled circle marked ORIGINAL. Hollow circles marked COUNTERIMAGES. A broken circle marked SETTLED. The categories had survived every software revision because no operator had needed a fourth.

Ilyan opened the legend editor.

His conservator's seal granted permission to rename a color, amend a closure cause, or add a hazard note. It did not permit removal of ORIGINAL. That symbol belonged to founder-level constants.

He requested Commons authority. Mina signed.

The editor refused again.

Jo connected an engineering service pane and exposed the legend table as plain text. “Constants are labels wearing expensive coats.”

Glass objected to the two-signature change. Its temporary council required route notice. Oru circulated the proposed deletion across every functioning seam. Violet approved. Indigo sent a paired pulse. Amber's surgeon answered between cases. Reed waited until its pumps reached a low cycle. Sable spent six words of precious bandwidth: REMOVE. PRESERVE OLD LEGEND IN ARCHIVE. Glass added its multi-key seal. North's answer came in pieces each time its antenna crossed alignment.

“North consents to remove priority marker. Maintain navigational reference by event clock.”

The field cooled to forty-five kelvin while the permissions gathered. No response came from the unnamed trace. Its six-two-six rhythm continued beneath the map like someone knocking from a distant compartment.

Ilyan saved the old legend to Sable's offered archive path. Then he selected ORIGINAL.

The delete key asked for a replacement symbol.

He chose none.

On the smoked glass, the filled circle around Nacre disappeared. Eight route lines remained. Violet curved high through the field. Indigo flickered beside it. Amber held close to the old central vector, while Reed drifted below. Sable shone faintly at the edge. Glass carried a sharp stable mark. North appeared only when its rotation brought the seam home. Designated Nacre's green line remained bright, but brightness now described reserve, not rank.

Ilyan moved his hand away from the console.

For the first time since the Loom had drawn more than one ship, its legend contained no symbol for the one reality entitled to erase the rest.

Chapter 16 — The Captain's Other Self

<!-- POV: Captain Sava Or -->

<!-- SETUP: eight-route map; Alda's suppressed record; Sava's ration-secrecy wound -->

<!-- PAYOFF: no Eda survives; ledger released; captain's authority submitted to review -->

Sava waited while her own face weighed barley.

The seam image updated once every three seconds. Between frames, Violet's ration room held still around the other Sava: one hand beneath a scoop, gray hair tied close to her neck, captain's posture surviving in shoulders bent over a food scale. When motion resumed, grain fell into a paper ration sack and the figure became a worker again.

No uniform remained on her. She wore a green-belt apron over a pressure suit patched at both knees. A queue passed behind her through the ration room, people presenting family chits and receiving portions under amber lamps. Violet had assigned its former captain to the evening shift because anyone who had signed settlement orders now worked where every subtraction could answer back.

Sava sat alone in her office aboard designated Nacre. She had dismissed her aide and turned off every display except the seam pane. The mission seal still occupied the wall above her desk. Someone had polished its bronze edge that morning according to schedule.

"Your measure runs light," she said.

The other Sava looked at the scale. "Evaporation loss in storage. We adjusted the issue table."

"By whose authority?"

A pause crossed the seam before the next frame. "You asked for private contact, not an inspection."

Sava folded her hands. "Habit."

"I remember."

The reply hurt because no explanation followed it. Sava had spent eleven years training people to answer rank before content. Across the seam, the same training returned stripped of obedience.

Violet's former captain finished the ration sack and passed it to an old man. She signed his chit, then waved the next family forward. "You have nine minutes before our council closes this channel. I asked to finish shift while we spoke."

"You could have chosen a private room."

"I did that for nineteen days."

The seam sound clipped the end of the sentence, but Sava heard no ambiguity. After Correction 141, Violet's Sava had invoked emergency authority, suppressed the first settlement evidence, and ordered drive crews to preserve mission course. Unlike designated Nacre, her ship had no harvested reserve to support the claim. When crews learned she had hidden closure estimates, she resigned before they removed her.

"Did resignation help?" Sava asked.

“It stopped my orders.”

“That does not answer the question.”

“No. It did not restore heat, feed the injured, or persuade frightened people to trust one another. It made room for a council that could begin those jobs.”

The next ration chit belonged to a woman holding two sleeping children against her hips. The other Sava checked the household mark and added a spoonful of barley to the sack.

Sava noticed. “That exceeds your table.”

“Night-shift children receive a sleep allowance.”

“The adults could claim it twice.”

“They could.”

“You do not verify?”

“We verify totals. We do not post names of families who ask for the allowance.”

“Then fraud stays hidden.”

“So does hunger.”

Sava increased the seam gain without requesting engineering approval, reached the custody ceiling, and received a flat refusal from the chart room. Even here, in her own office, her command met a circuit built to say no.

She lowered the request.

“Violet publishes district consumption,” the other Sava said. “Everyone sees what remains. No one sees which child needed a second bowl.”

“That distinction invites favoritism.”

“It does. We rotate the workers and audit the bins. We expose the quantity and protect the petitioner.”

On Nacre, Sava had done the reverse. She had hidden route persistence totals from the Commons while collecting the names of everyone who accessed unauthorized data. The secrecy protected command from challenge, not frightened workers from punishment. Until the other version stated the difference, Sava had treated both acts as variations of one burden.

A new family approached the scale. The former captain filled their sack, spoke quietly, and touched two fingers to her brow in place of a salute. The family answered with the same gesture.

“They still salute you,” Sava said.

“They greet me.”

“You know the difference?”

“I know they can leave with their grain if they don't.”

Sava looked toward the closed office door. Hadran stood outside with two security officers, not because she had ordered them there but because he still believed a captain required a perimeter. Beyond them, Commons clerks waited to serve notices. Her schedule pane carried eleven unresolved command questions, each marked for multi-key custody. The ship continued to ask her what she wanted while removing her ability to make the wanting sufficient.

“Why did you agree to speak?” she asked.

The other Sava finished the queue and locked the grain bin. “To find out whether you intended an apology or intelligence gathering.”

“Which do you think?”

“You began with my scale.”

A ration worker took over the station. Violet's Sava washed barley dust from her hands in a narrow basin, dried them on the apron, and carried the seam tablet into a supply corridor. Shelves passed behind her in broken frames. Many stood empty.

Sava set a formal record request on her desk, then left it unsigned. “I want your account of resignation.”

“For what record?”

“Captain's private archive.”

“No.”

“The Commons inquiry, then.”

“If it opens to every route.”

“That proceeding has not been constituted.”

“Constitute it.”

The other Sava stopped beside a door marked COLD GRAIN. Her image held there through two frozen intervals. “You still ask questions as though procedure arrives before the people who need it.”

“Procedure prevents the strongest need from taking the room.”

“It also gives strength a uniform.”

Sava almost ended the call. Her hand moved toward the seam key and halted above it. The impulse had no tactical purpose. Ending the call would only restore silence under her control.

Instead she asked, “What did you tell your crew?”

“That settlement remained necessary until we understood the field.”

“So did I.”

“I said branch contact might carry a hostile control pattern.”

“Hadran made the same assessment.”

"I sealed the drive nave."

Sava pictured Violet's chart room under emergency amber, its security team taking stations at doors identical to hers. "I sealed our route records."

"I ordered the first two contact packets erased."

"I ordered Alda's ledger held."

The supply corridor went still between frames. When movement returned, the other Sava had lowered the tablet.

"You found it too?" she asked.

"Not before our Ilyan did."

"Ours found only fragments. Command archive damage took the founder layer during the hull strike."

Sava opened the sealed directory beside her console. Alda's ledger waited under four restrictions, each added after the first. Founder confidentiality. Mission stability. Active course hazard. Article 4 emergency custody. That final authority had expired, but the file remained closed because a restriction did not remove itself when its justification ended.

"Why have you not released it?" Violet's Sava asked.

"Verification remains incomplete."

"Does it prove command knew the rejected routes persisted?"

"Yes."

"Does it contain timing records from the first trial?"

"Yes."

"Does any expert dispute the maintenance cipher?"

"Teren disputes two damaged intervals."

"Do those intervals alter the conclusion?"

"No."

"Then verification has completed the part people need."

Sava heard her own voice in the questions, the tone she used when an officer buried a decision beneath secondary detail. Having it turned against her produced an almost physical exposure. She could not dismiss the method without dismissing half her command history.

"There are personal records," she said. "Messages from people who died. Names of founder staff whose descendants live aboard every route."

"Offer redaction for private matter. Publish the crime."

“Redaction will look like another concealment.”

“It may. Secrecy has spent the trust that could have made careful release easy.”

A barley cart rolled behind the other Sava. She pressed herself against the shelf to let it pass. The worker pushing it did not pause for rank.

Sava studied the familiar face under ration-room light. The two of them shared childhood in Drum One during the crop failure. Same bunk. Same father. Same locked projection office where adults hid the yield curve. Their lives had separated only thirteen days earlier, yet one now spoke as if years had passed.

“Do you remember the orange ration slips?” Sava asked.

The other Sava's mouth tightened. “Father folded his into birds.”

“He said the color meant the printers had run out of white.”

“He knew what orange meant.”

“No. We learned later.”

“He knew.”

Sava leaned closer to the seam pane. “How?”

“Because he gave me his third slip and told me not to show you.”

The office seemed to contract around the desk.

Their father had died in the fourth week of crop rationing. Sava remembered his jokes becoming slower, his belt moving inward two holes, the adults calling his fever unfortunate. She had known he gave portions away. She had believed he divided them evenly between his daughters.

“He told me he had eaten on sanitation shift,” Violet's Sava said. “I kept the slip under my mattress for two days. Then I used it because I was hungry.”

“You never told me.”

“You never told me he gave you his second.”

Sava had kept that slip inside a schoolbook until hunger overcame shame. Their father had hidden each gift from the other child. He had turned scarcity into two private mercies, each carrying a lie. The sealed projections let him pretend no public choice existed. By the time the numbers emerged, he could no longer stand through a work period.

“All these years,” Sava said, “I thought the danger came from the projections leaking too early.”

“The projections did leak. People panicked.”

“Three store rooms were forced.”

“And command had already moved grain to officer sections.”

“They said guards could not keep order while hungry.”

“They made hunger proof of why they should eat first.”

The other Sava's anger sharpened the shared face into someone Sava did not know. Then a frame broke, and only blocks of light remained. When the image returned, exhaustion had replaced anger.

“I used the same lesson on Violet,” she said. “I hid our reserve decline because I thought fear would make everyone seize it. Drive crews found my private table. They did not seize the reserve. They seized the firing links from me.”

“Jo did something similar here.”

“Did you thank them?”

“No.”

“Neither did I.”

For the first time, both women almost smiled.

A chime announced two minutes left in Violet's approved interval. Sava looked at the sealed ledger. The restrictions formed a neat vertical row. Each named a duty. Together they performed only one action: keeping evidence away from the people harmed by it.

Before she could open the file, her door sounded.

“Captain,” Ilyan said through the intercom. “I request thirty seconds before the route session ends.”

She admitted him.

He entered without his Guild coat, carrying a data pane against his chest. Hadran stepped aside in the corridor. Two Commons clerks attempted to follow, but Sava raised a hand.

“Leave the door open,” she told them. “The conversation will enter public custody afterward.”

Ilyan saw the other Sava on the seam. His attention fixed on the shared face, then moved away with deliberate courtesy.

“What do you need?” Sava asked.

“A search authorization.”

“For which record?”

“Correction 133 and all later wake packets containing Eda Rook.”

The name altered the room more than either captain's face had. Sava knew the pressure accident. Four dead in Drum Five, six years earlier. She also knew Ilyan had certified Correction 133 before it, and that grief could mistake chronology for a door.

“Current routes diverged at Correction 141,” she said.

“I know the date.”

“Eda died before that.”

“Sable holds older wake records.”

“Records, not surviving ships.”

His fingers whitened around the pane. “Alda's table shows some seams recurring after later corrections. A nested route might have persisted through repeated field loads.”

“Teren reviewed that possibility.”

“He reviewed it under command restriction.”

Sava opened the causal appendix and turned the display toward him. “I removed the restriction this morning. Every older wake lost field support at settlement. Later corrections carried no physical continuity from them. Sable's packets may contain Eda alive for nineteen days after Correction 133. They may contain messages she never sent here. They do not contain a living Eda now.”

Ilyan did not touch the appendix.

The other Sava spoke from Violet. “Our archive agrees. I asked the same question about people we lost before 141.”

He looked at her then. “You found none?”

“None.”

The seam chime marked one minute.

Sava could have softened the refusal. She could approve the name search and let Ilyan exhaust hope against files. Command often mistook delay for mercy. The habit had consumed years aboard both ships.

“I will authorize recovery of every record naming her,” she said. “I will not describe that recovery as rescue.”

Ilyan's face changed without moving. “Thank you.”

“Do not thank me for closing a door that physics already closed.”

“I am thanking you for saying which door.”

Sava looked at the unsigned request. “When did you first search for her?”

“Six days after the pressure accident.”

“You had no captain's access then.”

“I had trainee custody of Correction 133. I looked for any route where Drum Five pressure remained stable.”

“And found one.”

“Several. The forecast cards carried survival values. I knew doctrine said no one lived them.” His gaze moved to Alda's open directory. “I wanted doctrine to be wrong in only the way that helped me.”

Sava remembered the old access alerts. A trainee conservator had called sealed route cards too often, and Assurance had classified the behavior as bereavement fixation. No officer had asked what the cards seemed to promise. They had treated his grief as a security weakness and left the false premise untouched because it kept

him at work.

“Command saw those searches,” she said. “My predecessor's office received the flags.”

Ilyan's expression hardened. “What did they do?”

“Nothing recorded beyond monitoring.”

“They knew persistence evidence existed. They watched me search for Eda inside it and let me keep certifying corrections.”

Sava did not defend people dead or retired. “Yes.”

“What would you have done?”

The truthful answer arrived without mercy. “Before this week, I would have called you privately, told you no branch contained her, and kept the basis sealed.”

“So I would have had to trust the office that watched me.”

“Yes.”

“Would you have removed me from duty?”

“If you continued searching.”

His eyes closed briefly. “I did continue. I learned to search without leaving the correction number in the request.”

Sava heard both the confession and the evidence against her kind of command. Secrecy had not removed his hope. It had trained hope to evade records.

“Submit every hidden search with your recovery request,” she said.

“That could cost my seal.”

“It could.”

“You are asking me to expose misconduct while you release the document that explains it.”

“I am placing my conduct under review for the same reason. Explanation does not cancel responsibility. Concealment does not preserve it either.”

Ilyan lifted the data pane from her desk, opened its attachment list, and added a private archive key. Twenty-three old searches appeared under his name. He stared at the total before signing them into Commons custody.

“Eda would have mocked the phrasing on half of these,” he said.

“What phrasing?”

“Pressure-stability variance inquiry. Counterroute personnel continuity. I could not type my sister's name.”

“Type it now.”

He changed the request title to RECORDS OF EDA ROOK. The words remained on Sava's desk after he sent the file.

"The recovery will include her," Sava said. "It will not make her current."

"I understand."

This time she believed he understood the sentence, though neither of them yet knew whether understanding could survive hearing Eda's voice in an old packet.

He left the amended request on the desk and stepped back.

Violet's Sava untied her ration apron. "My council needs the ledger before the next reserve session."

Sava removed the first restriction. Founder confidentiality vanished from the file. She removed mission stability and active course hazard. The expired Article 4 seal resisted because Hadran had attached a security concurrence. She called him into the doorway.

"Withdraw your concurrence."

He looked past her at the open file. "I cannot recommend release."

"I did not ask for recommendation."

"Then dismiss the seal over my objection. The record should show it."

Sava considered ordering him to remove it. That would turn disagreement into obedience and leave the objection hidden inside command. She selected OVERRIDE WITH DISSENT and entered his name.

The restriction cleared.

Beyond the open office door, one Commons clerk began reading the release notice aloud to people gathered in the corridor. No officer ordered them to move or leave.

Alda's ledger opened across the public pane: timing values, maintenance intervals, voices reduced to notes by frightened hands. Sava routed the complete evidentiary layer to the Commons, every contacted ship, and the Guild. Personal messages entered a temporary privacy queue governed by route representatives rather than her office.

The seam timer reached zero. Violet's Sava remained for one final frame, apron folded over her arm.

"You will have to do more than publish," she said.

"I know."

"Do you?"

Sava opened the command-review docket waiting beneath the ledger. It asked whether the captain submitted a subordinate action, a department, or another officer. None of the fields anticipated the captain placing her own emergency conduct under jurisdiction.

She entered CAPTAIN'S OFFICE, CORRECTIONS 141 AFTERMATH, FULL AUTHORITY REVIEW. Oru's clerk seal appeared almost at once. Violet added its council mark before the seam closed.

Sava stood, reached above her desk, and lifted the bronze mission seal from its wall hooks. Dust marked a clean oval where it had hung for eleven years. She set it beside the open docket rather than behind her chair.

When the office door released, the Commons clerks did not wait for permission to enter.

Chapter 17 — The Famine Calculation

<!-- POV: Teren Vale -->

<!-- SETUP: eight-route map; divided authority; Nessa's two-sky timing game; debris tide -->

<!-- PAYOFF: sustaining all eight by current method fails; Sable collapses; shared equation issued -->

Teren removed the word passengers from the model and watched three ships starve.

The physics did not care what he called them. Eight route domains occupied the projection above the Commons well, each carrying a Nacre, a population, a damaged reserve bank, and a Pelagos approach vector. Yet the old model counted one ship with passengers and seven field liabilities. It fed the designated route first. Every solution it offered looked orderly because the deaths had already entered under another heading.

He changed all eight objects to ships.

Red spread across the ration forecast.

Terraces in the Commons well had been cleared for the calculation session. Oru occupied the clerk's table below. Jo sat near the engineering cutoffs with a portable capacitor board across their knees. Ilyan stood inside the Guild boundary, listening to eight clock feeds through a single earpiece. Sava had taken a witness seat rather than the captain's dais.

Route images ringed the chamber. Violet held the largest pane. Indigo appeared as a classroom wall with Nessa's timing ladder still drawn in chalk. Amber showed only its hospital delegate's hands. Reed sent text to preserve pump current. Glass displayed three council members at separate tables. North came and went with its rotation. Sable held a narrow line at the edge, too faint for continuous image.

Teren expanded the forecast.

"Equal stabilization first," he said. "Not because I recommend it. Because equality in input exposes inequality in consequence."

A curve rose from each route. The Loom could keep every domain open by feeding phase pressure from designated Nacre's correction reserve. Violet needed less now that Jo's shared-capacitor procedure had carried it beyond day nineteen. Indigo's field geometry remained clean but its heating margin had narrowed. Amber could spare field power or hospital power, not both. Reed's pumps consumed nearly every flexible watt. Sable's outer braid leaked charge into hull scars. Glass retained strong reserve under political lock. North lost efficiency each time rotation pulled its seam away from alignment.

Jo studied the curve. "Your equal feed assumes our bank supplies every loss."

"Yes."

"Then stop calling it equal."

Teren changed the heading to IDENTICAL ALLOCATION FROM DESIGNATED NACRE.

The red ration line moved no differently.

He translated the result before shame could make him bury it under equations. "Imagine eight homes with separate cupboards. One cable runs between their heaters. Our house controls the cable, but drawing current for

another house still cools our own food store. Equal heater time does not create equal meals.”

“How long?” Mina asked from the second terrace.

“Until what?”

“Until people miss meals.”

Teren had prepared six definitions of failure. Calorie reduction. Protein substitution. Seed-stock consumption. Medical triage. Growth impairment among children. Population decline. Mina's question required him to choose the point a household would use.

“First compulsory reduction aboard designated Nacre in eleven days,” he said. “Amber already rations below our proposed first reduction. Reed reaches seed-stock use in twenty-three to thirty-one days, depending on pump repairs. Indigo follows inside two months. Under this allocation, sustained famine begins across at least three ships before stable settlement around Pelagos.”

The word famine altered the chamber. People who tolerated percentages looked at one another when hunger acquired a name.

Amber's hospital delegate spoke. “Your count excludes our current restricted diet.”

“It does.”

“Then famine has begun somewhere already.”

Teren amended the starting condition. Red moved backward to the present.

Oru asked him to open the population layer. The old forecast grouped every person under a single caloric average, useful for freight and dishonest for hunger. Children, pressure crews, pregnant people, elders, and patients did not meet shortage at the same rate. Neither did ship districts. A five-percent reduction in an officer galley removed choice. The same cut in Drum Six removed the final protein allotment from families already trading meal chits.

Teren separated designated Nacre by age, work, and current access. The average remained survivable. Four terraces watched a thousand smaller lines fall below it.

Mina pointed to the lower green-belt districts. “Those households supply food labor. Your model gives them less because their present draw reads lower.”

“It extrapolates recorded consumption.”

“Recorded consumption fell after the correction. People skipped meals while the galley queues doubled.”

“Why did ration control not log the missed issue?”

“Because no one issues what a person does not request.”

He added queue abandonment and meal-chit transfers. The famine threshold moved two days closer.

Sava spoke from her witness seat. “Officers receive a readiness supplement during course emergencies.”

“I included it.”

“Remove command staff from that supplement during nonphysical duty.”

Several heads turned. Teren changed the rule. The saved food delayed the first compulsory cut by only six hours, but the lower-district child line rose above immediate deficiency.

“Keep the change,” Sava said.

Oru entered it as a provisional ration motion rather than a captain's order. The distinction took thirty seconds longer. Teren marked those seconds too. Governance consumed time, but command secrecy had consumed people without displaying a clock.

Amber sent a patient distribution. Its average calorie figure looked adequate until Teren separated burn recovery, surgical infection, and ordinary households. Preserving sterile power forced the kitchen to serve cold meals. More food spoiled because heat could not reach it safely. A watt withdrawn for seam stabilization cost Amber twice: once at the surgical bay, again at the meal line.

Reed's numbers produced another shape. Water loss pushed residents toward dried stores, which then required more water to prepare. The obvious substitution accelerated the shortage. Reed's pump crews had already prohibited it, but the shared model still recommended dried grain because founder tables assumed water remained cheap.

Teren deleted that assumption.

Indigo supplied class-garden yields. Nessa's school plot contributed almost nothing to total calories, yet closing its lamps removed fresh greens from children whose stored diet lacked them. A model optimizing only energy would darken the garden first. A model tracking deficiency kept two lamp rows alive.

“What do you call that in the equation?” Nessa asked.

“A micronutrient constraint.”

“We call it lunch.”

Teren renamed the display field LUNCH REQUIREMENT and heard a few tired laughs in the Commons well. The humor did not reduce the curves. It made clear which parts had previously belonged to language no household used.

His teachers had loved models because corrections looked clean inside them. Add a term. Refine an input. Reduce uncertainty. No equation displayed the embarrassment of an old man changing a heading after a drive engineer pointed out that the heading lied.

“Second branch,” he said. “Support according to reserve loss.”

The Loom directed more phase pressure toward Sable, Reed, and North. Glass received almost none. Amber received less than its hospital needed. Violet held. Designated Nacre's correction bank fell below safe maneuver margin in six days.

North crossed alignment long enough to object. “Your support pulse increases our rotation error.”

Teren checked the transfer orientation. The caller had found a coupling his model treated as negligible. Every phase injection reached North at a different attitude, adding a small torque. After six hundred pulses, small became lethal.

He removed North from automatic support and assigned manual timing.

“Do not remove us,” North said. “Model our condition.”

“You cannot maintain receive lock.”

“We maintain a clock. Send the schedule. We choose which passes to accept.”

Teren restored North with local refusal authority. The forecast improved by two percent and became harder to administer. That difficulty felt healthier than the previous simplicity.

Glass asked why its strong reserve could not subsidize Amber.

“Your phase banks match because all ships diverged from one correction event,” Teren said. “A voluntary draw could cross Glass to Amber through the common braid, but the transfer loses energy at both seams. For every unit Amber receives, Glass spends one and a fraction. Designated Nacre spends a smaller amount holding the junction.”

“Can the junction rotate?”

“Not under present control architecture.”

“Can you alter it?”

The answer wanted to become a lecture. Teren held himself to the mechanism. “Possibly. The Loom assigns one receiver before it solves phase pressure. Every equation carries that receiver as a fixed point. Removing the fixed point makes the stack unstable.”

“Unstable meaning inaccurate or impossible?”

“We have treated those words as synonyms for one hundred nineteen years.”

A murmur moved through the terraces.

Oru lifted a finger. “Treat them separately now.”

Teren marked the question unresolved.

He opened the third branch: preserve a single clustered arrival corridor. Pelagos glowed above the Commons floor, surrounded by the debris tide mapped from long-range instruments. The tide contained charged dust, nickel-iron bodies, dead comet fragments, and smaller matter visible only through its effect on starlight. Nacre's original plan cut one narrow corridor through it, using settlement reserve for course corrections faster than ordinary thrust could provide.

Eight ships could not occupy that corridor together.

Teren set them nose to tail with the minimum safe separation. The line extended beyond the corridor's stable width. North's rotation doubled its clearance. Reed needed room to protect its damaged water baffle from lateral acceleration. Sable could not tolerate the sharp entry correction. Amber's hospital required a lower thrust ramp. Glass refused any sequence that placed its drive under designated command. Indigo's school and green-belt heating systems could survive the burn, but its field capacitors could not recharge before capture.

Ilyan spoke without removing the earpiece. “Offset arrival times.”

Teren ran the schedule. “The debris tide moves. A corridor cleared for the first ship partly closes before the third. By ship five, the required correction exceeds current reserve.”

“Widen it.”

“Each widening burn consumes what keeps the seams stable.”

“Then stagger by days.”

“North can wait. Amber cannot maintain braking reserve for the longest delay. Reed's food margin worsens under extended transit. Sable does not have days to spend.”

The faint Sable line pulsed once, acknowledging without voice.

Teren divided the arrival paths. One route toward Lattice orbit. Two toward Trojan rubble clusters. One captured by Thren. Others entered long transfer paths rather than immediate harbor. The field showed enough room, but every path required its own correction logic. The Loom had never been built to grant several routes independent safe outcomes. It calculated alternatives, then fed one.

“The sky has space,” Nessa said from Indigo. “Your line does not.”

Several adults turned toward her pane. She stood on a classroom stool so her face reached the camera. Behind her, children had drawn two arcs across the timing board. One began at a blue chalk sun. The other began at a yellow one. Their crossing points carried paired marks.

Teren enlarged the classroom image. “Show me.”

Nessa looked offscreen for permission. A teacher nodded from the edge of the frame.

“In two-sky, nobody owns the first beat,” she said. “We used to copy Indigo's rhythm onto the other board. Then the other class always came second, even when they sent before us. So we put a bell at both ends.”

She tapped blue and yellow chalk marks at the same apparent moment. Delay made one arrive after the other. The children corrected by measuring half the disagreement from each side, not by forcing either clock to surrender its starting point.

“That requires symmetric delay,” Teren said.

“It requires writing down both delays.”

“Seam phase does not stay symmetric under load.”

“Then write down both loads.”

Her impatience resembled Jo's, but Jo's impatience carried bolts and burns. Nessa treated his abstraction as an adult's refusal to finish arithmetic.

Teren copied the game into the calculation stack. The system rejected two receiver fields. He bypassed the interface and wrote the pair directly into a scratch model. Blue received yellow while yellow received blue. Each side accepted half the timing error. At low power, the relation held.

Jo leaned over their capacitor board. “Do not mistake a timing toy for a maneuver.”

“I am not.”

“You have the look.”

“What look?”

“The one physicists get before a repair crew loses sleep.”

Teren reduced the model to two ships. A shared phase gradient formed without naming either receiver. It oscillated, grew, and collapsed after four seconds. He added damping from Jo's Violet stabilization procedure. The relation held for nine seconds.

Violet offered its live seam values. Glass refused until it saw the scratch code. Amber contributed hospital power limits. Reed sent pump-cycle windows. North provided rotation passes. Indigo sent both halves of the children's timing board. Sable remained quiet.

With three ships, the scratch model held for two seconds. Four produced conflicting phase loops. Teren found that every loop still passed through designated Nacre as hidden reference even after he removed its receiver tag. Founder code had embedded priority below the visible equation.

He exposed the constant on the public display.

ORIGIN_WEIGHT: 1.0000.

All other routes carried zero.

Sava rose halfway from her witness seat, then sat again. “Can it be changed?”

“Yes.”

“What happens?”

“I do not know.”

Hadran, posted near the lower door, said, “Then nobody changes it on a live system.”

Teren agreed. “Not yet.”

The security chief seemed almost disappointed by the lack of argument.

Teren set every origin weight to one. The model created eight receivers and tore itself apart. He set them all to zero. Without a receiver, phase reserve dispersed as heat. Jo supplied a third option: each ship could act as receiver for its own local gradient while refusing the shared harvest path.

The solution began, then failed when their predicted courses converged inside the debris corridor.

“Separate the destinations,” Ilyan said.

Teren did. Seven paths opened and one struck the tide.

“Eight,” Lin corrected from Violet. “You drew your ship toward Lattice and left ours on its current line.”

Teren found the inherited assumption. Even after origin weights reached zero, the plotting layer had assigned Nacre's old destination to route zero. He deleted the destination and required every ship to provide one.

The system had room for eight paths now. It lacked enough phase reserve to place them there.

A warning tone sounded from Sable.

Not the shared maneuver bell. Three descending pulses, followed by a long interval. Jo came to their feet.

“Sable, identify alarm.”

The route answered in text because voice gain had failed.

> OUTER RESERVE CASCADE. BANKS FOUR THROUGH NINE BELOW HOLD. ESTIMATED CONTROL WINDOW: FIFTY-ONE MINUTES.

Teren's model still listed Sable's last reported reserve at eleven point eight percent. The new packet showed seven, then six point four as another bank isolated. Hull scoring had opened a leakage path faster than their earlier report predicted.

“Stop their seam draw,” Hadran said.

Sable's archivist sent a second line.

> DO NOT CLOSE RECEIVE BAND. ARCHIVE TRANSFER PREPARED.

Jo checked the engineering board. “Passive receipt costs them less than voice, but not nothing.”

“Can designated Nacre hold them?” Sava asked.

Teren ran the immediate rescue branch. Stabilizing Sable at current leakage required nearly half the correction reserve within an hour. Reed's pumps would lose shared support. North would lose timing assistance. Indigo's heat margin would fall below its school and garden load. The transfer might buy Sable six hours. It could not repair scored braid across a seam.

“How much for a narrow hold?” Ilyan asked.

“Enough for the archive?”

“Enough for people to choose what to send.”

Teren reduced the target. Twelve minutes of phase stability. Then twenty. At thirty-one, Reed's support failed. At twenty-four, designated Nacre retained one emergency correction. Amber volunteered reserve but its hospital battery line turned red. Glass offered more, conditional on Sable accepting no command claim. Violet offered the remainder from its stabilized bank.

Sable replied after a long delay.

> HOLD RECEIVE ONLY. DO NOT EMPTY OTHER SHIPS INTO OUR DAMAGE.

Mina gripped the terrace rail. “They are refusing rescue.”

“They are refusing this rescue,” Oru said.

The distinction mattered. Teren entered Sable's condition instead of translating it into surrender.

Jo built a receive-only hold from three contributors. Designated Nacre maintained the junction. Violet

supported phase timing. Glass absorbed part of the return noise. Sable kept local control and could break the circuit. The arrangement gained nineteen minutes at current leakage.

“Use it,” Sava said, then looked toward Oru. “I request that we use it.”

Oru called the three contributing route consents. Each approved. Jo closed the witness circuit.

Sable's line brightened by a fraction. Compressed packets began arriving too quickly for the public pane to open. Names, file hashes, household logs, maintenance tables, classroom records, and route fragments entered the custody queue. The amount exceeded the archive estimate Sable had sent during first contact.

Teren watched their reserve decline while the model continued to offer red branches. Preserve one ship by consuming others. Preserve the strongest and call weakness unfortunate. Keep all seams open until food failed. Cluster everyone into a corridor that could fit only by treating ships as points. Every answer smuggled a privileged receiver into either physics or language.

Nessa tapped the two chalk suns again.

“What if neither side catches the other?” she asked.

“The phase gradient becomes motion if we can stop it becoming heat,” Teren said.

“Motion where?”

“Different places.”

“That sounds like the board.”

He looked at the paired arcs. Each child kept a sky. Neither drawing needed to occupy the other's paper. The game synchronized disagreement rather than erasing it.

Teren opened a blank field equation. Instead of assigning an origin weight, he left a variable for every ship. Instead of one arrival constraint, he created eight. The system highlighted the empty terms as errors.

He sent the equation to Violet, Indigo, Amber, Reed, Sable, Glass, North, and designated Nacre. No ship received a completed solution. Each received the same missing fields: local reserve, survivable thrust, acceptable arrival body, refusal conditions, and a phase contribution that could not nominate another ship to pay.

“Session notice,” Oru said. “State the question for the record.”

Sable's packet stream crossed the first quarter mark. Its remaining control window fell below forty minutes.

Teren placed both hands on the clerk's rail. “Find one maneuver equation in which every route supplies its own boundary and no route serves as the privileged receiver.”

Eight copies left the Commons well. On the public map, eight clocks marked receipt at different times. One by one, the route panes began returning blank forms with their own numbers written in.

Chapter 18 — The Museum Opens

<!-- POV: Ilyan Rook -->

<!-- SETUP: Sable's reserve collapse; wake archive transfer; public custody -->

<!-- PAYOFF: museum opened; Sable closes; silence recorded as death -->

Ilyan found the museum inside a packet labeled KITCHEN CALIBRATION.

Sable had compressed its archive into the maintenance grammar the Loom ignored most easily. Oven temperatures held name indexes. Pipe-diameter tables carried dates. A list of acceptable grain moisture encoded voice fragments from routes settled decades before. Alda Quill had hidden her first wake ledger in the same kind of ordinary work, and Sable's archivists had widened that hiding place until it could hold millions of small records.

The transfer continued across the receive band. Each packet struck Ilyan's custody pane, verified against Sable's clock, and split into three copies. Commons received one. Violet held another. Designated Nacre's archive took the third under a seal no captain could close alone.

Above the console, Sable's route line pulsed at eleven-second intervals.

Ten seconds.

Twelve.

Its clock had begun to wander.

"Packet forty-one has a broken hash," Mina said from the Commons station.

Ilyan opened the damaged layer. The visible table described replacement handles for galley steam valves. Beneath it, thirty-two personal names ended in a block of noise.

"Request repeat?" he asked.

Jo answered from the rail. "A resend costs their braid."

The Sable archivist had left voice contact open at minimum gain. Breathing and room noise arrived in thin compressed strips. Somewhere near the pickup, people moved crates. Someone coughed. A child asked whether a light could be turned back on and received an answer too faint to resolve.

Ilyan keyed text.

> PACKET 41 INCOMPLETE. RESEND COST NOT JUSTIFIED WITHOUT YOUR CONSENT.

Sable replied after two clock pulses.

> NO RESEND. CROSS-INDEX NAMES IN PACKETS 7 AND 29.

He found the duplicates. Packet seven contained a sanitation roster from rejected Route Ochre after Correction 62. Packet twenty-nine held a classroom attendance correction from the same ship. Thirty-one names returned. The missing thirty-second appeared only as a family initial and a lunch debt.

Ilyan sent the discrepancy.

> KEEP BOTH FORMS. DO NOT INFER FULL NAME.

The instruction stopped him. Archive practice rewarded completion. Conservators restored damaged names from neighboring records, corrected spelling drift, and chose the most likely identity when two fields disagreed. Sable asked him to preserve uncertainty as part of the record. A person called M. Varo in one file and Little M in another could not be repaired into certainty for the convenience of a catalog.

He marked both forms.

“Next packet carries an opening key,” Mina said.

The file arrived under a bread-cycle schedule. Its hidden header contained one word: MUSEUM. Beneath it, Sable had written terms for public custody.

No ranking by usefulness.

No separation of crew records from children's records.

No claim that a surviving counterpart continued the person who died.

No correction of names without a source trail.

No closure described as silence, decay, or failed telemetry.

At the bottom, eight route seals marked consent. Sable's seal came first only because its clock had authored the packet. The other seven ships had added custody marks during the transfer.

Oru read the terms from the Commons well over a public channel. People aboard designated Nacre could hear them in tram stations, galley queues, pressure offices, and green-belt paths. The words also crossed every active seam. No one received a complete live sound at the same instant, but each route's clock attached the delays.

Ilyan opened the first drawer.

No physical cabinet moved. The museum existed as an archive architecture, but Sable had designed its entry like a room. A visitor encountered one ordinary record at a time. The system refused searches for heroism, command rank, or dramatic death until the visitor had crossed a page of meals, repairs, jokes, and corrections written by people who expected another day.

The first item came from Route Blue-Four after Correction 18.

> LATE MEAL CHIT: Naro Pell owes one bowl lentil mash to Sami Olt. Interest not permitted because Sami burns porridge.

A second field contained Sami's reply.

> Disputed. It burned once. The pot had political objections.

Dry laughter reached the chart room from the public conduit. It ended quickly, not from shame but because another item had opened.

A pressure work order followed. Someone named Aven Dorr requested a longer wrench for a valve placed

behind a support no adult hand could reach. The supervisor denied fabrication and advised rotating the wrist. Three days later, Aven filed the same request in larger type. The final note reported that the supervisor had tried the rotation, trapped his sleeve, and approved two wrenches.

Then came a classroom correction from Route Ochre.

> M. Varo present. Attendance board showed absent because she answered from under the seed table during alarm practice.

Little M's lunch debt sat beside it without any claim that the two names matched.

Ilyan's task required only checksum verification. He still found himself reading every line. Timing lived inside all of them. A meal borrowed and expected back. A tool request repeated until someone listened. A child marked present after a board declared otherwise. The records carried clocks more human than the field intervals he had used to prove inhabitants existed.

Sable's line missed a pulse.

Jo looked up. "Bank Seven dropped."

The route's remaining control estimate changed from twenty-two minutes to fourteen.

"Can Glass absorb more return noise?" Sava asked from the rear witness row.

"Already above their agreed ceiling."

"Violet?"

"Lin offers another two percent. Sable refuses."

Sava did not turn the refusal into an order. She gripped the rail and watched the archive advance.

Thirty-eight percent.

The number described packets transferred, not lives preserved. Ilyan hid the progress bar from the public view. Sable had prioritized the archive rather than sending everything. No percentage could say how much remained in rooms whose power failed before cataloging.

A voice packet opened without request.

The archivist sounded closer to the pickup now. "Nacre, confirm public access beyond your officer class."

Mina answered. "Commons channel shows fourteen thousand active receivers aboard designated Nacre. Route receipts continue. No rank filter."

"Confirm raw custody, not a memorial edit."

Ilyan displayed the checksums and witness seals. "Raw packets remain intact. The visitor layer cannot overwrite them."

"Confirm Sable appears as a source ship, not a rejected route tag."

He looked at the header. "Confirmed. Sable. Civic source. Independent clock."

The archivist breathed out. Background voices moved around her.

“Open all transferred drawers,” she said.

Oru asked, “Before privacy review?”

“We completed local consent where possible. Private medical layers remain sealed. Messages marked family-only remain keyed to recipient names. Everything else opens now.”

“Some people named may have counterparts living on other ships.”

“They may object later. We will not have later.”

The word entered the chamber without emphasis.

Oru called a custody vote among the eight route marks. Reed delayed while its pumps cycled. North answered across three alignment passes. Amber required confirmation that medical seals held. Glass checked each signature. Indigo's council asked Nessa's class whether children's ordinary records should open with adult work logs. Their answer arrived as a chalked sentence: WE WERE THERE AT THE SAME TIME.

Violet consented. Designated Nacre's Commons followed. Sable supplied the final local key.

The museum opened.

Across the chart-room glass, drawers unfolded by correction number. Not a list of eighteen million estimated dead. Sable had refused that false completeness. The archive held what survived: 8,214 names with source trails, 31,006 unresolved initials, 440,000 work and household records, and timing fragments from wakes too damaged to identify. The smaller archive count did not revise Teren's estimate; his model counted generated populations, while the museum could name only records that survived. Some corrections had three records. Others had thousands. Empty intervals remained visible.

People began searching.

A bakery worker in Drum Two found a recipe from a route where his grandmother had tried salt in a sweet yeast culture and written an apology to the entire night shift. A pressure technician found her own trainee code attached to a branch report she had never filed here. A child on Indigo discovered another version of a classroom rule and objected that both versions misspelled the same teacher's name. Glass located a petition signed by three ancestors of its temporary council, all demanding command publish a coolant shortage.

The first dispute reached Ilyan before the laughter over the burned porridge had ended. A sanitation worker aboard designated Nacre claimed a Route Blue-Four diary under his own archive identity. The writer shared his name, birth mark, parents, and work assignment through Correction 18. After divergence, the diary recorded nineteen days the living worker had never experienced.

“Those are my words if the route copied me,” he said over the public channel.

“They came from a person with your prior history,” Ilyan answered. “The museum cannot assign his later words to you.”

“He had my family.”

“He also had his own clock.”

“I want the file under my name.”

The worker did not sound greedy. He sounded bereaved by a life that resembled his closely enough to trigger possession. Ilyan understood the reflex. For years he had looked at route cards and imagined any surviving Eda would belong to the grief he carried.

Oru joined the channel. “You may attach a counterpart statement. The source record keeps Blue-Four identity.”

“What am I supposed to write?”

“Only what you know.”

After a long interval, the worker attached six words: I remember none of these days.

The museum displayed his statement beside the diary without merging them. Visitors could read the shared history, the divergence, and the living counterpart's refusal to claim memory he did not possess.

A second conflict came from Violet. Lin's council found command orders bearing his old name and rank. Some visitors had begun indexing them under Lin, while others used Ilyan Rook of Violet. He requested a source note.

“Records written before my renaming should remain unchanged,” he said. “The catalog must not use that history to revoke the name I chose after it.”

Ilyan added VIOLET SOURCE NAME AT TIME OF RECORD and CURRENT CIVIC NAME as separate fields. The change propagated to every route. Indigo asked for the same distinction for children whose names had changed after Correction 141. Glass required that political offices receive time fields too, so a former captain's order could not borrow authority from a later council.

Each request slowed the archive. None counted as delay to the transfer because Sable kept sending raw packets underneath their work. Ilyan began to understand why its archivists called the structure a museum rather than a ledger. A ledger sought balance. This place allowed two labels to remain side by side when forcing agreement would destroy evidence.

From Amber came a harder request. A surgeon found an old wake note naming a patient whose counterpart now lay in Amber's surgical bay. The historical note contained a diagnosis the living patient had not disclosed to family. Sable's privacy key sealed medical detail, but the existence of the note still appeared in search results.

“Hide the index from public name search,” Amber said. “Preserve it for the patient.”

Oru asked whether that would violate Sable's opening terms.

The archivist answered from the failing route. “No usefulness ranking. Privacy remains a living claim. Seal it.”

Ilyan created a recipient key that no captain, family member, or museum clerk could open without Amber's patient. The record stayed counted but unread. Its closed drawer appeared among the open ones, labeled by source and consent status rather than diagnosis.

Other claims followed. A person on Reed denied a wake joke attributed to his counterpart. A teacher on Indigo corrected the age attached to a child's drawing. Violet identified a burial list that used emergency nicknames because the local identity server had failed. North could contribute only during alignment, so its corrections

arrived as unfinished phrases and completed themselves over several passes.

Through all of it, the raw source remained untouched.

The public channel filled with requests, identifications, disputes, and corrections. Ilyan slowed the intake so no claim could alter source material. He created side records for living witnesses. Every addition carried the witness's route and clock.

Mina sent him a private note.

> EDA ROOK: THREE HITS. DO YOU WANT HOLD?

His hand stopped above the console.

The search authorization from Sava had entered Commons custody an hour earlier. He had expected days of processing, enough time to make grief procedural. Sable's indexes found the name without caring whether he had prepared himself.

He opened only the citations.

One belonged to a pressure inspection on Route Ash after Correction 133. Eda had signed a seal check, added a complaint about a stripped lamp fitting, and returned the work order. Another carried her name in a meal rotation. The third consisted of four timing knocks attached to an unidentified maintenance channel.

No message addressed him. No hidden life extended toward the present. Route Ash had persisted nineteen days, six hours, then closed with everyone aboard. Eda had lived those days while his designated counterpart studied route tables. Six years later, designated Eda died in Drum Five. Both facts fit without giving either death a replacement.

Ilyan marked the records OPEN UNDER SOURCE TERMS.

Mina did not send comfort. She added her witness seal beside his.

Sable missed two clock pulses.

The route line returned dimmer. Its archivist spoke in fragments.

"Local lights failing. We have moved to the chart batteries."

Jo opened a reply. "We can send another hold from designated reserve. Estimate nine minutes."

"For archive or hull?"

"Archive. We cannot reach your damaged braid."

"No."

"Violet and Glass both consent."

"Keep course reserve. Use receive band while our clock continues."

Ilyan heard someone behind the archivist ask how much remained. Her answer did not cross clearly. The packet stream accelerated.

Fifty-one percent.

A compressed classroom wall arrived from a route called Brass. Children had listed questions for Pelagos beneath a drawing of two moons. Did ocean water taste like tank leaks? Could birds get lost without ceilings? Who decided where a road ended? The final question had no punctuation.

A galley notice from Green-Eight banned a private joke involving algae paste and the captain's shoes. Sable's catalog preserved the joke's absence because the notice mattered even when nobody could recover what caused it.

A maintenance table carried Alda's hidden interval marks copied by an unknown engineer seventy years after her death. Beside them someone had written, IF THIS MEANS PEOPLE, WHY DOES THE BELL STOP?

Sable's control estimate fell below six minutes.

"Archive source requests a final live drawer," Mina said.

The packet contained Sable's own current records. Until then, its archivists had treated their ship as custodian rather than exhibit. Now ration notes, braid repairs, council disputes, class lists, and burial records entered beside the older wakes. A cafeteria board apologized for serving the same cold grain three meals in a row. A drive technician had drawn a crude picture of the scored hull and labeled the largest breach NOT FIXED BY OPTIMISM. Someone had filed the former captain's resignation under PERSONNEL, then cross-indexed it under WEATHER.

The last record showed a blank clock form.

"Explain the form," Ilyan said.

Sable's archivist answered after thirteen seconds. "Our final local time goes there. Do not estimate it from your receipt."

"You still have packets queued."

"The queue sends while control holds."

"Give us names of everyone in the chart room."

"They are in packet seventy-two."

"Tell us what you need."

"Witness the clock."

Ilyan switched every available sensor to passive receipt. Field pickup. Rail vibration. Seam light. Audio compression. Jo verified that no challenge pulse left designated Nacre. Glass and Violet did the same. Indigo stopped its timing game so Sable's signal occupied the band alone.

One clock moved at the edge of the route glass.

The museum queue reached sixty-three percent.

Sable sent a time mark.

Another followed eleven point four seconds later.

Then twelve point one.

The archivist read names, not as a complete population but as people present beside her. A braid engineer. Two archive clerks. A Commons witness. Three students who had carried storage tiles from the school. The ration worker who knew which cold grain note belonged in the current drawer. A security officer holding the chart-room door open because its latch had failed.

“Do not turn us into the only names,” she said. “The packets hold more.”

“We understand,” Oru replied.

“No. But preserve the instruction.”

Ilyan entered it word for word.

A hard tremor crossed the Sable pickup. Voices broke apart. The line brightened once as a reserve bank failed into the seam, then dropped below normal gain.

Jo said, “They have one bank.”

“Can we catch the surge?” Sava asked.

“Catching it would draw the closure faster.”

Nobody moved.

Sable's final packet header appeared. It contained no maintenance disguise, only the ship name and an empty timestamp field. The archivist's voice came through under heavy distortion.

“Clock mark follows. Record local, then receipt.”

A pulse crossed the seam.

Ilyan's console marked SABLE LOCAL 17:43:08.119 and NACRE RECEIPT 17:43:09.247. Violet, Indigo, Amber, Reed, Glass, and North added their receipt times as delay allowed.

The next expected pulse did not come.

Jo watched the field load. “Wait.”

Ilyan waited through twice Sable's longest recent interval. The chart room held pumps, breathing, and the faint public channel from thousands of open museum drawers. None came from Sable.

The Loom dimmed its line. A gray system label formed beside the last coordinate.

ECHO DECAY.

Ilyan selected the words and opened the classification field. Founder constants resisted deletion, but the map now answered to route and Commons seals. Oru approved the change. Jo confirmed that Sable carried no recoverable field load. Lin witnessed from Violet. Other ships added their clocks when their delays permitted.

Ilyan removed the false label.

He entered DEATH RECORDED: SABLE. CAUSE: ROUTE RESERVE COLLAPSE FOLLOWING CORRECTION 141 DAMAGE. LOCAL CLOCK AND MULTIROUTE RECEIPTS ATTACHED.

The museum remained open. Its transferred drawers did not brighten the dead route line or complete the packets that had failed to cross. Forty thousand unresolved files ended inside Sable. Ilyan preserved the missing range in the catalog rather than closing the gap.

On the map, seven ships remained.

Sable's trace had gone dark, but its final clock occupied a fixed place beside them. Ilyan set the record to require a witness acknowledgment from every future chart shift. The first acknowledgment came from designated Nacre. Violet followed. Then Indigo, Amber, Reed, Glass, and North marked the death at their own delayed times.

Only after the seventh receipt did Ilyan close Sable's channel.

The silence stayed in the log under its proper name.

Interlude III — Names From the Wake

<!-- REGISTER: curated records from Sable's museum packet -->

<!-- PAYOFF: ordinary lives preserved through Sable's final timestamp -->

Canteen Ledger, Drum Two, Table 14

Mera Venn owed Davi Sol one spoon of quince paste, payable after the next fruiting cycle. Davi disputed the unit. He said Mera had taken two spoons and offered as evidence the clean crescent at the bottom of his cup.

Mera answered that a cup could not testify against its washer.

Witnesses: Pilar Ame, Jun Set, and a child who signed only with a drawing of three teeth.

Resolution postponed because the shift bell rang.

Pump Four Repair Note

Technician Olan Mire replaced the intake seal with stock cut for a smaller housing. The proper ring had split when Sable crossed the charged fold. Olan warmed the substitute in a broth kettle, stretched it over a gauge cone, and held it under compression while the pump cycled.

The repair leaked one drop every seventeen minutes. Someone placed a blue cup beneath it. The night crew called the cup Assistant Pump Four and entered it on the inspection roster for eleven days.

When the proper ring arrived, apprentice Kesh Iri asked whether the assistant would be dismissed.

Olan wrote: PROMOTED TO CANTEEN.

Classroom Correction, Green Belt School 6

Question: If Sable and another ship begin from the same correction, which one keeps the true history?

Initial answer submitted by Tomi Pell, age nine: the one that gets to Pelagos.

Correction submitted by Tomi Pell, age nine years and four minutes: getting somewhere does not alter where you came from.

Teacher's mark: ACCEPTED.

Additional note from the student: You should have put that in the question.

Laundry Claim 88-441

A gray work shirt reached Drum Five laundry with the name SAEL ORU stitched inside the collar. No person by that name appeared on Sable's active register. The shirt showed six years of cuff repairs, lime dust at both shoulders, and a square patch cut from a child's blanket.

Laundry posted the shirt for reclamation. During the first week, seven people touched the patch and left it hanging.

On day eight, Vina Oru claimed it for her father. She had removed his name from the register after the hull breach because she could not bear the red closure mark. Laundry restored the entry from a paper shift card found in the pocket.

Sael Oru, pressure mason, twenty-six years aboard, preferred his tea cold.

Claim granted to Vina. Record retained at her request.

Message Draft, Unsent

Ralo,

I ate the pears before they softened. I know you kept them for the arrival table. You may list this among my established defects. If you still intend to speak to me after we reach Lattice, I will grow you six trees and let you choose where none of them belong.

I did not leave because of the quarrel. I left because Bulkhead Nine needed hands.

There should have been time to make those sentences arrive in a better order.

Galley Substitution, First Meal

The recipe called for twelve grams of salt per kettle. Cook Hessa Brell used nine after the condenser report warned of falling water reserve. Fifty-one diners returned bowls for correction. Thirty-two added salt. Nineteen asked for pepper instead. One person drank the broth without complaint and later filed a six-paragraph objection through the Commons queue.

Hessa printed the objection and pinned it beside the ladle.

Her reply, written beneath it: YOU HAD THREE CHANCES TO SPEAK WHILE I WAS IN THE ROOM.

Personal Tool Register

Belonging to Iv Sorn, rail maintenance:

One torque driver with the insulation chewed by Iv Sorn.

One gauge key with the insulation chewed by somebody Iv Sorn continued to blame.

Three black clips, four silver clips, and a red clip that opened no approved fastener aboard Sable.

A paper square folded around two tomato seeds.

Iv's daughter stated that the red clip opened the locked drawer where her parent hid birthday sweets. The inventory officer tested the claim. The drawer opened.

The clip classification changed from UNKNOWN to DOMESTIC OVERRIDE. Iv's daughter requested the sweets as inherited maintenance stock. The inventory officer approved half.

Commons Minute, Partial

Delegate Ren Vask moved to suspend arguments about who had caused the reserve shortfall until the pumps stopped losing pressure.

Delegate Mari Uln objected that blame did not require pressure.

Ren answered that lungs did.

The motion passed. Mari joined the pump crew after adjournment and worked beside Ren for six hours without withdrawing the objection.

Garden Work Order

A patch of beans in Belt Three had grown around a damaged light standard. Children tied the vines to the inactive frame and named each climbing stem after a destination feature copied from mission maps: Harbor, Lattice, Rain, East Island, and Home.

The stem called Home produced no flowers. East Island climbed into the live lamp and scorched two leaves. Rain spread sideways onto the path, where boots damaged it twice.

Gardener Sal Kito refused an instruction to clear the obstruction. She moved the path marker eighteen centimeters and wrote that navigation had already taken enough from the beans.

Revised path accepted.

Medical Addendum, Birth Record 19-S

Nim Ari arrived during the second reserve alarm. Parent and child required heat that the ward had been ordered to surrender. Nurse Dalen Roe disconnected the corridor display and used its supply to warm the incubator.

Security requested the name of the person responsible.

Dalen gave them the child's.

Nim remained alive for eleven days. The record includes no photograph because every available image channel carried engineering diagrams. A heel print survives in violet antiseptic on the back of the nurse's meal chit.

Private Joke Archive, Voluntary Submission

Prompt: Preserve one joke no future person will understand.

Submission by Ara Fen: Tell Jo that the starboard algae has apologized.

Submission by Ko Senn: Correction, port algae. Starboard algae has retained counsel.

Submission by Jo Fen, no relation claimed: Neither side can afford counsel. The legal moss has eaten it.

Curator's note: four hundred twelve people marked this exchange as important. No explanation accompanied the marks.

Shift Handoff, Cold Storage

Bay Six held embryos, grain stock, fungal cultures, and a tin containing five wedding rings recovered after accidents. Keeper Ilo Mar transferred duty to Keeper Sen Har with the following verbal note:

“Temperature has climbed two tenths. Rack C sticks at the lower hinge. The barley assay needs another hour. Also, Pava's ring belongs to her husband, not the memorial office. He will say he cannot take it. Ask twice.”

Sen Har asked twice.

The ring left the bay on a cord around Emet Pava's wrist.

Route Museum Intake

Sable's curators did not arrange the packet by rank, cause of death, or usefulness to the mission. They placed a child beside a pump seal, a lost shirt beside a galley complaint, and a birth record beside a joke whose algae required counsel.

The intake index reserved fields for names omitted from damaged rosters. Blank fields remained searchable. Where a voice survived without a speaker, the curators entered cadence, room tone, and any other person heard nearby.

One file contained nine seconds of scraping, a cough, and someone saying, "Not that bolt." The speaker never supplied a name. The museum retained the correction.

Clock Closure

At the end of the packet, the image narrowed to Sable's route clock. Frost covered the lower corner. The digits lost brightness from left to right while a hand outside the frame kept writing names into the intake field.

The final legible entry read: Curator Esen Drail, still receiving.

Sable final timestamp: 127:221:18:04:11.006.

Chapter 19 — The Parliament of Elsewheres

<!-- POV: Ilyan Rook -->

<!-- SETUP: seven surviving ships; refusal of merger; disputed designated-route authority -->

<!-- PAYOFF: civic standing precedes engineering debate; rotating order removes the center frame -->

Oru Pell began by turning off the captain's seal.

It occupied the center of the route glass, a white mission mark over the seven live seams. The seal had appeared whenever the chart room hosted a formal command session, even when the command under discussion concerned algae allocation or tram hours. Ilyan had stopped noticing it years ago. Now its removal left a pale circle in the glass, bright around the edges and empty inside.

“Clerk,” Captain Or said from the lower rail, “that mark authenticates the channel.”

“The clocks authenticate the channels,” Oru replied. “The seal identifies one authority. We have not agreed that it governs this room.”

Sava's hands closed behind her back. Her command jacket carried no mission pin. She had surrendered it for review three days earlier, but the chart-room system still obeyed her voice without asking whether the rest of them consented.

Ilyan stood at the conservator's station with seven timing columns open. A pulse reached Violet, returned across delay, and took its place beneath the local clock. Indigo followed. Amber arrived with a roughened edge from its failing coolant braid. Reed, Glass, and North answered in sequence. The seventh column belonged to the ship under his feet, which the Loom continued to label DESIGNATED NACRE whenever nobody was watching the legend.

Jo had covered that label with black tape.

Seven wall frames carried seven chart rooms. Their images arrived at different rates, never fast enough to disguise distance. Lin faced them from Violet beneath a patched ventilation tube. Indigo's delegation sat in a classroom because its Commons well had lost two power trunks. Nessa Quill occupied a rear desk with her hands folded under her chin, present as a technical witness rather than a delegate. Reed sent four people crowded around one functioning camera. Glass appeared overexposed along every metal edge. North's frame stuttered once each second. Amber showed three silhouettes behind fogged glass and no name placards.

Oru placed a paper resolution on the local console.

“Before anyone proposes a maneuver,” they said, “we establish who may refuse it.”

The sentence crossed the seams. Violet received it first. North heard it last, nine tenths of a second later. Oru waited for all six receipt lights before continuing. They had brought a clerk's patience into a room trained to treat delay as failure.

The first clause recognized each surviving route-ship as a civic jurisdiction continuous from the mission charter and altered by its own post-correction history. The second recognized every person aboard each ship as living mission population. The third denied any route authority to settle, merge, copy, extract, or extinguish another without that ship's consent.

At merge, Lin raised one finger.

Oru stopped. "Violet holds the floor."

"No," Lin said through a channel striped with interference. "Violet asks whether we possess the floor. That should not depend on your answer."

A few people in the local gallery shifted. Oru looked toward Ilyan, not for permission but for a timing status.

"All channels remain open," he said. "Violet can transmit without our grant."

"Then record that," Lin said. "I object to merger appearing as an act that consent could make safe. Consent does not preserve two pattern histories when matching structures occupy one field state. We will not authorize our own replacement and call the replacement rescue."

Teren Vale stood behind Jo at the engineering rail. He leaned toward his microphone, then caught himself. Oru had not recognized him.

"Do you propose different wording?" they asked Lin.

"Each ship retains bodily and civic continuity. No agreement may require person-merger or ship-merger."

Oru wrote by hand. The camera above their desk sent the revision to every frame as moving ink rather than finalized text. Ilyan watched copies of the line appear with uneven delay. On Violet, Lin bent to read. In Indigo's classroom, the delegate passed a printed strip to Nessa. Glass lost three words to a compression fault and requested them again.

North did not acknowledge the clause.

Its delegate sat alone before a steel bulkhead marked with frost. The frame identified her only as NORTH COMMONS SPEAKER. She had refused a personal name on the shared channel until standing could protect the register from seizure.

"North holds a procedural objection," she said.

Oru's pencil waited.

"You have coupled recognition with authority to decide a maneuver. We accept the former. We reject any schedule that forces the latter today."

"The resolution contains no maneuver."

"It creates a body for one. Our food pumps lost another two percent during this session's setup. If we bind ourselves to your meeting order, a majority can spend our reserve before we have translated the engineering packet."

Jo opened their mouth. Oru lifted the pencil, and Jo remained silent.

"What delay does North require?" Oru asked.

"Thirty ship-hours after a complete packet arrives in our local archive. Not after Nacre marks it sent."

The local gallery produced its first audible protest. Thirty hours consumed reserve no ship had to spare. Sable's

final clock still occupied a dark corner of Ilyan's display, disconnected from the live columns but not removed. He could hear Teren's breath quicken behind him.

Oru did not argue about scarcity. "North requests a receipt-based clock. Clerk records."

"Clerk also records that North's survival may not fit inside it," Teren said.

Every route frame showed some version of his interruption. On Reed, his mouth moved before the sound. Amber received his voice with the middle flattened into static. The delay made one breach of order happen seven times.

Oru turned their chair toward him. "Doctor Vale, which delegation authorized you to spend North's deliberation?"

"No delegation. The reserve curve authorizes urgency."

"Curves do not hold standing here."

Teren colored. His shame usually drove him into explanation; this time he lowered his head and stepped away from the rail.

Ilyan checked North's receipt clock. The complete engineering packet had not arrived. Two checksum segments remained queued behind the museum archive, because the routes had refused to give Sable's names lower transmission priority than maneuver models. North's thirty hours had not begun.

He sent that fact to Oru's private pane.

They read it and addressed North. "Your requested interval starts after those two segments clear. If any route transmits a revision, the affected section gets its own receipt time. Does that preserve your objection without placing every document behind a new thirty-hour wall?"

North's frame held still through one of its regular stutters. "It preserves the objection."

"Noted. It does not yet settle it."

Oru moved to the next clause.

Amber interrupted before they spoke.

The center silhouette came closer to its camera. Condensation hid the upper half of the person's face. "Who appointed Nacre to chair?"

No local voice answered. The question entered the chart room without volume and altered every posture inside it. Sava remained at the lower rail. Mina Rook, seated among the Commons observers, stopped writing. Oru set their pencil beside the paper.

"The Commons Assembly elected me clerk before contact with your route," they said. "No body containing Amber appointed me chair."

"Yet you control recognition, wording, receipt, and the sequence of speakers."

"Yes."

“You removed a seal and kept its chair.”

Ilyan heard Jo exhale through their nose. Oru had constructed the meeting around standing and still left one old geometry intact: the designated ship spoke first because its equipment opened the channels. Its clerk decided when delay counted as listening.

Oru asked Ilyan to display the transmission queue.

He hesitated. The queue included channel health, power costs, and the priority switches Jo had made to keep North audible. It also showed the local master clock distributing turn tokens to the other ships. Every token originated aboard Nacre.

“Full display?” he asked.

“Full.”

Ilyan expanded it across the pale circle where the captain's seal had been. Six outgoing grant lines spread from the local clock. The diagram looked less like a conversation than a hand with six leashes.

Amber's silhouette said, “There.”

“I see it,” Oru replied.

Lin leaned close to Violet's camera. “Violet proposes that the clerk rotate.”

“Among ships or among delegates?” Reed asked.

“Among clocks.”

Glass sent a burst of laughter. Its delegate had a dry voice and a bandage over one ear. “Good. Let a cesium braid endure politics.”

Oru turned to Jo. “Can the floor token originate from a different route after each recognized speaker?”

“It can originate everywhere or nowhere,” Jo said. “The seam controller already receives phase receipts. Replace the grant with a rotating order agreed in advance. Each route knows which clock follows which. If one channel fails, the next route waits one full cycle and proceeds.”

“Who sets the first position?” Amber asked.

Nessa lifted her hand in Indigo's rear row. The delegate glanced back, then moved aside so she could reach the classroom microphone.

“Use the last digits of Sable's final clock,” she said. “Nobody here chose them.”

The room went quiet without becoming empty. Ilyan kept the dead timestamp on his own screen because deleting it felt like accepting the old classification. He had not expected its numbers to govern anything living.

He entered the last seven nonrepeating digits. Jo mapped them against route checksum order. The result began with Glass, then North, designated Nacre, Reed, Indigo, Violet, and Amber. Oru read the sequence aloud. Each ship confirmed only its own place and the place that followed it.

“Does any route object to this rotating order for procedure?” Oru asked.

Amber answered last. “We do not object. We have not accepted your resolution.”

“Those remain separate questions.”

Jo armed the distributed token. A small ring appeared around Glass's frame. The Glass delegate looked startled by it, then smiled with one side of his mouth.

“Glass recognizes the clerk from Nacre for the limited purpose of reading Clause One,” he said.

The ring moved to North.

“North recognizes the clerk for the same limited purpose, subject to our receipt condition.”

Nacre's position came next. Oru looked toward Sava. Command habit tugged every eye to her, including Ilyan's. She did not step forward.

Mina spoke for the Commons delegation. “Nacre recognizes Oru Pell as recording clerk, not presiding sovereign.”

Reed confirmed. Indigo confirmed. Violet confirmed. Amber allowed the token to circle its frame for three breaths before speaking.

“Amber recognizes the recording function. Amber reserves judgment on every substantive clause.”

The ring returned to Glass.

Oru picked up their pencil. “Clause One, under distributed recognition.”

They read the text again. This time the words traveled through a sequence no local console could shorten. Each route repeated the final six words before passing the floor. The process took eleven minutes. Navigation staff around Ilyan grew restless; clerks in the gallery settled into attention. Delay stopped resembling damage once everyone had to inhabit it.

The standing clause passed without dissent. North marked its maneuver reservation. Amber added a note that recognition of population did not imply confidence in Nacre's engineering account. Oru entered both beneath the adopted text.

Clause Two required more work. Reed wanted stored embryos counted within each jurisdiction. Glass asked whether people in medically induced suspension retained an individual refusal right. Indigo insisted that classrooms receive notices in language children could challenge. Violet demanded that emergency councils formed after Correction 141 receive the same recognition as inherited Commons bodies.

North used its turn to ask who could bind children to a route decision. The speaker's image froze with her mouth open, then resumed after the missing frames arrived.

“Parents already consent to emergency maneuvers for households,” Sava said from the lower rail.

The floor token had not reached Nacre. Oru did not rebuke her. They entered the interruption with its exact timing and passed it onward as part of the public record.

When Indigo received the floor, its delegate moved the classroom microphone to Nessa's desk. “Captain Or says parents already make this choice,” he said. “Our witness asks to answer.”

Nessa sat up. "They make choices on a ship they can stay on afterward. If you choose which ship counts, a child cannot grow old enough to disagree."

Sava stepped closer to the seam frame. Ilyan recognized the increased care in her voice, each word fitted like a door seal. "No child can assume course authority. That remains true on every route."

"I didn't ask for course authority," Nessa said. "I asked you not to spend my later answer now."

The classroom behind her held nine empty desks. Several students had gone to assist in the garden after Indigo cut another lamp circuit. Their absence made the room look staged, though Ilyan could see soil on the floor and a patched sleeve hanging from one chair. Nessa waited with the blunt patience children reserved for adults who had substituted rank for reply.

Oru asked North whether a future-refusal clause addressed its concern.

"Only if survival preserves a future," the speaker said. "Write that delegates may authorize emergency risk but may not erase the represented population's later civic claims."

Reed objected that any maneuver could erase future claims by killing people. Glass argued that the clause needed to distinguish accepted risk from designed extinction. Teren sent Oru a technical phrase, then withdrew it before recognition. Ilyan saw the canceled text flash in the clerk's private queue: probabilistically disproportionate termination. Oru would have rejected it even if Teren had spoken. Nobody aboard a failing ship should have to parse six syllables to learn whether another route intended to close them.

Lin proposed simpler language. "No route may choose a result that requires another route not to exist."

Amber's delegate answered from behind renewed condensation. "Your own survival requires fuel our ship may need. Every allocation could end with one of us absent."

"Then no route may build its plan on that absence."

"That I can sign."

Oru wrote both lines into the margin and asked the routes to compare them locally. For twelve minutes the parliament fell into seven separate conversations. None crossed the seams except room tone: a chair scraping on Violet, pages turning on North, a child coughing in Indigo's classroom. Amber's audio carried a pump knock at irregular intervals. Ilyan marked the sound before he could stop himself and measured its worsening cadence.

Sava joined him at the conservator's station. "Can Amber hear that pump from their delegation room?"

"If it belongs to the room."

"Ask."

"Their local conversation remains private."

Her jaw set. An earlier Sava would have opened a safety override and called the intrusion care. This one watched Amber's frame until its conference indicator cleared.

When the rotating order reached Nacre, she used the public floor. "Amber, your audio carries a mechanical strike every eight seconds. We offer a drive engineer if you want one."

Jo glanced at her. The offer could supply instructions, not hands. Sava had kept the limit inside the sentence.

Amber's delegate listened to something beyond her camera. "Coolant bypass check valve. We know. Your engineer may review after this clause."

"Under your authority," Sava said.

Amber acknowledged with one lowered hand.

The seven local conferences converged on Lin's second wording. Oru attached North's future-claims sentence as a separate protection for people represented by emergency councils. Nessa asked that children appear in the explanatory note instead of vanishing into population. The clerk added them without converting her request into a symbol.

Oru did not compress the objections into one agreeable phrase. They let each alter the document. Ilyan watched the resolution accumulate marks until the page looked repaired rather than drafted. Seven scripts occupied its margins. Some additions arrived as block capitals, some as printed tape, one from Reed as a photographed grease-pencil line on a food crate.

During the fourth rotation, Amber returned to merger.

"If two ships voluntarily choose structural combination to conserve reserve, does your prohibition prevent them?"

Teren looked toward Lin. Ilyan felt the old professional desire for a clean answer. The physics supplied one: matching patterns entering the same state destroyed distinguishable histories. A new structure might continue afterward. No measurement could call it either predecessor without discarding what had differed.

Lin answered when Violet's turn arrived. "You may authorize destruction aboard your own ship under whatever law your people accept. You may not call the result our survival. Violet will not enter such a covenant, and nobody may preserve us by replacing us."

Amber's center silhouette wiped a clear patch in the condensation. A woman with a bruised temple looked through. "You speak as if we are eager to die incorrectly."

"I speak as someone whom Nacre offered to save as information."

"And you hear every question as Nacre's question."

Lin's face tightened. Because Ilyan knew that face, he saw the reply form before Lin chose against it. Shared habits survived divergence in the small muscles around the mouth.

"You are right," Lin said. "Violet withdraws the implication."

Amber did not thank him. Its delegate said, "Keep the prohibition. Replace voluntary with informed and reversible wherever the act permits reversal."

Oru added the words.

By the sixth hour, the route glass carried no map. Every available plane held text, faces, receipt clocks, or engineering queues. Pelagos remained only as a gold point in a small navigation pane near Ilyan's hand. The ships had spent reserve speaking before they had chosen how to move. That cost troubled him. The alternative had already cost Sable its remaining time.

Oru called the standing resolution for confirmation under the rotating order.

Glass adopted. North adopted with its delay reservation. Nacre adopted through Commons. Reed adopted. Indigo adopted. Violet adopted. Amber waited until condensation reclaimed the patch over its delegate's face.

“Amber adopts standing,” she said. “Amber does not recognize a designated chair, a designated truth, or a designated obligation to trust.”

“Recorded without amendment,” Oru said.

The ring completed its circuit and disappeared.

Sava approached the clerk's desk after the receipt lights steadied. “You now have a parliament,” she said quietly.

“We have standing,” Oru replied. “A parliament begins when refusal changes what happens next.”

They asked Ilyan to restore the route frames for the first engineering session. The software offered its familiar arrangement: Nacre large at the center, six smaller seams around it. He rejected the template. A second layout placed the captain's feed in the middle. He rejected that too.

Jo opened a raw geometry panel and assigned equal dimensions to all seven frames. The display could not fit them in one row. Ilyan curved them around the suspended glass, each image facing the empty circle left by the seal.

Oru took their clerk's seat at the edge of Nacre's frame. Glass would speak first when debate resumed. North's receipt clock continued counting toward readiness. Amber sent no farewell, only a clean checksum for the adopted resolution.

Ilyan saved the configuration under a new title: SEVEN-SHIP SESSION.

When the chart room lights lowered, no ship held the center frame.

Chapter 20 — The Vote No One Owned

<!-- POV: Captain Sava Or -->

<!-- SETUP: seven-ship standing; rotating order; disputed sole mission command -->

<!-- PAYOFF: five approvals, North's delay condition, Amber's abstention, revised rescue terms -->

Sava removed the command key from her wrist before the vote began.

The strip of black alloy had lived against her pulse for eleven years. It opened the drive nave, the captain's archive, the sealed stores, and every emergency channel aboard Nacre. During Article 4 it had also opened other people's records without notice. She placed it on Oru Pell's desk between a water cup and the seven-ship standing resolution.

Oru looked at the key. "The Commons did not ask for that."

"You are about to."

"I would rather ask after the terms exist."

"Terms acquire a different meaning while one person can override them."

The route glass showed seven equal frames. Glass held the first position in the rotating order; North followed; Nacre came third. Sava's image appeared inside one seventh of the ring, no larger than the bruised Amber delegate's face or Lin's patched chart room on Violet. The empty center displayed the proposition in Oru's hand.

No ship would claim designation privilege during a shared maneuver. No route would accept a correction funded by another route's involuntary closure. Each ship would preserve its own civic continuity, accept an independently calculated arrival body, and expose every reserve transfer to the other six. Execution would require all seven authorizations.

The text avoided any promise that they already possessed a maneuver. Teren had a symmetric release model and several pages of warnings. Ilyan had Alda Quill's blank course line. Nessa's timing game had supplied a phase relation no adult engineer had found. None of that yet amounted to a course through the debris tide.

Oru called the session to procedure.

"We vote today on governing principles," they said. "Approval does not authorize field load. Rejection does not revoke standing. A condition attaches only to the route stating it unless the other delegations adopt it. Abstention preserves participation in later drafting."

Sava listened for the pressure under each sentence. Oru had written the rules to prevent a vote from becoming a weapon against its minority. That protection also denied her the old comfort of a clean command result. No majority could tell her whether to risk forty-two thousand and six people. No captain could decide for the people behind the other frames.

Glass received the first floor token.

Its delegate sat with the bandage removed from his ear. A line of fresh stitches climbed into his hair. "Glass approves the principles," he said. "We attach a request that damage forecasts use local material tolerances, not

Nacre's hull averages.”

“Recorded as a drafting request,” Oru replied.

The token moved to North.

Its Commons speaker had supplied her name after the standing resolution protected the route register: Eren Vey. Frost still marked the bulkhead behind her. She placed both hands where the camera could see them.

“North approves only if no execution vote occurs until thirty ship-hours after our verified receipt of a complete course and complete failure model.”

Teren shifted at the engineering rail. Sava felt his objection before he voiced it, because she shared it. North's reserves might not survive the full delay. Every hour spent translating models widened the chance that there would be no North left to consent.

Oru asked, “Does North record approval with a delay condition, or rejection pending delay?”

“Approval with condition. If the condition cannot be met, you do not possess our approval.”

The clerk entered a gold mark rather than white. One approved. One conditional.

Nacre's token lit.

Mina Rook rose from the local Commons seats. The Assembly had argued for nine hours over who could speak for the ship. Sava remained captain. Mina carried the identity resolution. Jo represented engineering custody. Ilyan held the route record. None possessed the whole vote.

Mina read the Commons decision. “Nacre approves the principles, provided Captain Or's course authority applies only to this ship and cannot designate another route for settlement.”

Every frame showed Sava receiving the condition. The delay repeated the moment six times: her still face on Violet, her still face on Indigo, another on Amber where compression erased the line of her mouth. She had once believed duplicated images weakened authority. Now they prevented her from choosing which audience counted.

Oru said, “Captain Or, the condition concerns your office. Do you accept it?”

Sava had prepared a statement titled CONTINUITY OF MISSION AUTHORITY. Four hundred twelve words waited on her wrist pane, reviewed by the captain's counsel. She read the title once, closed the file, and left its narrow emergency exception unopened.

Her father had once divided a bowl of grain mash into unequal portions and told her he had already eaten. She had accepted the larger share because he used an adult voice and because hunger made obedience resemble trust. Three days later she found his untouched ration chits under a pipe clamp. He had controlled the information so she could survive. He had also denied her the chance to refuse the cost.

Sava closed the prepared statement.

“I accept that my command ends at Nacre's hull,” she said. “I retain duty to its crew, stores, drive, and safety. I relinquish sole mission command over any shared route maneuver. My key will remain in Commons custody whenever a field instruction can alter another ship.”

The chart room did not react at once. Her words had to cross six seams before anyone could know whether she had left herself an escape clause.

Lin spoke out of order from Violet. "Does that include authority to close a seam?"

"If closure affects only our equipment, I may order it for immediate safety. If closure strands another route or alters a shared maneuver, no."

"Who decides which kind it is?"

"Engineering custody and the route record, subject to Commons review." She looked toward Jo and Ilyan. "Not me alone."

Jo's expression gave her no pardon. They nodded once to indicate that the answer matched the proposed custody rule.

Mina cast Nacre's approval. Oru entered a second white mark.

Reed came next. Its delegation had moved from the food bay to a pressure shelter after a seam-power fault. Crates formed the table. A flour measure served as a microphone stand.

"Reed approves," its speaker said. "We request that independent arrival not mean abandonment after the maneuver. Shared warnings continue until every route clock returns."

Oru marked the request for rescue terms.

Indigo approved from the classroom. Its delegate spoke first, but Nessa held up the paper where her classmates had redrawn the two-sky game. Neither sky started first. Every player kept a separate score. A round ended only when both sides heard the closing knock.

"Indigo asks the same thing as Reed," the delegate said. "A ship does not count as separate only when separation makes rescue inconvenient."

A fourth white mark appeared.

Violet's turn reached Lin.

Behind him, his chart room had lost one ceiling strip. People moved through the darker section carrying ration canisters. Violet's emergency council stood beyond camera range, but Sava could hear their argument in low bursts. Some wanted the principles to include restitution from Nacre. Others feared that any shared field gave the designated ship another chance to harvest them.

Lin rested his hand on the console in the same place Ilyan often did. "Violet approves civic continuity, independent arrival, transparent transfer, and seven-party execution. Violet rejects any implied duty to preserve Nacre's mission name or command structure."

"No such duty appears," Oru said.

"It appears in every system label."

Jo had covered one label with tape. Thousands remained inside the Loom stack, the archive, and the drive controls. Sava could not remove them by statement.

“Then record the rejection,” she said. “Our software carries claims this vote does not.”

Lin looked toward her frame. “Violet records that Captain Or acknowledged the distinction.”

He cast the fifth approval.

Amber held the last position.

The bruised delegate had cleared the condensation from her camera. Her name strip read MARA SEN, COMMONS AND COOLANT. A pump struck every eight seconds behind her, the fault Sava had heard in the previous session. Jo had reviewed the valve remotely and sent three possible repairs. Amber had not yet attempted any. Its remaining coolant reserve sat eleven percent below Nacre's emergency floor.

Mara Sen read the proposition without looking at the other frames.

“Amber abstains.”

Oru waited. “Would Amber state grounds for the drafting record?”

“We are asked to renounce settlement reserve while carrying damage that the designated ship avoided through Correction 141. We are asked to accept an independent arrival body before anyone can show that our cooling plant survives the debris passage. We are also asked to trust open reserve accounting authored on Nacre's instruments.”

Teren requested the floor for technical response. Amber's turn had not ended. Oru denied him with a small movement of the hand.

Mara continued. “Five ships approve because the principles protect them from harvest. North approves if time protects its deliberation. Amber needs protection from becoming the route everyone recognizes and nobody can afford to carry.”

Sava looked at Amber's coolant graph. Under the old doctrine, the answer would have come ready-made. Settlement mode assigned loss to the less favorable ship and returned reserve to the designated line. Amber's damage had helped purchase the cleaner hull under Sava's feet. Now fairness could not repair the condenser or refill the tanks.

Oru asked, “What term would permit approval?”

“Guaranteed rescue allocation before symmetric release.”

Glass objected as soon as the token completed its circuit. “A guarantee from whose reserve?”

“From all routes according to capacity.”

North said, “Our capacity reaches zero before your repair window.”

Mara's bruised temple tightened. “Then your condition asks us to wait while our coolant boils away.”

Eren Vey did not soften. “Our condition prevents a complete model from arriving after we are dead and being called consent.”

Two protections met in the center and failed to become compatible. Sava could have ordered Nacre to commit reserve. The ship held more than Amber because Correction 141 had taken from Amber's line. Yet a unilateral

promise would restore the hierarchy she had just relinquished: the wealthy route granting life under terms it could later enforce.

She requested Nacre's next turn.

While the token traveled, Jo pulled Amber's cooling schematic beside the reserve tables. "A continuous transfer wastes phase energy," they said quietly. "We can offer a matched pulse after each of their pump cycles. They would control the request. Every supplying ship could set a floor."

"Would it guarantee the repair?" Sava asked.

"No. Hardware does not recognize guarantees."

"What can you state?"

"That no ship has to descend below its declared life-support floor. Amber receives in measured pulses. Each pulse requires Amber's request and the supplier's confirmation. If one route drops out, the schedule recomputes without treating its absence as fuel."

Teren leaned over the schematic. "The recomputation must remain independent of settlement mode."

Jo gave him a flat look. "That sentence should not need saying."

"It does."

Sava studied the proposed floors. Command training taught that reserves remained real only when a captain could deny their use. Shared rescue inverted the lesson. A promise remained real only when each participant could stop giving and the receiver could still see what stopping meant.

Nacre received the floor.

Sava addressed Amber. "Nacre offers correction reserve to your cooling repair under a recipient-called pulse schedule. The offer carries no claim on your vote. You set each request. We set our survival floor in public. Every transfer remains visible to all seven routes."

Mara's answer came quickly. "That gives Nacre a veto over every pulse."

"Yes."

"Then it guarantees nothing."

"It guarantees that refusal cannot hide. It also prevents me from purchasing your vote with a promise my ship cannot examine."

Amber's frame went dark for local conference.

The parliament moved to rescue language while Amber deliberated. Reed proposed a duty to assist within published survival floors. Glass added that damage inherited from settlement must receive priority without creating ownership of the assisted ship. North required all technical packets to include local-unit conversions and receipt clocks. Indigo demanded that a route declining aid retain standing.

Oru assembled the terms in the shared margin:

1. Each ship published a life-support floor calculated by its own engineering deck and open to challenge.
2. Rescue transfers occurred only after a recipient request and a supplier confirmation.
3. Prior settlement damage received priority in scheduling.
4. Aid created no command claim, vote claim, naming claim, or debt payable through later extinction.
5. A failed or withdrawn participant could not be used as involuntary reserve.

The list took two hours to cross every archive. Sava watched the receipt times rather than the local clock. North found a unit error in Glass's reserve floor. Violet found an old settlement flag hidden in the transfer routine. Jo disabled the routine and made all seven engineering decks verify its checksum. Amber remained in conference.

Jo then built a dry test with no field load. Seven columns appeared on the route glass, each beginning at a different reserve level. The test assigned Amber one request pulse and asked every supplier to publish what it could release without crossing its floor.

Glass offered first. Its answer arrived in local units based on a capacitor bank rebuilt after a hull fire. North could not compare the number until Teren translated it into phase tension. Reed offered half as much but attached a pressure forecast showing how the loss would shorten food-storage cooling. Indigo's contribution looked small until Nessa's teacher pointed out that its classroom lamps already ran four hours below mission minimum.

Sava watched the numbers become inhabited. Under command accounting, the offered total would have occupied one line. Here each fraction carried a pump, a darkened room, or a ration shelf behind it.

Nacre's engineering floor left the largest margin. Mina saw it and leaned toward Sava. "Commons will ask why we keep that much."

"Because our hull entered Correction 141 with less damage."

"That explains possession. It does not explain the floor."

Sava opened the local schedule. Drive reserve, hospital reserve, seed-bank cooling, maneuver margin. The last category still used the founder-era requirement for a single designated arrival. It retained enough energy to recover from a guidance error without asking another route for warning.

Jo tapped the line. "That margin assumes everyone else can be spent."

"Not explicitly."

"Machines do not need explicit motives."

Reducing the margin would place Nacre below the safety standard Sava had enforced throughout her captaincy. Keeping it would ask Amber to trust a standard built from Amber's earlier loss. She requested a failure comparison from Teren.

He projected two cases. With the old margin, Nacre could survive one additional blind correction alone. With the shared floor, no single ship could absorb the same error, but any four could exchange enough timing data to prevent it. The second case depended on routes remaining communicative. Sava had treated dependence as a fault because mission doctrine promised one hull at arrival.

"Set our floor to the shared case," she said.

Mina did not move. "Captain's order?"

"Recommendation from the captain. Commons decides the reserve category. Engineering confirms that the floor protects present life support."

Jo examined the heat tables. "Present life support holds. Solo recovery does not."

"Publish both statements."

The Assembly delegates conferred in the gallery. No seam delay protected Sava from their faces. Two opposed the reduction. One demanded a crew-wide notice before any transfer. Mina argued that a dry test required an honest number, not an authorized expenditure. Commons approved publication of the lower floor for modeling and reserved the actual release vote.

Nacre's available offer nearly doubled.

Amber's conference light flickered off, then on again. Mara had seen the revised column but did not reopen the public channel. Instead Amber's engineering deck sent a private receipt to Jo: MODEL FLOOR ACCEPTED FOR TESTING, NO TRANSFER AUTHORIZED.

Jo entered a simulated request. Each supplier received the same need and answered from its own clock. The offers arrived over twenty-three seconds. A settlement routine would have taken reserve according to Loom preference before the first route could refuse. The revised architecture waited for all declared responses, including zeroes, then displayed the shortfall without filling it from a silent source.

The first total missed Amber's repair need by six percent.

Nobody proposed taking the difference from North. Glass lowered its offer after discovering heat damage in the reconstructed bank. Reed found one more pulse by postponing a grain-freezer cycle. Indigo held its number unchanged. Violet shifted reserve from a memorial display circuit after Lin obtained local council consent. Nacre's Commons added a conditional second pulse, payable only if Amber's check valve opened during the first.

The test reached full repair energy without converting any route's failure into supply.

"That schedule takes nineteen minutes longer," Teren said.

Jo saved the result. "Then the coolant waits nineteen minutes or the people revise the floors. The machine does not choose whom to empty."

Sava heard no triumph in their voice. A test could show that terms operated under one set of numbers. It could not promise that fear would leave them intact under field load. Still, Amber had remained on the engineering channel long enough to watch every route refuse concealment.

During the wait, Oru returned Sava's command key.

She did not take it. "The vote remains open."

"Your relinquishment has entered the record."

"The custody term has not."

Oru turned the strip between two fingers. "Commons custody cannot mean my pocket."

Mina brought a transparent seal envelope from the Assembly desk. Jo signed across one edge. Ilyan signed the other. Sava placed her thumbprint below theirs, then slid the key inside. Mina sealed it and fixed the envelope to the public evidence rail.

The alloy hung in view of every frame.

Amber's image returned.

Mara Sen had removed the name strip from her chest and placed it on the console, as if even a name had become one item too many to carry. "Amber accepts the revised rescue terms for continued engineering work," she said. "Our abstention on the principle vote stands."

Oru asked, "Will Amber publish its survival floor and join the pulse test?"

"Yes."

"Will Amber participate in blank-course calculation?"

"We will provide damage and debris data. Participation does not promise execution."

"Recorded."

North requested a complete revised packet. Its thirty-hour clock reset only for the rescue section and any course dependencies touching it. Teren objected that a full reset consumed too much time. Eren Vey demanded a line-by-line dependency map instead. For once, Teren answered the question beneath the demand. He agreed to mark every altered equation so North could preserve receipt time on unchanged sections.

Oru displayed the principle vote.

Five white marks formed an uneven arc: Glass, Nacre, Reed, Indigo, Violet. North's gold mark carried its delay condition. Amber's frame held a clear abstention and an open engineering channel.

Sava counted them twice, a captain's habit that no change in jurisdiction would remove. The result offered no permission to maneuver. It offered no excuse to treat dissent as absence. Her ship remained hers to command within its hull, and even that command now had witnesses holding the key.

Jo sent the first coolant-pulse test pattern to Amber for local approval. Mara altered the interval by four tenths of a second and sent it back. No energy crossed yet.

On the evidence rail, Sava's black command key turned slowly inside its clear envelope while seven clocks measured the time before anyone could use it.

Chapter 21 — The Course That Stayed Blank

<!-- POV: Ilyan Rook -->

<!-- SETUP: Alda's blank ledger line; Nessa's symmetric game; Amber's open engineering channel -->

<!-- PAYOFF: a 117-second blind course authored by all routes; seven signatures fill the line -->

Ilyan pinned Alda Quill's blank line over the Loom's brightest forecast.

The sheet carried one hundred nineteen years of fold marks. Founder-era maintenance columns ran down both sides in brown ink: capacitor temperature, braid tension, bell delay, operator initials. Between Trial Seven and Trial Eight, Alda had ruled a narrow course field and left it empty. Every other field contained a vector or the word ABORT. This one held only a beginning clock, an ending clock, and space wide enough for a hand to refuse certainty.

Teren stood beneath it with chalk on both sleeves. “The duration matches nothing in the trial schedule.”

“It matches her hidden timing table,” Ilyan said.

“One hundred seventeen seconds appears as a service interruption.”

“Because command did not search maintenance records for navigation.”

The seven route frames curved around the chart room. Lin watched from Violet. Nessa occupied Indigo's classroom beside a board covered with two-sky game marks. Reed, Glass, and North sent engineering teams rather than civic delegates. Amber's frame carried Mara Sen and two coolant technicians, but its debris archive remained closed. Oru sat at the edge of Nacre's image to record receipts. No frame occupied the center.

Ilyan removed the current forecast from public view.

The room changed when Pelagos vanished. Operators looked toward the empty glass as if the star had gone somewhere physical. In its place he opened seven local velocity traces. They began at the same ancestral correction and diverged by small amounts that the old Loom would have treated as errors. Violet carried a port bias from its damaged drum. Indigo ran warm along the outer braid. North had reduced spin. Amber's damaged cooling plant shifted its center of mass each time a condenser loop flooded.

“We need one course,” Teren said, “but not one vector.”

Nessa frowned at him through the seam. “Then stop calling it one.”

He accepted the correction. “We need seven courses sharing one interval.”

“Better.”

Ilyan heard Lin laugh softly after the delay. The sound matched a laugh he might have given two weeks earlier, then ended on a different breath.

Teren drew the release relation across the route glass. Settlement mode normally predicted every route through a correction, ranked them, and strengthened one output by closing the rest. The proposed symmetric field could not rank. Each ship had to surrender its designation claim at the same phase instant, allowing the tension between domains to separate into seven velocity changes.

That much came from Nessa's game. It did not tell them where those changes pointed.

"The Loom will try to forecast during release," Teren said. "If any route supplies a complete preferred trajectory, the stack may rebuild a designated solution around it."

Jo leaned over the field controls. "We can blind the prediction layer."

"For how long?"

"Long enough to cross the phase inversion. Too long, and we enter the debris tide without warning."

Ilyan looked up at Alda's clocks. "117 seconds."

Nobody answered immediately. The number did not become safe because a founder had hidden it.

Lin enlarged Violet's copy of the ledger. "She left a course field, not a shutdown instruction."

"She left both clocks," Ilyan said.

"She may have recorded how long the first route stayed audible."

Teren compared the service column to the decoded wake ledger. "Trial Seven's rejected route answered for one hundred seventeen seconds before command dumped the field. Alda might have marked persistence, not a maneuver."

Ilyan felt the old temptation to turn an artifact into permission. He had spent his career making records carry only what custody could support. Alda's blank line proved she had withheld a course. It did not prove she had designed theirs.

"Then the interval belongs to the evidence," he said. "The course does not."

Oru entered the distinction in the session record.

Nessa tapped her board. "In our game, the dark round lasts until every player sends a return knock. You don't decide the time first."

"The field decides part of it," Teren said. "Phase inversion reaches stability no earlier than ninety-four seconds. Debris exposure becomes unacceptable above one hundred twenty-three."

"So 117 fits."

"It fits. It does not follow."

"Adults use different words for the same almost."

Teren nearly smiled. "Often."

They began with the clocks.

Ilyan sent a seven-pulse challenge through the seams. Each route returned it from its own cesium braid without correction to Nacre time. The replies spread across 1.41 seconds. Jo built a phase ring from the differences. Nessa assigned each ship the next clock in the rotating order, so no route certified its own offset.

Glass checked North. North checked Nacre. Nacre checked Reed. Reed checked Indigo. Indigo checked

Violet. Violet checked Amber. Amber checked Glass.

The first ring failed closure by nine thousandths of a second.

Teren blamed seam compression. Lin blamed Nacre's pulse shape. An Amber technician pointed toward Glass's clock drift but did not open the supporting diagnostic. The argument consumed eleven minutes before Nessa asked whether everyone had counted the same bell.

They had not. Nacre began at field-load tone. Reed began at the relay strike that followed it. Old procedure treated both as one event because local separation measured less than a hundredth of a second. Across seven verifications, the difference prevented closure.

Ilyan changed the instruction: count the sound you receive, not the event you assume sent it.

The second ring closed within six ten-thousandths.

Jo saved the timing braid without naming a master clock.

Next came direction.

No route could publish a complete course. If Nacre calculated seven trajectories, the Loom might treat its result as designation. If each ship calculated only itself, small errors could send two routes toward the same debris corridor. Teren proposed dividing the model into partial constraints that carried no usable trajectory alone.

Reed received mass and thrust response. Glass received seam phase and capacitor discharge. North received life-support limits during blackout. Indigo received timing order. Violet received route-separation geometry. Nacre received Pelagos capture boundaries. Amber had the debris tide.

Mara Sen looked at the assignment on her frame. "You gave us the data only we possess."

"We did not give it to you," Ilyan said. "Correction 141 did."

Amber's damage had forced its sensors to remain on the charged-dust fold six minutes longer than Nacre's. Its cooling failure put the ship on a canted route with an unobstructed view along the tide's underside. The same injury that made Amber least favorable under settlement doctrine had produced the only complete parallax sweep.

Mara's face hardened. "You need the record of what your correction did to us."

"Yes."

"And if we withhold it?"

"No complete course."

Teren moved toward his microphone. Ilyan raised a hand. This answer belonged to the record, not persuasion.

Mara conferred with someone outside Amber's frame. When she returned, the debris archive remained closed. "Amber will provide obstacle exclusions, not raw sensor history."

"Can exclusions complete your portion?" Teren asked.

“They can tell you where no ship may go.”

“They cannot reveal a corridor.”

“Then perhaps you should not have spent ours.”

The session held on that sentence.

Ilyan opened Nacre's archive for Correction 141 and sent Amber the full settlement accounting: reserve received, hull loss avoided, cooling damage displaced into Amber's line. Teren had already published the totals. This file included command timings, operator notes, and every automatic preference score. Ilyan included his own certification from the night of the correction, unsigned at the end but complete through the route closures he had once expected to approve.

Mara watched the transfer bar fill. “What is this meant to purchase?”

“Nothing. Your archive belongs in the account of ours.”

“You already know the numbers.”

“We did not preserve where they came from.”

Amber's receipt clock marked the file. Mara did not open its debris archive.

Work continued around the missing vector.

Reed supplied a matrix describing how each ship's loaded mass answered a common phase shove. Its food shortages had forced meticulous cargo inventories, which made its figures cleaner than Nacre's. Glass mapped capacitor discharge from patched banks and found that symmetry did not require equal energy, only equal relinquishment of claim. North converted its delay condition into a technical limit: no guidance change during blackout could assume a packet not yet acknowledged. Indigo built the timing order from children's return knocks and added a rule requiring every silent clock to appear as silence, not zero.

Violet handled separation geometry. Lin placed seven course rays on his route glass, each without a final bearing. He adjusted minimum distances until no pair could enter the same wake during the blind interval.

“You kept Nacre farthest from Violet,” Ilyan said.

“I kept the two heaviest phase histories apart.”

“That sounds like physics spoken defensively.”

“It can be both.”

Ilyan examined the distances. Lin had allowed Violet to pass closer to Indigo and Glass, routes whose field damage made their responses less predictable. Moving Nacre outward reduced collision risk but increased its debris exposure.

“Your geometry spends our hull,” he said.

“Your capture boundary spends our velocity.”

Neither accusation invalidated the figures. Ilyan sent Violet's matrix to Reed for blind verification. Reed

removed route names, reran the separations, and returned the same spacing within tolerance.

Teren assembled the partial constraints in a sandbox disconnected from the Loom. The pieces refused to produce a course. They formed a hollow shape: seven departure states, seven survival floors, a common blackout interval, and an empty region where debris data should have cut navigable channels.

On Nacre's frame, the shape resembled Alda's blank line widened into glass.

They tested false Amber vectors to see what information the model leaked. One caused a route collision at eighty-six seconds. Another preserved separation but sent North beyond Pelagos capture. A third fit every local constraint and crossed a debris filament visible only in Amber's exclusion list. Each failure appeared without revealing which valid bearing might remain.

"Good," Mara said after the third test.

Teren stared at her frame. "Good?"

"You cannot reconstruct our raw map from the constraints."

"No."

"You cannot calculate Amber's course without Amber."

"No route can calculate its own course without the others."

Mara looked past him toward Sava. The captain stood at the lower rail with no command key on her wrist. She had not spoken during the archive exchange.

Amber opened its sensor history.

The packet arrived in strips because its raw sweep exceeded the seam's safe bandwidth. Ilyan watched charged particles resolve into a moving wall. The tide did not behave like a fixed field. Pelagos radiation charged its finest grains in waves, causing filaments to braid and loosen over hours. Nacre's forward view had flattened that motion. Amber's canted parallax revealed a narrow interval when seven separated bodies could pass through different gaps.

Mara marked the lower filament. "Our coolant plume perturbed this edge. Remove the plume, and your forecast lies."

Jo added Amber's vent cycle to the model. "Can you hold the same release pattern during field load?"

"For ninety seconds. After that the check valve may chatter."

"The course needs 117 seconds."

"Then the model must include chatter."

Teren asked for its frequency range. Amber supplied it. Glass adjusted capacitor phase so valve vibration could not couple into the seam ring. Reed shifted one mass response to preserve distance. Violet narrowed the heaviest pair's crossing by half a degree. Nacre moved its Pelagos capture boundary outward, accepting a longer transfer after the tide.

No person held the whole calculation.

Ilyan could see Nacre's capture limits and the receipt timings. Lin could see separations and the constraints returned from Reed, but not Amber's raw debris map. Amber could see its parallax sweep and the partial phase solution, but not Glass's bank details. Teren's sandbox joined encrypted constraint proofs rather than full route states. It could answer whether all seven courses satisfied all seven limits. It could not display any course until each route unlocked its own result.

Oru asked what happened if one route lied.

The question annoyed Teren. Ilyan heard it in the force with which he set down his chalk. "A false constraint can make the proof fail."

"Can it make a false course pass?"

"Not if the proof system works as designed."

"Your qualifier carries most of the room."

Teren rubbed chalk from his thumb. "Each route verifies the constraints it receives. Nobody verifies hidden source data."

Mara said, "So Amber could move a debris filament and steer another ship into it."

"Or Nacre could falsify Pelagos capture," Lin added.

"Violet could distort separation geometry," Ilyan said.

North's Eren Vey spoke over a weak channel. "Then seven-party authorship creates seven sabotage points."

Sava remained at the lower rail. "It also removes one sabotage point with six victims."

The frame ring carried her sentence around its delays. She did not ask them to trust her because she had relinquished command. Ilyan saw what her restraint cost: a captain trained to close uncertainty now had to let suspicion remain distributed.

Oru requested an adversarial rehearsal. Each route would alter one private input inside a bounded copy. The others had to detect inconsistency without learning the source value.

Glass changed capacitor capacity. The proof rejected its own phase result before the false figure left local storage. Reed altered loaded mass and triggered a mismatch against its published thrust response. Indigo shifted one timing pulse; North's receipt chain caught the change. Violet narrowed a separation distance beyond the shared hull envelope, and Nacre's capture test exposed the impossible crossing.

Nacre's turn came fifth.

Ilyan held the local copy of Pelagos boundaries. The sandbox asked him to alter one figure. He moved Lattice's capture edge inward by the smallest margin that could improve Nacre's arrival while preserving a plausible sensor error. His hand paused over confirmation.

From Violet, Lin followed Ilyan's hand. "You chose a useful lie."

"A detectable one, I hope."

"Those categories have overlapped before."

Ilyan confirmed the change.

The proof traveled. Reed accepted it. Indigo accepted it. Violet's separation check passed. Amber rejected the boundary because its star-fix history placed the same edge farther out. The difference measured less than Nacre's old tolerance.

Mara highlighted the margin. "Your instrument standard would have let this through."

"Our standard assumes one ship carries the calibrated reference."

"Change it."

Ilyan replaced local authority with a cross-route interval. Any capture boundary had to include independent observations from three ships, and no two could derive from matching sensor damage. The rule widened uncertainty. It also removed the advantage he had given Nacre.

North altered a life-support floor next. Indigo caught a reserve dependency that the proof layer missed. Amber shifted a debris filament; Glass's phase map found that the new corridor demanded impossible discharge. On the seventh test, Amber reduced its own coolant plume and made its route look easier to carry. Reed's mass response detected the missing vent load.

Every lie left a shape in somebody else's constraint.

"Not proof against sabotage," Oru said after the last rejection.

"No," Teren replied. "Proof that sabotage requires breaking a published relation or persuading six routes to accept a new one."

Nessa leaned toward her microphone. "In the game, a player can still knock wrong."

"What happens then?" Oru asked.

"The next sky doesn't answer. Everybody sees whose turn stayed open."

Jo added an open-turn rule to the hardware covenant. A missing or inconsistent contribution could halt release before field load. Once blackout began, no route could revise the shared interval. That protection increased the danger of delay, especially for North, but removed a covert path for changing the course after others committed.

Ilyan asked each route to generate a local stop signal. The signals used different hardware: Glass closed a patched relay, Reed dropped a weighted knife switch, Indigo pressed two classroom keys at once, North broke an optical line, Violet opened a manual jaw, and Amber vented a measured coolant flash across its seam pickup. Nacre's stop ran through Jo's black service rail rather than the captain's key.

Seven tests halted the sandbox seven different ways.

Only after those refusals worked did Oru authorize a complete proof run.

They ran the first complete proof.

Failure at second 104. Indigo's warm braid drifted beyond Glass's phase tolerance.

Nessa's class adjusted the return-knock spacing. The children did not alter the physics; they saw that two

intervals everyone had treated as equal occupied different places in the pattern. Teren converted their correction into a nonlinear phase wait.

The second proof failed at second 116.

North's reduced spin moved a sensor mast into its own warning cone. North proposed retracting the mast. Jo rejected the timing; the motor could not lock before blackout. Reed instead redistributed a ballast transfer already scheduled for food cooling. North's center shifted by four centimeters. The warning cleared.

A third proof completed all 117 seconds.

The sandbox produced seven sealed packets. Each carried one ship's burn timing, attitude limits, and independent exit bearing. None opened until the other six checksum receipts arrived. The route glass displayed no chosen line. It showed seven locked fields around a blank center.

Teren sat down on the floor.

"Does it work?" Nessa asked.

"It satisfies the model."

"That isn't what I asked."

"It is what I can answer."

Jo ran a hardware rehearsal with capacitors isolated from the field. Bells sounded across the seven seams in rotating order. A dark interval followed, simulated rather than enacted. At its end, each route released a dummy exit packet. All seven clocks returned. Amber's virtual check valve chattered through the final twenty-seven seconds without breaking phase.

Ilyan compared the rehearsal against Alda's sheet. Her blank course field still had no vector because their result could not fit into one. He placed a transparent scan over the paper and divided the field into seven receipt cells.

Oru objected. "That alters the record."

"The overlay remains separate. Alda keeps her blank."

"What does the new line certify?"

"Authorship, not prediction."

One by one, the route delegations received the overlay. Glass signed its constraint proof. North signed subject to its execution delay, now preserved by dependency receipts. Nacre's Commons signed beside Sava's limited captaincy mark. Reed signed with the food-bay grease pencil reproduced as a digital stroke. Indigo's delegate signed for the ship; Nessa added a small two-sky diagram as technical witness. Violet signed LIN without route numbering.

Amber came last.

Mara Sen held the overlay beside Correction 141's accounting file. "Our signature confirms that our data formed the course. It does not authorize field load."

Oru read the clause back to every route. "Confirmed."

Mara signed.

The seven sealed course packets remained closed. No guidance command entered the Loom. North's clock continued toward its required receipt interval. Amber's coolant pulse test still awaited physical execution. The parliament had built a way through, not yet chosen to take it.

Ilyan lowered Alda's paper from the glass. The old field stayed empty beneath his fingertips. On the transparent record above it, seven signatures occupied the blank line.

Chapter 22 — The Unchoice

<!-- POV: Lin -->

<!-- SETUP: seven signed course packets; mutual suspicion; Jo's distributed hardware custody -->

<!-- PAYOFF: all seven relinquish designation privilege; predictive guidance goes dark for 117 seconds -->

Lin cut Violet's designation wire with a kitchen knife.

The wire ran under the chart console in a yellow sheath stamped PRIMARY ROUTE SENSE. Violet carried no proper isolation tool for it. That set had fused inside Drive Junction Three on day six, and every replacement aboard the ship belonged to a circuit currently keeping someone warm. Jo Malk watched through the seam while Lin sawed at the sheath with the serrated blade used for ration loaves.

“Not the white pair,” Jo said.

“I know.”

“You moved the camera.”

“Because my shoulder blocks the terminal if I don't.”

“Your shoulder also blocks the white pair.”

Lin shifted until Jo could see his hands. Across the low-frame image, Nacre's engineer occupied a narrow service bay under their own Loom. Copper braid surrounded them like roots stripped of soil. Six smaller windows showed the other routes at matching access points. Reed had removed half a wall. Glass used a mirror to expose its designation contact. Indigo's maintenance crew worked from a ladder while Nessa held the camera and supplied unwelcome commentary about everyone's grip.

The knife reached copper. Lin stopped.

Violet's local console still called the wire primary. It had called his ship rejected, then persistent, then dependent. Changing the labels in public records had not altered the current flowing through them. When the shared field took load, this one conductor could still tell settlement mode to present Violet as a candidate for extraction.

“Cut,” Jo said.

Lin severed it.

The yellow indicator on Violet's console went out. A red fault opened beside it: DESIGNATION SOURCE ABSENT. No alarm bell followed. Jo had disabled the automatic repair call because seven ships producing the same fault at once would flood every engineering deck.

Nessa leaned into Indigo's camera. “Ours says we broke navigation.”

“For once the error message gives a useful summary,” Jo replied.

“Did we break it correctly?”

“Check continuity to the local field jaw.”

Indigo's technician read the meter. Glass followed. Reed, North, Amber, and Nacre reported in rotating order. Violet came last on this circuit. Lin touched his meter leads to the cut ends. Open across designation. Closed across local control.

"Violet confirms," he said.

Seven red faults remained lit.

The parliament convened with its machinery exposed.

Oru Pell appeared at the edge of Nacre's service-bay frame, knees pressed against a braid cabinet. No formal chamber could show what the vote governed as clearly as this cramped place. Sava stood behind Jo without her command key. Ilyan occupied a side console, one hand resting near the seven sealed course packets. Teren held the field equations on a portable pane.

Oru read the operational proposition.

Each route would relinquish designation privilege at the same agreed phase instant. Settlement mode would receive no preferred ship and no permitted source for involuntary reserve. Shared tension would separate into seven independent corrections defined by the collectively authored packets. Every route accepted a distinct Pelagos arrival body. Predictive guidance would remain disabled for 117 seconds. Any missing authorization before field load would stop the maneuver for all. Any withdrawal after release would affect only the withdrawing ship's local hardware.

The proposition needed a name for the record.

Teren had called it symmetric phase release. Oru's draft used relinquishment maneuver. Nessa's class called it the round where nobody started first.

Lin looked at the red fault on Violet's console. All route doctrine had framed survival as a choice: this line or those lines, one arrival or many closures, mission success or waste. Even rescue proposals had offered another choice between merger and abandonment. The seven exposed wires did not choose a winner. They removed the mechanism that demanded one.

"Unchoice," he said.

Ilyan's image sharpened as Nacre granted the word more bandwidth. "Say what you mean."

"We have spent one hundred forty-one corrections choosing which ship receives reality. This does not choose all of us. No machine can promise that. It refuses the privilege of choosing one at the expense of the others." Lin looked toward Oru. "Call it the unchoice."

North received the floor before the clerk answered. Eren Vey said, "A refusal can still kill us."

"Yes."

"Then the name must not imply innocence."

"It implies no such thing."

Amber's Mara Sen spoke when the token reached her. "It also cannot imply that abstention has become impossible."

Oru wrote beneath the proposition: UNCHOICE, a simultaneous relinquishment of designation, not a claim of equal outcome.

The name passed seven clocks and returned without objection.

Jo began the hardware covenant.

They held up a ceramic bridge no longer than a finger. Seven versions had been fabricated from local stock. Each bridge replaced the cut designation wire with a circuit that could carry only three states: present, relinquished, stop. It could not name another route. It could not request settlement reserve. Every ship installed its own bridge and published a live continuity trace to the seams.

“Why three states?” Lin asked. “Present and stop should cover it.”

“Relinquished has to remain distinct from disconnected,” Jo said. “A broken wire cannot count as consent.”

Violet's bridge had been cut from an oven regulator. One edge still carried a flour burn. Lin seated it between the yellow leads and tightened the terminal with the knife tip. The console changed its fault: DESIGNATION SOURCE RELINQUISHED. That wording had not existed in its table yesterday.

Jo waited while every route installed its version.

North's bridge reported excess resistance. Its maintenance team scraped frost corrosion from a contact and tried again. Amber's first ceramic piece cracked under the clamp. Glass sent a revised pressure limit; Amber cut a second bridge and seated it by hand. Nacre finished early and left its trace visible without urging anyone else to hurry.

Lin watched Nacre's green continuity line. He did not trust it.

The fear had occupied him since the blank course proof. Nacre could wait until Violet and the other routes relinquished, then withdraw before its own phase release. Settlement mode would find one designation source remaining. The old privilege might rebuild itself around Ilyan's ship and take the tension everyone else had surrendered.

Violet's emergency council had named that possibility more plainly. Nacre could kill them again with cleaner records.

Lin requested the floor.

“If one bridge reports relinquished earlier than another, can a still-designated ship receive the phase surge?”

Jo answered from under the braid cabinet. “For less than a millisecond, yes.”

The council behind Lin began speaking at once. He muted Violet's room channel before the overlap crossed the seam.

“How much can Nacre take in that interval?” he asked.

“Nacre, no more than any other route still present.”

“That avoids the name.”

“It avoids pretending I know which route will lag.”

“Nacre controls the field-load bell.”

“Not after this covenant.”

Lin looked toward Ilyan. “Do you believe that?”

Ilyan kept his hand beside the sealed packets. “I believe Jo built it that way.”

“Not what I asked.”

The seam delay gave Ilyan time to evade. His face showed the answer arrive and fail inspection.

“No,” he said. “I believe Violet could use the same interval to reverse the surge.”

The accusation crossed every frame.

Lin's council fell silent behind the mute. A reverse surge had been proposed during the chart-room seizure. Violet engineers, furious after learning how settlement reserve moved, had calculated a way to push the stored tension back toward Nacre. Jo had blocked the attempt. Lin had not ordered it, but he had understood the desire well enough to delay condemning it.

“You think we will take revenge,” he said.

“I think someone aboard Violet may call it restitution. I know someone aboard Nacre may call withdrawal safety. A covenant that depends on our better names has already failed.”

Lin hated the fairness of the answer. Ilyan had not asked Violet for more trust than he offered.

Oru let the rotating token complete its circuit before speaking. “Can the millisecond interval be removed?”

Teren shook his head. “Physical switches do not close simultaneously across separate clocks.”

Nessa held up her game board. “Then don't close one switch.”

Jo turned toward her frame. “Explain.”

“In the dark round, every player sends half a knock. You hear a full knock only after the next sky answers.”

“A split trigger,” Teren said.

Jo was already drawing. Each route bridge could issue only half of its release condition. The next route in rotating order would issue the other half after verifying relinquishment. No ship could release itself. No ship could release the ship that verified it, because the pairs traveled opposite directions through the phase ring.

Lin traced the order. Glass carried half of North. North carried half of Nacre. Nacre carried half of Reed. Reed carried half of Indigo. Indigo carried half of Violet. Violet carried half of Amber. Amber carried half of Glass.

“Violet still verifies Amber,” Mara said.

“And Indigo verifies Violet,” Nessa replied.

“Children should not be allowed to sound pleased about custody architecture,” Jo said.

“Adults should have built a better one.”

The exchange drew one tired laugh from Reed. Then work resumed.

Jo rewired the bridge logic through verbal instructions and line diagrams. No mass, hand, or tool crossed a seam. Each route shaped its own ceramic and copper. Nacre could not touch Violet's circuit; Violet could not reach Nacre's. Trust took the form of seven people exposing local work to hostile inspection.

Lin removed his bridge and cut a second notch. His hands shook when the knife met ceramic. He had spent nineteen days learning to perform certainty for frightened people. Children watched his face during ration announcements. Council members watched whether he sat before arguments ended. The person who had once been Ilyan Rook aboard every route now needed the other Ilyan to inspect a circuit designed to prevent mutual murder.

He reseated the bridge.

"Violet ready for cross-check."

Indigo received Violet's continuity trace. Nessa's maintenance teacher read the values, then made Lin repeat the meter placement because his first image hid one contact. The second reading matched. Indigo issued no approval yet.

Violet checked Amber. Mara showed both sides of its replacement bridge. The second ceramic piece carried a hairline mark near the clamp.

"That line moved," Lin said.

"It came from cutting."

"Release pressure and show it again."

Mara stared through the camera. Amber's coolant plant lost useful minutes while Lin questioned a scratch. She released the clamp. The line widened.

"Crack," Jo said.

Amber fabricated a third bridge.

The cross-check traveled around the ring. Glass found heat drift in North's contact. North caught a polarity error on Nacre's stop line. Sava ordered no correction; Jo remade the connection under Commons observation. Nacre found Reed's half-trigger tied to a shared pump circuit. Reed isolated it with a manual relay. Violet accepted Amber's third bridge. Indigo accepted Violet's after a second continuity test.

All seven traces held.

Jo refused to call the covenant ready.

"Continuity proves the wires connect," they said. "It does not prove they fail safely."

They loaded a dry field into local shunts, too weak to move any ship and strong enough to make the bridge logic choose. Each route received a different fault to introduce. The assignments remained private until the test began.

North went first by withholding its half-trigger from Nacre. Reed's bridge saw no complete release and held. Nacre's local display reported WAITING ON NORTH rather than filling the absence with an internal

command.

“Old settlement logic would have bypassed after two seconds,” Teren said.

Jo pointed at the timer. “Let it reach three.”

At two seconds, a hidden recovery branch tried to open. North's polarity block stopped it, but the attempt appeared on every continuity trace. Jo swore and pulled Nacre's bridge from the dry circuit.

Sava watched the unauthorized branch remain red on her local pane. “What invoked it?”

“Founder-era loss protection.”

“Protection for whom?”

“The route still marked present.”

Lin heard his council react behind him. Nacre's cut wire had not removed every route preference. A century of safeguards lay beneath the named system, ready to rebuild hierarchy from failure.

Jo traced the branch to a relay under Nacre's prediction stack. Matching relays existed aboard every route. The relevant code carried no route name. It compared signal strength and treated the strongest remaining source as authority.

“Another designation mechanism,” Ilyan said.

“A fallback,” Teren replied, then winced at his own reflex.

Lin leaned toward the seam. “A fallback into what?”

Teren read the current path. “Into settlement ranking.”

Jo ordered every route to isolate the relay physically. On Violet it sat behind a plate welded shut after a fire. Lin fetched a hand drill. Two council members held the camera and lamp while he cut through the welds. Metal filings clung to his sleeves. The relay beneath carried a simple brass arrow pointing toward PRIMARY.

He broke the arrow off before disconnecting the coil.

Seven routes reported isolation. Jo repeated North's withheld-trigger test. The dry field waited without assigning authority. At five seconds, it discharged into each ship's local shunt.

Reed introduced a false stop next. Every bridge froze before half-signals moved. Indigo reversed its test polarity; Violet's verifier rejected it. Glass dropped continuity during a simulated release, and Amber's half remained unissued. Nacre attempted to assert a local command through the captain's bus. The signal reached Jo's service rail and died there because Sava's key remained sealed in Commons custody.

Violet's assigned fault came sixth.

Lin opened a test control labeled REVERSE PHASE. The option had survived Jo's block during the mutiny because Violet's engineers claimed they needed it for capacitor protection. In a live field, the command could drive tension against the rotating order and toward Nacre.

Ilyan saw the control through the camera. His face did not change.

“Ready?” Lin asked.

“Indigo verifies you,” Jo said. “Proceed.”

Lin pressed the control.

Violet's bridge accepted the local request. Indigo withheld its half. The reverse command went nowhere. Then a second path lit beneath it, a maintenance bypass added during Violet's first days of survival. Lin recognized the initials beside the patch. One of his council engineers had built it after learning what settlement had done.

The bypass sought raw field access.

Jo shouted for local isolation. Lin reached under the console and pulled the copper strap with both hands. It held. The kitchen knife lay beside the designation wire. He jammed its blade beneath the strap and twisted until the fastener split.

The dry field collapsed into Violet's shunt.

No surge crossed the seam. Indigo's half-trigger remained dark.

Behind Lin, Councilor Aven said, “That path protected us.”

“It gave us a way to strike another ship after consent,” Lin replied.

“It gave us leverage.”

“Against children on Nacre who never touched a route control.”

Aven stepped away from the console. He did not withdraw the objection.

Lin held the broken strap where every route could see it. “Violet reports an undisclosed bypass. It has been physically severed.”

Ilyan released a breath. “Nacre records the severance.”

Amber ran the final fault by requesting rescue reserve during release. Its bridge kept the request on the separate pulse circuit and refused to mix aid with phase tension. Mara confirmed that Amber could not turn a promised transfer into claim over another route.

The dry test ended with seven new defects entered in public view. Jo updated the circuit definition but did not erase the failed version. Oru attached both to the covenant, along with the names of the people who found each hidden path.

Only then did Jo mark the hardware ready.

Oru asked each delegation to state what its bridge could not do.

Glass: “It cannot accept settlement reserve.”

North: “It cannot release before verified receipt.”

Nacre, through Mina: “It cannot designate another route or restore sole command.”

Reed: “It cannot treat a failed clock as fuel.”

Indigo: "It cannot start first."

Violet's turn arrived. Lin looked at Ilyan's frame. "It cannot reverse the surge into another ship."

Amber said last, "It cannot turn rescue into a debt claim."

Jo sealed the circuit definitions. Oru attached them to the operational proposition.

North's thirty-hour condition had expired nineteen minutes earlier. Eren Vey confirmed receipt of the complete course, failure model, rescue terms, and every changed dependency. Amber's coolant repair had succeeded under the recipient-called pulse schedule. Mara reported stable temperature without converting the report into gratitude.

Nothing procedural remained incomplete.

That frightened Lin more than delay.

The sealed course packet waited on Violet's console. Opening it would reveal only Violet's bearing. The other ships' exits remained inaccessible, and no course could bend toward another after blackout began. Violet's council had authorized him to cast the operational key if every hardware condition passed. Four members had voted no. Two had demanded Nacre transfer reserve before commitment. One had asked Lin whether sharing a past with Ilyan made betrayal easier to hear.

He did not know.

Oru called the seven authorizations in rotating order.

Glass consented to field load and independent exit.

North consented under its completed receipt condition.

Nacre's authorization required three voices. Mina gave Commons consent. Jo accepted engineering custody. Sava spoke last.

"As captain of this ship only, I authorize Nacre's local risk under the adopted course."

No mission command followed the sentence.

Reed consented. Indigo's delegate consented while Nessa and her class held the two timing keys. Violet's token lit.

Lin unmuted his council. The members stood behind him in the damaged chart room. Beyond the open hatch, people had gathered along the service rail. They could not see Nacre except through Violet's glass. They could see him.

"Violet relinquishes designation," he said. "We authorize our course. We do not authorize our erasure, our merger, or revenge performed in our name."

He inserted the operational key.

Amber consented last.

Seven ships displayed RELINQUISHED.

Jo armed the split triggers. Half-signals moved around the ring and stopped at their verifying routes. Every frame showed a local hold light. If Nacre withdrew now, Reed could not release. If Violet reversed polarity, Indigo's half would fail. Nobody possessed a complete trigger for itself.

Ilyan opened Nacre's course packet. Lin opened Violet's. A bearing appeared on his private console, angled below the shared corridor and away from Indigo's warm braid. He saw no destination name, only burn timing and a capture envelope. The course offered survival without reunion.

"Clock check," Jo said.

Seven replies returned in sequence.

"Field jaws."

Seven local crews confirmed.

"Prediction isolation."

Lin placed his hand over Violet's guidance breaker. Across the frames, six other hands reached six different controls. Nessa and her teacher shared Indigo's pair of keys. Sava did not touch Nacre's breaker; Ilyan held one side while Mina held the other.

The maneuver bell struck once on Violet.

Through the seams came six related bells, separated by distance and clock. No tone claimed to begin the sound. Jo counted the phase ring closed.

"Release on receipt," they said.

Glass sent half of North. North answered Nacre. The condition moved through Reed, Indigo, Violet, and Amber. Lin received Indigo's half-trigger and felt Violet's bridge warm under the console.

Nacre's frame remained present.

Ilyan looked directly through the low-rate image. "Still here."

Lin sent Amber's half.

"Still here," he answered.

The ring closed.

Seven guidance breakers opened.

Violet's forecast line vanished. Pelagos disappeared. Debris warnings froze and then went black. A plain mechanical counter took their place, disconnected from prediction and already falling from 117 seconds.

For the first time since Violet had begun, no chosen route shone ahead of Lin.

Chapter 23 — Every Ship Falls

<!-- POV: Jo Malk -->

<!-- SETUP: seven-ship unchoice; 117-second blind course; Jo's hardware covenant -->

<!-- PAYOFF: every clock returns; seven distinct arrival vectors -->

The thrust collar kicked sideways before Jo could lock the third brace.

Their shoulder struck the hot casing. The torque driver left their hand, crossed the service trench, and caught beneath a coolant pipe. Around the drive nave, amber lamps flattened every face into planes of sweat and shadow. Nobody looked toward the dead prediction panels. They had agreed to fly without them for 117 seconds. Looking would not shorten the dark.

“Collar Three has lateral load,” Jo called.

A mechanic at the lower rail read back, “Three, lateral.”

“Not a warning. A load. Give me the jaw positions.”

Numbers came from three stations, quick and uneven. Two jaws held. One had closed farther than its command by the width of a fingernail. Under ordinary guidance, the Loom would have modeled that error across a thousand continuations, chosen the least damaging correction, then taken what it needed from every other ship. Now the predictive glass showed only a black field crossed by seven timing marks.

Jo reached into the trench. Their fingertips found the torque driver as the deck shifted again. This motion did not belong to commanded thrust. Nacre had begun to roll.

“Bridge, drive,” they said into the command loop. “We have rotational onset, starboard, increasing.”

Captain Or answered without delay. “Magnitude?”

“Still measuring.”

“Then tell me what you can do.”

Jo set the driver into the brace socket. “Keep the ship from teaching itself a new shape.”

The first blind-course bell had sounded forty-one seconds earlier. Each route had released designation privilege at the same synchronized count. Field tension no longer drained toward Nacre. It spread through seven independent hulls, seven drive assemblies, seven sets of frightened hands. The maneuver had opened with no chosen receiver. Every spare differential had to become motion somewhere.

Jo leaned their full weight onto the driver. The brace advanced half a turn and stopped. Heat came through their gloves.

A pulse struck the timing rail above the nave. Indigo. Another followed from North, then Reed. Their intervals arrived through the seams as narrow light and sound, nothing with mass, nothing that could turn a wrench. Each pulse said only that a clock continued somewhere else.

Glass missed its slot.

Amber came late by nine hundredths of a second.

Violet held exactly.

Nacre's own answering pulse went out while the deck rolled under Jo's boots.

"Fifty-three seconds," the lower mechanic said.

Nobody had asked for the count. Jo let it stand. In a room full of failing alignments, an unsolicited number could still serve as a handhold.

The collar kicked a second time. This blow carried enough force to peel frost from a feed line. White flakes scattered through the trench and melted against the casing. Jo saw the failure then: Jaw C had not merely overclosed. Its retaining pin had bent under asymmetric phase release. If the pin snapped, the jaw would cross the collar face and cut the drive's remaining vector into a tumbling moment.

"Kill C," Jo said.

The control mechanic looked up. "Hard isolate?"

"Physical. The command bus can't pull a bent pin."

"Collar balance will go outside covenant."

"It already left."

The mechanic slapped the isolation cover aside. Two crew hauled the red lever down. Jaw C's status vanished from the local board. The other jaws took its load, and the casing gave a deep metal complaint that Jo felt through their ribs.

Captain Or's voice returned. "Drive, chart reports one-point-eight degrees per second and rising."

"We removed one jaw. Give us ten seconds."

"You have six before the forward farms exceed their lateral limit."

Sava sounded more formal with each added danger. Jo had once mistaken that for calm. Now they heard the fear inside the complete sentences and exact units.

"Then warn the farms," Jo said. "I won't invent four seconds."

They crawled over the casing toward the manual trim wheel. Two mechanics joined them without instruction. The wheel required three bodies when pressure opposed it. Jo took the upper spoke. Leth Naro, whose left hand had three shortened fingers from a seal accident, braced below. On the far side, Pavi Sen wedged one boot against the rail.

"Counterroll trim," Jo said. "Half mark. No more."

They pulled together.

The wheel moved with the reluctant smoothness of a bolt taking damaged threads. Somewhere above them, habitat restraints would be catching carts, sleepers, vats, classroom furniture. Forty-two thousand and six people had entrusted their balance to a ring of metal that now needed three engineers and an uncertain half

mark.

At sixty-four seconds, the roll stopped increasing.

At sixty-seven, it began to slow.

The drive crew did not cheer. Pavi held the wheel. Leth watched the collar temperature. Jo listened to the seven pulse slots repeat on the rail.

A farm controller broke into the drive loop. "Forward nutrient tanks are walking in their cradles."

"Walking how?" Jo asked.

"Two centimeters downhill on every residual turn. Restraint Twelve has pulled one floor bolt."

The tanks held most of Drum One's next harvest in suspension. A split tank would not kill anyone during the maneuver. It would simply move the cost into the following months, when scattered ships had no settled supply chain and every seedling carried a schedule.

"Do you have people in the gallery?"

"Four. We can brace Twelve."

"Don't put bodies downhill of that tank. Shift fluid to the aft pair."

"Transfer pumps tripped on field interference."

Jo pictured the farm gallery from maintenance plans: six long tanks mounted across a curve, service pipes under grated paths, manual bypass wheels at each end. Engineers had designed the tanks for axial thrust and drum rotation. Nobody had asked them to tolerate a blind sideways fall created by seven ships declining to kill one another.

"Open the gravity drains between Twelve and Fourteen," Jo said. "Use the inspection crossline."

"That line carries rinse water."

"Today it carries lunch."

"The crossline has no food certification."

"Neither does the deck."

Pavi made another breath of laughter. On the loop, the controller stopped arguing and began giving orders to their own crew. Jo heard boots on grating, a wheel refusing to turn, then a younger voice calling for a longer bar.

The ship rolled through another tenth of a degree. Jo watched the load trace and translated it into liquid moving through an improvised pipe. "Open Fourteen first," they said. "Then Twelve. If you open the high tank against a closed receiver, you will burst the rinse seal."

"Fourteen opening." Metal knocks crossed the controller's microphone. "It has paint in the thread."

"Every manual fitting has paint in the thread. Break it."

A grunt, two curses, then the hollow rush of fluid answered. Tank Twelve's lateral load eased on Jo's board. The pulled floor bolt stayed pulled, but its strain stopped climbing.

"Flow established," the controller said. "We are losing nutrient concentration into the rinse pocket."

"How much?"

"Unknown until we spin up a sampler."

"Then you lost an unknown amount and saved the rest. Mark the pocket. Nothing goes to waste processing."

The controller repeated the instruction. A few days earlier, waste processing would have meant eventual feed into the Loom's correction reserve calculations, another clean label at the end of a dirty chain. Now it meant only a tank crew with a note and several hours of unpleasant filtering. Jo preferred a repair whose consequences remained where hands could reach them. Somewhere in Drum One, four growers were holding a pipe open by hand while the stars changed their address together.

"Drive," Sava said, "roll below half a degree per second."

"Farm Twelve holding," Jo replied. "Keep habitat restraints until we cross zero."

"Agreed."

Jo heard no surprise in the captain's acceptance of advice from a controller three decks away. Command had not vanished during the blind course. It had become crowded with people who possessed facts Sava could not own.

Glass had returned.

Amber tightened its delay.

Indigo sent two extra taps between assigned marks.

Jo frowned. "Chart, identify Indigo variance."

Oru Pell answered from the shared operations loop rather than Navigation. Authority had gained unfamiliar voices during the last three days. "Indigo reports timing control damage. Their school auxiliary has taken the phase count."

"School auxiliary?"

A child's voice crossed the seam in fragments, compressed until each consonant clicked. "We have both skies. Keep your turn."

Nessa Quill did not sound like an oracle. She sounded annoyed that adults had made a simple game difficult.

Jo pulled themselves upright. Their bruised shoulder objected. "Indigo, drive on Nacre. State your board."

"Main sequencer lost two slots," Nessa said. A faint adult voice prompted something behind her, too compressed to distinguish. "We can see Violet's lamp and yours. We put the missing turn between."

The two-sky game had begun as children copying lights through delayed seam frames. Neither side went first; each watched two clocks and placed its move where both allowed it. Teren had turned that pattern into field

mathematics. During the blind course, Indigo's main phase sequencer had failed, and the children had taken the count back from the machine built from their game.

"Keep doing exactly that," Jo said.

"We are."

The answer made Pavi laugh once, a short breath against the wheel.

Seventy-nine seconds.

A new vibration entered the collar. It did not kick or scrape. The whole ring began to hum beneath hearing, a pressure Jo knew from overloaded capacitors. They put a glove against the casing. The pulse traveled up their radius bone and settled in the elbow.

"Field reserve moving," Leth said.

"It shouldn't have a receiver."

"It doesn't. Look."

Seven columns climbed on the local board. Nacre's remained nearly level. Reed and Glass gained slight vector potential. North spent reserve in coarse bursts. Indigo oscillated around zero as the improvised phase count landed. Violet's column fell in a clean, frightening line.

Jo opened the seam circuit to Violet engineering. "Lin's ship, you are shedding reserve."

Static broke around a reply. A woman named Aven Rusk had signed Violet's side of the hardware covenant. Jo knew her only through twelve hours of technical arguments and one shared joke about isolation jaws designed by people who never had to reach them.

"We see it," Aven said.

"Close your phase vent."

"Can't. Release jaw fused open."

A matching failure, but with opposite consequence. Nacre trapped rotational load. Violet spilled reserve into shared phase tension, where the unchoice converted it into independent movement across all routes. If Violet continued shedding at that rate, it might finish the maneuver with no margin for Pelagos capture.

"Can you cut the jaw feed?" Jo asked.

"Not without losing seam timing."

"Cut it."

"Nessa needs our lamp."

Jo looked at the rail. Indigo's two extra taps held their place between Violet and Nacre. Remove Violet's signal, and the child's improvised sequence would lose one of its skies.

"Eighty-eight seconds," the lower mechanic announced.

Captain Or spoke over the bridge loop. "Jo, route spread has begun. We need Indigo's count until predictive release."

"Violet needs reserve after release."

"I know."

The old system would have solved this by deciding which ship deserved the remaining energy. It would have named one vessel safe and rendered the others useful. Jo could feel that answer waiting in the machinery, not as intent but as a worn channel into which every crisis tended to fall.

They opened all-route engineering. "We need a timing lamp for Indigo. Violet has to cut theirs."

North answered first. "Our seam phase won't hold their near slot."

"Amber?"

"Too much delay variation."

Glass reported damage to its transmitter braid. Reed could carry the signal, but its slot lay on the wrong side of Indigo's missing count.

Aven's voice came back from Violet. "We can hold ten more seconds."

"You have nineteen to guidance."

"Then we hold nineteen."

"No." Jo gripped the rail as Nacre's roll passed through level and began half a degree in the other direction. "The covenant forbids one ship paying for shared control."

"The covenant also says don't lose the count."

"It says we solve that together."

Silence on seven channels consumed one second.

Then Nessa said, "Move your turns."

Jo lowered their head toward the speaker. "Explain."

"You keep putting the missing place where Violet can reach it. Put it where Reed can."

The route slots had seemed fixed because the blank course depended on their shared opening order. Yet the field needed only the intervals now, not the names attached to each position. Jo saw the repair as soon as Nessa stated it. No route owned a slot. Reed could transmit Indigo's second reference if every other ship shifted one count without disturbing the total phase pattern.

"Teren?" Jo called.

His answer came thin with distance from the causal gallery. "She is correct. Rotate the pulse identities one position. Preserve spacing."

"Risk?"

“Misassignment during the change.”

“Household version.”

“Seven people changing seats in the dark while carrying a full basin.”

“Useful at last.”

Jo assigned the rotation. North took Nacre's slot. Nacre moved to Amber's. Amber crossed to Glass. Glass inherited Reed's old mark, and Reed took the place beside Indigo. Violet needed only to signal the transition, then cut its fused vent.

Each route acknowledged with one pulse. The sequence started at Nessa's next count.

Ninety-nine seconds.

North moved early. Indigo held.

Nacre moved late. The children corrected the space with two quick taps.

Amber and Glass crossed cleanly. Reed entered beside Indigo with a pulse broader than Violet's but stable.

“Now,” Jo told Aven.

Violet's timing lamp went dark.

For a fraction of a second, nothing replaced it. Jo heard the drive hum deepen. The route columns shivered. Every old instinct demanded that they seize a preferred line and pour the uncertainty away.

Reed pulsed.

Indigo answered.

The pattern closed without Violet.

Aven reported, “Feed cut. Reserve loss slowing.”

Jo released a breath they had not budgeted. “Keep enough for capture.”

“We intend to arrive.”

At 105 seconds, Nacre's collar temperature crossed yellow. Leth called it. Pavi adjusted the trim wheel by a quarter mark. Jo ordered water mist across the casing, and steam rolled through the service trench. The dead prediction panels stayed black.

At 109 seconds, debris struck the outer shield. The impacts arrived first as scattered grains, then as a hard rushing hiss through the ship's bones. A larger fragment rang somewhere forward. Damage reports broke across the command loop: torn antenna web, three punctures sealed, one farm lamp rail down, no pressure loss in inhabited space.

At 112 seconds, Glass lost its pulse.

The seven marks on the timing rail became six.

Nobody spoke the ship's name. Jo watched its empty position pass once. Indigo held the missing interval instead of filling it. The children understood that a lost signal did not grant permission to take its turn.

At 114 seconds, Glass came back with a ragged double strike.

“Transmitter arc,” its engineer said. “Clock holds.”

The nave answered with a sound too disordered for cheering. A wrench hit a rail. Someone laughed. Someone else told them both to shut up and keep count.

At 116 seconds, the drive hum climbed beyond Jo's bones into audible pitch. Seven reserve columns bent away from one another. Lines that had shared a corridor began separating across the local board, not predicted now but measured after motion had already happened.

The final blind-course bell sounded.

Prediction returned as a burst of white across every panel.

Crew flinched from the sudden light. The Loom flooded the board with trajectories, hazard cones, spin values, reserve margins. For one terrible moment, its old display tried to rank the routes. GREEN appeared beside Nacre. SETTLEMENT RECOMMENDED opened below the six diverging traces.

Jo drove their fist into the physical inhibit before the recommendation could acquire a command path.

“No settlement access,” they said.

The board protested with red borders. Jo kept the inhibit down until Leth locked it mechanically with a pin.

“Chart,” Captain Or said. “Report survival.”

Voices took the route names in rotating order.

North: clock present, coarse spin, hull intact.

Reed: clock present, reserve stable, transfer trajectory shallow.

Glass: clock present, transmitter damaged, Trojan approach viable.

Indigo: clock present. Nessa's class sent one unnecessary victory tap.

Violet: clock present, low reserve, Lattice capture still possible with assistance.

Amber: clock present, long-period transfer, debris clear.

Nacre: Jo looked around the drive nave before answering. The collar remained hot. Jaw C hung dead behind its isolation plate. Steam condensed on the black rails and ran in narrow lines toward the drains. Pavi and Leth still held the trim wheel, though the ship no longer rolled.

“Nacre clock present,” Jo said. “Rotational control recovered. Settlement physically inhibited.”

The shared corridor no longer existed. Pelagos navigation drew seven courses across the board. One bent toward Lattice. Two reached for the Trojan rubble fields. One took a long path near Thren. The remaining lines entered transfer orbits whose years and resupply costs had not yet resolved. None occupied the center because no center could contain them.

Jo retrieved the torque driver from the trench. Its handle carried a new gouge.

On the timing rail, the clocks repeated in their altered order. Indigo kept Reed beside it. Violet's pulse arrived faint but punctual. Glass carried a burr from its damaged braid. Each signal reached Nacre through a seam and went no farther than light or sound. That limitation no longer resembled failure. It marked seven places where work would continue beyond Jo's hands.

Captain Or asked for the route solution.

Jo read the heading on the restored display. The Loom had labeled the diverging lines as errors because its vocabulary possessed no field for a shared survival that separated.

They cleared the label and entered a new one.

ARRIVAL VECTORS: SEVEN.

The chart accepted the words. Across the nave, seven clocks kept time while every ship fell toward a different harbor.

Chapter 24 — The Convoy Without an Original

<!-- POV: Captain Sava Or -->

<!-- SETUP: seven surviving ships; dispersed trajectories; relinquished designation privilege -->

<!-- PAYOFF: Convoy of Elsewheres formed; Nacre becomes Common Route; mission seal removed -->

Sava rejected the command packet before its sender had finished signing it.

The packet hung above the chart table in provisional amber, addressed to CONVOY COMMANDER OR. Its routing field listed all seven ships. Beneath that line, North had attached debris clearances, Violet had attached reserve needs, and Reed had requested a common priority schedule for seam power. No one had voted Sava into the office. The network had found the oldest intact command certificate and built a throne out of metadata.

She pressed REJECT and selected the reason field marked WRONG JURISDICTION.

The system asked for a superior authority.

“None,” Sava said.

Ilyan looked up from the opposite rail. He had slept for perhaps an hour since predictive guidance returned. A bruise darkened the heel of one hand where the blind passage had thrown him against a console. “It won't accept that.”

“It will record it.”

She entered SEVEN VESSELS, SEVEN COMMANDS. The packet dissolved, but its requests remained in their separate queues. Rejecting an office did not cool Violet's capacitors or give Glass a new transmitter braid.

Beyond the suspended route glass, repair crews had replaced two shattered panels with clear safety board. The improvised material showed no course traces. Through it, Sava could see the black service rail and the pale composite wall behind the Loom, ordinary ship structure where an authoritative universe used to glow.

Oru Pell came through the chart-room hatch carrying a slate, three seal strips, and a coil of white cord. “You rejected North's packet.”

“I rejected the routing assumption.”

“North sent it because they need an answer within forty minutes.”

“North has a captain.”

“North's captain wants to know whether your vessel can share shield forecasts.”

Sava opened that request and authorized the data. “Then the answer requires Navigation, not a convoy commander.”

Oru tied one end of the white cord to the chart rail. They stretched it across the central operator's position and fixed it at the far side. During the Parliament, a rotating order had kept any ship from occupying the center frame. The cord now made that arrangement physical aboard Sava's own vessel.

“You understand someone must coordinate,” Oru said.

“Coordination does not require someone above it.”

“That sentence will be tested before first meal.”

“It has already survived a debris tide.”

Ilyan hid a tired smile by turning toward the glass.

Sava called the damage board. Nacre had lost twelve exterior antenna webs, one farm restraint, six meters of shield skin, and no people. Thirty-one injuries remained under treatment; four required surgery. The population counter still read 42,006. For eleven years, that number had entered every decision she made. It did not shrink because six other counters now held equal claim to survival.

On the shared route display, the ships continued to separate. Violet and Nacre held the nearest pair of vectors, though even their light delay widened by the minute. Indigo's path bent toward the far side of Lattice. Glass and Reed approached different Trojan clusters. North's coarse trajectory passed near Thren before any later transfer. Amber had accepted a long-period orbit that would carry its people away from direct settlement for years.

Each course demanded help from ships that could not share cargo through a seam. Forecasts, instructions, warnings, and promises could cross. Food and metal could not.

Oru placed the slate on the rail. “Provisional assembly in eleven minutes. One delegation per ship. Local captains attend for operational commitments.”

Sava scanned the agenda. Reciprocal distress priority. Seam scheduling. Shared hazard observations. Recognition of local law. Noninterference. A name.

The final item had gathered forty-two suggestions.

“Strike United Nacre,” she said.

“Already struck by six delegations.”

“Nacre Family?”

“Lin threatened to end his connection.”

“Reasonable.”

Oru touched the slate and enlarged three finalists. Common Fleet implied common command. Seven Routes excluded Sable and every closed wake whose records had made the vote possible. The third proposal had come from a maintenance worker on Glass and acquired endorsements from every ship within an hour.

CONVOY OF ELSEWHERE.

Sava read it twice. Convoy promised company without pretending proximity. Elsewheres refused a center, though its grammar carried enough sadness to keep triumph from hardening.

“I will not endorse a name before the delegations meet,” she said.

Oru gathered the slate. "You just endorsed it with your face."

"My face holds no standing."

"That must come as a relief after eleven years."

The meeting opened through six seam frames and one local table. Compressed images gave each remote delegation a narrow rectangle of monochrome light. Their motions reached Sava after small, differing delays. Nobody occupied the center frame. Oru stood behind the white cord and called the rotating order.

North spoke first. Its captain wanted immediate assurance that every ship would forward debris observations without charging reserve credit. Reed asked whether such assurance created liability when a damaged sensor supplied a false track. Glass needed transmitter components but could receive only fabrication patterns until physical rendezvous became possible. Amber demanded language recognizing a ship's right to choose a distant harbor without losing convoy membership.

Sava listened with her command slate closed.

On Nacre, officers had once brought her disagreements after reducing them into options. The captain selected one, recorded her reasoning, and carried the liability. Here, selecting for everyone would repeat the privilege they had just surrendered. Yet silence could perform the same theft if other ships waited for an authority she declined to exercise.

When Oru reached her slot, Sava stood.

"My vessel will forward all verified debris observations at no reserve charge," she said. "Unverified tracks will carry sensor confidence, age, and source. We can fabricate Glass's braid couplings if we meet within transfer range. Current geometry does not permit that before Lattice insertion. We will send the pattern and test tolerances now."

Glass's engineer lifted two fingers in acknowledgment.

"For Amber," Sava continued, "membership cannot depend on destination. If it does, we have renamed the arrival corridor and learned nothing."

Amber's frame froze for half a second, then recovered. Its delegate said, "Will your captaincy bind your own ship to those terms?"

"Yes."

"Will it bind ours?"

"No."

That smaller word cost more. Sava felt the absence of a mission-wide command path like a missing stair under her foot.

North's captain leaned toward their camera. "Then who orders aid when two ships claim the same seam reserve?"

"No one orders both," Sava said. "The ships disclose need. The rotating delegation allocates shared channel time under rules we adopt together. Each captain decides what their vessel can supply."

“That invites delay.”

“It requires consent.”

“Debris does not wait for consent.”

“No. That is why verified warnings forward automatically. People do not.”

Oru noted the distinction. Teren, seated behind Nacre's delegation, began to explain reserve arbitration and stopped when Mina touched his sleeve. Sava noticed his restraint because it looked painful.

The delegates worked through first meal. Seam power received priority by imminent hazard, then medical instruction, then navigation, then civic traffic. No ship could lose speaking access because it lacked reserve credit. A distress call opened every listening seam but granted no authority to alter another vessel's systems. Shared predictions carried the assumptions that produced them. Every rule had a failure edge. Instead of hiding those edges, Oru made the assembly name them.

Before the language reached a vote, North's frame flashed with a local alarm. Its captain turned away. Figures crossed behind the camera, one carrying a pressure hood. The seam compressed their shouts into hard, clipped bursts.

Oru paused the order. “North, do you declare distress?”

No answer came for six seconds.

Sava's hand moved toward the general command key and stopped above the rail. The key could reach only Common Route now. Even before the unchoice, it could never have operated North's hardware; the old display had allowed captains to mistake modeled obedience for reach.

North's image cleared. “Thermal rise in our aft shield pocket,” its captain said. “Possible embedded debris. We need exterior density from anyone who crossed our phase line.”

Glass had crossed nearest. Its transmitter damage made the reply break apart.

Sava opened Common Route's shield history. Her Navigation staff had no exact match, but Amber's long vector placed its sensors behind North's wake. Amber had not finished reviewing the provisional automatic-warning clause.

“Amber holds the useful angle,” Sava said when Oru reached her under emergency order. “Request their raw scatter field, minutes twenty-three through twenty-five.”

Amber's delegate replied, “Our captain has not authorized automatic release.”

“Then ask.”

“Delay to command is fourteen seconds each way.”

North's alarm continued without regard for procedure. Its thermal pocket climbed three degrees on the shared board.

Sava looked at Oru. In the old chart room, an emergency gave her permission to move first and justify later. Here, the rule under debate had no legal force, yet the people awaiting it already carried heat against their hull.

Oru said, "The rotating chair can recognize a temporary offer. It cannot compel."

Amber's delegate closed their eyes briefly, listening to a local channel. "We release the raw field as our delegation's civic act, pending captain review. Mark it unauthorized for maneuver use."

The data crossed. Ilyan and three route conservators worked from separate ships, their cursors visible on the same copied chart without sharing a machine. They found a fan of nickel grains spreading behind Amber's course. North's shield pocket had caught one dense knot. Another would arrive in nine minutes.

"North can yaw four tenths and let it pass," Ilyan said. "No prediction stack required."

North's captain checked the geometry. "That turn exposes our damaged radiator."

"Glass has a narrower option," Lin said from Violet's frame. He laid a revised avoidance line beside Ilyan's. "Two tenths now, reverse after the knot. Higher control cost, lower thermal risk."

Sava studied both. Her training ranked the cleaner course and prepared an order. North's captain instead called their own engineering chief into view. They argued for twenty seconds, then selected Lin's line with a reduced reversal.

The maneuver began on North alone. Every other ship watched through delayed telemetry. Its thermal trace peaked, held, then fell. The second debris knot passed beyond its radiator by three kilometers.

Amber's captain appeared after the danger ended. "I confirm release," she said. "Next time my delegate need not wait for me when life faces immediate hazard. Record the limit at raw observation. No thrust command."

Oru wrote the language into the clause. Sava watched a rule acquire authority from use rather than from her signature. The distinction left no ceremonial place for a captain to stand, but it delivered the warning. North's cooling trace remained visible beside the adopted text, a practical annotation no clerk could polish away.

At the second-hour mark, Violet's frame brightened. Lin sat in a classroom with route charts pinned behind him. His council had lost its usual chamber when the unchoice damaged two pressure doors. Children passed occasionally behind his chair, pretending not to watch.

"Violet accepts the distress rules," he said. "We reserve judgment on the archive clause."

The clause required each ship to preserve the records of closed routes. Nacre's draft referred to a common custody vault.

"What do you object to?" Oru asked.

"Common custody on which ship?"

No one answered.

Sava opened the clause and saw how quickly grammar rebuilt a capital. One vault would need one physical location. That location would outlive its provisional label and become the place from which authenticity flowed.

"Replicated custody," she said. "Checksums on every ship. Full copies where reserve permits. No master archive."

Lin looked toward her frame after the seam delay. "Including the Captain's sealed records?"

Sava understood the capital letter. "Including mine."

"Violet accepts with that change."

The last command archive she had released contained Alda Quill's ledger and evidence of one hundred nineteen years of concealment. Other sealed records remained: security decisions, private injury projections, ration contingencies. Some deserved privacy. None deserved immunity from the law of the ship that had produced them.

"Local review will determine personal protections," Sava added. "Secrecy cannot travel under the label of convoy security without a seven-ship vote."

Hadran Coil stood by the chart-room hatch. He no longer wore the security command band that had opened every door during the seizure. His face did not change, but he lowered his eyes to the slate where he now took public minutes.

The assembly reached the question of command near midday.

North nominated Sava as provisional coordinator for thirty-one days. Amber supported a shorter term. Reed proposed Oru. Glass proposed a rotating captaincy matching the speaking order. Indigo's delegation asked why the convoy needed a person rather than a schedule.

Sava waited until her vessel's turn.

"I decline convoy command," she said.

North's captain began speaking before her delayed image finished. "You have the intact staff and the longest command continuity."

"I also commanded the vessel that received every prior settlement."

"That history does not erase competence."

"No. It limits jurisdiction." Sava placed both hands on the closed slate. "I remain captain of this ship while its Commons permits me. I will act for its safety. I will not convert that duty into rank over six populations that never appointed me."

Amber's delegate asked, "If a joint maneuver becomes necessary?"

"Then we appoint authority for that maneuver, with scope and end time."

"And if there is no time?"

"Each captain acts. Shared warnings travel automatically. We accept that coordination can fail without solving the risk by making one ship sovereign."

The frames held seven versions of delay. Sava watched reactions arrive out of order: North's frustration, Reed's whispered side discussion, a nod from Glass. Lin looked down, perhaps at a local message, then back toward the camera.

"Violet supports no permanent convoy command," he said. "Rotating coordination by task."

Indigo agreed. Amber revised its support. By the time North's image caught up, the proposal had enough votes

to become the provisional rule.

Oru marked the term at twelve days, renewable by one delegation from every ship. Sava felt relief and fear take equal space inside her. Relief did not prove the choice safe. Fear did not prove it wrong.

They moved to the name.

Glass's maintenance worker appeared briefly beside its delegate, an older woman with a burned strip down one sleeve. She explained that a convoy had formed whenever separate repair crews carried the same damaged part down different passages. Elsewhere, aboard ship, usually meant a place beyond current responsibility. The proposed name reversed that habit.

"No original?" North's delegate asked.

"No rear ship either," the worker said. "Just bad spacing."

Dry laughter crossed the frames in staggered waves.

All seven delegations approved the Convoy of Elsewheres. Oru entered the name into the shared header. For the first time, the network accepted a collective field without assigning a primary vessel beneath it.

One naming problem remained local.

Nacre's Commons had opened a shipwide ballot during the assembly. Keeping Nacre would preserve the name carried from Sol, but it also preserved the habit of letting one vessel stand for every route. Sava had argued against forced renaming. Lin's ship would remain Violet because its people had made that tag into a civic name rather than a rank. Indigo wanted Indigo. North had declined every alternative in four words: WE KNOW WHERE NORTH IS.

Her own ship's ballot closed at the end of the meeting.

Oru read the result privately first, as procedure required. "Fifty-nine percent for Common Route. Twenty-eight for keeping Nacre. The remainder split among twelve proposals, including one obscene suggestion from drive engineering."

"Jo's?"

"They deny authorship with suspicious precision."

Sava looked through the clear replacement panel at the bare wall. "Certify it."

Oru opened the local record. The ship that had carried one name for 127 years became Common Route by vote of the people aboard it. Nacre remained in historical fields, on birth records, hull plates, old songs, and the mission seal fixed to the chart-room door. A new name did not acquit the old one.

"Captain," Ilyan said, "the route registry needs your command certificate."

She signed as CAPTAIN, COMMON ROUTE. The certificate rejected the entry once because its ship identifier expected NACRE. Jo, joining by audio from the drive nave, told the registry what they thought of inherited validation. Ilyan altered the public label without erasing the historical identifier, and the second attempt held.

By evening, the first convoy schedule crossed all seams. North sent a debris warning to Reed. Violet transferred

a capture model to Indigo. Amber offered long-period observations that no inner ship would possess. Glass transmitted a damaged-braid pattern so Common Route could test a replacement before any physical meeting. The ships could not pass tools, medicine, or hands through the glass. They could make distance less ignorant.

Sava ended her shift at the chart-room hatch.

The old mission seal occupied the center of the door: a metal Nacre flying alone toward a seven-pointed star, one course line engraved beneath its keel. Every captain since departure had touched it before a correction. Sava had done so 141 times.

A fabrication worker waited with a heat pad and two lifting blades. "Commons order," he said. "Remove and archive. No destruction."

Sava placed her palm against the cold metal once more. Not for luck. She checked whether the seal had loosened under the blind-course roll.

"Proceed."

The worker warmed the adhesive and worked a blade beneath the lower edge. Ilyan steadied the door. Oru held an archive sleeve open. When the final bond released, the seal came away without ceremony, leaving a paler oval and four empty bolt holes.

Sava took the door handle. No emblem waited above her hand. On the other side, the chart room held seven moving lines and no command position at their center.

She opened the unsealed door and reported for the next watch as captain of one ship.

Chapter 25 — Arrivals in Plural

<!-- POV: Ilyan Rook -->

<!-- SETUP: seven arrival vectors; distributed convoy; Common Route and Violet remain separate -->

<!-- PAYOFF: scheduled seams replace merger; first plural harbor map published -->

Ilyan removed the harbor circle for the fourth time.

The chart software restored it around Lattice before his hand left the glass. A generation mission carried one destination marker by design: a white ring large enough to contain the arriving ship, its service craft, its first orbital stores, and every approved descent corridor. Seven route traces now approached Pelagos. The software treated six as traffic conflicts.

He opened the map rules and found the field labeled PRIMARY ARRIVAL BODY.

“No,” he said.

Across the chart room, Mina looked up from a ration terminal. “Are you arguing with punctuation again?”

“Jurisdiction.”

“Does it argue back?”

“It keeps drawing a circle.”

Outside the repaired route glass, Pelagos filled more of the forward display than it had an hour earlier. The K-type star carried a warmer gold than the teaching images. Its light reached the chart room through filtered cameras and laid a narrow bar across the black rail. Lattice crossed below it, blue-gray and banded with weather. Farther out, Thren made a chipped white point near the edge of visibility.

Common Route would not reach either moon on its first pass.

Ilyan had known that since prediction returned. Knowing by vector and knowing by sight occupied different parts of him. For 127 years, the ship had spent food, sleep, hull, births, arguments, and 141 corrections to reach Lattice. Now the moon hung close enough for storm bands to show while Common Route bent onto a long-period transfer path that would carry it beyond immediate capture.

The alternative required three vessels to share one braking corridor. Their drive plumes would blind one another's sensors. Their first supply demands would fall on the same unbuilt orbital infrastructure. A clean arrival for Common Route meant collision risk for Violet and Indigo. Separation hurt because physics had no duty to make justice feel like reunion.

He reduced the automatic harbor circle to a point. The software expanded it again.

Jo entered carrying a replacement latch for the chart-room hatch. “Hit it with something.”

“That method suits your department.”

“It suits any machine that repeats a bad answer.”

Ilyan showed them the rule field.

Jo read it. "Delete primary."

"The schema requires a value."

"Put all seven."

"It accepts one object."

"Then the schema has volunteered for recycling."

They knelt beside the hatch before Ilyan could answer. The old mission seal had left four bolt holes in the door. Jo used two for the latch plate and ignored the others. Each turn of their driver carried into the rail as a neat metallic tick.

Mina crossed from the ration station. Her terminal displayed projected stores for the transfer path: algae protein remained sound; fresh beds would tighten; medical cultures required lower redundancy. The figures held no catastrophe, only months that Common Route had not planned to spend between arrival and harbor.

"Indigo can take Lattice's far capture band," she said. "Violet has the near polar line. We take the outer transfer and come back after their platforms can receive us."

"You say that as if we offered them a seat," Ilyan said.

"We did. They offered us one later."

"Fourteen months later."

"Then don't pack badly."

Mina had once challenged him for making family carry proof that mathematics should have supplied. Now she refused to make a moon prove they had arrived.

A pulse lit Violet's seam indicator. Lin had requested their scheduled chart session three minutes early, which meant his ship's clock still ran a fraction fast after reserve shedding. Ilyan adjusted the receive window instead of correcting him.

Lin's image opened at low frame. The classroom behind him had become a navigation annex. Paper charts covered the wall because Violet's display fabric had failed along one edge. Two people worked at a table under the old lesson lamps. Neither looked toward the seam.

"You still have one harbor," Lin said.

"The software has one harbor. I have a dispute."

"Your disputes often resemble software for the first hour."

Ilyan let the remark pass. They shared enough habits that offense sometimes arrived before he knew which version of himself had earned it.

Lin sent Violet's capture model. Its line curved around Lattice's northern hemisphere, close enough for weather instruments to resolve cloud shadows. Indigo's trajectory entered from the far side in a shallower orbit. The ships would not share an immediate platform, but service craft might bridge them after construction and consent. No mass could cross a seam. Everything physical would still require distance, fuel, docking rules, and

time.

“Your reserve?” Ilyan asked.

“Eight percent above capture minimum if Lattice's upper atmosphere matches the probe archive.”

“If it doesn't?”

“We take another orbit and lose two months of descent work.”

Common Route carried fourteen percent above transfer minimum. Before the unchoice, Ilyan would have described that difference as an available rescue margin. The debt engine had trained every officer to see unequal reserves as material waiting to move toward a chosen ship.

“We can send you updated atmosphere from our forward spectrometers,” he said.

“Send the measurements. Keep your reserve.”

“I did not offer reserve.”

“You leaned toward the old sentence.”

Ilyan looked down at his hands. Lin could still hear the pauses they had inherited together. “I offered weather.”

“Then I accept weather.”

The first data block crossed the seam. Common Route's instruments had sampled oxygen ratios, water vapor, high ice, and traces of salt lifted from Lattice's oceans into its upper air. The numbers carried a place no human aboard any route had touched. Ilyan watched Violet's checksum match.

Lin put the block into his local model. His capture margin rose by six tenths of a percent.

“Useful,” he said.

“That adjective lacks gratitude.”

“You used it on my dust fan yesterday.”

“I was grateful with precision.”

For a moment their delay held only two men trying not to smile across separate glass.

Then Lin enlarged the Pelagos system. “Our civic council rejected daily open seam.”

Ilyan's hand stopped above the chart. “Why?”

“Reserve, privacy, and the fact that half our schoolchildren have begun waiting beside the frame instead of doing lessons.”

“Common Route proposed six hours.”

“Common Route has more reserve.”

“We can pay the difference.”

“No.”

The refusal came without anger. That gave it more force.

Lin continued, “A permanent open seam turns Violet into a room attached to your chart room. We spent nineteen days proving we constitute a ship. We will not become your window.”

Ilyan had imagined continuous contact as restraint against loss. If the seam stayed open, Violet could not vanish into rumor. Lin could speak whenever a decision threatened to harden around him. Children on both ships could see that the other clock continued. Yet a window possessed a room on each side, and the person who controlled its frame chose what counted as view.

“What schedule do you propose?” Ilyan asked.

“Two navigation exchanges per day through capture. One civic session every third day. Emergency band always listening, transmit only on declared need. Personal messages queued by local rules.”

“That gives us twelve-hour gaps in your maneuver state.”

“It gives Violet twelve hours when no one elsewhere watches us arrive.”

Ilyan heard Eda's old pipe knock in memory and let it pass without answering. A signal could prove life. Continuous proof could become a demand made of the living.

“Agreed provisionally,” he said. “Review after first orbit.”

“By both ships.”

“Review requires both ships.”

Lin sent the schedule. Ilyan placed it beside Common Route's transfer windows. They did not align neatly. The map software highlighted unused gaps as inefficiency.

He left them unused.

By midday, every ship had submitted a harbor claim.

North selected a survey orbit around Thren. The ice moon offered water and sheltering cavities, but its distance from Lattice would turn every physical exchange into a voyage. North's captain described that isolation without apology. Their damaged radiator favored Thren's cold. Their Commons had voted to become the convoy's outer-system watch if later review confirmed the orbit.

Glass and Reed took opposing Trojan clusters. Neither cluster offered a harbor in the old sense. Rubble, nickel, carbon ice, and static-charged dust occupied volumes wide enough to contain nations. The ships would have to build docks from material that had never known a tool. Their separation protected both from competing for the same stable rocks.

Amber's long-period path carried it beyond Common Route's transfer arc. Its first circuit would take almost two years. In exchange, Amber would map the system's exterior debris and watch for hazards none of the inner ships could see early. Its delegates asked the map to show future return windows rather than a line that faded at the page edge.

Violet claimed Lattice's near polar orbit. Indigo claimed the far shallow band and the first descent study, subject to biological quarantine. Nessa appeared during the submission holding a class map in both hands. Her version showed two blue circles around Lattice, neither drawn first. The children's routes crossed only as dotted message lines.

Common Route received no immediate moon, cluster, or fixed orbit. Its path remained a transfer curve with a future Lattice approach if platforms, stores, and local consent allowed. Mina had the ration schedule. Jo had drive tolerances. Sava had signed the ship's acceptance without adding a command condition.

Ilyan opened the rejected common-harbor model once before deleting it. Seven traces converged on Lattice's northern capture gate. Their braking envelopes overlapped in a bright knot. Indigo entered Violet's sensor shadow. Common Route's exhaust crossed both ships' high atmosphere measurements. A small timing error by Glass turned three safe arcs into one field of blind drives.

The software treated each danger separately and proposed a priority order. First place reached orbit with full reserve. Second accepted a delayed burn. By fourth place, capture required reserve transfer from the ships behind it. The seventh line ended outside the system after spending its braking margin to protect the preceding six.

Mina stood beside him as the sequence played. "Show the first orbital store."

He added the probe-era platform assumptions. The planned store had never been built; generation missions expected the arriving ship to construct it. Seven populations reached for the same empty coordinate. Their projected water demand exceeded the first processing rig by a factor of five. Medical quarantine space failed before the third docking. Seed banks that could have remained separate entered one contamination chain.

"No collision in the first hour," Ilyan said. "Shortage by the second day. Shared life support failure if one culture carries the wrong fungus."

"And command?" Mina asked.

The model assigned queue control to the designated arrival vessel. Its selection rule still chose Common Route by inherited identifier, despite the renamed registry.

Ilyan cleared that field. Without a priority ship, the common harbor simulation stalled. All seven courses hovered at the gate, each physically possible alone and lethal in combination.

"Keep a copy," Mina said.

"Why preserve a plan no one will use?"

"So when someone on Lattice asks why we made their aunt wait fourteen months, we answer with more than principle."

He saved the model beside the ration projection. Separation needed no disguise as painless virtue. It took birthdays, shared meals, hands on the same rail, and the first descent from people who had expected those things all their lives. The collision model did not refund that cost. It showed what choosing one immediate reunion would charge everyone else.

Ilyan placed all seven claims on one field.

The display produced warning color until almost nothing else remained visible. COLLISION OF PRIMARY

OBJECTS. MULTIPLE DESTINATIONS. UNRESOLVED OWNERSHIP. INVALID HARBOR COUNT.

Oru, attending through the Commons link, read the errors over his shoulder. "At least it has stopped pretending this concerns punctuation."

"It cannot represent membership without co-location," Ilyan said.

"Can you?"

He looked at the lines. The convoy rules linked ships by warning, rescue, archive custody, and periodic communication. None required a shared coordinate. The old mission map used proximity as destiny: one ship, one moon, one settlement. A plural polity needed to remain legible while its people could not gather in one chamber.

"Not with a harbor circle," he said.

Ilyan built a new layer from communication windows. Scheduled seams became thin arcs between the moving routes. Emergency listening bands appeared as unfilled links, present but quiet. Physical transfer opportunities received a different mark, never crossing the current distance as though intent could move cargo. Future rendezvous windows showed uncertainty cones and fuel costs.

He added local-law boundaries around each ship rather than around territory. A person aboard Amber remained under Amber's civic jurisdiction while the ship moved through a path no one owned. Lattice orbit required later agreements with the people who would descend and build there. Thren did not become North's property because North arrived first. The Trojan clusters carried claims to constructed platforms, not to every stone within sensor range.

Oru watched the legend grow. "You are drafting law."

"I am assigning symbols."

"Those activities have shared a bunk for most of history."

The phrase reminded Ilyan of the archive shelves in his training vault. Categories had never waited inertly. A map gave some facts doors and left others against the wall.

Mina brought him a cup of tea that had cooled during transit from the Commons well. "Where do you put Sable?"

The seven living traces held the center of his work. Sable's final timestamp remained in the museum archive, copied on every ship. A conventional arrival map would omit any vessel that could not arrive.

Ilyan added historical wakes as a separate layer. He did not draw them approaching Pelagos. Sable had closed before the blind course, and the earlier routes had died across 141 settlements. Inventing arrival lines for them would turn record into consolation. Instead he placed their archive marks along the mission history, each linked to ordinary entries where records survived.

"What about unknown wakes?" Mina asked.

He left room in the legend. "Unconfirmed jurisdictions. No invented names."

"Better."

The map now reached backward without pretending restoration and forward without forcing convergence. It remained ugly. Lines accumulated around Lattice. Transfer cones crossed labels. Thren sat far enough out that the page needed a second scale. The seam schedule looked like careful stitching done by someone repairing cloth under poor light.

Jo examined it from the hatch. Their new latch clicked cleanly behind them. "Where do I hit this one?"

"Constructive criticism from engineering has concluded."

"Then why does Reed's physical transfer window look like a seam?"

Ilyan compared the marks. Jo had found a dangerous similarity. A reader could mistake scheduled communication for possible passage.

He changed all seam links to paired bell symbols, one at each transmitting ship. Physical trajectories kept continuous lines. No mark crossed from one vessel's hull into another without showing the distance between.

"Acceptable?" he asked.

Jo leaned close. "Make the bells offset."

"By how much?"

"Enough to show they answer. Not enough to look broken."

Ilyan set the second bell seven tenths of a symbol-width from the first. The old delay had measured 0.73 seconds, a small interval that opened the moral scale of the mission. He did not print the number. People who knew would hear it in the spacing.

The convoy reviewed the map at the next common seam window.

North requested a larger Thren inset. Glass corrected a Trojan label inherited from probe data. Reed objected to a resource estimate that treated rubble within its sensor cone as available inventory. Amber added its future return windows. Indigo's class asked for the two Lattice routes in different line textures because their oldest classroom printer had no color.

Lin studied the scheduled seam marks between Violet and Common Route.

"You gave our personal queue the same weight as navigation," he said.

"The line weight indicates reserve cost."

"It also tells people which connection matters."

Ilyan enlarged the legend. Lin had found another category pretending to carry no judgment. A medical warning used more reserve than a child's message, but reserve did not measure civic value.

"Separate scale," Ilyan said. "Technical cost in the schedule table. Equal mark on the public map."

Violet approved.

Sava spoke only when the review reached the title. The draft read PELAGOS ARRIVAL PLAN, followed by a

smaller line naming the Convoy of Elsewheres.

“Plan implies authority over future positions,” she said. “Some of these paths remain conditional.”

“Arrival chart?” Ilyan offered.

“Chart still suggests one observing ship.”

Nessa's voice arrived from Indigo. “Call it harbors.”

“North does not yet possess a harbor,” Oru said.

“Neither do we,” Nessa replied. “We have places to make them.”

Ilyan typed PLURAL HARBOR MAP.

The title did not make the map true. Every delegation still had to certify its own line. Every local Commons had to accept the symbols governing its people. The ships could still withdraw a schedule or change a path. That incompleteness no longer counted as a defect to hide.

Seven certification marks appeared in rotating order. None occupied a larger field. Sable's archive checksum sat beneath them, not as an eighth voter but as custody acknowledged.

Ilyan prepared the publication packet. The old system demanded an origin marker before release. He removed the requirement from the copied public schema, documented the change, and sent the patch to every route for inspection. Checksums came back one by one.

Lin's arrived last because Violet had entered the shadow of Lattice and needed to relay around Indigo. The delay stretched beyond its schedule. Ilyan listened to the chart room pumps, Jo's new latch settling in the door, and the distant tram moving through a ship that had reached Pelagos without reaching land.

Violet's bell appeared beside Common Route's.

MATCH, the checksum field reported.

Ilyan pressed publish.

The first plural harbor map opened on seven ships at seven positions across the system. Lattice carried two approaching lines. Thren waited beside North's wide survey arc. The Trojan clusters held Glass and Reed without closing around them. Amber and Common Route curved through transfer paths whose future endpoints remained open.

Between those separate courses, paired bells marked the hours when people had agreed to answer.

No circle returned around them.

Chapter 26 — The Last Correction

<!-- POV: Teren Vale -->

<!-- SETUP: debt engine exposed; settlement inhibited; plural harbor paths -->

<!-- PAYOFF: settlement mode removed; seven-key governance; ordinary-thrust adjustment creates no abandoned route -->

Teren loosened the first harvest coupler with both hands.

The tool slipped once on the superconducting collar and struck his thumb. Pain brightened under the nail. Jo Malk, braced across the service frame, watched without sympathy.

“Again,” they said.

“I know how threads work.”

“You know how field equations work. Threads have consequences.”

Teren reset the wrench. The collar had not moved in 119 years. Frost clouded its engraved maintenance marks, including Sen Quill's initials and three inspection stamps from apprentices now dead. Beneath the frost, a red line identified the conduit that had carried free-energy differential from settled routes into the designated ship.

The debt engine had never occupied a separate room. It lived in command logic, field geometry, transfer braid, and habits of explanation. Removing it required work in all four places. Teren could unbolt only one.

He pulled until the coupler gave with a short metal cry.

Jo caught the freed end before it struck the frame. No current crossed it. The unchoice had left settlement inhibited, and three mechanical locks now kept the harvest bus cold. Even so, Teren waited for a pressure change or secondary field tone. His body had spent too many years trusting status lights.

“Coupler One separated,” Jo called.

A Commons witness recorded the time. Another witness watched through a seam from Violet. Six remote status marks waited along the service display, one for every other living ship. Nobody needed to observe the bolt for physics. They observed because Teren had once allowed a machine to become pardonable through privacy.

Jo tied a yellow custody tag through the coupler eye. “Your thumb is bleeding.”

“It remains attached.”

“An ambitious standard for causal physics.”

They moved to the second collar.

Above the service frame, the Loom's prediction braid continued its cold circulation. It could still model dust, drive heat, orbital capture, and food trajectories without instantiating any route domain. Seam capacitors continued carrying scheduled light and sound between separate ships. The parts they removed belonged specifically to settlement: transfer couplers, privilege selectors, automatic extraction jaws, and the command bridge that turned a prediction into inhabited alternatives available for harvest.

Teren had spent half his career claiming those functions could not be separated safely. Jo had spent six days making him prove himself wrong.

“Second collar has reverse load scars,” he said.

“From Correction 141?”

“Older. Repeated.”

Jo leaned closer. The inner surface carried fine silver arcs where reserve had crossed against designed direction. Alda Quill might have seen similar marks after the first trial. Sen Quill certainly had. Teren's teachers had filed them under recovery noise, then taught him the noun until it felt like mechanism.

“Photograph before removal,” he said.

The Commons witness raised the recorder.

“No,” Jo said. “You do it.”

Teren looked at them.

“You are giving testimony in forty minutes. Start with what your hands touched.”

He took the recorder. Through magnification, the scars resembled river deltas cut into metal. Teren described their depth, direction, likely age, and relation to the transfer bus. He did not say the metal accused him. Metal had done exactly what people designed it to do.

After the image locked to the public archive, Jo placed the wrench in his hand again.

The second coupler rotated half a turn and jammed. Teren increased pressure until Jo caught his wrist.

“Stop proving you have shoulders.”

“The thread has seized.”

“The thread has a lock tab.”

A narrow strip of alloy lay folded into a recess behind the collar. Its position did not appear on the service diagram. Jo cleaned frost from it with a brush, exposing a stamped instruction: **REMOVE ONLY AFTER ROUTE SETTLEMENT CONFIRMED.**

Teren called up the founder drawing. The tab had no part number. Later revisions showed the same empty space. Someone had added a physical interlock after design approval, then omitted it from every teaching copy.

“Alda?” the witness asked.

“No evidence,” Teren said. “Do not attach a name because we want one.”

He photographed the tab in place. Jo measured it before unfolding the metal. The concealed strip had bitten into a sensor contact on every use, reporting full settlement before the coupler could turn. Removal crews following the official diagram would have found the collar immovable and assumed the bus formed one inseparable assembly.

Jo held the strip under the recorder. “This machine has been winning arguments by hiding a clip.”

“People hid it,” Teren said.

“Machines remain excellent storage for bad decisions.”

Once the tab came free, the coupler turned easily. A dark residue marked the contact. Teren sampled it for archive dating, capped both ends of the open line, and ran a continuity test across the cold bus. One path remained conductive.

“That cannot be right,” he said.

Jo took the meter. They tested each cap, then traced the remaining path into a narrow service conduit. The wire entered the prediction cabinet instead of the harvest rack.

Teren knew its purpose before they opened the panel. Settlement confirmation had fed the predictive stack as a training truth. Every closed route taught future models that designation succeeded. The stack never received outcomes from the people who lived after rejection, because their clocks vanished at settlement.

“Remove the training return,” he said.

Jo shook their head. “Not until you decide what replaces it. Prediction still needs measured outcomes.”

They followed the wire to a recorder board with 141 green indicators. Each correction had written one favored history back into the model. The result did not merely harvest routes. It trained the Loom to call harvested survival accurate.

Teren opened the revised architecture. “All route observations before closure should have entered here. We cannot recover them. Going forward, ordinary maneuvers feed measured results from the acting ship. Seam forecasts feed source-labeled observations from every consenting ship.”

“And if reports conflict?”

“The model retains conflict. No averaging without a stated method.”

The witness asked, “Who approves that change?”

“Every ship, before the prediction stack trains on shared data.”

They left the wire connected but disabled its write path with a transparent physical break. Jo mounted the removed bridge on the outside of the cabinet, where an inspector could see it without opening a panel. Teren wrote the replacement rule for seven-ship review and attached the hidden tab as evidence.

Only then did Jo return the wrench.

The testimony chamber occupied the Commons well, but most listeners could not fit inside it. Terraces filled with Common Route residents. Seam frames carried the six other ships in rotating order. Lattice shone on Violet's background display. North's feed arrived from the wide dark near Thren. Amber appeared only as voice during low-reserve intervals.

Teren stood at the lowest table with his injured thumb wrapped in seal cloth.

Oru Pell asked the first question. “Did you know rejected domains persisted?”

Teren had prepared seventeen careful sentences. Shame offered him all of them at once.

“I knew measurements contradicted durationless closure,” he said. “I accepted explanations that protected the program I served. I did not know the full nineteen-day window until Jo's test. I knew enough to ask harder questions years earlier, and I chose not to.”

The well held ventilation noise and small movements. Nobody supplied him with outrage large enough to hide inside.

A delegate from Reed asked, “When did you first see persistence evidence?”

“Forty-three years ago, during my apprenticeship. A residual metabolism estimate appeared in a settlement review. Sen Quill called it model contamination from the designated route.”

“Did you believe him?”

“I wanted his explanation to hold. I checked the derivation only inside the assumptions he gave me.”

“You are a causal physicist.”

“Yes.”

“Then you understood assumptions decide what a derivation can see.”

“Yes.”

The second answer left no place for technical refuge.

Violet's witness came forward in her frame. Aven Rusk still wore an engineering coverall marked by the blind maneuver. “How many corrections did you certify after that review?”

“Seventy-two.”

“And how many settlements?”

“The archive cannot recover a complete count of routes. Conservative reconstruction exceeds eighteen million person-instances across the full program.”

“I asked for your certifications.”

“Seventy-two corrections. Each included multiple settlements.”

Aven nodded once and stepped back.

Teren explained the machinery when asked. He used the household comparison he had avoided during early hearings. The Loom made seven full meals, served one table, then burned six occupied rooms to heat the chosen plate. The comparison remained inadequate because rooms did not plead, build schools, or alter their own ration rules. He stated its failure before anyone else had to.

A child from Indigo submitted a written question through her delegation: WHY DID ADULTS CALL PEOPLE ENERGY?

Teren read the line twice.

“We did not say people,” he answered. “That omission allowed us to call what remained energy. Alda Quill recorded a child and a second bell. Later reports removed the child, then the bell, then the clock. By the time I learned the system, equations contained only reserve.”

Nessa's voice entered from off frame. “That is how. Why?”

Because arrival mattered. Because teachers carried authority. Because millions remained invisible while one population stood near enough to count. Because Teren had mistaken explanation for innocence.

“We wanted the ship we could see to survive,” he said. “We accepted a category that made harming the others sound impossible. I accepted it because changing the category would have made my work criminal and our safety uncertain.”

“What happens to you?” Nessa asked.

Oru intervened gently. “Consequences remain under review.”

Teren looked toward Indigo's frame. “I lose unilateral Loom access. I continue under supervised repair while your ships decide further responsibility. I do not get to pay once and call the account closed.”

The child considered that. “Good.”

No one laughed.

Glass's delegate requested the next question. “Why do you remain on the removal crew?”

The answer seemed obvious to Teren, which warned him to examine it. “I know the architecture.”

“So did the people who concealed it.”

“Yes.”

“Expertise created your authority. Your failures now make the same expertise your defense against removal.”

Murmurs moved across the local terraces. Teren looked toward Jo. They gave him no rescue.

“I should not supervise myself,” he said. “Engineering controls physical work. Commons witnesses custody. Remote observers verify every test. My role continues because specific knowledge can reduce immediate risk, not because knowledge grants me ownership.”

“How long does that exception last?”

He had no date prepared. The repair plan assumed his participation through the first full orbital cycle, then treated later access as an administrative question.

“It should end when others can perform the work without me,” he said.

“That answer lets you decide when others know enough.”

“Yes. It does.” Teren opened the training schedule on his slate. Twelve physicists across four ships had begun reviewing Loom architecture. Most carried fragments. None yet understood the entire causal stack, a concentration once celebrated as senior mastery. “Set the condition externally. Require three independent crews to pass an adversarial inspection designed by the ships that suffered settlement. I may teach and answer. I do

not grade.”

Reed's delegate asked, “Why three?”

“Two can reproduce one shared error.”

“Three can reproduce it too.”

“They can. Add public test design and minority reports.”

Oru entered the proposal as a recommendation, not a decision. Glass reserved its approval until local review. Teren felt the difference between contributing an answer and having it accepted.

A man in the Common Route terrace stood when Oru recognized him. He introduced himself as Pell An, pressure maintenance, older brother to one of the technicians lost with Eda Rook six years earlier. His grief did not come from settlement, yet he had lived aboard the ship whose survival Teren's work protected.

“If you had exposed this forty years ago,” Pell said, “would Nacre have reached Pelagos?”

“I cannot know.”

“Would millions be alive?”

“Many routes would not have been instantiated. Some later people might never have existed in any form. We cannot compute a clean replacement history.”

Pell's jaw tightened. “That sounds like another assumption fence.”

“It can become one. The fact I can state without model work concerns completed harm: settlement killed conscious people. Counterfactual rescue numbers should not reduce that fact or enlarge it for rhetoric.”

“Then what good is your calculation?”

“It tells us which future mechanisms carry the same danger. It cannot sentence the past into a total we find emotionally adequate.”

Pell sat without forgiving him.

Oru read the preliminary restrictions into the record. Teren would hold no solo credential, enter no causal service space without two named witnesses, publish every architecture note to replicated custody, and recuse himself from final authorization on the training-return replacement he had proposed. Further penalties remained open under each ship's legal review.

The terms sounded small against eighteen million. Making them sound larger would only repeat his habit of using scale to hide mechanism.

Sava Or testified after him about command archives and suppressed evidence. Jo described physical settlement paths. Ilyan entered the wake ledger, Sable's museum checksum, and the blank-course signatures. The hearing did not produce a verdict by midday. It produced custody: facts carried by seven archives, questions assigned dates, authority divided before anyone could seal the room.

When Teren left the Commons well, two former students waited on the upper path. One turned away. The other handed him a list of corrections from her own tenure and asked where to begin checking them.

“Start with every harmless noun,” he said.

She did not thank him.

Five settlement couplers lay on the causal-gallery floor by evening.

Jo had arranged them on insulated mats rather than stacking them. Each retained its archive tag. The open bus ended in capped collars that could not accept a field connection without visible fabrication work. Teren knelt at the command bridge and removed the privilege selector's physical key barrel.

The barrel fit inside his palm. Captains had turned versions of it before each correction, authorizing the Loom to select one receiver after predictive review. The shape looked disappointingly ordinary: seven pins, a worn grip, one serial number.

Sava watched from outside the witness line. She had surrendered the key that morning.

“Do you want it archived with the mission seal?” Teren asked.

“No,” she said. “Archive it with the couplers. It belongs to the mechanism, not to command.”

He tagged it accordingly.

Jo opened the replacement governance panel. Seven empty authorization fields crossed its surface. One credential belonged to each ship. None could activate settlement because the command paths no longer existed. The keys governed changes to seam topology, high-energy prediction modes, and any future attempt to attach new field hardware.

“State the rule,” the Commons witness said.

Teren faced the recorder. “Seven-key authorization requires one current public credential from every member ship. A key can approve inspection or bounded operation. It cannot delegate its vote to another ship. Any missing key leaves the requested mode cold.”

“What if one ship loses communication?”

“Existing prediction and scheduled seams continue. No structural change proceeds.”

“What if delay makes unanimity dangerous?”

“Emergency ordinary navigation remains under each local captain. The Loom may advise without instantiating routes.”

“What if all seven authorize a new settlement bus?”

Jo answered before Teren could. “Then they still need to fabricate it while the rest of us watch.”

The witness recorded both replies.

Credentials arrived in rotating order. North first, then Reed, Glass, Indigo, Violet, Amber, and Common Route. Each key opened one field and no more. When the seventh mark settled, the panel enabled a low-energy prediction test.

Teren loaded a known debris pattern. The Loom modeled six possible ordinary burns as mathematics only. No route glass brightened. No secondary clocks appeared. Reserve draw stayed inside the prediction braid.

“Test one clean,” Jo said.

“Run adverse input.”

They gave it sensor contradiction, missing mass, and an impossible capture demand. The old stack would have answered uncertainty by widening route instantiation. The revised stack returned INSUFFICIENT BASIS and listed the data it lacked.

Jo tapped the message. “A machine admitting ignorance. Keep that one.”

Teren ran seam continuity. Violet's scheduled bell crossed first. Indigo answered from Lattice's far capture band. Reed sent a timing tone from its Trojan approach. The links consumed reserve on both sides and moved no matter. Their limits held.

The final test requested settlement.

The panel returned MODE ABSENT.

Not disabled. Not awaiting authority. Absent.

Teren removed his hands from the console.

An alarm opened before anyone could sign the test report.

Amber's exterior watch had detected a dark rubble sheet crossing Common Route's transfer path. The fragments carried little metal and reflected poorly. At current heading, the ship would enter the sheet in nine hours. Shielding could tolerate the smaller grains, but three bodies near the center exceeded the repaired forward plates.

Ilyan sent the observation to the causal gallery. “We need a lateral change of eighty meters per second before second watch.”

The number would once have opened a correction protocol.

Sava arrived at the command rail. “Prediction options?”

Teren loaded Amber's data. The Loom calculated ordinary thrust windows without creating domains. Six curves appeared as forecasts, each labeled by assumptions and sensor confidence. The safest required a twenty-three-minute drive burn. It would lengthen Common Route's transfer by eleven days and consume four percent of maneuver fuel.

Jo read the curves. “Drive can give twenty-three minutes. Collar Three needs inspection after.”

Mina joined through Commons audio. “Rations absorb eleven days. Fresh beds do not enjoy it.”

“Medical?” Sava asked.

“Stable if cooling holds.”

Oru confirmed that local mission law left ordinary hazard avoidance with the captain, subject to public record.

No seven-ship vote applied. Amber had supplied warning, not command.

Sava looked at Teren. "Could better prediction reduce the burn?"

He understood the question beneath it. Could the removed machinery purchase comfort without anyone naming the price?

"Better sensors might," he said. "Route instantiation would offer more outcomes by making people live them. That path no longer exists."

"Then we burn."

The command went to drive.

Common Route turned by ordinary thrust. Pumps changed pitch through the hull. Acceleration pressed Teren's feet into the gallery floor, gentle compared with causal shear and impossible to mistake for theory. The maneuver bell struck once at load.

He waited.

No second bell answered through an unlisted seam. The scheduled route clocks continued at their own intervals. Violet sent capture weather. Indigo reported a clean far orbit. Amber refined the rubble edge. Nothing settled so Common Route could move.

On the prediction glass, the ship's measured line shifted away from the dark sheet. The five removed couplers remained on their mats. Seven authorization marks watched over a Loom that could warn, calculate, and connect but could no longer choose a population to burn.

Teren opened the navigation record. Its template offered CORRECTION 142.

He changed the heading to ORDINARY THRUST ADJUSTMENT and attached Amber's warning, Jo's drive report, the public authority note, and the fuel cost. In the field once reserved for settled-route count, he entered NO ABANDONED ROUTE.

The burn ended. One bell sounded above the gallery, complete without an answer.

Epilogue — A Map That Does Not Close

<!-- POV: Nessa Quill -->

<!-- SETUP: ordinary route records; living convoy; plural harbor map -->

<!-- PAYOFF: Museum founded on Lattice; delayed bell received as a neighbor's message -->

Nessa hung the blank edge of the map first.

The paper had come down from Indigo in a pressure tube and still curled toward its old shape. She fixed its left corners to the museum wall, smoothed the weather layer across the middle, and left the right side unsecured. Morning air from the open shutters lifted it with every salt gust.

“You missed two pins,” said the school curator.

“They are not missing.”

“Then why are they in your mouth?”

Nessa took the pins from between her lips. At fourteen, she had acquired strong views about questions with visible answers. “Amber sends its revised transfer line tomorrow. Reed may move its fabrication orbit. If I pin the edge, we will tear it taking the map down.”

The curator inspected the empty margin. “Leave enough room.”

“That depends on what they say.”

Three years had passed since Indigo reached Lattice's far capture band. The first descent camp now occupied a basalt shelf above a gray-green sea. Storm grass held the paths in place. Rain collection cloth stretched between the low buildings. The museum stood nearest the landing field, built from local stone and two cargo shells that no longer needed to fly.

Its full name crossed the entrance beam in seven sets of lettering: Museum of Abandoned Routes.

Nessa had objected to abandoned when the committee chose it. The people had not abandoned their ships. Other people had made machinery abandon them. Oru Pell had argued that keeping the old word visible gave every guide a duty to explain its direction. Nessa still disliked it, which the committee recorded beside her vote.

Inside, no room attempted to contain eighteen million faces. One gallery held Sable's compressed museum. Another preserved Alda Quill's timing tables beside the maintenance instructions that had hidden them. Most displays carried ordinary records: a lunch debt settled with two dried pears, a recipe corrected in three hands, a pressure-work order with an insult written beneath the official diagnosis, a child's unfinished apology.

Living ships occupied the eastern room. Violet sent classroom attendance, council arguments, weather notes, and one weekly list of things nobody considered historic yet. North contributed ice-core catalogues from Thren. Glass and Reed transmitted fabrication mistakes from opposite Trojan clusters. Common Route had reached its later Lattice approach, then sent crews to build the high platform that earlier arrivals could not have shared safely. Amber remained in transfer again, carrying outer-system observations toward a new survey station.

The archive never used one master copy. Every convoy ship held checksums. Full records moved where reserve and local consent permitted. When versions differed, the museum showed the difference instead of choosing a clean ancestor.

Nessa climbed down from the stool. A class from the lower storm school waited outside with wet boots and assignment slates. Their teacher asked them to wipe the basalt mud before entering. They did, mostly.

One child stopped at the map. “Which line started first?”

“None of these,” Nessa said. “They start at different places.”

“But which ship came from the real one?”

The teacher looked embarrassed. Nessa did not. Children found old categories the way they found sharp scraps in construction dirt.

"All seven carried a continuous ship," she said. "Sable did too, until its clock stopped. The routes before Sable carried people as long as they lasted."

The child pointed to the open margin. "What goes there?"

"Messages we have not received."

"From who?"

"We will know when they sign them."

That answer satisfied nobody, which made it useful.

The class entered the ordinary-records gallery. Their assignment asked each child to choose one item without a famous name. One picked a Sable repair note about a leaking soup valve. Another chose a Violet attendance sheet because three pupils had drawn fish in the margin. The first child lingered over a wake recipe that ended halfway through the oven temperature.

"Did they finish eating it?" she asked.

"We don't know," Nessa said.

"Then why keep the recipe?"

"Because somebody expected to."

The child copied the unfinished line onto her slate. She left room beneath it without being told.

A tone sounded from the museum receiver. The class turned together. Scheduled civic traffic had ended an hour earlier, and emergency calls used a harsher pitch.

Nessa crossed to the small route glass set into the wall. Its seam indicator showed Violet. Orbital geometry had carried the ship behind Lattice during its transmission window, so the message had taken relays through Indigo's old platform and Common Route's high station.

The first bell arrived clipped by distance.

A pause followed.

Then a second tone answered, late enough for the children to hear two separate strikes.

The receiver opened Violet's packet: a new school map, signed by Lin's classroom and addressed to Nessa's. Its right margin contained three unfilled lines.

"Is it broken?" the child asked.

"No," Nessa said. "It came a long way."

She set Violet's map beside their own. The two blank edges lifted in the sea air without matching exactly.

No one closed them.